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# Lie Groups

— *EYNTKA* Series —

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# *Contents*

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|---------------------------------------------------------------------------------------------------|-----------|
| <b>Contents</b>                                                                                   | <b>3</b>  |
| <br>                                                                                              |           |
| <b>I Lie Groups and Their Lie Algebras</b>                                                        | <b>1</b>  |
| <br>                                                                                              |           |
| <b>1 Matrix Lie Groups and the Classical Groups</b>                                               | <b>3</b>  |
| 1.1 The General Linear Group as a Manifold . . . . .                                              | 4         |
| 1.2 The Classical Groups . . . . .                                                                | 7         |
| 1.3 The Matrix Exponential and Logarithm; One-Parameter Subgroups . . . . .                       | 15        |
| 1.4 The Lie Algebra of a Matrix Group . . . . .                                                   | 24        |
| 1.5 $SU(2)$ , $SO(3)$ , and the Double Cover $SU(2) \rightarrow SO(3)$ . . . . .                  | 32        |
| <br>                                                                                              |           |
| <b>2 Lie Groups and Their Lie Algebras</b>                                                        | <b>41</b> |
| 2.1 Lie Groups: Definition, Examples, and the Category of Lie Groups . . . . .                    | 42        |
| 2.2 Left-Invariant Vector Fields and the Lie Algebra . . . . .                                    | 47        |
| 2.3 One-Parameter Subgroups and the Exponential Map . . . . .                                     | 54        |
| 2.4 The Differential of a Homomorphism and the Naturality of the Exponential . . . . .            | 60        |
| 2.5 The Adjoint Representations $\text{Ad}$ and $\text{ad}$ ; the Bracket Intrinsically . . . . . | 67        |
| <br>                                                                                              |           |
| <b>3 The Lie Group–Lie Algebra Correspondence</b>                                                 | <b>75</b> |
| 3.1 Lie Subgroups and Lie Subalgebras . . . . .                                                   | 76        |
| 3.2 The Closed Subgroup Theorem . . . . .                                                         | 81        |
| 3.3 Homomorphisms and Lie’s Integration Theorems . . . . .                                        | 88        |

|           |                                                                                 |            |
|-----------|---------------------------------------------------------------------------------|------------|
| 3.4       | Simply-Connected Groups and Lie's Third Theorem . . . . .                       | 95         |
| 3.5       | The Baker-Campbell-Hausdorff Formula . . . . .                                  | 102        |
| <b>4</b>  | <b>Homogeneous Spaces, Actions, and Covering Groups</b>                         | <b>111</b> |
| 4.1       | Group Actions and the Quotient Manifold $G/H$ . . . . .                         | 112        |
| 4.2       | Transitive Actions and Homogeneous Spaces . . . . .                             | 120        |
| 4.3       | Examples: Spheres, Projective Spaces, Grassmannians, and Flag Manifolds . . . . | 125        |
| 4.4       | Covering Groups, the Fundamental Group, and Discrete Central Kernels . . . . .  | 131        |
| 4.5       | Fiber Bundles from Subgroups and the Isotropy Representation . . . . .          | 136        |
| <b>II</b> | <b>The Structure of Lie Algebras</b>                                            | <b>145</b> |
| <b>5</b>  | <b>Nilpotent and Solvable Lie Algebras</b>                                      | <b>147</b> |
| 5.1       | Ideals, Quotients, and the Derived Series . . . . .                             | 147        |
| 5.2       | The Lower Central Series and Nilpotency . . . . .                               | 155        |
| 5.3       | Engel's Theorem . . . . .                                                       | 160        |
| 5.4       | Lie's Theorem . . . . .                                                         | 166        |
| 5.5       | The Radical and the Nilradical . . . . .                                        | 172        |
|           | <b>Index</b>                                                                    | <b>179</b> |
|           | <b>Bibliography</b>                                                             | <b>183</b> |

## **Part I**

# **Lie Groups and Their Lie Algebras**



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# *Matrix Lie Groups and the Classical Groups*

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In the middle of the nineteenth century Galois had shown that the solvability of a polynomial equation is controlled by a finite group of symmetries: the permutations of the roots preserving every algebraic relation among them. Sophus Lie set out to build the analogue for differential equations, where the natural symmetries form not a finite set but a continuous family, the rotations of the plane, the translations along a flow, the scalings carrying one solution of an equation to another. The object governing such a family is a group that is at the same time a smooth manifold, so that one may differentiate along it, and the interplay between the group law and the differentiable structure is the source of the theory's power. Lie's own goal, a theory of integration by continuous symmetry parallel to Galois's theory of solvability, proved harder to carry through; but the objects he introduced organize a large part of modern mathematics, from the harmonic analysis of functions on a symmetric space, through the gauge fields of physics and the automorphic forms at the center of number theory, to the classification of the algebraic groups.

This book approaches Lie theory from its analytic and differential-geometric side: a Lie group is a manifold carrying a compatible group law, and the aim is to understand such groups, their infinitesimal counterparts (the Lie algebras), and their representations. The parallel algebraic development, in which the same groups are studied as varieties over an arbitrary field, is the subject of the companion volume [3]; the two meet at the theory of root systems in Part II.

The present chapter takes the most concrete route in. Before a Lie group is defined in the generality of an abstract manifold, the task of the next chapter, the classical groups are met as what they are at the surface: sets of matrices cut out by explicit polynomial or analytic constraints. The general linear group, and inside it the orthogonal, unitary, and symplectic groups, are exhibited directly as smooth manifolds, and the two objects around which the whole theory turns, the matrix exponential and the Lie algebra, are produced by differentiating matrix-valued functions rather than by invoking

any abstract apparatus. The reward for this concreteness is that when the general theory arrives in the next chapter it finds a fully worked family of examples already in place. The chapter closes with the group  $SU(2)$  and its two-to-one relation with the rotation group of three-space, the example carried through the rest of the book.

## 1.1 The General Linear Group as a Manifold

The starting object of the whole subject is the set of all  $n \times n$  matrices over a field, and the invertible ones among them. Throughout this chapter  $\mathbb{F}$  denotes either the real field  $\mathbb{R}$  or the complex field  $\mathbb{C}$ , and  $M_n(\mathbb{F})$  denotes the set of  $n \times n$  matrices with entries in  $\mathbb{F}$ . Addition and scalar multiplication make  $M_n(\mathbb{F})$  an  $\mathbb{F}$ -vector space of dimension  $n^2$ , and recording a matrix by its  $n^2$  entries identifies it with  $\mathbb{F}^{n^2}$ ; matrix multiplication makes it moreover an associative algebra with unit the identity matrix  $I$ . The identification with  $\mathbb{F}^{n^2}$  equips  $M_n(\mathbb{F})$  with a topology and a smooth structure, those of a finite-dimensional vector space, and every construction below is made against this fixed background.

What singles out the invertible matrices is that they, and only they, admit a two-sided multiplicative inverse; they form a group. The point of this section is that this group is at the same time a smooth manifold, and that the manifold structure is not imposed by hand but inherited for free from the ambient space of all matrices. This is the first place where the two structures a Lie group must carry, that of a group and that of a manifold, sit together, and it is the ambient stage on which every other group in the chapter appears.

### Definition 1.1.1: General Linear Group

Let  $\mathbb{F}$  be  $\mathbb{R}$  or  $\mathbb{C}$ . The *general linear group* of rank  $n$  over  $\mathbb{F}$  is the group of invertible matrices

$$GL_n(\mathbb{F}) = \{A \in M_n(\mathbb{F}) \mid \det A \neq 0\}$$

under matrix multiplication, with identity element the identity matrix  $I$  and the inverse of  $A$  its matrix inverse  $A^{-1}$ .

That the condition  $\det A \neq 0$  characterizes invertibility is the content of the cofactor formula: a matrix has a two-sided inverse if and only if its determinant is a unit of  $\mathbb{F}$ , and every nonzero element of a field is a unit. The determinant is a polynomial in the  $n^2$  entries of  $A$ , so it is a continuous, indeed smooth, function  $\det: M_n(\mathbb{F}) \rightarrow \mathbb{F}$ . The set  $\mathbb{F} \setminus \{0\}$  is open in  $\mathbb{F}$ , and  $GL_n(\mathbb{F})$  is its preimage under this continuous map. An open subset of a smooth manifold is itself a smooth manifold of the same dimension, an *open submanifold*, with charts obtained by restricting the ambient charts (see [2]). The general linear group therefore acquires a manifold structure at no cost.

**Proposition 1.1.2: The General Linear Group is an Open Submanifold**

The group  $\mathrm{GL}_n(\mathbb{R})$  is an open subset of  $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$ , hence a smooth manifold of dimension  $n^2$ . The group  $\mathrm{GL}_n(\mathbb{C})$  is an open subset of  $M_n(\mathbb{C}) \cong \mathbb{R}^{2n^2}$ , hence a smooth manifold of real dimension  $2n^2$ . In each case the group operations

$$m: \mathrm{GL}_n(\mathbb{F}) \times \mathrm{GL}_n(\mathbb{F}) \rightarrow \mathrm{GL}_n(\mathbb{F}), \quad (A, B) \mapsto AB, \quad \iota: \mathrm{GL}_n(\mathbb{F}) \rightarrow \mathrm{GL}_n(\mathbb{F}), \quad A \mapsto A^{-1}$$

are smooth.

**Proof:**

Openness and the resulting manifold structure were established above:  $\mathrm{GL}_n(\mathbb{F}) = \det^{-1}(\mathbb{F} \setminus \{0\})$  is the preimage of an open set under the continuous map  $\det$ , hence open, and an open subset of the vector space  $M_n(\mathbb{F})$  is a smooth manifold of the same dimension. Over  $\mathbb{R}$  this dimension is  $n^2$ ; over  $\mathbb{C}$ , identifying each complex entry with its real and imaginary parts identifies  $M_n(\mathbb{C})$  with  $\mathbb{R}^{2n^2}$ , so the real dimension is  $2n^2$ .

For the multiplication, the  $(i, j)$  entry of  $AB$  is  $\sum_{k=1}^n A_{ik}B_{kj}$ , a polynomial in the entries of  $A$  and of  $B$ ; a map into  $\mathbb{F}^{n^2}$  whose every component is a polynomial in the coordinates is smooth, so  $m$  is smooth. For the inversion, Cramer's rule gives

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$$

where the *adjugate*  $\operatorname{adj}(A)$  has as its entries the  $(n-1) \times (n-1)$  signed minors of  $A$ , themselves polynomials in the entries of  $A$ . On  $\mathrm{GL}_n(\mathbb{F})$  the denominator  $\det A$  never vanishes, so each entry of  $A^{-1}$  is a ratio of a polynomial by a nowhere-zero polynomial, hence a smooth function of the entries of  $A$ . Over  $\mathbb{C}$  the same formulas exhibit the two maps as smooth in the real and imaginary parts of the entries. Thus  $m$  and  $\iota$  are smooth, completing the proof.

A manifold that is also a group whose multiplication and inversion are smooth is precisely what the next chapter will call a *Lie group*, and proposition 1.1.2 is the statement that  $\mathrm{GL}_n(\mathbb{F})$  is one. Every group met in this chapter is a subgroup of some  $\mathrm{GL}_n(\mathbb{F})$ , and the general linear group is the ambient object inside which the classical groups will be cut out; for that reason its own geometry is worth settling first. The most basic geometric invariant, the number of connected pieces, already distinguishes the real from the complex case.

**Proposition 1.1.3: Connected Components of the General Linear Group**

The group  $\mathrm{GL}_n(\mathbb{C})$  is path-connected. The group  $\mathrm{GL}_n(\mathbb{R})$  has exactly two connected components, namely

$$\mathrm{GL}_n^+(\mathbb{R}) = \{A \in \mathrm{GL}_n(\mathbb{R}) \mid \det A > 0\}, \quad \mathrm{GL}_n^-(\mathbb{R}) = \{A \in \mathrm{GL}_n(\mathbb{R}) \mid \det A < 0\},$$

each of which is open and path-connected. The component  $\mathrm{GL}_n^+(\mathbb{R})$  is a normal subgroup of index two.

**Proof:**

The argument rests on one fact from linear algebra: every matrix of determinant 1 is a product of *transvections*, the matrices  $I + \lambda E_{ij}$  with  $i \neq j$  and  $\lambda \in \mathbb{F}$ , where  $E_{ij}$  is the matrix with a single 1 in position  $(i, j)$  and zeros elsewhere. This is the statement that Gaussian elimination reduces a determinant-one matrix to the identity using only the row operation that adds a multiple of one row to another. Given any  $A \in \mathrm{GL}_n(\mathbb{F})$ , set  $D = \mathrm{diag}(\det A, 1, \dots, 1)$ ; then  $D^{-1}A$  has determinant 1, so

$$A = D \cdot S, \quad D = \mathrm{diag}(\det A, 1, \dots, 1), \quad S \in \mathrm{SL}_n(\mathbb{F}) \quad (1.1)$$

with  $S = D^{-1}A$  a product of transvections.

*Step 1: each factor is joined to the identity by a path.* A single transvection  $I + \lambda E_{ij}$  is joined to  $I$  by  $t \mapsto I + t\lambda E_{ij}$  for  $t \in [0, 1]$ , a path lying in  $\mathrm{GL}_n(\mathbb{F})$  because each of these matrices has determinant 1. A product of transvections is therefore joined to  $I$  by the pointwise product of the corresponding paths, which again lies in  $\mathrm{GL}_n(\mathbb{F})$ ; so every  $S \in \mathrm{SL}_n(\mathbb{F})$  is path-connected to  $I$  within  $\mathrm{SL}_n(\mathbb{F})$ . The diagonal factor  $D = \mathrm{diag}(d, 1, \dots, 1)$  with  $d = \det A$  is handled by moving  $d$ : for any path  $d: [0, 1] \rightarrow \mathbb{F} \setminus \{0\}$  from  $d(0) = d$  to  $d(1) = 1$ , the matrices  $\mathrm{diag}(d(t), 1, \dots, 1)$  form a path in  $\mathrm{GL}_n(\mathbb{F})$  from  $D$  to  $I$ .

*Step 2: the complex case.* When  $\mathbb{F} = \mathbb{C}$  the punctured plane  $\mathbb{C} \setminus \{0\}$  is path-connected, so a path from  $d = \det A$  to 1 as required in Step 1 always exists. Concatenating the path from  $A = DS$  to  $S$  (moving  $D$  to  $I$ ) with the path from  $S$  to  $I$  shows that every  $A \in \mathrm{GL}_n(\mathbb{C})$  is path-connected to  $I$ . Hence  $\mathrm{GL}_n(\mathbb{C})$  is path-connected.

*Step 3: the real case splits according to the sign of the determinant.* When  $\mathbb{F} = \mathbb{R}$  the set  $\mathbb{R} \setminus \{0\}$  is the disjoint union of the two open rays  $\mathbb{R}_{>0}$  and  $\mathbb{R}_{<0}$ , and  $\det: \mathrm{GL}_n(\mathbb{R}) \rightarrow \mathbb{R} \setminus \{0\}$  is continuous, so the two preimages  $\mathrm{GL}_n^+(\mathbb{R})$  and  $\mathrm{GL}_n^-(\mathbb{R})$  are disjoint, nonempty, open, and cover  $\mathrm{GL}_n(\mathbb{R})$ . No connected set meets both, since  $\det$  would then take a value of each sign and, being continuous, the value 0 in between, which is impossible on  $\mathrm{GL}_n(\mathbb{R})$ . It remains to see that each piece is path-connected. If  $\det A > 0$ , then in (1.1) the scalar  $d = \det A$  lies in  $\mathbb{R}_{>0}$ , which is path-connected, so the path of Step 1 from  $D$  to  $I$  may be taken with  $d(t) > 0$  throughout; the concatenated path from  $A$  to  $I$  then satisfies  $\det(\mathrm{diag}(d(t), 1, \dots, 1)S(t)) = d(t) > 0$  at every time, so it stays in  $\mathrm{GL}_n^+(\mathbb{R})$ . Thus  $\mathrm{GL}_n^+(\mathbb{R})$  is path-connected. For  $\mathrm{GL}_n^-(\mathbb{R})$ , fix the reflection  $J = \mathrm{diag}(-1, 1, \dots, 1)$ , which has determinant  $-1$ . Given  $A$  with  $\det A < 0$ , the matrix  $J^{-1}A$  has determinant  $> 0$ , hence is joined to  $I$  by a path  $\gamma$  in  $\mathrm{GL}_n^+(\mathbb{R})$ ; then  $t \mapsto J\gamma(t)$  joins  $J$  to  $A$  and has determinant  $-\det \gamma(t) < 0$  throughout, so it lies in  $\mathrm{GL}_n^-(\mathbb{R})$ . Hence  $\mathrm{GL}_n^-(\mathbb{R})$  is path-connected as well.

Finally,  $\mathrm{GL}_n^+(\mathbb{R})$  is the kernel of the composite homomorphism  $\mathrm{GL}_n(\mathbb{R}) \xrightarrow{\det} \mathbb{R} \setminus \{0\} \xrightarrow{\mathrm{sgn}} \{\pm 1\}$ , hence a normal subgroup; its two cosets are  $\mathrm{GL}_n^+(\mathbb{R})$  and  $\mathrm{GL}_n^-(\mathbb{R}) = J \cdot \mathrm{GL}_n^+(\mathbb{R})$ , so it has index two. This exhibits the two path-connected pieces as the two components, completing the proof.

The disconnectedness of  $\mathrm{GL}_n(\mathbb{R})$  is the matrix expression of orientation: the sign of the determinant records whether a linear automorphism of  $\mathbb{R}^n$  preserves or reverses orientation, and no continuous deformation through invertible maps can turn one into the other without passing through a singular matrix. Over  $\mathbb{C}$  this obstruction dissolves, because a single complex scalar can be routed around the origin; the punctured line  $\mathbb{C} \setminus \{0\}$  is connected where  $\mathbb{R} \setminus \{0\}$  is not. The same split will reappear for the orthogonal group in the next section, where  $\mathrm{SO}(n)$  plays the role of the identity component.

**Example 1.1.4: The Groups  $\mathrm{GL}_1$  and  $\mathrm{GL}_2(\mathbb{R})$** 

For  $n = 1$  a matrix is a scalar and invertibility means nonvanishing, so  $\mathrm{GL}_1(\mathbb{F}) = \mathbb{F}^\times$ , the multiplicative group of the field. Here proposition 1.1.3 recovers a familiar picture:  $\mathrm{GL}_1(\mathbb{R}) = \mathbb{R}^\times$  is the disjoint union of the two half-lines  $\mathbb{R}_{>0}$  and  $\mathbb{R}_{<0}$ , while  $\mathrm{GL}_1(\mathbb{C}) = \mathbb{C}^\times$  is the connected punctured plane.

For  $n = 2$  the group  $\mathrm{GL}_2(\mathbb{R})$  is a four-dimensional manifold, the open subset of  $M_2(\mathbb{R}) \cong \mathbb{R}^4$  where  $ad - bc \neq 0$  for the matrix  $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ . Its identity component  $\mathrm{GL}_2^+(\mathbb{R})$  contains every rotation  $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ , of determinant 1, and every positive scaling. A reflection such as  $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ , of determinant  $-1$ , lies in the other component  $\mathrm{GL}_2^-(\mathbb{R})$  and cannot be connected to the identity without its determinant passing through zero.

Two features of  $\mathrm{GL}_n(\mathbb{F})$  separate it from the groups studied in the rest of the chapter and are worth recording now. It is *noncompact*: it contains the matrices  $\mathrm{diag}(t, 1, \dots, 1)$  for all  $t \neq 0$ , an unbounded family, so  $\mathrm{GL}_n(\mathbb{F})$  is an unbounded subset of  $M_n(\mathbb{F}) \cong \mathbb{F}^{n^2}$  and cannot be compact. It is also *dense* in  $M_n(\mathbb{F})$ : any matrix can be perturbed to an invertible one, since the determinant of  $A - tI$  is a nonzero polynomial in  $t$  and so vanishes for only finitely many  $t$ . The classical groups introduced next trade this openness for the opposite geometry: cut out by equations rather than inequalities, several of them are compact, and it is compactness that will make their representation theory in Part III so tractable.

**Exercise 1.1.1**

1. Show that  $\mathrm{GL}_n(\mathbb{F})$  is dense in  $M_n(\mathbb{F})$ : given any  $A \in M_n(\mathbb{F})$ , prove that  $A - tI$  is invertible for all but finitely many  $t \in \mathbb{F}$ , and deduce that every matrix is a limit of invertible ones.
2. Prove that  $\mathrm{GL}_n(\mathbb{F})$  is not compact, in two ways: by exhibiting an unbounded sequence in it, and by showing it is a nonempty proper open subset of the connected space  $M_n(\mathbb{F})$  and hence not closed.
3. Determine the center of  $\mathrm{GL}_n(\mathbb{F})$ . Show that a matrix commuting with every element of  $\mathrm{GL}_n(\mathbb{F})$  (equivalently, with every transvection  $I + E_{ij}$ ) is a scalar multiple  $\lambda I$  of the identity with  $\lambda \in \mathbb{F}^\times$ .
4. Prove that  $\det: \mathrm{GL}_n(\mathbb{F}) \rightarrow \mathbb{F}^\times$  is a surjective group homomorphism with kernel  $\mathrm{SL}_n(\mathbb{F})$ , and conclude that  $\mathrm{SL}_n(\mathbb{F})$  is a normal subgroup with  $\mathrm{GL}_n(\mathbb{F})/\mathrm{SL}_n(\mathbb{F}) \cong \mathbb{F}^\times$ .
5. Using proposition 1.1.3, show that  $A \mapsto \mathrm{sgn}(\det A)$  induces an isomorphism  $\mathrm{GL}_n(\mathbb{R})/\mathrm{GL}_n^+(\mathbb{R}) \cong \{\pm 1\}$ , and that  $\mathrm{GL}_n^+(\mathbb{R})$  is exactly the set of products of transvections and positive scalings.

## 1.2 The Classical Groups

The general linear group is cut out of the space of all matrices by an *inequality*,  $\det A \neq 0$ , which is why it is open. The groups of present interest are cut out instead by *equations*: they are the subgroups of  $\mathrm{GL}_n$  preserving some piece of structure, a determinant normalization, a symmetric or Hermitian inner product, or a skew form. Preserving structure is a closed condition, so these subgroups are closed rather than open, and several of them are compact. Each is presented here as the solution set of a system of polynomial (or, in the complex case, real-analytic) equations, and the

uniform tool that converts such a solution set into a smooth manifold is the regular value theorem. Recall the statement in the form used throughout: if  $F: M \rightarrow N$  is a smooth map of manifolds and  $c \in N$  is a *regular value*, meaning that the differential  $DF_p$  is surjective at every  $p \in F^{-1}(c)$ , then  $F^{-1}(c)$  is an embedded submanifold of  $M$  of dimension  $\dim M - \dim N$ , with tangent space  $T_p F^{-1}(c) = \ker DF_p$  at each of its points (see [2]). For each classical group the recipe is the same: write the group as  $F^{-1}(c)$  for an explicit  $F$  into a vector space of “constraint values”, check surjectivity of  $DF$  along the group by a single substitution, and read off the dimension. The closed-subgroup theorem of chapter 1 will later subsume all of these computations under one statement, that every closed subgroup of a Lie group is a Lie subgroup; the point of doing them by hand now is to make the manifold structure, and above all the dimension, of each principal example explicit and concrete.

### Definition 1.2.1: Special Linear Group

The *special linear group* is the kernel of the determinant,

$$\mathrm{SL}_n(\mathbb{F}) = \{A \in \mathrm{GL}_n(\mathbb{F}) \mid \det A = 1\}$$

the group of matrices of determinant 1.

The determinant is the simplest structure a matrix can preserve, namely volume with orientation, and the special linear group is the first application of the regular value recipe.

### Proposition 1.2.2: The Special Linear Group is a Submanifold

$\mathrm{SL}_n(\mathbb{R})$  is a closed embedded submanifold of  $\mathrm{GL}_n(\mathbb{R})$  of dimension  $n^2 - 1$ , and  $\mathrm{SL}_n(\mathbb{C})$  is a closed embedded submanifold of  $\mathrm{GL}_n(\mathbb{C})$  of real dimension  $2(n^2 - 1)$  (complex dimension  $n^2 - 1$ ).

#### **Proof:**

The determinant restricts to a smooth map  $\det: \mathrm{GL}_n(\mathbb{F}) \rightarrow \mathbb{F}^\times$ , and  $\mathrm{SL}_n(\mathbb{F}) = \det^{-1}(1)$ . The derivative of the determinant at  $A$  is computed from the expansion

$$\det(A + tX) = \det(A) \det(I + tA^{-1}X) = \det(A)(1 + t\mathrm{tr}(A^{-1}X) + O(t^2))$$

valid for  $A$  invertible, where the middle equality factors out  $A$  and the last uses  $\det(I + tB) = 1 + t\mathrm{tr}(B) + O(t^2)$ . Differentiating at  $t = 0$  gives Jacobi’s formula

$$D \det_A(X) = \det(A) \mathrm{tr}(A^{-1}X)$$

As  $X$  ranges over  $M_n(\mathbb{F})$  the quantity  $\mathrm{tr}(A^{-1}X)$  ranges over all of  $\mathbb{F}$ , for instance  $X = AE_{11}$  gives  $\mathrm{tr}(E_{11}) = 1$ ; hence  $D \det_A$  is surjective onto  $\mathbb{F}$  at every  $A \in \mathrm{GL}_n(\mathbb{F})$ , so every value, in particular 1, is regular. Therefore  $\mathrm{SL}_n(\mathbb{F}) = \det^{-1}(1)$  is an embedded submanifold, closed because  $\{1\}$  is closed, of dimension  $\dim \mathrm{GL}_n(\mathbb{F}) - \dim \mathbb{F}$ . Over  $\mathbb{R}$  this is  $n^2 - 1$ ; over  $\mathbb{C}$ , where  $\det$  is a map of real manifolds onto  $\mathbb{C} \cong \mathbb{R}^2$ , the real codimension is 2, giving real dimension  $2n^2 - 2$ , completing the proof.

The next three groups preserve an inner product. Over  $\mathbb{R}$  the relevant structure is a symmetric bilinear form; over  $\mathbb{C}$ , a Hermitian form; and the groups preserving them are the orthogonal and unitary

groups. These are the compact classical groups, and the mechanism producing their compactness, that an isometry sends an orthonormal basis to an orthonormal one and so has bounded entries, is visible already in the proofs.

### Definition 1.2.3: Orthogonal and Special Orthogonal Groups

The *orthogonal group* is the group of real matrices preserving the standard inner product on  $\mathbb{R}^n$ ,

$$O(n) = \{A \in GL_n(\mathbb{R}) \mid A^\top A = I\}$$

equivalently the matrices whose columns form an orthonormal basis. The *special orthogonal group* is the subgroup of determinant 1,

$$SO(n) = O(n) \cap SL_n(\mathbb{R}) = \{A \in O(n) \mid \det A = 1\}$$

the group of rotations of  $\mathbb{R}^n$ .

The relation  $A^\top A = I$  forces  $(\det A)^2 = \det(A^\top) \det(A) = 1$ , so an orthogonal matrix has determinant  $\pm 1$  and the condition  $\det A = 1$  picks out one half. The orthogonal group is the prototype for the regular value computation, carried out here in full detail.

### Proposition 1.2.4: The Orthogonal Group is a Compact Submanifold of Dimension $\binom{n}{2}$

$O(n)$  is a compact embedded submanifold of  $M_n(\mathbb{R})$  of dimension  $\binom{n}{2} = \frac{n(n-1)}{2}$ . It is disconnected: the determinant takes the values  $\pm 1$  on it, and the two clopen pieces  $\det^{-1}(1) = SO(n)$  and  $\det^{-1}(-1)$  are each a compact submanifold of the same dimension.

#### Proof:

Let  $\text{Sym}_n(\mathbb{R}) = \{S \in M_n(\mathbb{R}) \mid S^\top = S\}$  be the space of symmetric matrices, a real vector space of dimension  $\frac{n(n+1)}{2}$ , and define

$$F: M_n(\mathbb{R}) \rightarrow \text{Sym}_n(\mathbb{R}), \quad F(A) = A^\top A$$

which lands in  $\text{Sym}_n(\mathbb{R})$  because  $(A^\top A)^\top = A^\top A$ , and is smooth because its entries are polynomials. By construction  $O(n) = F^{-1}(I)$ .

*Step 1: I is a regular value.* Fix  $A \in O(n)$ . The differential of  $F$  at  $A$  is

$$DF_A(X) = X^\top A + A^\top X$$

as one sees by expanding  $F(A + tX) = A^\top A + t(X^\top A + A^\top X) + t^2 X^\top X$  and differentiating at  $t = 0$ . To see that  $DF_A$  is onto  $\text{Sym}_n(\mathbb{R})$ , substitute  $X = AZ$  and use  $A^\top A = I$ :

$$DF_A(AZ) = (AZ)^\top A + A^\top (AZ) = Z^\top A^\top A + A^\top AZ = Z^\top + Z$$

Given any symmetric  $S$ , taking  $Z = \frac{1}{2}S$  yields  $Z^\top + Z = S$ ; since  $X \mapsto Z = A^{-1}X$  is a bijection of  $M_n(\mathbb{R})$ , the map  $DF_A$  is surjective. Hence  $I$  is a regular value.

*Step 2: the manifold structure and dimension.* By the regular value theorem  $O(n) = F^{-1}(I)$  is

an embedded submanifold of  $M_n(\mathbb{R})$  of dimension

$$\dim M_n(\mathbb{R}) - \dim \text{Sym}_n(\mathbb{R}) = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2} = \binom{n}{2}$$

*Step 3: compactness.* The set  $O(n)$  is closed in  $M_n(\mathbb{R})$ , being  $F^{-1}(I)$  for continuous  $F$ . It is bounded: if  $A^T A = I$  then the diagonal entries of  $A^T A$  give  $\sum_i a_{ij}^2 = 1$  for each column  $j$ , so the Frobenius norm satisfies  $\|A\|^2 = \sum_{i,j} a_{ij}^2 = n$ . A closed bounded subset of  $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$  is compact by the Heine–Borel theorem.

*Step 4: the two pieces.* The determinant is continuous on  $O(n)$  and takes only the values  $\pm 1$ , both attained ( $I$  has determinant 1 and  $\text{diag}(-1, 1, \dots, 1)$  has determinant  $-1$ ), so  $\det^{-1}(1) = \text{SO}(n)$  and  $\det^{-1}(-1)$  are disjoint, nonempty, and both open and closed in  $O(n)$ . Each is a submanifold of dimension  $\binom{n}{2}$ , being open in  $O(n)$ , and each is compact, being closed in the compact space  $O(n)$ . That  $\text{SO}(n)$  is connected, so that these two pieces are exactly the connected components, is proposition 1.2.10 below, completing the proof.

The complex analogue replaces the transpose by the conjugate transpose and the symmetric form by a Hermitian one. Write  $A^* = \overline{A}^T$  for the conjugate transpose.

#### Definition 1.2.5: Unitary and Special Unitary Groups

The *unitary group* is the group of complex matrices preserving the standard Hermitian inner product on  $\mathbb{C}^n$ ,

$$U(n) = \{A \in \text{GL}_n(\mathbb{C}) \mid A^* A = I\}$$

the matrices whose columns form a Hermitian-orthonormal basis. The *special unitary group* is the subgroup of determinant 1,

$$\text{SU}(n) = U(n) \cap \text{SL}_n(\mathbb{C}) = \{A \in U(n) \mid \det A = 1\}$$

The relation  $A^* A = I$  gives  $|\det A|^2 = \overline{\det A} \det A = \det(A^*) \det(A) = 1$ , so a unitary matrix has determinant on the unit circle, and  $\text{SU}(n)$  is the fibre of the determinant over  $1 \in S^1$ .

#### Proposition 1.2.6: The Unitary and Special Unitary Groups are Compact Submanifolds

$U(n)$  is a compact embedded submanifold of  $M_n(\mathbb{C}) \cong \mathbb{R}^{2n^2}$  of real dimension  $n^2$ . Its subgroup  $\text{SU}(n)$  is a compact embedded submanifold of real dimension  $n^2 - 1$ .

#### Proof:

Let  $\text{Herm}_n = \{H \in M_n(\mathbb{C}) \mid H^* = H\}$  be the space of Hermitian matrices. This is a *real* vector space (it is not closed under multiplication by  $i$ ) of real dimension  $n^2$ : a Hermitian matrix has  $n$  real diagonal entries and  $\binom{n}{2}$  free complex entries above the diagonal, contributing  $n + 2\binom{n}{2} = n + n(n-1) = n^2$  real parameters. Define

$$F: M_n(\mathbb{C}) \rightarrow \text{Herm}_n, \quad F(A) = A^* A$$

a smooth map (its entries are real-polynomial in the real and imaginary parts of the entries of

$A$ ) with  $U(n) = F^{-1}(I)$ .

*Step 1:  $I$  is a regular value.* For  $A \in U(n)$  the differential is  $DF_A(X) = X^*A + A^*X$ , and the substitution  $X = AZ$  with  $A^*A = I$  gives

$$DF_A(AZ) = (AZ)^*A + A^*(AZ) = Z^*A^*A + A^*AZ = Z^* + Z$$

Given Hermitian  $H$ , taking  $Z = \frac{1}{2}H$  gives  $Z^* + Z = H$ , so  $DF_A$  is surjective onto  $\text{Herm}_n$ ; hence  $I$  is a regular value and  $U(n)$  is an embedded submanifold of real dimension  $2n^2 - n^2 = n^2$ .

*Step 2: compactness.* As with the orthogonal group,  $U(n) = F^{-1}(I)$  is closed, and the columns of a unitary matrix are unit vectors, so  $\|A\|^2 = n$  and  $U(n)$  is bounded; it is therefore compact by Heine–Borel.

*Step 3: the special unitary group.* By the regular value theorem the tangent space at  $A \in U(n)$  is  $T_A U(n) = \ker DF_A = \{AZ \mid Z^* + Z = 0\}$ , that is  $A \cdot \mathfrak{u}(n)$  where  $\mathfrak{u}(n)$  is the space of skew-Hermitian matrices. Consider the smooth map  $\det: U(n) \rightarrow S^1$ . Its derivative at  $A$ , along a tangent vector  $X = AZ$  with  $Z$  skew-Hermitian, is by Jacobi's formula

$$D \det_A(AZ) = \det(A) \operatorname{tr}(A^{-1}AZ) = \det(A) \operatorname{tr}(Z)$$

A skew-Hermitian matrix has purely imaginary diagonal, so  $\operatorname{tr}(Z) \in i\mathbb{R}$ , and  $\operatorname{tr}(Z)$  takes every value in  $i\mathbb{R}$  (for example  $Z = \operatorname{diag}(it, 0, \dots, 0)$ ). Thus  $D \det_A$  maps  $T_A U(n)$  onto  $\det(A) \cdot i\mathbb{R}$ , which is exactly the tangent line to  $S^1$  at the point  $\det(A)$ . Hence  $\det: U(n) \rightarrow S^1$  is a submersion, 1 is a regular value, and  $SU(n) = \det^{-1}(1)$  is an embedded submanifold of  $U(n)$  of dimension  $n^2 - 1$ . It is closed in the compact group  $U(n)$ , hence compact, as we sought to show.

The last family preserves a skew-symmetric form. Its notation is a persistent source of confusion, so the two groups sharing the name “symplectic” are fixed side by side, together with a convention.

#### Definition 1.2.7: Symplectic Groups

Let  $J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix} \in M_{2n}(\mathbb{F})$ , the matrix of the standard skew form  $\omega(x, y) = x^T J y$  on  $\mathbb{F}^{2n}$ . The *symplectic group* is the group preserving  $\omega$ ,

$$\operatorname{Sp}(2n, \mathbb{F}) = \{A \in \operatorname{GL}_{2n}(\mathbb{F}) \mid A^T J A = J\}$$

The *compact symplectic group* is the group of quaternionic-unitary matrices,

$$\operatorname{Sp}(n) = \{A \in M_n(\mathbb{H}) \mid A^* A = I\}$$

where  $\mathbb{H}$  is the quaternions and  $A^* = \overline{A}^T$  is the quaternionic conjugate transpose.

**1.2.8 Convention: Two Meanings of “Symplectic Group”** The symbol  $\mathrm{Sp}(2n, \mathbb{F})$  denotes the (noncompact) group of  $2n \times 2n$  matrices over  $\mathbb{F}$  preserving the standard skew form; the symbol  $\mathrm{Sp}(n)$  denotes the (compact) group of  $n \times n$  quaternionic matrices preserving the standard quaternionic-Hermitian form. Both have dimension  $n(2n+1)$ , but they are different groups:  $\mathrm{Sp}(2n, \mathbb{R})$  is noncompact and  $\mathrm{Sp}(n)$  is compact. The compact group is a real form of the complex group  $\mathrm{Sp}(2n, \mathbb{C})$ ; concretely, under the standard identification  $\mathbb{H}^n \cong \mathbb{C}^{2n}$  one has  $\mathrm{Sp}(n) = \mathrm{U}(2n) \cap \mathrm{Sp}(2n, \mathbb{C})$ , the verification of which is a routine matter of unwinding the quaternionic structure, taken up where the compact group is needed in Part III. Throughout this book  $\mathrm{Sp}(2n, \mathbb{F})$  always carries an explicit field and even size  $2n$ ; the bare symbol  $\mathrm{Sp}(n)$  always means the compact group.

The noncompact group is another instance of the regular value recipe, with the skew-symmetric matrices as the space of constraint values.

**Proposition 1.2.9:** The Symplectic Group is a Submanifold of Dimension  $n(2n+1)$

$\mathrm{Sp}(2n, \mathbb{F})$  is a closed embedded submanifold of  $\mathrm{GL}_{2n}(\mathbb{F})$  of dimension  $n(2n+1)$  over  $\mathbb{R}$  (and complex dimension  $n(2n+1)$  over  $\mathbb{C}$ ). The compact group  $\mathrm{Sp}(n)$  is a compact subgroup of  $\mathrm{U}(2n)$ , hence compact, of dimension  $n(2n+1)$ .

**Proof:**

Let  $\mathrm{Skew}_{2n}(\mathbb{F}) = \{S \in M_{2n}(\mathbb{F}) \mid S^\top = -S\}$ , of dimension  $\binom{2n}{2} = n(2n-1)$ , and define  $F: M_{2n}(\mathbb{F}) \rightarrow \mathrm{Skew}_{2n}(\mathbb{F})$  by  $F(A) = A^\top J A$ ; this is skew-symmetric because  $J^\top = -J$ , and  $\mathrm{Sp}(2n, \mathbb{F}) = F^{-1}(J)$ . The differential at  $A$  is  $DF_A(X) = X^\top J A + A^\top J X$ , and for  $A \in \mathrm{Sp}(2n, \mathbb{F})$  the substitution  $X = AZ$  with  $A^\top J A = J$  gives

$$DF_A(AZ) = Z^\top A^\top J A + A^\top J A Z = Z^\top J + JZ$$

Given skew-symmetric  $S$ , set  $Z = -\frac{1}{2}JS$ . Using  $J^\top = -J$ ,  $J^2 = -I$ , and  $S^\top = -S$ , the transpose is  $Z^\top = -\frac{1}{2}S^\top J^\top = -\frac{1}{2}(-S)(-J) = -\frac{1}{2}SJ$ , so

$$Z^\top J + JZ = -\frac{1}{2}SJ^2 - \frac{1}{2}J^2S = \frac{1}{2}S + \frac{1}{2}S = S$$

Hence  $DF_A$  is surjective onto  $\mathrm{Skew}_{2n}(\mathbb{F})$ , the value  $J$  is regular, and  $\mathrm{Sp}(2n, \mathbb{F}) = F^{-1}(J)$  is a closed embedded submanifold of dimension

$$(2n)^2 - n(2n-1) = 4n^2 - 2n^2 + n = 2n^2 + n = n(2n+1)$$

For the compact group, the identification  $\mathbb{H}^n \cong \mathbb{C}^{2n}$  of box 1.2.8 realizes  $\mathrm{Sp}(n)$  as  $\mathrm{U}(2n) \cap \mathrm{Sp}(2n, \mathbb{C})$ , a closed subset of the compact group  $\mathrm{U}(2n)$ , hence compact; its dimension  $n(2n+1)$  is computed from its Lie algebra in chapter 1. This completes the proof.

It remains to settle the connectivity left open for  $\mathrm{SO}(n)$ ,  $\mathrm{U}(n)$ , and  $\mathrm{SU}(n)$ . Each is proved by the same idea: diagonalize, or bring to real normal form, and then contract the eigenvalues to 1 along a path that stays inside the group.

**Proposition 1.2.10: Connectivity of the Compact Classical Groups**

The groups  $U(n)$ ,  $SU(n)$ , and  $SO(n)$  are path-connected. Consequently  $O(n)$  has exactly two connected components,  $SO(n)$  and  $\det^{-1}(-1)$ .

**Proof:**

*Step 1: the unitary group.* A unitary matrix is normal, so by the spectral theorem it is unitarily diagonalizable:  $A = PDP^{-1}$  with  $P \in U(n)$  and  $D = \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n})$  for real  $\theta_1, \dots, \theta_n$ , the eigenvalues lying on the unit circle because  $A$  is unitary. The path  $D(s) = \text{diag}(e^{is\theta_1}, \dots, e^{is\theta_n})$  for  $s \in [0, 1]$  runs inside  $U(n)$  from  $D(1) = D$  to  $D(0) = I$ , and conjugating,  $s \mapsto PD(s)P^{-1}$  is a path in  $U(n)$  from  $A$  to  $I$ . Hence  $U(n)$  is path-connected.

*Step 2: the special unitary group.* For  $A \in SU(n)$  diagonalize as above; now  $\det A = \prod_j e^{i\theta_j} = 1$ , so  $\sum_j \theta_j \in 2\pi\mathbb{Z}$ , and subtracting a suitable integer multiple of  $2\pi$  from one of the angles (which does not change the corresponding eigenvalue) arranges  $\sum_j \theta_j = 0$  exactly. Then the same path  $D(s)$  satisfies  $\det D(s) = e^{is \sum_j \theta_j} = 1$  for all  $s$ , so it lies in  $SU(n)$ , and  $s \mapsto PD(s)P^{-1}$  joins  $A$  to  $I$  within  $SU(n)$ . Hence  $SU(n)$  is path-connected.

*Step 3: the special orthogonal group.* A matrix  $A \in SO(n)$  is real orthogonal, hence normal, and the real spectral theorem for orthogonal matrices provides  $Q \in SO(n)$  with  $Q^{-1}AQ$  block-diagonal, the blocks being  $2 \times 2$  rotations  $R(\theta_k) = \begin{pmatrix} \cos \theta_k & -\sin \theta_k \\ \sin \theta_k & \cos \theta_k \end{pmatrix}$  together with diagonal entries  $\pm 1$ ; the real eigenvalues  $\pm 1$  occur because they are the only real numbers on the unit circle, and the number of  $-1$  entries is even because  $\det A = 1$ , so those entries pair up into rotation blocks  $R(\pi) = \text{diag}(-1, -1)$ . Thus  $Q^{-1}AQ$  is a direct sum of rotation blocks  $R(\theta_1), \dots, R(\theta_m)$  and a  $+1$  identity block. Replacing each angle  $\theta_k$  by  $s\theta_k$  gives a path of block matrices in  $SO(n)$  from  $Q^{-1}AQ$  (at  $s = 1$ ) to  $I$  (at  $s = 0$ ), and conjugating by  $Q$  produces a path in  $SO(n)$  from  $A$  to  $I$ . Hence  $SO(n)$  is path-connected.

Since  $SO(n) = \det^{-1}(1)$  is path-connected and  $\det^{-1}(-1) = \text{diag}(-1, 1, \dots, 1) \cdot SO(n)$  is its homeomorphic image under left multiplication, both clopen pieces of  $O(n)$  from proposition 1.2.4 are connected, so they are precisely the two components of  $O(n)$ , completing the proof.

The dimensions and topological types of the classical groups are collected in fig. 1.1 for reference; the entries for  $O(n)$  and  $U(n)$  recur constantly in Part III, where the representation theory of the compact groups is developed. Every entry has been established above, with one exception: the connectedness of the two symplectic groups is recorded as a standard fact, to be proved from the polar decomposition of Part IV, where  $\text{Sp}(2n, \mathbb{R})$  and  $\text{Sp}(n)$  are analyzed alongside the other real forms.

| Group                         | Defining relation                | Field        | $\dim_{\mathbb{R}}$ | Topology              |
|-------------------------------|----------------------------------|--------------|---------------------|-----------------------|
| $\mathrm{GL}_n(\mathbb{R})$   | $\det \neq 0$                    | $\mathbb{R}$ | $n^2$               | open, 2 components    |
| $\mathrm{GL}_n(\mathbb{C})$   | $\det \neq 0$                    | $\mathbb{C}$ | $2n^2$              | open, connected       |
| $\mathrm{SL}_n(\mathbb{R})$   | $\det = 1$                       | $\mathbb{R}$ | $n^2 - 1$           | closed, connected     |
| $\mathrm{SL}_n(\mathbb{C})$   | $\det = 1$                       | $\mathbb{C}$ | $2(n^2 - 1)$        | closed, connected     |
| $\mathrm{O}(n)$               | $A^T A = I$                      | $\mathbb{R}$ | $\binom{n}{2}$      | compact, 2 components |
| $\mathrm{SO}(n)$              | $A^T A = I, \det = 1$            | $\mathbb{R}$ | $\binom{n}{2}$      | compact, connected    |
| $\mathrm{U}(n)$               | $A^* A = I$                      | $\mathbb{C}$ | $n^2$               | compact, connected    |
| $\mathrm{SU}(n)$              | $A^* A = I, \det = 1$            | $\mathbb{C}$ | $n^2 - 1$           | compact, connected    |
| $\mathrm{Sp}(2n, \mathbb{R})$ | $A^T J A = J$                    | $\mathbb{R}$ | $n(2n + 1)$         | closed, connected     |
| $\mathrm{Sp}(n)$              | $A^* A = I$ (over $\mathbb{H}$ ) | $\mathbb{H}$ | $n(2n + 1)$         | compact, connected    |

Figure 1.1: The classical groups, their defining relations, real dimensions, and topological types

The smallest orthogonal and unitary groups already exhibit the two shapes that recur throughout: a circle and its higher analogues.

**Example 1.2.11: The Circle Groups  $\mathrm{SO}(2)$  and  $\mathrm{U}(1)$**

The group  $\mathrm{SO}(2)$  consists of the plane rotations

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad \theta \in \mathbb{R}$$

and  $\theta \mapsto R(\theta)$  is a surjective homomorphism  $\mathbb{R} \rightarrow \mathrm{SO}(2)$  with kernel  $2\pi\mathbb{Z}$ , so  $\mathrm{SO}(2) \cong \mathbb{R}/2\pi\mathbb{Z} \cong S^1$ , a circle, of dimension  $\binom{2}{2} = 1$  as predicted. Likewise  $\mathrm{U}(1) = \{z \in \mathbb{C} \mid |z| = 1\} = S^1$  is the unit circle in  $\mathbb{C}$ , of real dimension 1, and  $z \mapsto \begin{pmatrix} \operatorname{Re} z & -\operatorname{Im} z \\ \operatorname{Im} z & \operatorname{Re} z \end{pmatrix}$  is an isomorphism  $\mathrm{U}(1) \xrightarrow{\sim} \mathrm{SO}(2)$  realizing the standard fact that multiplication by a unit complex number is a rotation of the plane. The next dimension up, the group  $\mathrm{SU}(2)$ , turns out to be a 3-sphere and is the subject of section 1.5.

**Exercise 1.2.1**

1. Verify directly that  $\mathrm{SO}(3)$  has dimension 3 and  $\mathrm{SU}(2)$  has dimension 3, and that  $\mathrm{O}(1) = \{\pm 1\}$  is zero-dimensional with two components, consistent with  $\binom{1}{2} = 0$ .
2. Show that  $\det: \mathrm{U}(n) \rightarrow S^1$  is a surjective group homomorphism with kernel  $\mathrm{SU}(n)$ , and deduce that  $\mathrm{U}(n)/\mathrm{SU}(n) \cong S^1$ .
3. Prove that the center of  $\mathrm{SU}(n)$  is the group of scalar matrices  $\zeta I$  with  $\zeta^n = 1$ , so that  $Z(\mathrm{SU}(n)) \cong \mu_n$ , the group of  $n$ th roots of unity.
4. Show that  $A \in \mathrm{Sp}(2n, \mathbb{R})$  if and only if  $A$  preserves the skew form  $\omega(x, y) = x^T J y$ , that is  $\omega(Ax, Ay) = \omega(x, y)$  for all  $x, y \in \mathbb{R}^{2n}$ , and conclude that  $\det A = 1$  for every symplectic matrix.
5. Prove that

$$\mathrm{SU}(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} \mid \alpha, \beta \in \mathbb{C}, |\alpha|^2 + |\beta|^2 = 1 \right\}$$

and deduce that  $\mathrm{SU}(2)$  is diffeomorphic to the unit sphere  $S^3 \subset \mathbb{C}^2 \cong \mathbb{R}^4$ .

### 1.3 The Matrix Exponential and Logarithm; One-Parameter Subgroups

The classical groups of the previous section were described statically, as solution sets of matrix equations. The subject acquires its dynamics from a single construction that turns a matrix into a motion: the exponential of a matrix  $X$  is a point of  $\mathrm{GL}_n(\mathbb{F})$ , and letting  $X$  vary linearly, through  $tX$  for  $t \in \mathbb{R}$ , sweeps out a curve through the identity whose velocity at time zero is  $X$  itself. This is the bridge between the linear world of  $M_n(\mathbb{F})$ , where one adds and takes brackets, and the curved world of the group, where one multiplies. Everything in the group theory of the following chapters is read off this bridge: the tangent space at the identity becomes a Lie algebra, and the exponential map carries its linear structure onto the group.

The construction copies the scalar power series  $e^x = \sum_{k \geq 0} x^k/k!$  verbatim, replacing the scalar  $x$  by a matrix  $X$  and scalar powers by matrix powers. The first task is to make sense of the resulting infinite sum of matrices, and the tool for that is the operator norm, whose submultiplicativity tames every convergence question in this section.

#### Definition 1.3.1: Operator Norm on $M_n(\mathbb{F})$

The *operator norm* of  $X \in M_n(\mathbb{F})$  is

$$\|X\| = \sup_{v \neq 0} \frac{\|Xv\|}{\|v\|} = \sup_{\|v\|=1} \|Xv\|$$

where  $\|v\|$  is the Euclidean norm on  $\mathbb{F}^n$ . It makes  $M_n(\mathbb{F})$  a normed vector space, and the supremum is finite and attained because the unit sphere of  $\mathbb{F}^n$  is compact and  $v \mapsto \|Xv\|$  is continuous.

Two properties of this norm carry the entire section. First, it dominates each matrix entry, so convergence in operator norm is the same as entrywise convergence and hence as convergence in  $M_n(\mathbb{F}) \cong \mathbb{F}^{n^2}$  with its usual topology; a sequence of matrices converges in operator norm if and only if each of its  $n^2$  entry-sequences converges. Second, and this is the property with no scalar analogue, the norm is *submultiplicative*:

$$\|XY\| \leq \|X\| \|Y\| \quad \text{for all } X, Y \in M_n(\mathbb{F})$$

This holds because  $\|XYv\| \leq \|X\| \|Yv\| \leq \|X\| \|Y\| \|v\|$  for every  $v$ ; iterating gives  $\|X^k\| \leq \|X\|^k$ , which is what forces the exponential series to converge as fast as its scalar shadow.

#### Definition 1.3.2: Matrix Exponential

For  $X \in M_n(\mathbb{F})$  the *matrix exponential* is the sum of the series

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!} = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots$$

where  $X^0 = I$  by convention. It defines a map  $\exp: M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$ .

For the definition to be legitimate the series must converge, and it must converge to a matrix depending smoothly on  $X$ ; both follow from the norm estimates just recorded.

**Proposition 1.3.3: Convergence and Smoothness of the Exponential**

For every  $X \in M_n(\mathbb{F})$  the series  $\sum_{k \geq 0} X^k/k!$  converges absolutely in  $M_n(\mathbb{F})$ , so  $\exp(X)$  is a well-defined matrix, and it satisfies the bound  $\|\exp(X)\| \leq e^{\|X\|}$ . Moreover the map  $\exp: M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$  is smooth, and its value at 0 is  $\exp(0) = I$ .

**Proof:**

The space  $M_n(\mathbb{F}) \cong \mathbb{F}^{n^2}$  is a finite-dimensional normed vector space, hence complete, so a series converges as soon as it converges absolutely; it therefore suffices to bound the norms of the terms. Submultiplicativity gives  $\|X^k\| \leq \|X\|^k$ , so

$$\sum_{k=0}^{\infty} \left\| \frac{X^k}{k!} \right\| = \sum_{k=0}^{\infty} \frac{\|X^k\|}{k!} \leq \sum_{k=0}^{\infty} \frac{\|X\|^k}{k!} = e^{\|X\|}$$

The right-hand series is the convergent scalar exponential of the real number  $\|X\|$ , so the series of norms converges; the matrix series converges absolutely, and its sum  $\exp(X)$  has norm at most  $e^{\|X\|}$  by the triangle inequality applied to the partial sums.

For smoothness, each partial sum  $S_N(X) = \sum_{k=0}^N X^k/k!$  is a polynomial map  $M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$ , hence smooth. On any ball  $\|X\| \leq r$  the tail is bounded uniformly by  $\sum_{k > N} r^k/k!$ , which tends to 0 as  $N \rightarrow \infty$  independently of  $X$ ; the same estimate applied to the series of any fixed partial derivative (itself an exponential-type series in the entries, dominated in the same way) shows that all partial derivatives converge uniformly on compact sets. A limit of smooth maps whose derivatives to every order converge uniformly on compact sets is smooth, so  $\exp$  is smooth. Setting  $X = 0$  leaves only the  $k = 0$  term,  $\exp(0) = I$ , completing the proof.

The bound  $\|\exp(X)\| \leq e^{\|X\|}$  is the running estimate of the section: it lets one differentiate the series term by term, rearrange it, and multiply two such series together, all justified by absolute convergence. The algebraic properties collected next are exactly those of the scalar exponential that survive the passage to matrices, and the one that does not survive, the additivity  $e^{x+y} = e^x e^y$ , is isolated afterward because its failure is the source of the bracket.

**Proposition 1.3.4: Algebraic Properties of the Exponential**

For all  $X \in M_n(\mathbb{F})$  and all  $s, t \in \mathbb{R}$  (or  $\mathbb{C}$ ) the matrix exponential satisfies:

1.  $\exp(0) = I$ , and  $t \mapsto \exp(tX)$  is a smooth curve in  $M_n(\mathbb{F})$  with  $\frac{d}{dt} \exp(tX) = X \exp(tX) = \exp(tX)X$ .
2.  $\exp((s+t)X) = \exp(sX)\exp(tX)$ ; in particular  $\exp(X)$  is invertible with inverse  $\exp(X)^{-1} = \exp(-X)$ , so  $\exp(X) \in \mathrm{GL}_n(\mathbb{F})$ .
3. For every  $g \in \mathrm{GL}_n(\mathbb{F})$ , conjugation is intertwined:  $\exp(gXg^{-1}) = g \exp(X) g^{-1}$ .
4. The determinant of the exponential is the exponential of the trace:  $\det \exp(X) = e^{\mathrm{tr} X}$ .

**Proof:**

We prove the four statements in turn; each rests on the absolute convergence of proposition 1.3.3, which permits term-by-term differentiation and the rearrangement of the double series obtained from a product.

1. That  $\exp(0) = I$  is proposition 1.3.3. Differentiating the series  $\exp(tX) = \sum_k t^k X^k / k!$  term by term, legitimate because the differentiated series  $\sum_k t^{k-1} X^k / (k-1)!$  converges uniformly on compact  $t$ -intervals by the same domination as before, gives

$$\frac{d}{dt} \exp(tX) = \sum_{k=1}^{\infty} \frac{t^{k-1} X^k}{(k-1)!} = X \sum_{j=0}^{\infty} \frac{t^j X^j}{j!} = X \exp(tX)$$

where the third equality reindexes  $j = k - 1$  and factors one  $X$  to the left; factoring it to the right instead gives  $\exp(tX)X$ , and the two agree because  $X$  commutes with every power of itself.

2. Fix  $X$  and consider  $f(t) = \exp((s+t)X)$  and  $h(t) = \exp(sX)\exp(tX)$  as functions of  $t$ . By part (1) both satisfy the same linear differential equation  $y'(t) = Xy(t)$  with the same initial value  $y(0) = \exp(sX)$ : for  $f$  this is the chain rule applied to part (1), and for  $h$  it is part (1) applied to the right-hand factor with  $\exp(sX)$  constant. The initial value problem  $y' = Xy$ ,  $y(0) = \exp(sX)$  has a unique solution (its difference from any other solution has derivative  $X$  times itself and vanishes at 0, hence vanishes identically, a point proved in full in theorem 1.3.10), so  $f = h$ ; that is  $\exp((s+t)X) = \exp(sX)\exp(tX)$ . Setting  $t = -s$  and using  $\exp(0) = I$  gives  $\exp(sX)\exp(-sX) = I$ , so  $\exp(sX)$  is invertible with the stated inverse; taking  $s = 1$  puts  $\exp(X)$  in  $\mathrm{GL}_n(\mathbb{F})$ .
3. Since  $(gXg^{-1})^k = gX^k g^{-1}$  for every  $k$ , and conjugation  $A \mapsto gAg^{-1}$  is a continuous linear map, it commutes with the convergent sum:

$$\exp(gXg^{-1}) = \sum_{k=0}^{\infty} \frac{(gXg^{-1})^k}{k!} = \sum_{k=0}^{\infty} \frac{gX^k g^{-1}}{k!} = g \left( \sum_{k=0}^{\infty} \frac{X^k}{k!} \right) g^{-1} = g \exp(X) g^{-1}$$

4. Consider  $\varphi(t) = \det \exp(tX)$ , a smooth real (or complex) function of  $t$ . By Jacobi's formula

for the derivative of the determinant (established in proposition 1.2.2), and part (1),

$$\varphi'(t) = D_{\exp(tX)} \det \left( \frac{d}{dt} \exp(tX) \right) = \det(\exp(tX)) \operatorname{tr}(\exp(tX)^{-1} X \exp(tX)) = \varphi(t) \operatorname{tr}(X)$$

where the last equality uses that trace is invariant under conjugation,  $\operatorname{tr}(\exp(tX)^{-1} X \exp(tX)) = \operatorname{tr}(X)$ . Thus  $\varphi$  solves the scalar equation  $\varphi' = (\operatorname{tr} X)\varphi$  with  $\varphi(0) = \det I = 1$ , whose unique solution is  $\varphi(t) = e^{t \operatorname{tr} X}$ . Setting  $t = 1$  gives  $\det \exp(X) = e^{\operatorname{tr} X}$ , completing the proof.

Part (2) is the statement that  $\exp$  lands in the group and, restricted to a single line  $\mathbb{R}X$ , is a homomorphism from the additive group  $\mathbb{R}$  to  $\operatorname{GL}_n(\mathbb{F})$ ; this is the germ of the one-parameter subgroups taken up at the end of the section. Part (4) already pays a concrete dividend: it identifies the trace-zero matrices as exactly those whose exponentials have determinant 1, so the exponential of the space of trace-zero matrices lands in  $\operatorname{SL}_n(\mathbb{F})$ , the first instance of the pattern by which the defining equation of a classical group is the linearization of the group condition. What part (2) does *not* say is that  $\exp(X + Y) = \exp(X) \exp(Y)$  for two different matrices  $X, Y$ ; that identity requires the two to commute, and its failure is the mechanism behind the entire theory.

#### Proposition 1.3.5: Exponential of a Sum of Commuting Matrices

Let  $X, Y \in M_n(\mathbb{F})$ . If  $XY = YX$  then

$$\exp(X + Y) = \exp(X) \exp(Y) = \exp(Y) \exp(X)$$

The commutativity hypothesis cannot be dropped: there exist  $X, Y$  with  $\exp(X + Y) \neq \exp(X) \exp(Y)$ .

#### Proof:

*Step 1: the commuting case.* Suppose  $XY = YX$ . Both series converge absolutely by proposition 1.3.3, so their product may be expanded as a double series and summed in any order, in particular grouped by the total degree  $m = j + k$ :

$$\exp(X) \exp(Y) = \left( \sum_{j=0}^{\infty} \frac{X^j}{j!} \right) \left( \sum_{k=0}^{\infty} \frac{Y^k}{k!} \right) = \sum_{m=0}^{\infty} \sum_{j+k=m} \frac{X^j Y^k}{j! k!} = \sum_{m=0}^{\infty} \frac{1}{m!} \sum_{j=0}^m \binom{m}{j} X^j Y^{m-j}$$

The inner sum is the binomial expansion of  $(X + Y)^m$ , which is valid *precisely because  $X$  and  $Y$  commute*: only then do the factors in  $(X + Y)^m$  reorder into powers of  $X$  times powers of  $Y$  with binomial coefficients. Hence the last sum is  $\sum_m (X + Y)^m / m! = \exp(X + Y)$ . The same computation with the roles of  $X$  and  $Y$  interchanged gives  $\exp(Y) \exp(X) = \exp(Y + X) = \exp(X + Y)$ , so all three agree.

*Step 2: a non-commuting counterexample.* Take the nilpotent matrices

$$X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

which do not commute, since  $XY = \operatorname{diag}(1, 0)$  while  $YX = \operatorname{diag}(0, 1)$ . As  $X^2 = Y^2 = 0$ , the

series truncate:  $\exp(X) = I + X = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $\exp(Y) = I + Y = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ , so

$$\exp(X) \exp(Y) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

On the other hand  $X + Y = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  satisfies  $(X + Y)^2 = I$ , so its exponential sums the even and odd parts to hyperbolic functions,

$$\exp(X + Y) = \cosh(1) I + \sinh(1) (X + Y) = \begin{pmatrix} \cosh 1 & \sinh 1 \\ \sinh 1 & \cosh 1 \end{pmatrix}$$

with  $\cosh 1 \approx 1.543$  and  $\sinh 1 \approx 1.175$ . These two matrices are unequal, so  $\exp(X + Y) \neq \exp(X) \exp(Y)$ , as we sought to show.

The gap between  $\exp(X + Y)$  and  $\exp(X) \exp(Y)$  measures the failure of  $X$  and  $Y$  to commute, and unwinding it to second order will produce the commutator  $[X, Y] = XY - YX$  in the next section; the Baker-Campbell-Hausdorff formula of Chapter 3 is the exact accounting of this gap to all orders. For now the counterexample serves as a warning label on part (2) of proposition 1.3.4: along a single line the exponential is a homomorphism, but the map on all of  $M_n(\mathbb{F})$  is not.

Having built  $\exp$ , one wants to invert it, both to recover  $X$  from  $\exp(X)$  and to introduce coordinates on the group near the identity. The inverse is the matrix logarithm, defined by the series for  $\log(1 + x)$  with the same substitution of a matrix for the scalar. Because that scalar series has radius of convergence 1, the matrix version converges only near the identity, which is exactly the regime needed.

#### Proposition 1.3.6: The Matrix Logarithm

For  $A \in M_n(\mathbb{F})$  with  $\|A\| < 1$  the series

$$\log(I + A) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} A^k = A - \frac{A^2}{2} + \frac{A^3}{3} - \dots$$

converges absolutely and defines a smooth map on the open unit ball  $\{A \in M_n(\mathbb{F}) \mid \|A\| < 1\}$ . It inverts the exponential where both are defined:  $\exp(\log(I + A)) = I + A$  for  $\|A\| < 1$ , and  $\log(\exp(X)) = X$  whenever  $\|\exp(X) - I\| < 1$ , in particular whenever  $\|X\| < \log 2$ .

#### Proof:

*Step 1: convergence and smoothness.* For  $\|A\| < 1$ , submultiplicativity gives  $\|A^k\| \leq \|A\|^k$ , so

$$\sum_{k=1}^{\infty} \left\| \frac{(-1)^{k+1}}{k} A^k \right\| \leq \sum_{k=1}^{\infty} \frac{\|A\|^k}{k} = -\log(1 - \|A\|) < \infty$$

so the series converges absolutely, and the same domination on the ball  $\|A\| \leq r < 1$  makes the convergence, together with that of every differentiated series, uniform on compact subsets; the limit is therefore smooth by the argument of proposition 1.3.3.

*Step 2: the two composites are formal inverses.* The scalar identities

$$\exp(\log(1+a)) = 1+a, \quad \log(\exp(x)) = x$$

hold as identities of formal power series in a single indeterminate, because  $\exp$  and  $\log$  are inverse as formal power series over  $\mathbb{Q}$ . Both composites  $\exp \circ \log$  and  $\log \circ \exp$ , applied to a *single* matrix  $A$  (respectively  $X$ ), only ever combine powers of that one matrix, which commute with one another; so every rearrangement used to verify the formal identity is a rearrangement of an absolutely convergent series of commuting matrices, valid by proposition 1.3.5 and Step 1. Substituting the matrix for the indeterminate therefore transports the two formal identities to genuine matrix identities on their domains of convergence:  $\exp(\log(I+A)) = I+A$  for  $\|A\| < 1$ , and  $\log(\exp(X)) = X$  whenever  $\|\exp(X) - I\| < 1$ .

*Step 3: the explicit domain.* If  $\|X\| < \log 2$  then, using  $\exp(X) - I = \sum_{k \geq 1} X^k/k!$  and  $\|X^k\| \leq \|X\|^k$ ,

$$\|\exp(X) - I\| \leq \sum_{k=1}^{\infty} \frac{\|X\|^k}{k!} = e^{\|X\|} - 1 < e^{\log 2} - 1 = 1$$

so the hypothesis  $\|\exp(X) - I\| < 1$  of Step 2 is met and  $\log(\exp(X)) = X$ , completing the proof.

The logarithm inverts the exponential on overlapping neighborhoods, and the two maps are smooth; what remains is to package this into the statement that  $\exp$  is a diffeomorphism near 0, which supplies the analytic charts on the group used throughout the book. The engine is the value of the differential of  $\exp$  at the origin, and it is the simplest possible value.

#### Theorem 1.3.7: The Exponential is a Local Diffeomorphism at the Origin

The differential of  $\exp: M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$  at 0 is the identity,  $D\exp_0 = \text{id}_{M_n(\mathbb{F})}$ . Consequently there are open neighborhoods  $U$  of 0 in  $M_n(\mathbb{F})$  and  $V$  of  $I$  in  $\text{GL}_n(\mathbb{F})$  such that

$$\exp: U \rightarrow V \quad \text{is a diffeomorphism, with inverse } \log: V \rightarrow U$$

#### Proof:

The differential  $D\exp_0$  acts on a tangent vector  $X \in M_n(\mathbb{F})$  by differentiating along the line  $t \mapsto tX$  through the origin:

$$D\exp_0(X) = \left. \frac{d}{dt} \right|_{t=0} \exp(tX) = \left. \frac{d}{dt} \right|_{t=0} \left( I + tX + \frac{t^2}{2}X^2 + \cdots \right) = X$$

the higher terms all carrying a factor  $t$  that vanishes at  $t = 0$ . Thus  $D\exp_0 = \text{id}$ , which is invertible. By the inverse function theorem ([2]),  $\exp$  restricts to a diffeomorphism from some open neighborhood  $U$  of 0 onto some open neighborhood  $V = \exp(U)$  of  $\exp(0) = I$ . Shrinking  $U$  if necessary so that  $\|X\| < \log 2$  on  $U$ , proposition 1.3.6 identifies the inverse diffeomorphism with  $\log$ , since  $\log(\exp(X)) = X$  there; and shrinking  $V$  so that  $\|A - I\| < 1$  on  $V$  makes  $\log$  defined on all of  $V$  with  $\exp(\log(A)) = A$ . This exhibits  $\exp: U \rightarrow V$  and  $\log: V \rightarrow U$  as mutually inverse diffeomorphisms, completing the proof.

This is the analytic foundation for everything that follows. Near the identity,  $\log$  is a coordinate chart on  $\mathrm{GL}_n(\mathbb{F})$  valued in the vector space  $M_n(\mathbb{F})$ , and  $\exp$  reads those coordinates back onto the group; the group multiplication, transported through  $\log$  into a neighborhood of 0, becomes the operation  $X * Y = \log(\exp X \exp Y)$  that Baker-Campbell-Hausdorff will expand in brackets. That  $D \exp_0 = \mathrm{id}$  is what makes the linearization exact to first order: to first order the group law near  $I$  is just addition in  $M_n(\mathbb{F})$ , and the correction terms are governed by the noncommutativity measured in proposition 1.3.5. The chart is local, valid only near  $I$ ; the question of how far  $\exp$  reaches globally, in particular whether it is onto, depends on the group and is settled group by group in later chapters (for the compact groups it is onto, established in Part III).

Before turning to the theorem that organizes all of this, it is worth computing the exponential in the cases where it can be summed by hand, both to make the definition concrete and to see the algebraic properties confirmed on examples.

**Example 1.3.8: Rotations, Nilpotents, and  $\det = e^{\mathrm{tr}}$**

1. *A plane-rotation generator.* Let  $X = \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}$  for  $\theta \in \mathbb{R}$ . Then  $X^2 = -\theta^2 I$ , so the powers cycle through  $I, X, -\theta^2 I, -\theta^2 X, \theta^4 I, \dots$ , and splitting the exponential series into even and odd powers gives

$$\exp(X) = \left( \sum_{m=0}^{\infty} \frac{(-1)^m \theta^{2m}}{(2m)!} \right) I + \left( \sum_{m=0}^{\infty} \frac{(-1)^m \theta^{2m+1}}{(2m+1)!} \right) \frac{X}{\theta} = \cos \theta I + \sin \theta \frac{X}{\theta} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

so  $\exp(X) = R(\theta) \in \mathrm{SO}(2)$  is the rotation by angle  $\theta$ . The generator  $X$  is skew-symmetric with trace 0, and indeed  $\det \exp(X) = \cos^2 \theta + \sin^2 \theta = 1 = e^0 = e^{\mathrm{tr} X}$ , as proposition 1.3.4(4) requires. This recovers, from the single generator  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , the entire circle group  $\mathrm{SO}(2)$  met in example 1.2.11.

2. *A nilpotent matrix.* Let  $N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ , the  $3 \times 3$  nilpotent Jordan block. Then  $N^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$  and  $N^3 = 0$ , so the series terminates after three terms:

$$\exp(tN) = I + tN + \frac{t^2}{2} N^2 = \begin{pmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}$$

a polynomial in  $t$ , not a transcendental function. The trace of  $N$  is 0 and  $\det \exp(tN) = 1$ , again matching part (4); the curve  $t \mapsto \exp(tN)$  is a one-parameter subgroup consisting of unipotent upper-triangular matrices.

3. *A diagonal check of  $\det = e^{\mathrm{tr}}$ .* For  $D = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$  the powers are  $D^k = \mathrm{diag}(\lambda_1^k, \dots, \lambda_n^k)$ , so summing entrywise gives  $\exp(D) = \mathrm{diag}(e^{\lambda_1}, \dots, e^{\lambda_n})$ . Its determinant is  $\prod_i e^{\lambda_i} = e^{\sum_i \lambda_i} = e^{\mathrm{tr} D}$ , confirming proposition 1.3.4(4) directly. Since every matrix over  $\mathbb{C}$  is conjugate to an upper-triangular one, with the eigenvalues on the diagonal, the conjugation equivariance of part (3) and the conjugation-invariance of both  $\det$  and  $\mathrm{tr}$  propagate this identity from the diagonal case to all of  $M_n(\mathbb{C})$ , which is the second proof of  $\det \exp X = e^{\mathrm{tr} X}$  promised in outline.

The three computations exhibit the two extreme behaviors of the exponential. When the generator has imaginary spectrum, as in (1), the exponential is periodic and traces a circle; when the generator

is nilpotent, as in (2), the exponential is polynomial and the orbit runs off to infinity. Every matrix group interpolates between these, and the interpolation is organized by the observation that each of these curves  $t \mapsto \exp(tX)$  is a homomorphism  $\mathbb{R} \rightarrow \mathrm{GL}_n(\mathbb{F})$ . The final result of the section is that these are the *only* such homomorphisms: the exponential does not just produce one-parameter subgroups, it produces all of them.

**Definition 1.3.9: One-Parameter Subgroup**

A *one-parameter subgroup* of  $\mathrm{GL}_n(\mathbb{F})$  is a smooth group homomorphism  $\gamma: \mathbb{R} \rightarrow \mathrm{GL}_n(\mathbb{F})$ ; that is, a smooth curve with

$$\gamma(s+t) = \gamma(s)\gamma(t) \quad \text{for all } s, t \in \mathbb{R}, \quad \gamma(0) = I$$

By proposition 1.3.4(2), for each fixed  $X$  the curve  $\gamma_X(t) = \exp(tX)$  is a one-parameter subgroup, with velocity  $\gamma'_X(0) = X$  at the identity by theorem 1.3.7. The content of the next theorem is the converse: prescribing the initial velocity  $X$  determines the whole homomorphism, so the assignment  $X \mapsto \exp(tX)$  is a bijection between  $M_n(\mathbb{F})$  and the set of one-parameter subgroups.

**Theorem 1.3.10: Classification of One-Parameter Subgroups**

Every one-parameter subgroup  $\gamma: \mathbb{R} \rightarrow \mathrm{GL}_n(\mathbb{F})$  has the form

$$\gamma(t) = \exp(tX), \quad X = \gamma'(0) \in M_n(\mathbb{F})$$

Conversely every  $X \in M_n(\mathbb{F})$  arises this way from a unique one-parameter subgroup, namely  $t \mapsto \exp(tX)$ . Thus  $X \mapsto (t \mapsto \exp(tX))$  is a bijection from  $M_n(\mathbb{F})$  onto the set of one-parameter subgroups of  $\mathrm{GL}_n(\mathbb{F})$ , with inverse  $\gamma \mapsto \gamma'(0)$ .

**Proof:**

*Step 1: a one-parameter subgroup solves a linear differential equation.* Let  $\gamma$  be a one-parameter subgroup and set  $X = \gamma'(0)$ . Differentiating the homomorphism relation  $\gamma(s+t) = \gamma(s)\gamma(t)$  with respect to  $s$  and then setting  $s = 0$  gives, since  $\gamma(0) = I$ ,

$$\gamma'(t) = \left. \frac{d}{ds} \right|_{s=0} \gamma(s+t) = \left. \frac{d}{ds} \right|_{s=0} (\gamma(s)\gamma(t)) = \gamma'(0)\gamma(t) = X\gamma(t)$$

So  $\gamma$  solves the linear matrix initial value problem

$$\gamma'(t) = X\gamma(t), \quad \gamma(0) = I \tag{1.2}$$

*Step 2: the exponential solves the same problem.* By proposition 1.3.4(1) the curve  $\eta(t) = \exp(tX)$  satisfies  $\eta'(t) = X\exp(tX) = X\eta(t)$  and  $\eta(0) = I$ , so  $\eta$  is also a solution of (1.2).

*Step 3: uniqueness of the solution.* It remains to show (1.2) has only one solution, so that  $\gamma = \eta$ . Suppose  $\gamma_1, \gamma_2$  both solve it, and set  $\delta(t) = \exp(-tX)\gamma_1(t) - \exp(-tX)\gamma_2(t) = \exp(-tX)(\gamma_1(t) - \gamma_2(t))$ . Using  $\frac{d}{dt} \exp(-tX) = -X\exp(-tX)$  from proposition 1.3.4(1) and the product rule, the derivative of  $u(t) = \exp(-tX)\gamma_i(t)$  is

$$u'(t) = -X\exp(-tX)\gamma_i(t) + \exp(-tX)X\gamma_i(t) = \exp(-tX)(-X + X)\gamma_i(t) = 0$$

where the middle step uses  $\exp(-tX)X = X \exp(-tX)$ , again from part (1). So each  $\exp(-tX)\gamma_i(t)$  is constant, equal to its value  $\exp(0)\gamma_i(0) = I$  at  $t = 0$ ; hence  $\exp(-tX)\gamma_i(t) = I$  for all  $t$ , giving  $\gamma_i(t) = \exp(tX)$  for both  $i$ . In particular the solution is unique, so  $\gamma(t) = \exp(tX)$ .

*Step 4: the bijection.* Step 1 through Step 3 show every one-parameter subgroup is  $t \mapsto \exp(tX)$  with  $X = \gamma'(0)$ , so the map  $X \mapsto (t \mapsto \exp(tX))$  is surjective onto one-parameter subgroups. It is injective because its inverse is the well-defined operation  $\gamma \mapsto \gamma'(0)$ : from  $\gamma(t) = \exp(tX)$  one recovers  $\gamma'(0) = X$  by theorem 1.3.7. Hence the assignment is a bijection with the stated inverse, completing the proof.

The theorem is the first crossing of the bridge announced at the start: it identifies a purely group-theoretic object, the smooth homomorphisms  $\mathbb{R} \rightarrow \mathrm{GL}_n(\mathbb{F})$ , with a purely linear one, the matrices  $X = \gamma'(0)$ . The matrix  $X$  is the *infinitesimal generator* of the flow  $\gamma$ , and the differential equation (1.2) is the statement that the group element moves, at each instant, by left translation of its current velocity  $X$ . This is the seed of the Lie algebra: the next section defines the Lie algebra of a matrix group  $G$  as the set of generators  $X$  whose one-parameter subgroups  $\exp(tX)$  stay inside  $G$ , and the classification theorem is what guarantees this set captures all the infinitesimal motions of  $G$ . The uniqueness argument of Step 3, that  $\exp(-tX)\gamma(t)$  is constant, recurs verbatim when the flow of a left-invariant vector field is constructed intrinsically in Chapter 2; here it is available by bare hands because everything lives in the linear space  $M_n(\mathbb{F})$ .

### Exercise 1.3.1

1. Show that  $\exp(X^\top) = \exp(X)^\top$  and  $\exp(X^*) = \exp(X)^*$  for all  $X \in M_n(\mathbb{F})$ . Deduce that if  $X$  is skew-symmetric ( $X^\top = -X$ ) then  $\exp(X) \in \mathrm{SO}(n)$ , and if  $X$  is skew-Hermitian ( $X^* = -X$ ) then  $\exp(X) \in \mathrm{U}(n)$ .
2. Show that  $\exp: M_2(\mathbb{R}) \rightarrow \mathrm{GL}_2(\mathbb{R})$  is not surjective by proving that no real  $X$  satisfies  $\exp(X) = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$ . (Hint:  $\det \exp X = e^{\mathrm{tr} X} > 0$  forces the image into  $\mathrm{GL}_2^+(\mathbb{R})$ ; then analyze the real Jordan form of a putative  $X$  with  $\exp(X)$  having the single eigenvalue  $-1$ .)
3. Let  $X = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$  be block-diagonal. Prove that  $\exp(X) = \begin{pmatrix} \exp(A) & 0 \\ 0 & \exp(B) \end{pmatrix}$ . Use this together with example 1.3.8(1) to write down  $\exp(tX)$  for  $X = \mathrm{diag}\left(\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}\right) \in \mathfrak{so}(4)$ , and identify its image as a one-parameter subgroup of  $\mathrm{SO}(4)$ .
4. For  $X, Y \in M_n(\mathbb{F})$  define  $\beta(t) = \exp(tX)Y \exp(-tX)$ . Compute  $\beta(0)$  and  $\beta'(0)$ , showing  $\beta'(0) = XY - YX$ . (This is the computation that will define the bracket in the next section; do it here by differentiating the product.)
5. Let  $\gamma: \mathbb{R} \rightarrow \mathrm{GL}_n(\mathbb{F})$  be a one-parameter subgroup with generator  $X = \gamma'(0)$ , and let  $G \subseteq \mathrm{GL}_n(\mathbb{F})$  be a closed subgroup. Prove that if  $\gamma(t) \in G$  for all  $t$  in some open interval around 0, then  $\gamma(t) \in G$  for all  $t \in \mathbb{R}$ .
6. Using the logarithm series of proposition 1.3.6, compute  $\log(R(\theta))$  for the rotation  $R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$  with  $|\theta|$  small, and verify that it equals the generator  $\begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}$  of example 1.3.8(1), consistent with  $\exp$  and  $\log$  being mutually inverse near  $I$ .

## 1.4 The Lie Algebra of a Matrix Group

The exponential map of section 1.3 turns a single matrix  $X$  into a one-parameter family  $t \mapsto \exp(tX)$  of group elements, a curve through the identity whose velocity at time zero is  $X$  itself. This section reverses the emphasis: rather than fix  $X$  and watch the curve, fix a matrix group  $G$  and ask which matrices  $X$  generate a one-parameter family that stays inside  $G$  for all time. The answer is a vector space, the *Lie algebra* of  $G$ , and it is the first appearance of the object around which the entire book is built. For each classical group of section 1.2 the space will be a concrete list of matrix conditions, and its dimension will match the manifold dimension already computed, so that the algebra recovers by pure linear algebra what the regular value theorem found by calculus.

The vector-space structure is only half the story. The set of matrices generating one-parameter subgroups of  $G$  carries a second operation, the commutator  $[X, Y] = XY - YX$ , and this bracket is what promotes the tangent space to a Lie algebra in the abstract sense. The bracket measures the failure of  $\exp(X)$  and  $\exp(Y)$  to commute, and its closure inside the tangent space is the algebraic shadow of the fact that  $G$  is closed under multiplication. By the end of the section the reader will have both the concrete matrix Lie algebras and the abstract definition they instantiate, together with the running example  $\mathfrak{su}(2)$ , whose bracket is the cross product on  $\mathbb{R}^3$ .

### Definition 1.4.1: Lie Algebra of a Matrix Group

Let  $G \subseteq \mathrm{GL}_n(\mathbb{F})$  be a matrix group. The *Lie algebra* of  $G$  is the set

$$\mathfrak{g} = \{X \in M_n(\mathbb{F}) \mid \exp(tX) \in G \text{ for all } t \in \mathbb{R}\}$$

of matrices whose associated one-parameter subgroup lies entirely in  $G$ .

The definition is engineered to be checkable: to decide whether  $X$  belongs to  $\mathfrak{g}$  one differentiates the defining equations of  $G$  along the curve  $\exp(tX)$  at  $t = 0$ . Before extracting those conditions for the classical groups, the first order of business is to see that  $\mathfrak{g}$  is a real vector space at all, which is not obvious from the definition: it asks about the exponential of  $tX$ , and the exponential is neither additive nor linear.

### Proposition 1.4.2: The Lie Algebra is a Real Vector Space

For any matrix group  $G \subseteq \mathrm{GL}_n(\mathbb{F})$ , the Lie algebra  $\mathfrak{g}$  of definition 1.4.1 is a real vector subspace of  $M_n(\mathbb{F})$ . It contains 0, and if  $X, Y \in \mathfrak{g}$  and  $\lambda \in \mathbb{R}$  then  $\lambda X \in \mathfrak{g}$  and  $X + Y \in \mathfrak{g}$ .

#### **Proof:**

The zero matrix lies in  $\mathfrak{g}$  because  $\exp(t \cdot 0) = I \in G$  for every  $t$ . For closure under real scalars, fix  $X \in \mathfrak{g}$  and  $\lambda \in \mathbb{R}$ ; then for every  $t \in \mathbb{R}$  the product  $t\lambda$  is again a real number, so

$$\exp(t(\lambda X)) = \exp((t\lambda)X) \in G$$

because  $X \in \mathfrak{g}$ . Hence  $\lambda X \in \mathfrak{g}$ . This uses only that the scalar is real; it is the reason  $\mathfrak{g}$  is a real vector space even when  $\mathbb{F} = \mathbb{C}$ .

Closure under addition is where the exponential's nonlinearity must be confronted. The tool is

the *Lie product formula*

$$\exp(X + Y) = \lim_{m \rightarrow \infty} \left( \exp\left(\frac{X}{m}\right) \exp\left(\frac{Y}{m}\right) \right)^m \quad (1.3)$$

valid for all  $X, Y \in M_n(\mathbb{F})$ , which expresses the exponential of a sum as a limit of products of exponentials, precisely the kind of expression a group is closed under.

*Step 1: proof of the Lie product formula.* Fix  $X, Y$  and set  $A_m = \exp(X/m) \exp(Y/m)$ . Expanding each factor to second order in the small parameter  $1/m$  gives

$$\exp\left(\frac{X}{m}\right) = I + \frac{X}{m} + O\left(\frac{1}{m^2}\right), \quad \exp\left(\frac{Y}{m}\right) = I + \frac{Y}{m} + O\left(\frac{1}{m^2}\right)$$

where the  $O(1/m^2)$  terms are matrices of operator norm bounded by a constant (depending on  $\|X\|, \|Y\|$ ) times  $1/m^2$ . Multiplying,

$$A_m = I + \frac{X + Y}{m} + O\left(\frac{1}{m^2}\right)$$

Since  $\|A_m - I\| \rightarrow 0$ , for  $m$  large the matrix  $A_m$  lies in the domain where the logarithm series of proposition 1.3.6 converges, and there  $\log(I + B) = B + O(\|B\|^2)$ ; applying this with  $B = A_m - I = \frac{X+Y}{m} + O(1/m^2)$  gives

$$\log A_m = \frac{X + Y}{m} + O\left(\frac{1}{m^2}\right)$$

Because  $\log$  and  $\exp$  are mutually inverse near  $I$  and  $0$  respectively (theorem 1.3.7), one has  $A_m = \exp(\log A_m)$ , and therefore

$$A_m^m = \exp(m \log A_m) = \exp\left(X + Y + O\left(\frac{1}{m}\right)\right)$$

As  $m \rightarrow \infty$  the argument  $X + Y + O(1/m)$  converges to  $X + Y$  in  $M_n(\mathbb{F})$ , and  $\exp$  is continuous, so  $A_m^m \rightarrow \exp(X + Y)$ . This is (1.3).

*Step 2: additive closure.* Suppose  $X, Y \in \mathfrak{g}$  and fix  $t \in \mathbb{R}$ . Applying (1.3) to the matrices  $tX$  and  $tY$ ,

$$\exp(t(X + Y)) = \exp(tX + tY) = \lim_{m \rightarrow \infty} \left( \exp\left(\frac{tX}{m}\right) \exp\left(\frac{tY}{m}\right) \right)^m$$

For each fixed  $m$ , the matrices  $\exp(tX/m)$  and  $\exp(tY/m)$  lie in  $G$ , because  $X, Y \in \mathfrak{g}$  and  $t/m$  is a real number; their product raised to the  $m$ th power is a product of elements of  $G$ , hence lies in  $G$ , a subgroup. The one-parameter subgroup  $\exp(t(X + Y))$  is thus a limit of elements of  $G$ . A matrix group  $G$  is defined by polynomial equations in the entries (the classical groups of section 1.2 are each  $F^{-1}(c)$  for a continuous  $F$ ), so  $G$  is closed in  $\mathrm{GL}_n(\mathbb{F})$ , and a limit of elements of  $G$  that remains invertible lies in  $G$ . Since  $\exp(t(X + Y))$  is invertible with inverse  $\exp(-t(X + Y))$ , it lies in  $G$ . As  $t$  was arbitrary,  $X + Y \in \mathfrak{g}$ , completing the proof.

The role of the Lie product formula is worth dwelling on. Addition of matrices is a linear operation with no group-theoretic meaning, yet (1.3) realizes  $\exp(X + Y)$  entirely through multiplication and limits, the two operations a closed subgroup respects. This is the mechanism that transfers the linear structure of  $M_n(\mathbb{F})$  onto the tangent space  $\mathfrak{g}$ : the sum  $X + Y$  is legitimate inside  $\mathfrak{g}$  precisely because its exponential can be manufactured from the exponentials of  $X$  and  $Y$  without ever leaving  $G$ . The

same interplay between the additive world of  $\mathfrak{g}$  and the multiplicative world of  $G$  drives the bracket computation below.

For the classical groups the definition of  $\mathfrak{g}$  unwinds into a linear condition on  $X$ , one per group, obtained by differentiating the defining relation. These conditions are the working description of the classical Lie algebras, and the point of the theorem is that they reproduce the dimensions of section 1.2 exactly.

### Theorem 1.4.3: The Classical Lie Algebras

The Lie algebras of the classical groups are the following matrix spaces, with the indicated real dimensions.

1.  $\mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$ , the full matrix algebra;  $\dim_{\mathbb{R}} \mathfrak{gl}_n(\mathbb{R}) = n^2$  and  $\dim_{\mathbb{R}} \mathfrak{gl}_n(\mathbb{C}) = 2n^2$ .
2.  $\mathfrak{sl}_n(\mathbb{F}) = \{X \in M_n(\mathbb{F}) \mid \text{tr} X = 0\}$ , the traceless matrices;  $\dim_{\mathbb{R}} \mathfrak{sl}_n(\mathbb{R}) = n^2 - 1$  and  $\dim_{\mathbb{R}} \mathfrak{sl}_n(\mathbb{C}) = 2(n^2 - 1)$ .
3.  $\mathfrak{so}(n) = \{X \in M_n(\mathbb{R}) \mid X^T = -X\}$ , the real skew-symmetric matrices;  $\dim_{\mathbb{R}} \mathfrak{so}(n) = \binom{n}{2}$ .
4.  $\mathfrak{u}(n) = \{X \in M_n(\mathbb{C}) \mid X^* = -X\}$ , the skew-Hermitian matrices;  $\dim_{\mathbb{R}} \mathfrak{u}(n) = n^2$ .
5.  $\mathfrak{su}(n) = \{X \in M_n(\mathbb{C}) \mid X^* = -X, \text{tr} X = 0\}$ , the traceless skew-Hermitian matrices;  $\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1$ .
6.  $\mathfrak{sp}(2n, \mathbb{F}) = \{X \in M_{2n}(\mathbb{F}) \mid X^T J + JX = 0\}$ , where  $J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$ ;  $\dim_{\mathbb{R}} \mathfrak{sp}(2n, \mathbb{R}) = n(2n + 1)$ .

In each case the dimension of  $\mathfrak{g}$  equals the manifold dimension of the corresponding group computed in section 1.2.

### Proof:

The argument is the same for each group: differentiate the defining relation of  $G$  along  $t \mapsto \exp(tX)$  at  $t = 0$ , using  $\frac{d}{dt}\big|_0 \exp(tX) = X$  and the derivative rules for the exponential from proposition 1.3.4.

1. *The general linear group.* Every  $X \in M_n(\mathbb{F})$  has  $\exp(tX) \in \text{GL}_n(\mathbb{F})$  for all  $t$ , since  $\exp(tX)$  is invertible with inverse  $\exp(-tX)$  by proposition 1.3.4. Hence  $\mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$ , of real dimension  $n^2$  over  $\mathbb{R}$  and  $2n^2$  over  $\mathbb{C}$ , matching  $\dim \text{GL}_n(\mathbb{F})$  from proposition 1.1.2.
2. *The special linear group.* The relation defining  $\text{SL}_n(\mathbb{F})$  is  $\det = 1$ . By the identity  $\det \exp(tX) = e^{t \text{tr} X}$  of proposition 1.3.4, the curve  $\exp(tX)$  lies in  $\text{SL}_n(\mathbb{F})$  for all  $t$  if and only if  $e^{t \text{tr} X} = 1$  for all  $t \in \mathbb{R}$ , which holds if and only if  $\text{tr} X = 0$ . Thus  $\mathfrak{sl}_n(\mathbb{F})$  is the space of traceless matrices. The trace is one linear equation on  $M_n(\mathbb{F})$ , and it is surjective onto  $\mathbb{F}$ , so its kernel has real codimension 1 over  $\mathbb{R}$  and real codimension 2 over  $\mathbb{C}$ , giving  $n^2 - 1$  and  $2(n^2 - 1)$ , matching proposition 1.2.2.
3. *The orthogonal and special orthogonal groups.* The relation is  $A^T A = I$ . Substituting  $A = \exp(tX)$  and using  $\exp(tX)^T = \exp(tX^T)$ , the curve satisfies  $\exp(tX^T) \exp(tX) = I$  for all  $t$ . Differentiating at  $t = 0$ , the product rule gives  $X^T + X = 0$ , so  $X$  is skew-

symmetric. Conversely, if  $X^\top = -X$  then  $tX^\top$  and  $tX$  commute (they are negatives), so  $\exp(tX^\top)\exp(tX) = \exp(t(X^\top + X)) = \exp(0) = I$  by proposition 1.3.5, and  $\exp(tX) \in O(n)$  for all  $t$ . A skew-symmetric matrix has zero diagonal and its below-diagonal entries are the negatives of those above, so its free parameters are the  $\binom{n}{2}$  strictly-upper entries; hence  $\dim \mathfrak{so}(n) = \binom{n}{2}$ , matching proposition 1.2.4. The determinant-one constraint cutting  $SO(n)$  out of  $O(n)$  imposes nothing further on the algebra: every skew-symmetric  $X$  already has  $\text{tr} X = 0$ , so  $\det \exp(tX) = e^{t\text{tr} X} = 1$  automatically, and  $\mathfrak{so}(n)$  is at once the Lie algebra of  $O(n)$  and of  $SO(n)$ , reflecting that the two groups share the identity component.

4. *The unitary group.* The relation is  $A^* A = I$ . With  $A = \exp(tX)$  and  $\exp(tX)^* = \exp(tX^*)$  (the conjugate transpose of  $\exp(tX)$  is  $\exp$  of the conjugate transpose, as the power series has real coefficients), the same differentiation gives  $X^* + X = 0$ , so  $X$  is skew-Hermitian; the converse is the commuting exponential identity as in (3). A skew-Hermitian matrix has purely imaginary diagonal entries ( $n$  real parameters) and  $\binom{n}{2}$  free complex entries above the diagonal ( $n(n-1)$  real parameters), the below-diagonal entries being determined by  $X^* = -X$ ; the total is  $n + n(n-1) = n^2$ , matching proposition 1.2.6.
5. *The special unitary group.* Here  $X$  must be skew-Hermitian and additionally satisfy the special condition. By (2) the determinant constraint  $\det \exp(tX) = 1$  forces  $\text{tr} X = 0$ , so  $\mathfrak{su}(n)$  is the space of traceless skew-Hermitian matrices. Tracelessness removes one real parameter from  $\mathfrak{u}(n)$ : a skew-Hermitian matrix has purely imaginary diagonal  $\text{diag}(i\theta_1, \dots, i\theta_n)$ , and  $\text{tr} X = i(\theta_1 + \dots + \theta_n) = 0$  is the single real equation  $\sum_j \theta_j = 0$ . Hence  $\dim \mathfrak{su}(n) = n^2 - 1$ , matching proposition 1.2.6.
6. *The symplectic group.* The relation is  $A^\top J A = J$ . With  $A = \exp(tX)$ , the condition  $\exp(tX)^\top J \exp(tX) = J$  for all  $t$ , differentiated at  $t = 0$ , yields  $X^\top J + JX = 0$ ; the converse again follows because  $X^\top J + JX = 0$  makes  $t \mapsto \exp(tX)^\top J \exp(tX)$  have vanishing derivative and value  $J$  at  $t = 0$ , hence constantly  $J$ . The condition  $X^\top J = -JX$  says exactly that  $JX$  is symmetric:  $(JX)^\top = X^\top J^\top = -X^\top J = JX$ , using  $J^\top = -J$ . The map  $X \mapsto JX$  is a linear bijection of  $M_{2n}(\mathbb{F})$  (with inverse  $Y \mapsto J^{-1}Y = -JY$ , since  $J^2 = -I$ ), so  $\mathfrak{sp}(2n, \mathbb{F})$  is linearly isomorphic to the space of symmetric  $2n \times 2n$  matrices, of dimension  $\binom{2n+1}{2} = \frac{2n(2n+1)}{2} = n(2n+1)$ , matching proposition 1.2.9.

In every case the linearized condition reproduces the dimension found by the regular value theorem in section 1.2, completing the proof.

The coincidence of dimensions is not an accident but the concrete face of a general principle. The Lie algebra  $\mathfrak{g}$  is the tangent space to  $G$  at the identity: differentiating a defining equation of  $G$  along a curve through  $I$  is exactly the operation the regular value theorem performs when it identifies the tangent space  $T_I G$  with  $\ker DF_I$ , and the two computations agree because they are the same computation. The exponential description of definition 1.4.1 and the tangent-space description of section 1.2 therefore define the same object; the next chapter will make the identification  $\mathfrak{g} = T_I G$  the definition of the Lie algebra of an abstract Lie group. One reads off from theorem 1.4.3 that the skew-symmetric matrices form the infinitesimal rotations and the skew-Hermitian matrices the infinitesimal unitaries: exponentiating a small skew matrix produces a rotation close to the identity, the linear-algebra content of “a rotation is generated by an angular velocity”.

So far  $\mathfrak{g}$  is only a vector space. The operation that makes it a Lie algebra is the commutator, and its

presence is forced by the same closure under multiplication that gave additive closure. If  $X, Y \in \mathfrak{g}$ , then conjugating the one-parameter subgroup of  $Y$  by an element of the one-parameter subgroup of  $X$  stays in  $G$ , and differentiating this conjugation produces a new element of  $\mathfrak{g}$ , which turns out to be the commutator.

**Proposition 1.4.4: The Lie Algebra is Closed under the Commutator**

Let  $G \subseteq \mathrm{GL}_n(\mathbb{F})$  be a matrix group with Lie algebra  $\mathfrak{g}$ . For all  $X, Y \in \mathfrak{g}$ , the commutator

$$[X, Y] = XY - YX$$

again lies in  $\mathfrak{g}$ . The bracket  $[-, -]$  is  $\mathbb{R}$ -bilinear, alternating (so that  $[X, X] = 0$  and  $[Y, X] = -[X, Y]$ ), and satisfies the Jacobi identity

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$$

for all  $X, Y, Z \in \mathfrak{g}$ .

**Proof:**

*Step 1: conjugation stays in the group.* Fix  $X, Y \in \mathfrak{g}$  and  $s \in \mathbb{R}$ , and consider the curve

$$c(s) = \exp(sX) Y \exp(-sX) \in M_n(\mathbb{F})$$

obtained by conjugating  $Y$ . For each real  $s$  the elements  $\exp(sX)$  and  $\exp(-sX) = \exp(sX)^{-1}$  lie in  $G$ , so conjugation by  $\exp(sX)$  is an automorphism of  $G$  carrying the one-parameter subgroup  $t \mapsto \exp(tY)$  to  $t \mapsto \exp(sX) \exp(tY) \exp(-sX)$ . This last curve lies in  $G$ , and by the conjugation equivariance of the exponential (proposition 1.3.4),

$$\exp(sX) \exp(tY) \exp(-sX) = \exp(tc(s)) \quad (1.4)$$

where  $c(s) = \exp(sX) Y \exp(-sX)$ . The left side lies in  $G$  for all  $t$ , so (1.4) shows  $\exp(tc(s)) \in G$  for all  $t$ ; that is,  $c(s) \in \mathfrak{g}$  for every  $s \in \mathbb{R}$ .

*Step 2: differentiate the conjugation.* The set  $\mathfrak{g}$  is a vector subspace of the finite-dimensional space  $M_n(\mathbb{F})$  (proposition 1.4.2), hence closed, and  $s \mapsto c(s)$  is a smooth curve lying in  $\mathfrak{g}$ . A closed subspace contains the derivative of any differentiable curve lying in it, so  $c'(0) \in \mathfrak{g}$ . Computing the derivative by the product rule, with  $\frac{d}{ds} \exp(sX) = X \exp(sX)$  and  $\frac{d}{ds} \exp(-sX) = -\exp(-sX)X$ ,

$$c'(s) = X \exp(sX) Y \exp(-sX) - \exp(sX) Y \exp(-sX) X$$

Evaluating at  $s = 0$ , where  $\exp(0) = I$ , gives

$$c'(0) = XY - YX = [X, Y]$$

Therefore  $[X, Y] \in \mathfrak{g}$ , which is the closure claim.

*Step 3: the algebraic identities.* Bilinearity of  $(X, Y) \mapsto XY - YX$  is inherited from the bilinearity of matrix multiplication:  $[aX + bX', Y] = a[X, Y] + b[X', Y]$  and likewise in the second slot, for  $a, b \in \mathbb{R}$ . The bracket is alternating because  $[X, X] = XX - XX = 0$ ; expanding  $[X + Y, X + Y] = 0$  and using bilinearity gives  $[X, Y] + [Y, X] = 0$ , so  $[Y, X] = -[X, Y]$ . For the Jacobi identity,

expand each of the three double brackets. Writing out  $[X, [Y, Z]] = X(YZ - ZY) - (YZ - ZY)X$  and the two cyclic analogues, the sum

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]]$$

consists of the twelve triple products  $XYZ, XZY, \dots$ , each appearing once with a plus sign and once with a minus sign; they cancel in pairs, leaving 0. This establishes all three identities, completing the proof.

The conjugation curve  $c(s)$  in the proof is the first appearance of the adjoint action, which the next chapter will develop into the representations  $\text{Ad}$  and  $\text{ad}$  that encode the bracket intrinsically. What matters now is the structural conclusion: the tangent space  $\mathfrak{g}$  is not just a vector space but is closed under an antisymmetric bilinear product satisfying the Jacobi identity. This package of data has a name, and it is the central algebraic object of the theory.

#### Definition 1.4.5: Abstract Lie Algebra

A *Lie algebra* over a field  $k$  is a  $k$ -vector space  $\mathfrak{g}$  equipped with a bilinear map

$$[-, -]: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}, \quad (X, Y) \mapsto [X, Y]$$

called the *bracket*, satisfying two axioms:

1. the bracket is *alternating*:  $[X, X] = 0$  for all  $X \in \mathfrak{g}$  (equivalently, over any field of characteristic  $\neq 2$ , antisymmetric:  $[Y, X] = -[X, Y]$ );
2. the bracket satisfies the *Jacobi identity*:

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$$

for all  $X, Y, Z \in \mathfrak{g}$ .

A *homomorphism* of Lie algebras is a  $k$ -linear map  $\varphi: \mathfrak{g} \rightarrow \mathfrak{h}$  preserving the bracket,  $\varphi([X, Y]) = [\varphi(X), \varphi(Y)]$ .

proposition 1.4.4 says precisely that the Lie algebra  $\mathfrak{g}$  of a matrix group, with the commutator bracket  $[X, Y] = XY - YX$ , is an abstract real Lie algebra in the sense of definition 1.4.5. The classical algebras  $\mathfrak{gl}_n, \mathfrak{sl}_n, \mathfrak{so}(n), \mathfrak{u}(n), \mathfrak{su}(n)$ , and  $\mathfrak{sp}(2n, \mathbb{F})$  are all real Lie algebras under the commutator, and one checks in each case that the bracket preserves the defining condition: for instance if  $X, Y$  are skew-symmetric then  $[X, Y]^T = (XY - YX)^T = Y^T X^T - X^T Y^T = YX - XY = -[X, Y]$ , so  $[X, Y] \in \mathfrak{so}(n)$ , consistent with proposition 1.4.4. The full matrix algebra  $\mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$  under the commutator is the ambient Lie algebra of which all the others are subalgebras, the exact analogue of  $\text{GL}_n(\mathbb{F})$  being the ambient group.

The abstract definition frees the bracket from its origin as a commutator of matrices. In Chapter 2 the Lie algebra of a general Lie group is constructed intrinsically, as the space of left-invariant vector fields under the Lie bracket of vector fields, with no matrices in sight, and definition 1.4.5 is the target of that construction: the intrinsic bracket will satisfy exactly these two axioms. The matrix Lie algebras of this chapter are then recovered as the special case where the group sits inside some  $\text{GL}_n(\mathbb{F})$ , and the intrinsic bracket restricts to the commutator. The reader who wants a preview should keep in mind that the abstract axioms are all that survive of the group structure at the

infinitesimal level, and that the entire structure theory of Parts II and beyond is carried out with the axioms alone.

The bridge between the group and its algebra is the exponential map. The defining property of  $\mathfrak{g}$  is that  $\exp$  sends it into  $G$ , and the local diffeomorphism of section 1.3 pins down the dimension of the correspondence.

#### Corollary 1.4.6: The Exponential Bridge and the Dimension Match

Let  $G \subseteq \mathrm{GL}_n(\mathbb{F})$  be a matrix group with Lie algebra  $\mathfrak{g}$ . Then  $\exp$  maps  $\mathfrak{g}$  into  $G$ , and the tangent space to  $G$  at the identity contains  $\mathfrak{g}$ :

$$\mathfrak{g} \subseteq T_I G$$

Suppose in addition that  $G$  is presented as a regular-value preimage  $G = F^{-1}(c)$  for a smooth map  $F$ , as each classical group of section 1.2 is. Then  $\mathfrak{g} = T_I G$ , the map  $\exp$  restricts to a diffeomorphism from a neighborhood of 0 in  $\mathfrak{g}$  onto a neighborhood of  $I$  in  $G$ , and in particular

$$\dim_{\mathbb{R}} \mathfrak{g} = \dim G$$

#### Proof:

That  $\exp(\mathfrak{g}) \subseteq G$  is immediate from definition 1.4.1: for  $X \in \mathfrak{g}$ , taking  $t = 1$  gives  $\exp X = \exp(1 \cdot X) \in G$ . For the inclusion into the tangent space, let  $X \in \mathfrak{g}$ . The curve  $t \mapsto \exp(tX)$  lies in  $G$  by definition 1.4.1, passes through  $I$  at  $t = 0$ , and has velocity  $X$  there by proposition 1.3.4; a velocity vector of a curve in  $G$  through  $I$  is by definition an element of  $T_I G$ , so  $X \in T_I G$ . Hence  $\mathfrak{g} \subseteq T_I G$ .

Suppose now  $G = F^{-1}(c)$  with  $c$  a regular value of a smooth  $F$ , so that  $T_I G = \ker DF_I$ . The reverse inclusion  $T_I G \subseteq \mathfrak{g}$  is the content of the discussion following theorem 1.4.3: differentiating the defining relation  $F = c$  along a curve in  $G$  through  $I$  shows that the linearized condition cutting out  $\mathfrak{g}$  is exactly  $\ker DF_I$ , so  $\mathfrak{g} = \ker DF_I = T_I G$ . With this equality in hand, by theorem 1.3.7 the map  $\exp$  is a diffeomorphism from a neighborhood  $U$  of 0 in  $M_n(\mathbb{F})$  onto a neighborhood of  $I$  in  $\mathrm{GL}_n(\mathbb{F})$ , with  $D\exp_0 = \mathrm{id}$ ; restricting to  $\mathfrak{g}$ , the differential  $D\exp_0 = \mathrm{id}$  carries  $\mathfrak{g} = T_0 \mathfrak{g}$  isomorphically onto  $\mathfrak{g} = T_I G$ , so the inverse function theorem applied within  $G$  makes the restriction  $\exp: U \cap \mathfrak{g} \rightarrow G$  a diffeomorphism onto a neighborhood of  $I$  in  $G$ . Two manifolds admitting a diffeomorphism between open sets have equal dimension, so  $\dim_{\mathbb{R}} \mathfrak{g} = \dim G$ , completing the proof.

The corollary is the quantitative form of the slogan that the Lie algebra is an infinitesimal copy of the group: near the identity,  $G$  looks exactly like its tangent space  $\mathfrak{g}$ , with  $\exp$  providing the chart. The inclusion  $\mathfrak{g} \subseteq T_I G$  holds for every matrix group and is the easy half, read off by differentiating  $\exp(tX)$ ; the reverse inclusion, which upgrades it to the equality  $\mathfrak{g} = T_I G$  and thereby forces the dimensions to match, is what requires the group to be presented as a regular-value preimage, and it is established above exactly for the classical groups of section 1.2. For a matrix group given only as an abstract embedded submanifold, with no defining map  $F$  at hand, the equality  $\mathfrak{g} = T_I G$  is still true but rests on the closed-subgroup theorem, proved in Chapter 3, which produces the exponential chart from the closedness of  $G$  alone. The dimension equality has already been verified group by group in theorem 1.4.3, but the corollary explains why it had to hold:  $\exp$  is a local chart from  $\mathfrak{g}$  to  $G$ , so the two spaces cannot have different dimensions. The global behavior of  $\exp$  is more delicate,

and whether it is surjective onto  $G$  depends on the group; section 1.5 will show  $\exp$  is onto for the compact group  $\mathrm{SU}(2)$ , while for  $\mathrm{GL}_n(\mathbb{R})$  it is not, as the negative-determinant matrices are never exponentials.

Everything comes together in the smallest nonabelian compact example,  $\mathfrak{su}(2)$ , the Lie algebra of the running-example group  $\mathrm{SU}(2)$ . Its bracket is the cross product, and this identification is the first instance of the dictionary the book exists to develop.

**Example 1.4.7: The Lie Algebra  $\mathfrak{su}(2)$  and the Cross Product**

By theorem 1.4.3,  $\mathfrak{su}(2)$  is the space of traceless skew-Hermitian  $2 \times 2$  complex matrices. A traceless skew-Hermitian matrix has purely imaginary diagonal summing to zero and off-diagonal entries  $b, -\bar{b}$ , so it has the form

$$\begin{pmatrix} ia & b \\ -\bar{b} & -ia \end{pmatrix}, \quad a \in \mathbb{R}, \quad b \in \mathbb{C}$$

a real three-dimensional space, consistent with  $\dim \mathfrak{su}(2) = 2^2 - 1 = 3$ . A convenient basis is built from the *Pauli matrices*

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

which are Hermitian and traceless; multiplying by  $-\frac{i}{2}$  produces traceless skew-Hermitian matrices. Set

$$X_k = -\frac{i}{2} \sigma_k, \quad k = 1, 2, 3$$

explicitly

$$X_1 = \frac{1}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad X_2 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad X_3 = \frac{1}{2} \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$$

Each  $X_k$  is traceless and satisfies  $X_k^* = -X_k$ , so  $X_1, X_2, X_3 \in \mathfrak{su}(2)$ , and they are linearly independent over  $\mathbb{R}$ , hence a basis.

The bracket relations are computed directly from the Pauli identity  $\sigma_j \sigma_k = \delta_{jk} I + i \sum_{\ell} \varepsilon_{j k \ell} \sigma_{\ell}$ , where  $\varepsilon_{j k \ell}$  is the sign of the permutation  $(j, k, \ell)$ . The commutator of Pauli matrices is therefore  $[\sigma_j, \sigma_k] = 2i \sum_{\ell} \varepsilon_{j k \ell} \sigma_{\ell}$ , and so

$$[X_j, X_k] = \left(-\frac{i}{2}\right)^2 [\sigma_j, \sigma_k] = -\frac{1}{4} \cdot 2i \sum_{\ell} \varepsilon_{j k \ell} \sigma_{\ell} = \sum_{\ell} \varepsilon_{j k \ell} \left(-\frac{i}{2} \sigma_{\ell}\right) = \sum_{\ell} \varepsilon_{j k \ell} X_{\ell}$$

In cyclic form,

$$[X_1, X_2] = X_3, \quad [X_2, X_3] = X_1, \quad [X_3, X_1] = X_2 \tag{1.5}$$

These are exactly the multiplication rules of the standard basis  $e_1, e_2, e_3$  of  $\mathbb{R}^3$  under the cross product:  $e_1 \times e_2 = e_3$ ,  $e_2 \times e_3 = e_1$ ,  $e_3 \times e_1 = e_2$ .

Define the linear map  $\Phi: \mathfrak{su}(2) \rightarrow \mathbb{R}^3$  by  $\Phi(X_k) = e_k$  on the basis and extending by linearity. It is a linear isomorphism of three-dimensional real vector spaces, and it preserves the bracket: on basis elements (1.5) matches the cross product term by term, and both operations are bilinear, so  $\Phi([X, Y]) = \Phi(X) \times \Phi(Y)$  for all  $X, Y$ . Therefore

$$\Phi: \mathfrak{su}(2) \xrightarrow{\sim} (\mathbb{R}^3, \times)$$

is an isomorphism of Lie algebras, where  $\mathbb{R}^3$  carries the cross product as its bracket. The cross product does satisfy the Jacobi identity, which is the familiar vector identity  $a \times (b \times c) + b \times (c \times a) + c \times (a \times b) = 0$ .

$a) + c \times (a \times b) = 0$ ; the isomorphism  $\Phi$  makes this identity and the Jacobi identity for  $\mathfrak{su}(2)$  the same statement. This identification is the infinitesimal form of the double cover  $SU(2) \rightarrow SO(3)$  taken up in section 1.5: the algebra  $(\mathbb{R}^3, \times)$  is also  $\mathfrak{so}(3)$  (the skew-symmetric  $3 \times 3$  matrices act on  $\mathbb{R}^3$  as  $v \mapsto \omega \times v$ ), so  $\mathfrak{su}(2) \cong \mathfrak{so}(3)$  even though  $SU(2)$  and  $SO(3)$  are not isomorphic as groups.

The isomorphism  $\mathfrak{su}(2) \cong \mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$  is a compressed statement of the entire local theory of these groups: two nonisomorphic groups share a Lie algebra, so they must agree near the identity and differ only in their global topology. That the infinitesimal object cannot distinguish  $SU(2)$  from  $SO(3)$ , while the groups themselves are distinguished by a factor of  $\mathbb{Z}/2$ , is the phenomenon section 1.5 analyzes in full, and it is the reason the passage from group to algebra loses information that only the topology of the group can supply.

### Exercise 1.4.1

1. Write down the standard basis of the skew-symmetric matrices  $\mathfrak{so}(3)$ , namely the three matrices  $L_1, L_2, L_3$  with  $(L_k)_{ij} = -\varepsilon_{kij}$ , and verify the bracket relations  $[L_1, L_2] = L_3$ ,  $[L_2, L_3] = L_1$ ,  $[L_3, L_1] = L_2$ . Conclude that  $\mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$ , and hence  $\mathfrak{so}(3) \cong \mathfrak{su}(2)$  as real Lie algebras.
2. Show that for the commutator bracket the trace gives a Lie algebra homomorphism  $\text{tr}: \mathfrak{gl}_n(\mathbb{F}) \rightarrow \mathbb{F}$  (with  $\mathbb{F}$  abelian, so  $[\lambda, \mu] = 0$ ), and deduce that  $\mathfrak{sl}_n(\mathbb{F}) = \ker \text{tr}$  is an ideal. Prove the direct sum decomposition  $\mathfrak{gl}_n(\mathbb{F}) = \mathfrak{sl}_n(\mathbb{F}) \oplus \mathbb{F}I$  of vector spaces, and determine the bracket of a scalar matrix  $\lambda I$  with an arbitrary  $X \in \mathfrak{gl}_n(\mathbb{F})$ .
3. Using  $\det \exp X = e^{\text{tr} X}$ , show directly from definition 1.4.1 that a matrix  $X \in M_n(\mathbb{R})$  lies in  $\mathfrak{sl}_n(\mathbb{R})$  if and only if  $\text{tr} X = 0$ , without invoking theorem 1.4.3. Give an explicit  $X \in \mathfrak{gl}_2(\mathbb{R})$  with  $\text{tr} X \neq 0$  and verify that  $\exp(tX) \notin \text{SL}_2(\mathbb{R})$  for some  $t$ .
4. For  $X \in \mathfrak{su}(2)$  with  $\Phi(X) = v \in \mathbb{R}^3$  under the isomorphism of example 1.4.7, show that  $X^2 = -\frac{1}{4}|v|^2 I$ , where  $|v|$  is the Euclidean norm. Deduce the closed form

$$\exp(X) = \cos\left(\frac{|v|}{2}\right) I + \frac{\sin\left(\frac{|v|}{2}\right)}{|v|/2} X$$

for  $v \neq 0$ , and confirm that  $\exp(X) \in SU(2)$ .

5. Verify that  $\mathfrak{sp}(2n, \mathbb{F})$  is closed under the commutator: if  $X^\top J + JX = 0$  and  $Y^\top J + JY = 0$ , show  $[X, Y]^\top J + J[X, Y] = 0$ . This confirms  $\mathfrak{sp}(2n, \mathbb{F})$  is a Lie subalgebra of  $\mathfrak{gl}_{2n}(\mathbb{F})$ , as proposition 1.4.4 predicts.
6. Show that  $\mathfrak{u}(n)$  is a *real* but not a complex vector subspace of  $M_n(\mathbb{C})$ : exhibit a skew-Hermitian matrix  $X$  for which  $iX$  is not skew-Hermitian. Explain how this is consistent with proposition 1.4.2, which asserts only that the Lie algebra of a matrix group is a real vector space.

## 1.5 $SU(2)$ , $SO(3)$ , and the Double Cover $SU(2) \rightarrow SO(3)$

Every structural feature that the later chapters develop in generality is already present, in fully computable form, in the smallest noncommutative compact group,  $SU(2)$ . This section studies it directly: as a manifold it is a 3-sphere, and its Lie algebra  $\mathfrak{su}(2)$  is three-dimensional, small enough

that the exponential map, the adjoint action, and the passage to  $SO(3)$  can all be written out with explicit matrices. The centerpiece is the double cover  $SU(2) \rightarrow SO(3)$ , a two-to-one surjective homomorphism that realizes  $SO(3)$  as the quotient  $SU(2)/\{\pm I\}$  and identifies  $SU(2)$  as the universal cover of the rotation group. This one map explains why a rotation of ordinary space cannot be tracked continuously by a single square root, why  $\pi_1(SO(3)) \cong \mathbb{Z}/2$ , and where the spin of quantum mechanics comes from. It is the first place in the book where the correspondence between a group and its Lie algebra does concrete work, and  $SU(2)$  becomes the running example that every part returns to: its maximal torus and Weyl group in the compact theory, its irreducible representations and their characters at the analytic summit, and its noncompact sibling  $SL_2(\mathbb{R})$  in the theory of real forms.

The starting point is the explicit description of  $SU(2)$  obtained at the end of the previous section, which realizes it as a sphere.

**Proposition 1.5.1:  $SU(2)$  is the 3-Sphere**

The special unitary group in dimension two is

$$SU(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} \mid \alpha, \beta \in \mathbb{C}, |\alpha|^2 + |\beta|^2 = 1 \right\}$$

and the map  $(\alpha, \beta) \mapsto \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}$  is a diffeomorphism from the unit sphere  $S^3 \subset \mathbb{C}^2 \cong \mathbb{R}^4$  onto  $SU(2)$ . In particular  $SU(2)$  is compact, connected, and simply connected.

**Proof:**

A matrix  $A \in U(2)$  has orthonormal columns; writing the first column as  $(\alpha, \beta)^T$  with  $|\alpha|^2 + |\beta|^2 = 1$ , the second column is a unit vector orthogonal to it, hence a unit scalar multiple  $\lambda(-\bar{\beta}, \bar{\alpha})^T$  with  $|\lambda| = 1$ . The determinant of the resulting matrix is  $\lambda(|\alpha|^2 + |\beta|^2) = \lambda$ , so the constraint  $\det A = 1$  defining  $SU(2)$  forces  $\lambda = 1$ , giving the displayed form. Conversely every matrix of that form is special unitary, as a direct check of  $A^*A = I$  and  $\det A = 1$  shows. The parametrization  $(\alpha, \beta) \mapsto A$  is thus a bijection from  $\{(\alpha, \beta) \in \mathbb{C}^2 \mid |\alpha|^2 + |\beta|^2 = 1\} = S^3$  onto  $SU(2)$ ; it is the restriction of a real-linear injection  $\mathbb{C}^2 \rightarrow M_2(\mathbb{C})$ , hence a smooth embedding, and its inverse (read off  $\alpha$  and  $\beta$  from the first column) is smooth, so it is a diffeomorphism.

The sphere  $S^3$  is compact, being closed and bounded in  $\mathbb{R}^4$ , and it is connected and simply connected:  $S^3$  minus a point is diffeomorphic to  $\mathbb{R}^3$  by stereographic projection, in which every loop contracts, and any loop in  $S^3$  misses a point because a continuous image of the interval cannot fill the three-sphere. These topological facts about spheres are established in [5]; they transfer to  $SU(2)$  through the diffeomorphism, completing the proof.

That  $SU(2)$  is a sphere is the geometric fact underlying everything that follows. Compactness is what will later force its representation theory to be completely reducible and its irreducible representations to be finite-dimensional; connectedness means it is generated by any neighborhood of the identity, so that its structure is determined by its Lie algebra; and simple connectedness is precisely the property that makes  $SU(2)$ , rather than  $SO(3)$ , the group corresponding to the shared Lie algebra  $\mathfrak{su}(2) \cong \mathfrak{so}(3)$  under the equivalence of Chapter 3. The three-sphere is also the unit group of the quaternions: writing a quaternion as  $q = \alpha + \beta j$  with  $\alpha, \beta \in \mathbb{C}$  identifies  $\{q \in \mathbb{H} \mid |q| = 1\}$  with the matrices above, so  $SU(2)$ ,  $S^3$ , and the unit quaternions are three names for one object.

The Lie algebra of  $SU(2)$  was identified in section 1.4 as the space of traceless skew-Hermitian  $2 \times 2$  matrices. Its concrete shape is worth having in view before exponentiating.

**Example 1.5.2: The Lie Algebra  $\mathfrak{su}(2)$  in Coordinates**

By theorem 1.4.3 the Lie algebra of  $SU(2)$  is

$$\mathfrak{su}(2) = \{X \in M_2(\mathbb{C}) \mid X^* = -X, \operatorname{tr} X = 0\}$$

the traceless skew-Hermitian matrices. Skew-Hermiticity forces the diagonal entries to be purely imaginary and the off-diagonal entries to be conjugate-negatives, and tracelessness makes the two diagonal entries opposite, so a general element is

$$X = \begin{pmatrix} ic & -\bar{b} \\ b & -ic \end{pmatrix}, \quad c \in \mathbb{R}, b \in \mathbb{C}$$

depending on the three real parameters  $c$  and  $\operatorname{Re} b, \operatorname{Im} b$ ; thus  $\dim_{\mathbb{R}} \mathfrak{su}(2) = 3 = 2^2 - 1$ , matching the dimension of  $SU(2)$  from proposition 1.2.6. A convenient basis is provided by the matrices

$$E_1 = \frac{1}{2} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad E_2 = \frac{1}{2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad E_3 = \frac{1}{2} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

which are  $\frac{1}{2i}$  times the Pauli matrices, and whose brackets are  $[E_1, E_2] = E_3$ ,  $[E_2, E_3] = E_1$ ,  $[E_3, E_1] = E_2$ , the cross-product relations of  $\mathbb{R}^3$ ; this is the identification  $\mathfrak{su}(2) \cong (\mathbb{R}^3, \times)$  recorded in example 1.4.7.

The determinant provides a natural quadratic form on this space, and its shape is what makes  $SU(2)$  act by rotations below. For  $X = \begin{pmatrix} ic & -\bar{b} \\ b & -ic \end{pmatrix} \in \mathfrak{su}(2)$ ,

$$\det X = (ic)(-ic) - (-\bar{b})(b) = c^2 + |b|^2 \tag{1.6}$$

a positive-definite quadratic form. Evaluating on the basis gives  $\det E_j = \frac{1}{4}$  for each  $j$ , and since  $\det(sX) = s^2 \det X$  for a scalar  $s$ , the form  $\det$  is  $\frac{1}{4}$  times the standard Euclidean squared norm in the coordinates dual to  $E_1, E_2, E_3$ ; equivalently  $\det X = -\frac{1}{2} \operatorname{tr}(X^2)$ , because for traceless  $X$  the Cayley-Hamilton relation gives  $X^2 = -(\det X)I$  and hence  $\operatorname{tr}(X^2) = -2 \det X$ . In either form the quadratic form is a scalar multiple of the trace form and is manifestly invariant under conjugation, the property that drives the double cover.

The exponential map sends  $\mathfrak{su}(2)$  into  $SU(2)$ , and because  $\mathfrak{su}(2)$  is three-dimensional the exponential admits a closed formula in the spirit of Euler's formula  $e^{i\theta} = \cos \theta + i \sin \theta$ . The mechanism is that every  $X \in \mathfrak{su}(2)$  satisfies a quadratic relation, so its powers cycle and the exponential series resums into cosine and sine.

**Proposition 1.5.3: The Exponential is Onto  $SU(2)$** 

For  $X \in \mathfrak{su}(2)$  set  $|X| = \sqrt{\det X}$ , so that  $X^2 = -|X|^2 I$ . Then

$$\exp X = \cos(|X|) I + \frac{\sin(|X|)}{|X|} X \quad (1.7)$$

interpreted as  $\exp 0 = I$  when  $X = 0$ . The map  $\exp: \mathfrak{su}(2) \rightarrow SU(2)$  is surjective.

**Proof:**

*Step 1: the quadratic relation.* By the Cayley-Hamilton theorem a traceless  $2 \times 2$  matrix  $X$  satisfies  $X^2 - (\operatorname{tr} X)X + (\det X)I = 0$ , which with  $\operatorname{tr} X = 0$  reads  $X^2 = -(\det X)I$ . For  $X \in \mathfrak{su}(2)$  the determinant is  $|X|^2 \geq 0$  by (1.6), so writing  $r = |X|$  gives  $X^2 = -r^2 I$ .

*Step 2: resumming the series.* Assume  $r > 0$ . The relation  $X^2 = -r^2 I$  gives  $X^{2k} = (-1)^k r^{2k} I$  and  $X^{2k+1} = (-1)^k r^{2k} X$  for all  $k \geq 0$ . Splitting the exponential series (absolutely convergent by definition 1.3.2) into even and odd powers,

$$\exp X = \sum_{k \geq 0} \frac{X^{2k}}{(2k)!} + \sum_{k \geq 0} \frac{X^{2k+1}}{(2k+1)!} = \left( \sum_{k \geq 0} \frac{(-1)^k r^{2k}}{(2k)!} \right) I + \left( \sum_{k \geq 0} \frac{(-1)^k r^{2k}}{(2k+1)!} \right) X$$

The first bracketed sum is  $\cos r$ ; the second is  $\frac{1}{r} \sum_k \frac{(-1)^k r^{2k+1}}{(2k+1)!} = \frac{\sin r}{r}$ . This is (1.7). For  $X = 0$  both sides equal  $I$ , and the right-hand side has the removable singularity  $\frac{\sin r}{r} \rightarrow 1$  as  $r \rightarrow 0$ , so the formula holds uniformly with the stated convention.

*Step 3: surjectivity.* Let  $A \in SU(2)$ ; by proposition 1.5.1 write  $A = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}$ . Then  $\operatorname{tr} A = \alpha + \bar{\alpha} = 2 \operatorname{Re} \alpha \in [-2, 2]$ , so there is  $r \in [0, \pi]$  with  $\cos r = \operatorname{Re} \alpha$ . If  $A = I$  then  $A = \exp 0$ ; if  $A = -I$  then  $\operatorname{Re} \alpha = -1$ ,  $r = \pi$ , and  $A = \exp(\pi Y)$  for any  $Y \in \mathfrak{su}(2)$  with  $|Y| = 1$ , since (1.7) gives  $\exp(\pi Y) = \cos \pi I + \sin \pi Y = -I$ . Otherwise  $\operatorname{Re} \alpha \in (-1, 1)$ , so  $\sin r > 0$ , and the matrix

$$X = \frac{r}{\sin r} (A - \operatorname{Re}(\alpha) I)$$

lies in  $\mathfrak{su}(2)$ : the matrix  $A - \operatorname{Re}(\alpha)I$  is traceless because  $\operatorname{tr} A = 2 \operatorname{Re} \alpha$ , and skew-Hermitian because its diagonal entries are  $\pm i \operatorname{Im} \alpha$  (purely imaginary) while its off-diagonal entries  $-\bar{\beta}, \beta$  are conjugate-negatives; scaling by the real factor  $r/\sin r$  preserves both properties. This  $X$  has  $|X| = r$ , since  $\det X = (r/\sin r)^2 \det(A - \operatorname{Re}(\alpha)I) = (r/\sin r)^2 (\operatorname{Im} \alpha)^2 + (r/\sin r)^2 |\beta|^2$  and  $\sin^2 r = 1 - \operatorname{Re}(\alpha)^2 = (\operatorname{Im} \alpha)^2 + |\beta|^2$  (using  $|\alpha|^2 + |\beta|^2 = 1$ ), giving  $\det X = r^2$ . Substituting into (1.7),

$$\exp X = \cos(r) I + \frac{\sin r}{r} \cdot \frac{r}{\sin r} (A - \operatorname{Re}(\alpha)I) = \operatorname{Re}(\alpha) I + A - \operatorname{Re}(\alpha) I = A$$

so  $A = \exp X$ . Every element of  $SU(2)$  is therefore an exponential, as we sought to show.

Formula (1.7) is the noncommutative avatar of Euler's formula: as  $t$  runs over  $\mathbb{R}$ , the curve  $t \mapsto \exp(tX)$

traces a great circle on the sphere  $SU(2)$ , periodic with period  $2\pi/|X|$ , and its image is the one-parameter subgroup generated by  $X$  from theorem 1.3.10. Surjectivity of the exponential is special to compact connected groups and is proved in general in Part III; for  $SU(2)$  it is visible by hand because the group is a sphere and every point lies on a great circle through the identity. The formula also makes the covering map computable: a rotation angle on the  $SU(2)$  side is  $|X|$ , and it corresponds to a rotation angle of  $2|X|$  on the  $SO(3)$  side, the source of the factor of two in the double cover.

**Example 1.5.4: One-Parameter Subgroups Through the Poles**

Take  $X = E_1 = \frac{1}{2} \text{diag}(i, -i)$ , so  $|E_1| = \sqrt{\det E_1} = \frac{1}{2}$  and  $tX = \frac{t}{2} \text{diag}(i, -i)$  has  $|tX| = |t|/2$ . Formula (1.7) gives

$$\exp(tE_1) = \cos\left(\frac{t}{2}\right)I + 2\sin\left(\frac{t}{2}\right)E_1 = \begin{pmatrix} e^{it/2} & 0 \\ 0 & e^{-it/2} \end{pmatrix}$$

directly recovering the diagonal exponential  $\exp \text{diag}(a, -a) = \text{diag}(e^a, e^{-a})$  with  $a = it/2$ . This one-parameter subgroup is the standard maximal torus of  $SU(2)$ , the diagonal circle, which the compact theory of Part III returns to. It has period  $4\pi$  in  $t$ , not  $2\pi$ : the parameter must run to  $4\pi$  before  $\exp(tE_1)$  returns to  $I$ , whereas the corresponding rotation of  $\mathbb{R}^3$  has period  $2\pi$ . The doubling is again the signature of the cover.

The group  $SO(3)$  is 3-dimensional, connected, and compact, with Lie algebra of the same dimension. Its Lie algebra  $\mathfrak{so}(3)$ , the real skew-symmetric  $3 \times 3$  matrices, is spanned by the infinitesimal rotations about the three coordinate axes,

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad L_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

whose brackets are  $[L_1, L_2] = L_3$ ,  $[L_2, L_3] = L_1$ ,  $[L_3, L_1] = L_2$ , the identical cross-product relations found for  $E_1, E_2, E_3$  in example 1.5.2. The linear map  $L_i \mapsto E_i$  is therefore an isomorphism of Lie algebras, so

$$\mathfrak{su}(2) \cong \mathfrak{so}(3) \cong (\mathbb{R}^3, \times) \tag{1.8}$$

all three being the three-dimensional real Lie algebra with the cross-product bracket. Concretely  $\mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$  via  $v \mapsto (w \mapsto v \times w)$ : the matrix  $L_v$  of “cross with  $v$ ” acts on  $w$  as  $v \times w$ , and  $L_{v \times v'} = [L_v, L_{v'}]$  recovers the Jacobi identity for the cross product. The isomorphism (1.8) predicts, by the correspondence of Chapter 3, a covering homomorphism between the associated connected groups. The rest of the section constructs it explicitly.

The construction is the adjoint action:  $SU(2)$  acts on its own Lie algebra by conjugation, and because  $\mathfrak{su}(2)$  carries the invariant quadratic form (1.6), this action lands in the orthogonal group of that form, which is a copy of  $SO(3)$ .

**Theorem 1.5.5: The Double Cover  $SU(2) \rightarrow SO(3)$**

For  $g \in SU(2)$  let  $\varphi(g)$  be the conjugation map

$$\varphi(g): \mathfrak{su}(2) \rightarrow \mathfrak{su}(2), \quad \varphi(g)(X) = gXg^{-1}$$

Identifying  $\mathfrak{su}(2)$  with  $\mathbb{R}^3$  via the basis  $E_1, E_2, E_3$ , with the Euclidean structure given by the form  $\det$  of (1.6), the assignment  $g \mapsto \varphi(g)$  is a homomorphism

$$\varphi: SU(2) \rightarrow SO(3)$$

It is surjective, and its kernel is exactly the center  $\{\pm I\}$ . Hence  $\varphi$  induces an isomorphism

$$SU(2)/\{\pm I\} \xrightarrow{\sim} SO(3)$$

and is a two-to-one covering map.

**Proof:**

*Step 1:  $\varphi(g)$  is a well-defined orthogonal transformation.* For  $g \in SU(2)$  and  $X \in \mathfrak{su}(2)$  the matrix  $gXg^{-1}$  is again traceless (trace is conjugation-invariant) and skew-Hermitian, since  $g^{-1} = g^*$  gives  $(gXg^{-1})^* = gX^*g^{-1} = -gXg^{-1}$ ; so  $\varphi(g)$  maps  $\mathfrak{su}(2)$  to itself, linearly, with inverse  $\varphi(g^{-1})$ . It preserves the form (1.6) because the determinant is conjugation-invariant,  $\det(\varphi(g)X) = \det(gXg^{-1}) = \det X$ . Thus  $\varphi(g)$  is a linear isometry of the Euclidean space  $(\mathfrak{su}(2), \det) \cong \mathbb{R}^3$ , that is  $\varphi(g) \in O(3)$ . That  $\varphi$  is a homomorphism is immediate:  $\varphi(gh)(X) = (gh)X(gh)^{-1} = g(hXh^{-1})g^{-1} = \varphi(g)\varphi(h)(X)$ , and  $\varphi(I) = \text{id}$ .

*Step 2: the image lies in  $SO(3)$ .* The group  $SU(2)$  is connected by proposition 1.5.1, and  $\varphi: SU(2) \rightarrow O(3)$  is continuous (its matrix entries are quadratic polynomials in the entries of  $g$ ), so the image  $\varphi(SU(2))$  is a connected subset of  $O(3)$ . The determinant  $\det \circ \varphi: SU(2) \rightarrow \{\pm 1\}$  is continuous on a connected space, hence constant, and it takes the value  $\det \varphi(I) = \det(\text{id}) = 1$ ; therefore  $\det \varphi(g) = 1$  for all  $g$ , so  $\varphi(SU(2)) \subseteq SO(3)$  and  $\varphi: SU(2) \rightarrow SO(3)$ .

*Step 3: the kernel is  $\{\pm I\}$ .* Suppose  $\varphi(g) = \text{id}$ , that is  $gXg^{-1} = X$  for all  $X \in \mathfrak{su}(2)$ , so  $g$  commutes with every traceless skew-Hermitian matrix. The matrices  $E_1, E_2, E_3$  span  $\mathfrak{su}(2)$  over  $\mathbb{R}$ , and together with  $I$  they span  $M_2(\mathbb{C})$  over  $\mathbb{C}$  (they are, up to scalars, the Pauli matrices, which with  $I$  form a basis of  $M_2(\mathbb{C})$ ); so  $g$  commutes with all of  $M_2(\mathbb{C})$ . An element commuting with every matrix is a scalar,  $g = \lambda I$ , and  $g \in SU(2)$  forces  $\lambda^2 = \det g = 1$ , hence  $\lambda = \pm 1$ . Thus  $\ker \varphi = \{\pm I\}$ , and since  $\pm I$  are the only scalars in  $SU(2)$ , this is exactly the center.

*Step 4: surjectivity.* Fix a unit vector  $u \in \mathfrak{su}(2)$ , so  $u^2 = -I$  by Step 1 of proposition 1.5.3. The one-parameter group  $s \mapsto \varphi(\exp(su))$  in  $SO(3)$  has derivative at  $s = 0$  equal to the operator  $\text{ad}(u)(X) = uX - Xu = [u, X]$  on  $\mathfrak{su}(2)$ , since  $\frac{d}{ds}\big|_0 \exp(su)X \exp(-su) = uX - Xu$  by proposition 1.3.4; hence  $\varphi(\exp(su)) = \exp(s \text{ad}(u))$  as maps of  $\mathfrak{su}(2)$ . Under the identification (1.8) the operator  $\text{ad}(u)$  is rotation generator about the axis  $u$ ; because the structure constants of  $E_1, E_2, E_3$  in the orthonormal frame of the form (1.6) are twice those of the geometric generators  $L_i$  (each  $E_j$  has  $\det E_j = \frac{1}{4}$ , i.e. Euclidean length  $\frac{1}{2}$ , while the  $L_i$  have length 1), the flow  $\exp(s \text{ad}(u))$  is the rotation of  $\mathbb{R}^3$  by angle  $2s$  about the axis  $u$ . Consequently, given  $R \in SO(3)$ ,

Euler's theorem writes  $R$  as rotation by some angle  $\theta$  about some unit axis  $u \in S^2 \subset \mathfrak{su}(2)$ , and then  $g = \exp(\frac{\theta}{2}u)$  satisfies  $\varphi(g) = \exp(\frac{\theta}{2}\text{ad}(u)) = \text{rotation by } \theta \text{ about } u = R$ . As  $R$  was arbitrary,  $\varphi$  is surjective.

*Step 5: the covering.* By Steps 2 through 4,  $\varphi: SU(2) \rightarrow SO(3)$  is a surjective homomorphism with kernel  $\{\pm I\}$ , so the first isomorphism theorem for groups ([1]) gives a group isomorphism  $SU(2)/\{\pm I\} \xrightarrow{\sim} SO(3)$ . Both sides are compact three-dimensional manifolds, and the induced map is a continuous bijection between compact Hausdorff spaces, hence a homeomorphism; its differential at the identity is the Lie algebra isomorphism (1.8) (Step 4 computed  $d\varphi_I = \frac{1}{2}\text{ad}$  under the frames, a linear isomorphism), so it is a diffeomorphism. Finally  $\varphi$  is two-to-one with fibers  $\{g, -g\} = g \cdot \ker \varphi$ , and since  $d\varphi_I$  is an isomorphism  $\varphi$  is a local diffeomorphism at  $I$ , spread to every point by left translation; a two-to-one local diffeomorphism between compact manifolds with discrete fibers is a two-sheeted covering map, completing the proof.

The factor of two is the arithmetic heart of the cover, and it is worth isolating. For a unit vector  $u \in \mathfrak{su}(2)$  the element  $\exp(\frac{\theta}{2}u)$  on the  $SU(2)$  side maps under  $\varphi$  to rotation by the full angle  $\theta$  on the  $SO(3)$  side: the group parameter is half the spatial rotation angle. This is exactly the discrepancy seen in example 1.5.4, where  $\exp(tE_1)$  had period  $4\pi$  while its image rotation of  $\mathbb{R}^3$  had period  $2\pi$ . Passing to  $-I$  at the halfway point  $t = 2\pi$  is the mechanism: the two elements  $\pm g$  of a fiber are the two lifts of a single rotation, and no continuous choice of lift exists over all of  $SO(3)$ .

The theorem says that  $SO(3)$  is  $SU(2)$  with antipodal points of the sphere  $S^3$  glued together, which is the standard model of three-dimensional real projective space. So  $SO(3) \cong \mathbb{R}P^3$  as a manifold: the rotation group of ordinary space is real projective three-space. The two preimages  $\pm g$  of a rotation  $R = \varphi(g)$  are the two ways to represent that rotation by a unit quaternion, differing by a sign, and the impossibility of choosing the sign continuously over all of  $SO(3)$  is the failure of  $SO(3)$  to be simply connected, computed next.

#### Corollary 1.5.6: The Fundamental Group of $SO(3)$

The map  $\varphi: SU(2) \rightarrow SO(3)$  is the universal cover of  $SO(3)$ , and

$$\pi_1(SO(3)) \cong \mathbb{Z}/2$$

The generator is any loop in  $SO(3)$  that lifts to a path in  $SU(2)$  from  $I$  to  $-I$ , such as the image under  $\varphi$  of  $t \mapsto \exp(tE_1)$ ,  $t \in [0, 2\pi]$ ; this loop is not contractible, but traversing it twice is.

#### Proof:

By proposition 1.5.1 the total space  $SU(2) \cong S^3$  is simply connected, and by theorem 1.5.5 the map  $\varphi$  is a covering with fiber the two-element group  $\{\pm I\}$ . A covering map with simply-connected total space is the universal cover of its base, and for a universal cover arising from a central subgroup (here  $\{\pm I\}$ ) the fundamental group of the base is the deck group, which is the kernel:  $\pi_1(SO(3)) \cong \ker \varphi = \{\pm I\} \cong \mathbb{Z}/2$ . These covering-space facts are proved in [5].

For the explicit generator, the path  $\gamma(t) = \exp(tE_1)$  for  $t \in [0, 2\pi]$  runs in  $SU(2)$  from  $\gamma(0) = I$  to  $\gamma(2\pi) = \text{diag}(e^{i\pi}, e^{-i\pi}) = -I$  by example 1.5.4, so  $\varphi \circ \gamma$  is a loop in  $SO(3)$  (it returns to  $\varphi(-I) = \varphi(I) = \text{id}$ ) whose lift  $\gamma$  has distinct endpoints  $I \neq -I$ . A loop whose lift to the universal cover is not closed is noncontractible, so  $\varphi \circ \gamma$  generates  $\pi_1(SO(3))$ . Doubling it,  $t \in [0, 4\pi]$ ,

produces a lift  $\gamma(t)$  from  $I$  back to  $\gamma(4\pi) = I$ , a closed lift, so the doubled loop is contractible; the generator therefore has order two, completing the proof.

The order-two element of  $\pi_1(SO(3))$  is the mathematics behind the physical fact that a continuous  $2\pi$  rotation of a rigid body is not homotopic to no rotation, while a  $4\pi$  rotation is: the “belt trick”, in which a  $360^\circ$  twist in a belt cannot be undone without moving the ends but a  $720^\circ$  twist can, is a demonstration of a noncontractible loop in  $SO(3)$  together with its contractible square. In quantum mechanics the same two-to-one relationship is why the state of a spin- $\frac{1}{2}$  particle acquires a sign under a  $2\pi$  rotation and returns to itself only after  $4\pi$ : the physical symmetry group is  $SO(3)$ , but the group acting on wavefunctions is its double cover  $SU(2)$ , and the sign is the nontrivial element of the kernel. The representations that see this sign, the ones that do not descend to  $SO(3)$ , are the half-integer-spin representations of  $SU(2)$  classified in Part III.

This section launches the running example that threads the rest of the book. The group  $SU(2)$  reappears in each part, always as the smallest place to see a new phenomenon in full: its diagonal circle is the maximal torus whose conjugates cover the group, and the reflection swapping  $\text{diag}(z, \bar{z})$  to  $\text{diag}(\bar{z}, z)$  is the Weyl group  $\mathbb{Z}/2$ , both taken up in the compact theory of maximal tori. Its irreducible representations are the symmetric powers of the standard two-dimensional representation, one in each dimension, and their characters are the functions  $\chi_n(\theta) = \sin((n+1)\theta)/\sin\theta$  that the Weyl character formula produces as its first nontrivial instance. Its noncompact sibling  $SL_2(\mathbb{R})$ , sharing the complexified Lie algebra  $\mathfrak{sl}_2(\mathbb{C})$  but replacing the sphere by a group with a hyperbolic plane inside it, is the running example of the theory of real forms and the Iwasawa decomposition in Part IV. Every one of these returns to the concrete matrix computations of this section.

### Exercise 1.5.1

1. Using (1.7), compute  $\exp(\theta E_3)$  explicitly as a  $2 \times 2$  complex matrix, and verify that  $\exp(2\pi E_3) = -I$  while  $\exp(4\pi E_3) = I$ . Interpret the period  $4\pi$  in terms of the double cover.
2. Show that the map sending the quaternion  $q = a + bi + cj + dk$  (with  $a, b, c, d \in \mathbb{R}$ ) to the matrix  $\begin{pmatrix} a+bi & c+di \\ -c+di & a-bi \end{pmatrix}$  restricts to an isomorphism from the group of unit quaternions  $\{q \mid |q| = 1\}$  onto  $SU(2)$ , and that under this identification  $\mathfrak{su}(2)$  is the space of purely imaginary quaternions  $bi + cj + dk$ .
3. Fix  $g = \exp(\frac{\theta}{2} E_3)$ . Compute  $\varphi(g)$  as a  $3 \times 3$  matrix in the basis  $E_1, E_2, E_3$  by evaluating  $gE_1g^{-1}$ ,  $gE_2g^{-1}$ , and  $gE_3g^{-1}$ , and confirm that  $\varphi(g)$  is the rotation of  $\mathbb{R}^3$  by angle  $\theta$  about the third axis.
4. Prove that the center of  $SU(2)$  is exactly  $\{\pm I\}$ , and deduce from theorem 1.5.5 that  $SO(3)$  has trivial center.
5. Show that there is no continuous group homomorphism  $\psi: SO(3) \rightarrow SU(2)$  with  $\varphi \circ \psi = \text{id}_{SO(3)}$ ; that is, the double cover does not split. Deduce that  $SU(2)$  and  $SO(3) \times \{\pm I\}$  are not isomorphic as Lie groups.
6. Explain why  $SU(2)$  and  $SO(3)$  are not isomorphic as Lie groups despite having isomorphic Lie algebras, and reconcile this with the principle that a Lie algebra determines its simply-connected group uniquely.



## 2

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# *Lie Groups and Their Lie Algebras*

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The previous chapter met the classical groups at the surface, as sets of matrices cut out by explicit equations, and produced the two objects on which the theory turns, the exponential map and the Lie algebra, by differentiating matrix-valued functions. Nothing in those constructions truly required the ambient matrices. The tangent space at the identity, the one-parameter subgroups, the bracket: each was a manifestation of the group law and the smooth structure alone, and each survives when the matrices are removed and only a manifold carrying a compatible group law remains. That manifold is a Lie group in the sense of this chapter, and the task now is to rebuild the machinery of chapter 1 intrinsically, on any Lie group, with no embedding into  $\mathrm{GL}_n$  presupposed.

The mechanism that replaces the matrix computations is left-invariance. A single tangent vector at the identity spreads, by left translation, to a vector field defined over the whole group; the vector fields arising this way are closed under the Lie bracket of vector fields, and this bracket makes the tangent space at the identity a Lie algebra  $\mathfrak{g}$ , the intrinsic counterpart of the matrix Lie algebra of chapter 1. The flows of these left-invariant fields are the one-parameter subgroups, and assembling them at time one defines the exponential map  $\exp: \mathfrak{g} \rightarrow G$ . Differentiating the action of the group on itself by conjugation produces the adjoint representations  $\mathrm{Ad}$  and  $\mathrm{ad}$ , and recovers the bracket in the form  $[X, Y] = \mathrm{ad}(X)Y$ . For a matrix group every one of these constructions returns the answer already computed by hand: the intrinsic exponential is the matrix exponential,  $\mathrm{Ad}(g)$  is conjugation  $X \mapsto gXg^{-1}$ , and  $\mathrm{ad}(X)$  is the commutator  $Y \mapsto XY - YX$ . The abstract theory is thus anchored at every step to the concrete picture of the first chapter, and the reader who keeps  $\mathrm{SU}(2)$  in view loses nothing by the ascent to generality.

## 2.1 Lie Groups: Definition, Examples, and the Category of Lie Groups

Chapter 1 studied groups of matrices: closed subgroups of  $\mathrm{GL}_n(\mathbb{F})$ , cut out by polynomial equations, whose smooth structure came for free from the ambient vector space  $M_n(\mathbb{F})$ . Every construction there was anchored in coordinates. This chapter rebuilds the same theory intrinsically. A Lie group will be defined as a manifold carrying a compatible group structure, with no reference to an ambient  $\mathrm{GL}_n(\mathbb{F})$ , and the tangent space at the identity will replace the concrete matrix Lie algebra. The payoff is generality: the theorems apply to groups that are not presented as matrix groups, and the constructions become functorial. The cost is that the smoothness which was automatic for matrix groups must now be built into the definition. This first section pays that cost, sets up the category of Lie groups, and records the first structural consequence of merging manifold and group, namely that a Lie group looks the same near every one of its points.

The unifying theme of the section is that the group structure of a Lie group is rigid in a way that an abstract group's is not: multiplication is a smooth map, so it can be differentiated, and the single operation of left translation by a group element is simultaneously a group-theoretic bijection and a diffeomorphism of manifolds. This dual nature is the source of every result below. Left translation transports the neighborhood of the identity to the neighborhood of any point, forcing the manifold to be homogeneous; it trivializes the tangent bundle, forcing parallelizability; and it preserves orientation, forcing orientability. Each of these is a geometric fact about  $G$  read off from a purely algebraic operation.

### Definition 2.1.1: Lie Group

A *Lie group* is a smooth manifold  $G$  that is also a group, such that the multiplication and inversion maps

$$m: G \times G \rightarrow G, \quad (g, h) \mapsto gh, \quad \iota: G \rightarrow G, \quad g \mapsto g^{-1}$$

are smooth, where  $G \times G$  carries the product smooth structure. The dimension of  $G$  as a Lie group is its dimension as a manifold.

The two smoothness requirements can be collapsed into one. Smoothness of  $m$  and  $\iota$  is equivalent to smoothness of the single map

$$\mu: G \times G \rightarrow G, \quad (g, h) \mapsto gh^{-1}$$

Indeed, if  $m$  and  $\iota$  are smooth then  $\mu = m \circ (\mathrm{id}_G \times \iota)$  is a composite of smooth maps, hence smooth. Conversely, given  $\mu$  smooth, inversion is recovered as  $\iota(g) = \mu(e, g)$ , the restriction of  $\mu$  to the smooth submanifold  $\{e\} \times G$ , so  $\iota$  is smooth; and then  $m(g, h) = \mu(g, \iota(h)) = \mu(g, h^{-1})$  is again a composite of smooth maps. The one-map formulation is the more economical to verify in examples, and it is the version used to check that a subset of a Lie group inheriting the two operations is itself a Lie group.

The manifold structure and the group structure are not independent data tacked together; the smoothness axiom forces them to interact. The first and most basic interaction is that the group elements act on  $G$  by diffeomorphisms. This is the intrinsic replacement for the observation, made throughout Chapter 1, that multiplication by a fixed matrix is a smooth invertible self-map of  $\mathrm{GL}_n(\mathbb{F})$  with smooth inverse, visible directly from proposition 1.1.2.

**Proposition 2.1.2: Translations and Conjugations are Diffeomorphisms**

Let  $G$  be a Lie group and  $g \in G$ . The *left translation*, *right translation*, and *conjugation* maps

$$L_g: G \rightarrow G, \quad h \mapsto gh, \quad R_g: G \rightarrow G, \quad h \mapsto hg, \quad C_g: G \rightarrow G, \quad h \mapsto ghg^{-1}$$

are diffeomorphisms of  $G$ , with inverses  $L_g^{-1} = L_{g^{-1}}$ ,  $R_g^{-1} = R_{g^{-1}}$ , and  $C_g^{-1} = C_{g^{-1}}$ . Consequently:

1.  $G$  is *homogeneous*: for any two points  $g, h \in G$  the diffeomorphism  $L_{hg^{-1}}$  carries  $g$  to  $h$ , so  $G$  looks the same near every point.
2.  $G$  is *parallelizable*: its tangent bundle  $TG$  is trivial,  $TG \cong G \times T_e G$ .
3.  $G$  is *orientable*.

**Proof:**

*Step 1: the translations are diffeomorphisms.* Fix  $g \in G$ . Left translation  $L_g$  is the composite

$$G \xrightarrow{h \mapsto (g, h)} G \times G \xrightarrow{m} G$$

of the smooth map  $h \mapsto (g, h)$  (smooth because its two components, the constant  $g$  and the identity, are smooth) with the smooth multiplication  $m$  of definition 2.1.1, so  $L_g$  is smooth. The group axioms give  $L_g \circ L_{g^{-1}} = L_{gg^{-1}} = L_e = \text{id}_G$  and likewise  $L_{g^{-1}} \circ L_g = \text{id}_G$ , so  $L_g$  is a bijection with inverse  $L_{g^{-1}}$ , which is smooth by the same argument. Hence  $L_g$  is a diffeomorphism. The identical argument, using  $h \mapsto (h, g)$  and the relation  $R_g \circ R_{g^{-1}} = R_{g^{-1}g} = \text{id}_G$ , shows  $R_g$  is a diffeomorphism with inverse  $R_{g^{-1}}$ . Note that  $C_g = L_g \circ R_{g^{-1}}$  is a composite of two diffeomorphisms, hence a diffeomorphism, with inverse  $C_{g^{-1}} = L_{g^{-1}} \circ R_g$ .

*Step 2: homogeneity.* Given  $g, h \in G$ , set  $k = hg^{-1}$ . Then  $L_k$  is a diffeomorphism by Step 1, and  $L_k(g) = (hg^{-1})g = h$ . A diffeomorphism carrying  $g$  to  $h$  identifies a neighborhood of  $g$  with a neighborhood of  $h$  compatibly with the smooth structure, so every point of  $G$  has a neighborhood diffeomorphic to a neighborhood of any other point. This is the assertion that  $G$  is homogeneous.

*Step 3: parallelizability.* For each  $g \in G$  the differential of left translation at the identity,

$$(dL_g)_e: T_e G \rightarrow T_g G$$

is a linear isomorphism, since  $L_g$  is a diffeomorphism carrying  $e$  to  $g$ . Define

$$\Psi: G \times T_e G \rightarrow TG, \quad (g, v) \mapsto (dL_g)_e(v)$$

The map  $\Psi$  is a bijection: for each  $g$  it restricts to the linear isomorphism  $(dL_g)_e: \{g\} \times T_e G \rightarrow T_g G$  onto the fiber over  $g$ . That  $\Psi$  and its inverse are smooth follows from the smoothness of  $m$ : the assignment  $(g, h) \mapsto (dL_g)_h$  is smooth because  $L_g$  is built from  $m$ , so  $\Psi$  is a smooth bundle isomorphism covering the identity of  $G$ . Its existence is exactly the statement that  $TG$  is trivial,  $TG \cong G \times T_e G$ , so  $G$  is parallelizable.

*Step 4: orientability.* A parallelizable manifold is orientable. Fix any basis  $v_1, \dots, v_n$  of  $T_e G$ , where  $n = \dim G$ . By Step 3 the vector fields  $g \mapsto (dL_g)_e(v_i)$  are smooth and, at each point  $g$ , their values  $(dL_g)_e(v_1), \dots, (dL_g)_e(v_n)$  form a basis of  $T_g G$ , since  $(dL_g)_e$  is a linear isomorphism.

This global smooth frame determines an orientation of  $G$ : declare a basis of  $T_g G$  positively oriented when it has positive determinant relative to the frame. The frame is smooth, so the orientation is consistent across overlaps, and  $G$  is orientable, completing the proof.

The proposition is short, but it already separates Lie groups from arbitrary manifolds. Homogeneity says  $G$  has no distinguished point apart from the identity, and even the identity is distinguished only by the group structure, not by the geometry: near  $e$  and near any other  $g$  the manifold is indistinguishable. This is why so much of Lie theory is local analysis at the identity, the tangent space  $T_e G$  being the germ from which the whole group is reconstructed. Parallelizability is the geometric engine of the entire book. The trivialization  $TG \cong G \times T_e G$  says that a tangent vector at  $e$  can be translated to a well-defined tangent vector at every point, producing a global vector field; these translated fields are the left-invariant vector fields of section 2.2, and their bracket is what makes  $T_e G$  a Lie algebra. Orientability then follows for free, which already rules out some manifolds as Lie groups: no even-dimensional real projective space  $\mathbb{RP}^{2k}$  can be a Lie group, since it is not orientable.

The abstraction is only as convincing as the examples it captures. The matrix groups of Chapter 1 are the first source, and every group named there is a Lie group in the sense of definition 2.1.1. The next example collects these together with the smallest new examples, the circle and its products, which are not naturally presented as matrix groups at all.

### Example 2.1.3: First Examples of Lie Groups

The following are Lie groups.

1. *The general linear group*  $\mathrm{GL}_n(\mathbb{F})$ , for  $\mathbb{F} = \mathbb{R}$  or  $\mathbb{C}$ . proposition 1.1.2 establishes exactly the content of definition 2.1.1:  $\mathrm{GL}_n(\mathbb{F})$  is an open submanifold of  $M_n(\mathbb{F})$ , and multiplication and inversion are smooth. Its dimension is  $n^2$  over  $\mathbb{R}$  and  $2n^2$  over  $\mathbb{C}$ . This is the ambient Lie group inside which the classical groups sit.
2. *The circle*  $S^1 = \mathrm{U}(1)$ . Regard  $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$  as the unit circle in  $\mathbb{C}$ , a one-dimensional smooth manifold, with group operation the multiplication of complex numbers. In the angle coordinate  $z = e^{i\theta}$  the product is  $e^{i\theta}e^{i\varphi} = e^{i(\theta+\varphi)}$  and the inverse is  $e^{-i\theta}$ , both smooth in  $\theta$ , so  $S^1$  is a one-dimensional abelian Lie group. As a group of  $1 \times 1$  unitary matrices it is exactly  $\mathrm{U}(1)$  of definition 1.2.5, and example 1.2.11 already identified  $\mathrm{SO}(2) \cong \mathrm{U}(1) \cong S^1$ .
3. *The tori*  $\mathbb{T}^n = (S^1)^n$ . The  $n$ -fold product of circles is a compact connected abelian Lie group of dimension  $n$ , with coordinatewise multiplication. Equivalently  $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ , the quotient of the additive group  $\mathbb{R}^n$  by the lattice  $\mathbb{Z}^n$ , with the quotient smooth structure. These are the maximal tori that control the structure theory of compact groups in Part III.
4. *Direct products*. If  $G$  and  $H$  are Lie groups, the product manifold  $G \times H$  with componentwise multiplication  $(g_1, h_1)(g_2, h_2) = (g_1g_2, h_1h_2)$  is a Lie group of dimension  $\dim G + \dim H$ ; multiplication and inversion are smooth because they are so in each factor. The tori are the special case  $G = H = S^1$  iterated.
5. *Discrete groups*. Any group  $\Gamma$  equipped with the discrete topology is a zero-dimensional Lie group: a 0-manifold is a countable set of points, and every map out of a discrete space is smooth, so  $m$  and  $\iota$  are automatically smooth. Finite groups and the additive group  $\mathbb{Z}$  are Lie groups in this way, of dimension 0.
6. *The classical groups*. The groups  $\mathrm{SL}_n(\mathbb{F})$ ,  $\mathrm{O}(n)$ ,  $\mathrm{SO}(n)$ ,  $\mathrm{U}(n)$ ,  $\mathrm{SU}(n)$ , and  $\mathrm{Sp}(2n, \mathbb{F})$  of section 1.2 are each a smooth embedded submanifold of some  $\mathrm{GL}_N(\mathbb{F})$ , cut out as a regular-value preimage, on which the ambient multiplication and inversion restrict to smooth maps.

Each is therefore a Lie group. They are moreover *Lie subgroups* of  $\mathrm{GL}_N(\mathbb{F})$ : embedded submanifolds that are also subgroups. That every closed subgroup of a Lie group is automatically an embedded Lie subgroup, with no separate verification of the manifold structure required, is Cartan's closed subgroup theorem, proved in Chapter 3; here the manifold structure was already supplied by the regular value theorem in section 1.2.

7. *The running example*  $\mathrm{SU}(2)$ . By proposition 1.5.1,  $\mathrm{SU}(2)$  is diffeomorphic to the 3-sphere  $S^3$ , and it is a compact, connected, simply connected Lie group of dimension 3. It is the smallest nonabelian compact Lie group and the thread this book follows from here.

The circle deserves a second look, because it exhibits a feature invisible in the matrix examples: a Lie group need not come with a preferred embedding into  $\mathrm{GL}_n(\mathbb{F})$ . The description  $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$  presents the torus as a quotient, with no matrices at all, and the two descriptions of  $S^1$ , as unit complex numbers and as  $\mathbb{R}/\mathbb{Z}$ , are related by the smooth surjective homomorphism  $\mathbb{R} \rightarrow S^1$ ,  $t \mapsto e^{2\pi it}$ , whose kernel is  $\mathbb{Z}$ . This homomorphism is the exponential map of the circle in disguise, and the fact that  $S^1$  is a quotient of its own tangent line by a discrete subgroup is the one-dimensional shadow of the covering-group theory of Chapter 4. That the same abstract Lie group admits several concrete presentations is precisely the freedom the intrinsic definition buys.

To organize maps between Lie groups one asks for the structure-preserving maps: those respecting both the group law and the smooth structure. Requiring smoothness of a group homomorphism is a strong condition, and Chapter 3 will show it is even automatic for continuous homomorphisms, so the class of morphisms is more robust than it first appears.

#### Definition 2.1.4: Homomorphism of Lie Groups

Let  $G$  and  $H$  be Lie groups. A *homomorphism of Lie groups* is a map  $\varphi: G \rightarrow H$  that is both a group homomorphism,  $\varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2)$  for all  $g_1, g_2 \in G$ , and a smooth map of manifolds. It is an *isomorphism of Lie groups* if it is in addition a bijection whose inverse  $\varphi^{-1}: H \rightarrow G$  is also a homomorphism of Lie groups, that is, a group isomorphism that is a diffeomorphism.

A homomorphism of Lie groups automatically commutes with the translation structure of proposition 2.1.2: from  $\varphi(gh) = \varphi(g)\varphi(h)$  one reads  $\varphi \circ L_g = L_{\varphi(g)} \circ \varphi$ , so  $\varphi$  intertwines left translation on  $G$  with left translation on  $H$ . This compatibility is what will force the differential  $d\varphi_e$  at the identity to determine  $\varphi$  on all of a connected  $G$ , the theme of section 2.4. The determinant furnishes the running example of a homomorphism:  $\det: \mathrm{GL}_n(\mathbb{F}) \rightarrow \mathbb{F}^\times$  is a group homomorphism, and it is smooth because the determinant is a polynomial in the matrix entries, so it is a homomorphism of Lie groups whose kernel is the Lie subgroup  $\mathrm{SL}_n(\mathbb{F})$  of definition 1.2.1. The exponential  $\mathbb{R} \rightarrow S^1$ ,  $t \mapsto e^{2\pi it}$ , and the inclusion  $\mathrm{SU}(2) \hookrightarrow \mathrm{GL}_2(\mathbb{C})$  are two further homomorphisms already in hand.

With objects and morphisms fixed, the class of Lie groups assembles into a category. Naming it is not idle bookkeeping: the entire program of the book is to compare this category with the category of Lie algebras, and the functor  $\mathrm{Lie}$  constructed in section 2.4 is the bridge between them.

### Proposition 2.1.5: The Category of Lie Groups

Lie groups and their homomorphisms form a category **LieGrp**: the composite of two Lie group homomorphisms is a Lie group homomorphism, the identity map of a Lie group is a homomorphism, and composition is associative. An isomorphism in this category, in the categorical sense of a morphism with a two-sided inverse morphism, is exactly an isomorphism of Lie groups in the sense of definition 2.1.4.

#### Proof:

If  $\varphi: G \rightarrow H$  and  $\psi: H \rightarrow K$  are Lie group homomorphisms, then  $\psi \circ \varphi$  is a group homomorphism (the composite of group homomorphisms) and a smooth map (the composite of smooth maps), hence a Lie group homomorphism. For each Lie group  $G$  the identity map  $\text{id}_G$  is a group homomorphism and is smooth, so it is a Lie group homomorphism, and it is a two-sided identity for composition. Associativity of composition holds because composition of set maps is associative. Thus **LieGrp** is a category.

For the identification of isomorphisms: a morphism  $\varphi: G \rightarrow H$  with a two-sided inverse morphism  $\varphi^{-1}: H \rightarrow G$  in **LieGrp** is a bijection whose inverse is a Lie group homomorphism, hence a group isomorphism that is a diffeomorphism, which is a Lie group isomorphism by definition 2.1.4. Conversely a Lie group isomorphism is by definition a homomorphism admitting an inverse homomorphism, which is a two-sided inverse morphism in **LieGrp**, completing the proof.

One structural feature of a Lie group has no analogue for a bare topological group and organizes much of what follows: the piece of  $G$  containing the identity is itself a group. Let  $G^0$  denote the connected component of  $G$  containing the identity  $e$ , the *identity component*. Because  $G$  is a manifold it is locally connected, so its connected components are open; in particular  $G^0$  is an open subset of  $G$ . It is a subgroup: the map  $(g, h) \mapsto gh^{-1}$  is continuous and sends  $G^0 \times G^0$ , a connected set containing  $(e, e)$ , into a connected subset of  $G$  containing  $ee^{-1} = e$ ; a connected set containing  $e$  lies in the component  $G^0$ , so  $gh^{-1} \in G^0$  for all  $g, h \in G^0$ , and  $G^0$  is a subgroup. It is normal: for any  $g \in G$  the conjugation  $C_g$  of proposition 2.1.2 is a homeomorphism fixing  $e$ , so it carries the component  $G^0$  onto the component of  $C_g(e) = e$ , which is again  $G^0$ ; thus  $gG^0g^{-1} = G^0$ . Being open and a subgroup,  $G^0$  has open cosets, so its cosets partition  $G$  into the connected components, and the quotient  $G/G^0$  is a discrete group, the *component group*. For  $\text{GL}_n(\mathbb{R})$  this recovers the two components of proposition 1.1.3, with  $G^0 = \text{GL}_n^+(\mathbb{R})$  and component group  $\mathbb{Z}/2$ ; for the connected classical groups of proposition 1.2.10 one has  $G^0 = G$ . The subgroup and quotient theory that turns  $G^0$  and  $G/G^0$  into the first tools for dissecting a Lie group is developed in full in Chapter 3.

### Definition 2.1.6: Identity Component

The *identity component*  $G^0$  of a Lie group  $G$  is the connected component of  $G$  containing the identity element  $e$ . It is an open normal subgroup of  $G$ , and the quotient  $G/G^0$  is the *component group*, a discrete group whose elements are the connected components of  $G$ .

The identity component isolates the part of  $G$  accessible to the infinitesimal methods of this book. Everything the Lie algebra sees lies in  $G^0$ : the exponential image, the one-parameter subgroups, and the neighborhood of the identity all sit inside the connected piece, and section 2.3 will show that a connected Lie group is generated by any neighborhood of  $e$ , so  $G^0$  is generated by exponentials. The passage from  $G^0$  to  $G$  is the passage from the differential-geometric to the discrete: it is governed by

the finite or countable component group, not by the Lie algebra, and this is why the theorems of Parts II and III are stated for connected groups. Two Lie groups with the same Lie algebra can differ only in their component groups and their fundamental groups, the discrete invariants the tangent space cannot detect, exactly as  $SU(2)$  and  $SO(3)$  share the Lie algebra  $\mathfrak{su}(2) \cong \mathfrak{so}(3)$  of example 1.4.7 while differing by the factor  $\mathbb{Z}/2$  of theorem 1.5.5.

### Exercise 2.1.1

1. Let  $G$  and  $H$  be Lie groups. Prove in full that the product manifold  $G \times H$ , with the componentwise group operation, is a Lie group of dimension  $\dim G + \dim H$ , by exhibiting the map  $((g_1, h_1), (g_2, h_2)) \mapsto (g_1 g_2^{-1}, h_1 h_2^{-1})$  as smooth and invoking the one-map criterion following definition 2.1.1. Identify the tangent space  $T_{(e,e)}(G \times H)$  with  $T_e G \oplus T_e H$ .
2. Let  $G$  be a Lie group with identity  $e$  and inversion  $\iota: G \rightarrow G$ . Using that  $m(g, \iota(g)) = e$  is constant, differentiate at  $e$  to prove that the differential of inversion at the identity is minus the identity,

$$d\iota_e = -\text{id}_{T_e G}$$

(Hint: the differential of  $m$  at  $(e, e)$  is  $dm_{(e,e)}(v, w) = v + w$  under the identification  $T_{(e,e)}(G \times G) \cong T_e G \oplus T_e G$ ; establish this first from  $m(g, e) = g$  and  $m(e, h) = h$ .)

3. Let  $\varphi: G \rightarrow H$  be a homomorphism of Lie groups. Prove that  $\varphi$  has constant rank, that is, the rank of the differential  $d\varphi_g: T_g G \rightarrow T_{\varphi(g)} H$  is the same for every  $g \in G$ . (Hint: use the intertwining relation  $\varphi \circ L_g = L_{\varphi(g)} \circ \varphi$  and that translations are diffeomorphisms, proposition 2.1.2, to relate  $d\varphi_g$  to  $d\varphi_e$  by isomorphisms.)
4. Prove that  $\mathbb{T}^n = (S^1)^n$  is a compact, connected, abelian Lie group of dimension  $n$ , and that the map  $\mathbb{R}^n \rightarrow \mathbb{T}^n$ ,  $(t_1, \dots, t_n) \mapsto (e^{2\pi i t_1}, \dots, e^{2\pi i t_n})$ , is a surjective homomorphism of Lie groups with kernel  $\mathbb{Z}^n$ . Conclude that  $\mathbb{T}^n \cong \mathbb{R}^n / \mathbb{Z}^n$  as groups.
5. Let  $G$  be a Lie group. Prove that the identity component  $G^0$  is open in  $G$ , and that it is closed. Deduce that the connected components of  $G$  are exactly the cosets  $gG^0$ , that each is open and closed, and that  $G^0$  is the smallest open subgroup of  $G$ . (Hint: an open subgroup is closed because its complement is a union of its nontrivial cosets, each of which is open.)
6. Using proposition 2.1.2, explain why the even-dimensional real projective spaces  $\mathbb{RP}^{2k}$  and the Klein bottle carry no Lie group structure, while the odd-dimensional spheres  $S^1$  and  $S^3$  do. Which single property from proposition 2.1.2 does the obstruction rest on, and why is that property alone not enough to guarantee a manifold is a Lie group?

## 2.2 Left-Invariant Vector Fields and the Lie Algebra

Chapter 1 built the Lie algebra of a matrix group  $G \subseteq GL_n(\mathbb{F})$  by hand, as the space of matrices  $X$  whose one-parameter subgroups  $t \mapsto \exp(tX)$  stay inside  $G$ , and identified it with the tangent space  $T_I G$ . That construction leaned on the ambient matrix ring at every step: the exponential series, the commutator  $XY - YX$ , the trace. A general Lie group carries no ambient ring, so the same object must be built from the manifold alone. The raw material is the tangent space  $T_e G$  at the identity, a vector space of the right dimension but with no bracket in sight. The task of this section is to manufacture the bracket intrinsically, from nothing but the group multiplication and the differential

calculus of [2].

The mechanism is left translation. A single tangent vector  $v \in T_e G$  can be transported to every point  $g$  of the group by pushing it forward along the diffeomorphism  $L_g: h \mapsto gh$ , producing a vector field on all of  $G$  that is, by construction, insensitive to the group's own left multiplication. Such *left-invariant* vector fields form a vector space canonically isomorphic to  $T_e G$ , and, because the Lie bracket of two of them is again one, they form a Lie subalgebra of all vector fields. Transporting that bracket back to  $T_e G$  turns the identity tangent space into a Lie algebra in the abstract sense of definition 1.4.5, with no choice of coordinates, no exponential, and no matrices. This is the definition of the Lie algebra of a Lie group that the rest of the book uses, and its first job will be to reproduce, for  $G = \mathrm{GL}_n(\mathbb{F})$ , exactly the commutator of Chapter 1.

### Definition 2.2.1: Left-Invariant Vector Field

Let  $G$  be a Lie group with identity  $e$ , and for  $g \in G$  let  $L_g: G \rightarrow G$ ,  $L_g(h) = gh$ , denote left translation, a diffeomorphism by proposition 2.1.2. A vector field  $X$  on  $G$  is *left-invariant* if it is  $L_g$ -related to itself for every  $g \in G$ , that is,

$$(L_g)_* X = X \quad \text{for all } g \in G$$

Written out at a point, this is the condition

$$d(L_g)_h(X_h) = X_{gh} \quad \text{for all } g, h \in G$$

Setting  $h = e$  shows that a left-invariant field is determined by its value at the identity through

$$X_g = d(L_g)_e(X_e) \tag{2.1}$$

The set of left-invariant vector fields on  $G$  is written  $\mathfrak{g}$ .

The defining equation says that the flow of  $X$  commutes with left multiplication: transporting a tangent vector from  $h$  to  $gh$  by the differential of  $L_g$  lands on the value the field already assigns at  $gh$ . A left-invariant field is thus rigid data, pinned down everywhere by its single value  $X_e$  at the identity through (2.1). This is the first sign that  $\mathfrak{g}$  will be no larger than  $T_e G$ . The reverse direction, that *every*  $v \in T_e G$  arises this way from a genuine smooth vector field, is the substance of the isomorphism below, and it is where the smoothness of the group operations enters.

### Example 2.2.2: Left-Invariant Fields on the Circle

Take  $G = S^1$ , realized as the unit complex numbers under multiplication, with identity  $e = 1$ . The tangent space  $T_1 S^1$  is one-dimensional, spanned by the velocity  $v = i$  of the curve  $\theta \mapsto e^{i\theta}$  at  $\theta = 0$ . Left translation by  $g = e^{i\alpha}$  is  $L_g(e^{i\theta}) = e^{i(\alpha+\theta)}$ , whose differential rotates tangent vectors by  $\alpha$ ; pushing  $v = i$  forward from 1 to  $g$  gives the velocity  $ig$  of the curve  $\theta \mapsto e^{i(\alpha+\theta)}$  at its start. In the angular coordinate  $\theta$ , this is the constant coordinate field

$$X = \frac{\partial}{\partial \theta}$$

the unit tangent that circulates the circle at constant angular speed. It is nowhere zero, so a single left-invariant field trivializes the tangent bundle of  $S^1$ ; the general form of this phenomenon, parallelizability, appears in (2.3) below. Every left-invariant field on  $S^1$  is a constant real multiple of  $\partial/\partial\theta$ , matching  $\dim S^1 = 1$ .

The isomorphism between left-invariant fields and  $T_e G$  is the load-bearing statement of the section. It has two parts. The evaluation map  $X \mapsto X_e$  is manifestly linear and injective by (2.1); the work is in the inverse, showing that the field  $g \mapsto d(L_g)_e v$  produced from a tangent vector  $v$  is smooth, not just a set-theoretic assignment of vectors. Once the isomorphism is in place, the bracket of vector fields descends to  $T_e G$ , and the abstract Lie algebra materializes.

### Theorem 2.2.3: Left-Invariant Fields and the Bracket on $T_e G$

Let  $G$  be a Lie group.

1. The evaluation map

$$\varepsilon: \mathfrak{g} \rightarrow T_e G, \quad X \mapsto X_e$$

is a linear isomorphism from the space of left-invariant vector fields to the tangent space at the identity. Its inverse sends  $v \in T_e G$  to the field  $\tilde{v}$  defined by  $\tilde{v}_g = d(L_g)_e v$ , which is smooth.

2. If  $X$  and  $Y$  are left-invariant, so is their Lie bracket  $[X, Y]$ . Hence  $\mathfrak{g}$  is a Lie subalgebra of the Lie algebra of all vector fields on  $G$ .
3. Transporting the bracket through  $\varepsilon$  makes  $T_e G$  a Lie algebra: for  $u, v \in T_e G$ , set

$$[u, v] := [\tilde{u}, \tilde{v}]_e$$

This bracket is bilinear, alternating, and satisfies the Jacobi identity, so  $T_e G$  is a Lie algebra in the sense of definition 1.4.5.

#### Proof:

*Step 1: evaluation is a linear isomorphism.* The map  $\varepsilon$  is linear because the value of a sum or scalar multiple of vector fields at  $e$  is the corresponding sum or scalar multiple of values. It is injective: if  $X_e = 0$  then (2.1) gives  $X_g = d(L_g)_e(0) = 0$  for every  $g$ , so  $X$  is the zero field. For surjectivity, fix  $v \in T_e G$  and define  $\tilde{v}_g = d(L_g)_e v$ . This assignment satisfies the left-invariance identity: for  $g, h \in G$ , the chain rule applied to  $L_{gh} = L_g \circ L_h$  gives  $d(L_{gh})_e = d(L_g)_h \circ d(L_h)_e$ , whence

$$d(L_g)_h(\tilde{v}_h) = d(L_g)_h(d(L_h)_e v) = d(L_{gh})_e v = \tilde{v}_{gh}$$

so  $\tilde{v}$  is left-invariant provided it is a smooth vector field. Smoothness is the one point that uses more than the chain rule, and it is where the smoothness of multiplication enters. Let  $m: G \times G \rightarrow G$  be the smooth multiplication. The value  $\tilde{v}_g = d(L_g)_e v$  is the image of  $v$  under the partial differential of  $m$  in its second argument at  $(g, e)$ ; a smooth map has smooth partial differentials, so  $g \mapsto \tilde{v}_g$  is a smooth section of  $TG$ . Concretely, choose a smooth curve  $c: (-\epsilon, \epsilon) \rightarrow G$  with  $c(0) = e$  and  $c'(0) = v$ ; then

$$\tilde{v}_g = \left. \frac{d}{dt} \right|_0 m(g, c(t)) = \left. \frac{d}{dt} \right|_0 g c(t)$$

and the right side depends smoothly on  $g$  because  $m$  is smooth and the  $t$ -derivative of the smooth family  $g \mapsto m(g, c(\cdot))$  is smooth in  $g$  (see [2] for smooth dependence of derivatives on parameters). Thus  $\tilde{v} \in \mathfrak{g}$  and  $\varepsilon(\tilde{v}) = \tilde{v}_e = d(L_e)_e v = v$ , so  $\varepsilon$  is surjective with inverse  $v \mapsto \tilde{v}$ , hence a linear isomorphism.

*Step 2: the bracket of left-invariant fields is left-invariant.* The input is the naturality of the Lie bracket under smooth maps, from [2]: if  $\varphi: M \rightarrow N$  is smooth,  $X_1$  on  $M$  is  $\varphi$ -related to  $Y_1$  on  $N$ , and  $X_2$  is  $\varphi$ -related to  $Y_2$ , then  $[X_1, X_2]$  is  $\varphi$ -related to  $[Y_1, Y_2]$ . Apply this with  $\varphi = L_g$  and  $M = N = G$ : a left-invariant field  $X$  is by definition  $L_g$ -related to itself, and likewise  $Y$ , so  $[X, Y]$  is  $L_g$ -related to  $[X, Y]$ . As this holds for every  $g \in G$ , the bracket  $[X, Y]$  is left-invariant. The bracket of vector fields is already bilinear, alternating, and satisfies the Jacobi identity on all vector fields ([2]); restricting to the subspace  $\mathfrak{g}$ , closed under the bracket by what was just shown, exhibits  $\mathfrak{g}$  as a Lie subalgebra.

*Step 3: transport to  $T_e G$ .* By Step 1 the map  $\varepsilon$  is a linear isomorphism, so  $[u, v] := [\tilde{u}, \tilde{v}]_e$  is a well-defined bilinear operation on  $T_e G$ , the image of the bracket on  $\mathfrak{g}$  under  $\varepsilon$ . Transporting a Lie bracket through a linear isomorphism preserves each of its axioms: bilinearity, the alternating property  $[u, u] = [\tilde{u}, \tilde{u}]_e = 0$ , and the Jacobi identity, since each is a multilinear identity that  $\varepsilon^{-1}$  carries from  $\mathfrak{g}$  to  $T_e G$  and  $\varepsilon$  carries back. Therefore  $(T_e G, [-, -])$  satisfies the two axioms of definition 1.4.5, completing the proof.

The theorem is the intrinsic replacement for the entire vector-space and bracket apparatus of Chapter 1. Evaluation at  $e$  collapses the infinite-dimensional space of left-invariant fields onto the finite-dimensional  $T_e G$  with no loss, and the bracket of fields, which a priori knows about the whole manifold, is completely recorded by its value at a single point. That value is the intrinsic bracket. The construction is canonical: it required choosing neither coordinates nor a curve representing each tangent vector (the curve  $c$  in Step 1 was an auxiliary device whose  $t$ -derivative is coordinate-free), so the Lie-algebra structure on  $T_e G$  belongs to the group and to nothing else.

#### Definition 2.2.4: The Lie Algebra of a Lie Group

Let  $G$  be a Lie group. Its *Lie algebra* is the tangent space at the identity,

$$\mathfrak{g} = \text{Lie}(G) := T_e G$$

equipped with the bracket  $[u, v] = [\tilde{u}, \tilde{v}]_e$  transported from left-invariant vector fields by theorem 2.2.3. Via the isomorphism  $\varepsilon$  of that theorem,  $\text{Lie}(G)$  is identified with the Lie algebra of left-invariant vector fields, and the two descriptions are used interchangeably.

Because  $\varepsilon$  is a linear isomorphism onto  $T_e G$ , the Lie algebra has the dimension of the group:

$$\dim \mathfrak{g} = \dim T_e G = \dim G \quad (2.2)$$

Two notations for the same object are now in play, and both are useful. As a tangent space,  $\mathfrak{g} = T_e G$  is where velocities of curves through  $e$  live and where the exponential map of section 2.3 will land its derivative. As a space of left-invariant fields,  $\mathfrak{g}$  is where the bracket is visibly a Lie bracket of vector fields, with its Jacobi identity inherited rather than recomputed. The identification  $\varepsilon$  lets one pass freely between the pointwise picture and the global-field picture, and an element of  $\mathfrak{g}$  should be read as both at once: a tangent vector  $X_e$  and the entire field  $X$  it generates.

A basis of  $\mathfrak{g}$  has a geometric consequence worth recording immediately. If  $v_1, \dots, v_n$  is a basis of  $T_e G$ , then the left-invariant fields  $\tilde{v}_1, \dots, \tilde{v}_n$  are, at every point  $g$ , the images  $d(L_g)_e v_1, \dots, d(L_g)_e v_n$  of a basis under the isomorphism  $d(L_g)_e$ , hence a basis of  $T_g G$ . They therefore form a *global frame*

$$T_g G = \text{span}_{\mathbb{R}}\{(\tilde{v}_1)_g, \dots, (\tilde{v}_n)_g\} \quad \text{for every } g \in G \quad (2.3)$$

which trivializes the tangent bundle,  $TG \cong G \times \mathbb{R}^n$ . Every Lie group is thus *parallelizable*, the property already asserted in proposition 2.1.2, and (2.3) is its proof: left translation spreads any frame at  $e$  to a frame everywhere. This is a strong restriction on which manifolds can be groups. The circle  $S^1$ , the torus  $\mathbb{T}^n$ , and every Lie group are parallelizable, whereas the even-dimensional sphere  $S^2$  is not: the hairy-ball theorem states that every continuous vector field on  $S^2$  vanishes at some point, so  $S^2$  admits no nowhere-zero field, let alone a global frame, and therefore carries no Lie group structure.

The next property makes the exponential map of the following section possible at all. A one-parameter subgroup will be the integral curve of a left-invariant field through the identity, and for it to be defined for every real time, the field must be complete. On a general manifold an integral curve can run off to infinity in finite time; left-invariance forbids this, because the flow near the identity can be translated to reproduce the flow near any point, extending every integral curve indefinitely.

### Proposition 2.2.5: Left-Invariant Fields Are Complete

Every left-invariant vector field  $X$  on a Lie group  $G$  is complete: for each  $g \in G$  the maximal integral curve of  $X$  starting at  $g$  is defined on all of  $\mathbb{R}$ . Consequently  $X$  generates a global flow  $\theta: \mathbb{R} \times G \rightarrow G$ , and  $\dim \mathfrak{g} = \dim G$ .

#### Proof:

*Step 1: integral curves are translates of one another.* Let  $\gamma: (a, b) \rightarrow G$  be an integral curve of  $X$ , so  $\gamma'(t) = X_{\gamma(t)}$ , and fix  $g \in G$ . The left-translated curve  $\sigma(t) = g\gamma(t) = L_g(\gamma(t))$  is again an integral curve of  $X$ . Indeed, differentiating and using the chain rule,

$$\sigma'(t) = d(L_g)_{\gamma(t)}(\gamma'(t)) = d(L_g)_{\gamma(t)}(X_{\gamma(t)}) = X_{g\gamma(t)} = X_{\sigma(t)}$$

where the third equality is the left-invariance of  $X$  from definition 2.2.1. Thus  $L_g$  carries integral curves of  $X$  to integral curves of  $X$ , defined on the same time interval.

*Step 2: extend the flow near  $e$  to all time.* By the existence and uniqueness theorem for integral curves [2], there is an  $\epsilon > 0$  and an integral curve  $\gamma: (-\epsilon, \epsilon) \rightarrow G$  with  $\gamma(0) = e$ . Suppose, for contradiction, that the maximal integral curve  $\gamma_g$  of  $X$  with  $\gamma_g(0) = g$  had a finite upper endpoint  $T$  for its domain. Choose a time  $t_0$  with  $T - \epsilon < t_0 < T$ , and set  $h = \gamma_g(t_0)$ . By Step 1 applied to the base curve  $\gamma$ , the translate  $t \mapsto h\gamma(t) = L_h(\gamma(t))$  is an integral curve of  $X$  defined on  $(-\epsilon, \epsilon)$  with value  $h$  at  $t = 0$ . By uniqueness of integral curves it agrees with  $\gamma_g(t_0 + t)$  wherever both are defined, and it extends  $\gamma_g$  to times up to  $t_0 + \epsilon > T$ , contradicting the maximality of  $T$ . The same argument rules out a finite lower endpoint. Hence  $\gamma_g$  is defined on all of  $\mathbb{R}$ , so  $X$  is complete.

*Step 3: the global flow and the dimension.* A complete vector field generates a global flow  $\theta: \mathbb{R} \times G \rightarrow G$ ,  $\theta(t, g) = \gamma_g(t)$ , smooth by the smooth dependence of solutions on initial conditions [2]. Finally  $\dim \mathfrak{g} = \dim T_e G = \dim G$  because  $\varepsilon$  of theorem 2.2.3 is a linear isomorphism, restating (2.2), which completes the proof.

The completeness of every left-invariant field is what will let the exponential map be defined on all of  $\mathfrak{g}$ , not just on a neighborhood of the origin: for each  $X \in \mathfrak{g}$  the integral curve through the identity runs for all time, and its value at  $t = 1$  is  $\exp X$  in section 2.3. On a general manifold no such global object exists, because a vector field's integral curves may escape in finite time; the group structure removes this obstruction entirely. The proof extracts the mechanism cleanly: the flow of

a left-invariant field is determined near the identity and then copied to every other point by left translation, so there is no room for finite-time blowup. This homogeneity of the flow is the analytic counterpart of the algebraic fact that a group looks the same at every point.

It remains to check that the intrinsic construction reproduces the concrete Lie algebra of Chapter 1 when  $G = \mathrm{GL}_n(\mathbb{F})$ . This is the compatibility that justifies calling the abstract  $T_e G$  the Lie algebra: it must specialize to the matrix commutator, or the two chapters would be describing different objects.

**Proposition 2.2.6: The Intrinsic Lie Algebra of  $\mathrm{GL}_n(\mathbb{F})$**

Let  $G = \mathrm{GL}_n(\mathbb{F})$ , an open subset of  $M_n(\mathbb{F})$  by proposition 1.1.2, so that its tangent space at the identity is canonically

$$T_I \mathrm{GL}_n(\mathbb{F}) = M_n(\mathbb{F}) = \mathfrak{gl}_n(\mathbb{F})$$

the space of theorem 1.4.3. For  $X \in M_n(\mathbb{F})$  the associated left-invariant vector field  $\tilde{X}$  is given at  $A \in \mathrm{GL}_n(\mathbb{F})$  by

$$\tilde{X}_A = AX \tag{2.4}$$

that is,  $\tilde{X}$  is the linear vector field  $A \mapsto AX$  on  $\mathrm{GL}_n(\mathbb{F})$ .

**Proof:**

Because  $\mathrm{GL}_n(\mathbb{F})$  is open in the vector space  $M_n(\mathbb{F})$ , its tangent space at any point  $A$  is canonically  $M_n(\mathbb{F})$  itself, and a tangent vector at  $A$  is the velocity of a curve of matrices through  $A$ . Left translation  $L_A(B) = AB$  is the restriction to  $\mathrm{GL}_n(\mathbb{F})$  of the linear map  $B \mapsto AB$  on  $M_n(\mathbb{F})$ . The differential of a linear map is the map itself under this canonical identification, so

$$d(L_A)_I: M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F}), \quad Z \mapsto AZ$$

Applying (2.1) with  $g = A$  and  $X_e = X \in T_I \mathrm{GL}_n(\mathbb{F}) = M_n(\mathbb{F})$  gives  $\tilde{X}_A = d(L_A)_I(X) = AX$ , which is (2.4). This is smooth in  $A$ , confirming directly that  $\tilde{X}$  is a genuine vector field, as theorem 2.2.3 guarantees in general, completing the proof.

The left-invariant field of  $X$  on  $\mathrm{GL}_n(\mathbb{F})$  is the matrix version of the constant field on the circle in example 2.2.2: it assigns to each group element  $A$  the tangent vector  $AX$ , spreading the single matrix  $X = \tilde{X}_I$  over the whole group by left multiplication. With (2.4) in hand the intrinsic bracket can be identified with the commutator. The Lie bracket of two vector fields on an open set of  $M_n(\mathbb{F})$  is computed from their directional derivatives: writing the fields as  $M_n(\mathbb{F})$ -valued functions  $A \mapsto AX$  and  $A \mapsto AY$ , the bracket is

$$[\tilde{X}, \tilde{Y}]_A = D(\tilde{Y})_A(\tilde{X}_A) - D(\tilde{X})_A(\tilde{Y}_A)$$

where  $D(\tilde{Y})_A$  is the derivative of  $A \mapsto AY$ , namely the linear map  $Z \mapsto ZY$ . Substituting  $\tilde{X}_A = AX$  and  $\tilde{Y}_A = AY$ ,

$$[\tilde{X}, \tilde{Y}]_A = (AX)Y - (AY)X = A(XY - YX)$$

Evaluating at  $A = I$  gives  $[\tilde{X}, \tilde{Y}]_I = XY - YX$ . Through the identification  $\varepsilon$  of theorem 2.2.3, the intrinsic bracket on  $T_I \mathrm{GL}_n(\mathbb{F}) = \mathfrak{gl}_n(\mathbb{F})$  is therefore the commutator

$$[X, Y] = XY - YX \tag{2.5}$$

recovering exactly the matrix bracket of proposition 1.4.4. The value  $[\tilde{X}, \tilde{Y}]_A = A(XY - YX)$  at every  $A$ , not just at  $I$ , is itself the left-invariant field of the commutator, a computation that reconfirms Step 2 of theorem 2.2.3 in this case. The full, coordinate-free explanation of why the intrinsic bracket agrees with the commutator, expressed through the adjoint representation, is deferred to section 2.5, where it appears as theorem 2.5.4; the direct computation above is enough for now to see that the intrinsic Lie algebra of  $\mathrm{GL}_n(\mathbb{F})$  is  $\mathfrak{gl}_n(\mathbb{F})$  with its commutator, and a matrix subgroup  $G \subseteq \mathrm{GL}_n(\mathbb{F})$  inherits the bracket of its ambient  $\mathfrak{gl}_n(\mathbb{F})$ .

For the running example, the identification closes cleanly. The intrinsic Lie algebra of  $\mathrm{SU}(2)$  is  $T_I\mathrm{SU}(2)$ , and this tangent space is precisely the space of traceless skew-Hermitian matrices of example 1.4.7: a tangent vector at  $I$  is the velocity  $c'(0)$  of a curve  $c$  in  $\mathrm{SU}(2)$  through  $I$ , and differentiating the relations  $c(t)^*c(t) = I$  and  $\det c(t) = 1$  at  $t = 0$  returns  $X^* + X = 0$  and  $\mathrm{tr} X = 0$ . By (2.5) the intrinsic bracket on  $T_I\mathrm{SU}(2)$  is the commutator, so the intrinsic Lie algebra of  $\mathrm{SU}(2)$  is  $\mathfrak{su}(2)$  with the bracket of example 1.4.7, and in particular it is the cross product on  $\mathbb{R}^3$ . The abstract machine of this section, applied to  $\mathrm{SU}(2)$ , returns the concrete algebra Chapter 1 computed by hand.

### Exercise 2.2.1

1. On the  $n$ -torus  $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ , with group operation addition of coordinates modulo 1 and identity  $e = 0$ , show that the constant coordinate fields  $\partial/\partial\theta_1, \dots, \partial/\partial\theta_n$  are left-invariant, and that they form a basis of the Lie algebra  $\mathrm{Lie}(\mathbb{T}^n)$ . Compute all brackets  $[\partial/\partial\theta_i, \partial/\partial\theta_j]$  and conclude that  $\mathrm{Lie}(\mathbb{T}^n)$  is abelian, that is,  $[u, v] = 0$  for all  $u, v$ .
2. Let  $G$  be a Lie group of dimension  $n$  and let  $v_1, \dots, v_n$  be a basis of  $T_e G$ . Prove in detail that the left-invariant fields  $\tilde{v}_1, \dots, \tilde{v}_n$  form a global frame, using that  $d(L_g)_e$  is a linear isomorphism for every  $g$ . Deduce that  $TG$  is trivial, and explain why this shows the even-dimensional spheres  $S^{2k}$  with  $k \geq 1$  carry no Lie group structure.
3. For  $G = \mathrm{GL}_n(\mathbb{F})$  and  $X \in M_n(\mathbb{F})$ , verify by a direct differential computation that the field  $\tilde{X}_A = AX$  of proposition 2.2.6 is left-invariant: show that  $d(L_B)_A(\tilde{X}_A) = \tilde{X}_{BA}$  for all  $A, B \in \mathrm{GL}_n(\mathbb{F})$ . Then take  $X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  in  $\mathfrak{gl}_2(\mathbb{R})$  and write out  $\tilde{X}_A$  for a general  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ .
4. A vector field  $X$  on  $G$  is *right-invariant* if  $(R_g)_*X = X$  for all  $g$ , where  $R_g(h) = hg$ . State and prove the right-invariant analogue of part (1) of theorem 2.2.3: evaluation at  $e$  is a linear isomorphism from right-invariant fields to  $T_e G$ , with inverse  $v \mapsto \hat{v}$  where  $\hat{v}_g = d(R_g)_e v$ . For  $G = \mathrm{GL}_n(\mathbb{F})$ , compute  $\hat{X}_A$  and contrast it with the left-invariant field of proposition 2.2.6.
5. Show that a connected Lie group  $G$  is abelian if and only if every left-invariant vector field is also right-invariant. (Left and right translations always commute by associativity; for the harder direction the exponential map of section 2.3 is used to conclude that the elements moved by these flows are central and generate  $G$ , so a clear argument reducing the claim to the commuting of left and right translations is what is asked for.)
6. Using only the left-invariant fields  $\tilde{X}_A = AX$  on  $\mathrm{GL}_2(\mathbb{R})$  and the coordinate expression for the Lie bracket of vector fields, recompute  $[\tilde{X}, \tilde{Y}]_I$  for  $X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ , and confirm it equals the commutator  $XY - YX$  of proposition 1.4.4.

## 2.3 One-Parameter Subgroups and the Exponential Map

chapter 1 produced the exponential of a matrix group by summing a power series and produced the one-parameter subgroups of  $\mathrm{GL}_n(\mathbb{F})$  as the curves  $t \mapsto \exp(tX)$ ; the classification theorem 1.3.10 showed these are all of them. Neither the series nor the classification made any use of the ambient matrices beyond the linear differential equation  $\gamma' = X\gamma$  that a homomorphism  $\mathbb{R} \rightarrow \mathrm{GL}_n(\mathbb{F})$  must satisfy. On a general Lie group there is no power series, but there is still a differential equation: a smooth homomorphism  $\gamma: \mathbb{R} \rightarrow G$  is determined by its velocity  $X = \gamma'(0)$ , because  $X$  generates a left-invariant vector field whose flow line through the identity is  $\gamma$ . This section builds the intrinsic exponential map out of that flow, transferring every property of the matrix exponential to the abstract setting, and then verifies that for a matrix group the two exponentials coincide.

The linchpin is the observation, recorded in theorem 2.2.3 and proposition 2.2.5, that a tangent vector  $X \in \mathfrak{g} = T_e G$  determines a left-invariant vector field  $X^L$  whose flow exists for all time. Because the field is left-invariant, its flow interacts with the group law in the simplest possible way: translating a flow line by a group element produces another flow line. That single fact turns the integral curve through the identity into a homomorphism, and it is what the first theorem establishes.

### Theorem 2.3.1: One-Parameter Subgroups and the Lie Algebra

Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g} = T_e G$ . A *one-parameter subgroup* of  $G$  is a smooth homomorphism  $\gamma: \mathbb{R} \rightarrow G$ , that is, a smooth curve with  $\gamma(s+t) = \gamma(s)\gamma(t)$  for all  $s, t \in \mathbb{R}$  (so in particular  $\gamma(0) = e$ ). For each  $X \in \mathfrak{g}$  let  $X^L$  be the left-invariant vector field with  $X_e^L = X$  (definition 2.2.1) and let  $\gamma_X$  be the integral curve of  $X^L$  through  $e$ . Then:

1.  $\gamma_X$  is a one-parameter subgroup of  $G$ , defined for all  $t \in \mathbb{R}$ , with  $\gamma_X'(0) = X$ .
2. Every one-parameter subgroup  $\gamma$  of  $G$  equals  $\gamma_X$  for  $X = \gamma'(0)$ .

Consequently  $X \mapsto \gamma_X$  is a bijection from  $\mathfrak{g}$  onto the set of one-parameter subgroups of  $G$ , with inverse  $\gamma \mapsto \gamma'(0)$ .

#### Proof:

The two statements are proved in turn.

1. *The integral curve is a homomorphism.* By proposition 2.2.5 the field  $X^L$  is complete, so its integral curve  $\gamma_X: \mathbb{R} \rightarrow G$  through  $\gamma_X(0) = e$  is defined for all  $t \in \mathbb{R}$ ; it satisfies

$$\gamma_X'(t) = X_{\gamma_X(t)}^L, \quad \gamma_X(0) = e \quad (2.6)$$

and  $\gamma_X'(0) = X_e^L = X$  is the velocity at the identity.

Fix  $s \in \mathbb{R}$  and consider the two curves

$$\alpha(t) = \gamma_X(s+t), \quad \beta(t) = \gamma_X(s)\gamma_X(t) = L_{\gamma_X(s)}(\gamma_X(t))$$

The claim is  $\alpha = \beta$ ; since  $s$  is arbitrary, this claim is exactly the homomorphism property. Both curves start at the same point:  $\alpha(0) = \gamma_X(s)$  and  $\beta(0) = \gamma_X(s)\gamma_X(0) = \gamma_X(s)e = \gamma_X(s)$ . Both satisfy (2.6). For  $\alpha$  this is the chain rule applied to the time-translation

$t \mapsto s+t$ :  $\alpha'(t) = \gamma'_X(s+t) = X_{\gamma_X(s+t)}^L = X_{\alpha(t)}^L$ . For  $\beta$  this is where left-invariance enters. Writing  $g = \gamma_X(s)$  and using that  $L_g$  is a diffeomorphism (proposition 2.1.2),

$$\beta'(t) = \frac{d}{dt} L_g(\gamma_X(t)) = (dL_g)_{\gamma_X(t)}(\gamma'_X(t)) = (dL_g)_{\gamma_X(t)}(X_{\gamma_X(t)}^L)$$

The defining property of a left-invariant field is that it is  $L_g$ -related to itself,  $(dL_g)_h(X_h^L) = X_{gh}^L$  for all  $g, h$  (definition 2.2.1); applying this with  $h = \gamma_X(t)$  turns the last expression into  $X_{g\gamma_X(t)}^L = X_{\beta(t)}^L$ . Thus  $\beta$  also solves (2.6), now with initial point  $\gamma_X(s)$ .

Both  $\alpha$  and  $\beta$  are integral curves of  $X^L$  through the common initial point  $\gamma_X(s)$  at  $t = 0$ . By the uniqueness of integral curves of a smooth vector field ([2]),  $\alpha = \beta$  for all  $t$ ; that is,  $\gamma_X(s+t) = \gamma_X(s)\gamma_X(t)$  for all  $s, t \in \mathbb{R}$ . Hence  $\gamma_X$  is a smooth homomorphism  $\mathbb{R} \rightarrow G$ , a one-parameter subgroup, with  $\gamma'_X(0) = X$ .

2. *Every one-parameter subgroup is an integral curve.* Let  $\gamma: \mathbb{R} \rightarrow G$  be a one-parameter subgroup and set  $X = \gamma'(0) \in T_e G = \mathfrak{g}$ . Differentiating  $\gamma(s+t) = \gamma(s)\gamma(t) = L_{\gamma(s)}(\gamma(t))$  with respect to  $t$  and evaluating at  $t = 0$ , where  $\gamma(0) = e$  and  $\gamma'(0) = X$ ,

$$\gamma'(s) = \left. \frac{d}{dt} \right|_{t=0} L_{\gamma(s)}(\gamma(t)) = (dL_{\gamma(s)})_e(\gamma'(0)) = (dL_{\gamma(s)})_e(X) = X_{\gamma(s)}^L$$

the last equality being the definition of the left-invariant field  $X^L$  generated by  $X$ . So  $\gamma$  solves (2.6) with  $\gamma(0) = e$ , the same problem  $\gamma_X$  solves. Uniqueness of integral curves gives  $\gamma = \gamma_X$ .

Parts (1) and (2) show  $X \mapsto \gamma_X$  is a surjection onto the one-parameter subgroups with a two-sided inverse  $\gamma \mapsto \gamma'(0)$ : part (1) gives  $\gamma'_X(0) = X$ , and part (2) gives  $\gamma_{\gamma'(0)} = \gamma$ . Hence the assignment is a bijection with the stated inverse, completing the proof.

The theorem is the intrinsic form of the classification theorem 1.3.10, and the parallel is exact. There, a one-parameter subgroup of  $\mathrm{GL}_n(\mathbb{F})$  solved the linear equation  $\gamma' = X\gamma$  and was pinned down by its initial velocity; here it solves the equation  $\gamma' = X_{\gamma}^L$  of an integral curve and is pinned down the same way. The matrix version is the special case in which the left-invariant field  $X^L$  is the explicit vector field  $A \mapsto AX$  and the flow can be summed as a power series. What has been removed is any need for that series: the group law forces the integral curve to be a homomorphism, and completeness forces it to run for all time, so the correspondence between generators and one-parameter subgroups holds on any Lie group. Evaluating each subgroup at a single time now packages the whole correspondence into one map.

### Definition 2.3.2: The Exponential Map

Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g}$ . The *exponential map*  $\exp: \mathfrak{g} \rightarrow G$  sends  $X \in \mathfrak{g}$  to the value at time 1 of its one-parameter subgroup:

$$\exp(X) = \gamma_X(1)$$

where  $\gamma_X$  is the one-parameter subgroup of theorem 2.3.1 with  $\gamma'_X(0) = X$ .

The definition compresses the entire family  $\{\gamma_X\}_{X \in \mathfrak{g}}$  into a single map on  $\mathfrak{g}$ , and the family is

recovered from it by rescaling. For  $X \in \mathfrak{g}$  and  $t \in \mathbb{R}$ , the curve  $s \mapsto \gamma_X(ts)$  is a one-parameter subgroup with velocity  $t\gamma'_X(0) = tX$  at  $s = 0$ ; by the uniqueness in theorem 2.3.1 it is  $\gamma_{tX}$ , so  $\gamma_{tX}(s) = \gamma_X(ts)$ . Taking  $s = 1$  gives

$$\exp(tX) = \gamma_{tX}(1) = \gamma_X(t) \quad (2.7)$$

so that  $t \mapsto \exp(tX)$  is exactly the one-parameter subgroup  $\gamma_X$ , and  $\exp$  traces out the whole flow as  $t$  varies. This identity is the workhorse of every property below: it turns statements about the family of flows into statements about the single map  $\exp$ .

The next proposition collects the properties inherited from (2.7). They are the intrinsic counterparts of the algebraic properties of the matrix exponential in proposition 1.3.4, and the proofs are shorter because the analytic labor has already been done in building the flow.

**Proposition 2.3.3: Properties of the Exponential Map**

Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g}$  and exponential map  $\exp: \mathfrak{g} \rightarrow G$ . For all  $X \in \mathfrak{g}$  and all  $s, t \in \mathbb{R}$ :

1.  $\exp(0) = e$ .
2.  $\exp((s+t)X) = \exp(sX)\exp(tX)$ , and consequently  $\exp(-X) = \exp(X)^{-1}$ ; the curve  $t \mapsto \exp(tX)$  is the one-parameter subgroup  $\gamma_X$  with velocity  $\frac{d}{dt}\big|_0 \exp(tX) = X$ .
3.  $\exp: \mathfrak{g} \rightarrow G$  is smooth.

**Proof:**

*Part (1).* The one-parameter subgroup generated by  $X = 0$  is the constant curve  $\gamma_0 \equiv e$ , since the zero vector field has the constant integral curve through  $e$ ; hence  $\exp(0) = \gamma_0(1) = e$ .

*Part (2).* By (2.7),  $t \mapsto \exp(tX) = \gamma_X(t)$  is the one-parameter subgroup  $\gamma_X$ , so the homomorphism property  $\gamma_X(s+t) = \gamma_X(s)\gamma_X(t)$  of theorem 2.3.1 reads

$$\exp((s+t)X) = \exp(sX)\exp(tX)$$

Setting  $t = -s$  and using part (1) gives  $\exp(sX)\exp(-sX) = \exp(0) = e$ , so  $\exp(-sX) = \exp(sX)^{-1}$ ; taking  $s = 1$  gives  $\exp(-X) = \exp(X)^{-1}$ . The velocity statement is  $\frac{d}{dt}\big|_0 \exp(tX) = \gamma'_X(0) = X$ .

*Part (3).* Smoothness of  $\exp$  is the smooth dependence of a flow on its initial data, applied to a single well-chosen field. Consider the Lie group  $G$  together with its Lie algebra  $\mathfrak{g}$ , and form on the product manifold  $\mathfrak{g} \times G$  the vector field  $V$  whose value at  $(X, g)$  is  $(0, X_g^L)$ : at each point it is the zero vector in the  $\mathfrak{g}$ -direction and the left-invariant field  $X^L$  evaluated at  $g$  in the  $G$ -direction. The assignment  $(X, g) \mapsto X_g^L$  is smooth in both variables, because  $X_g^L = (dL_g)_e(X)$  depends smoothly on  $g$  through the smooth multiplication of  $G$  and linearly, hence smoothly, on  $X$ ; so  $V$  is a smooth vector field on  $\mathfrak{g} \times G$ . Its integral curve through  $(X, e)$  is  $t \mapsto (X, \gamma_X(t))$ , since the first coordinate is constant and the second solves (2.6). This field is complete, as each  $X^L$  is complete by proposition 2.2.5. The flow of a complete smooth vector field is smooth jointly in time and initial condition ([2]), so the time-1 flow  $(X, e) \mapsto (X, \gamma_X(1)) = (X, \exp X)$  is smooth; projecting to  $G$  shows  $X \mapsto \exp(X)$  is smooth, completing the proof.

Part (2) is the reason the map deserves the name “exponential”: restricted to a single line  $\mathbb{R}X$  in  $\mathfrak{g}$  it converts addition of real multiples of  $X$  into multiplication in  $G$ , exactly as the scalar exponential converts addition into multiplication. The restriction to a line is essential. The full map  $\exp$  is not a homomorphism from  $(\mathfrak{g}, +)$  to  $G$  unless  $G$  is abelian, because  $\exp(X)\exp(Y)$  and  $\exp(X + Y)$  disagree already at second order when  $X$  and  $Y$  fail to commute, precisely the failure of  $\exp(A + B) = \exp(A)\exp(B)$  recorded for matrices in proposition 1.3.5. What survives universally is the one-parameter law of part (2), and it is enough to make  $\exp$  a chart near the identity.

The value of the differential of  $\exp$  at the origin is what promotes the one-parameter law into a local coordinate system. As with the matrix exponential in theorem 1.3.7, this differential is the identity, and the reason is the same one-line computation: differentiating  $t \mapsto \exp(tX)$  at  $t = 0$  returns  $X$ .

#### Theorem 2.3.4: The Exponential is a Local Diffeomorphism

Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g}$ . Identifying  $T_0\mathfrak{g}$  with  $\mathfrak{g}$  and  $T_eG$  with  $\mathfrak{g}$ , the differential of  $\exp$  at the origin is the identity,

$$d(\exp)_0 = \text{id}_{\mathfrak{g}}$$

Consequently there are open neighborhoods  $U$  of 0 in  $\mathfrak{g}$  and  $V$  of  $e$  in  $G$  such that  $\exp: U \rightarrow V$  is a diffeomorphism.

#### Proof:

The differential  $d(\exp)_0$  acts on a tangent vector  $X \in T_0\mathfrak{g} = \mathfrak{g}$  by differentiating  $\exp$  along any curve through 0 with velocity  $X$ ; the line  $t \mapsto tX$  is such a curve, so by (2.7) and proposition 2.3.3(2),

$$d(\exp)_0(X) = \left. \frac{d}{dt} \right|_{t=0} \exp(tX) = \gamma'_X(0) = X$$

under the identification  $T_eG = \mathfrak{g}$ . Thus  $d(\exp)_0 = \text{id}_{\mathfrak{g}}$ , which is invertible. Since  $\exp$  is smooth (proposition 2.3.3) with invertible differential at 0, the inverse function theorem ([2]) provides open neighborhoods  $U$  of 0 and  $V = \exp(U)$  of  $\exp(0) = e$  on which  $\exp$  is a diffeomorphism, completing the proof.

The theorem installs  $\exp$  as a chart on  $G$  centered at the identity, with  $\mathfrak{g}$  as the model vector space, so that near  $e$  the group is described by the linear data of its Lie algebra. This is the intrinsic version of the exponential chart of theorem 1.3.7, and it is what makes the Lie algebra a faithful record of the group near the identity. A first structural consequence is that a connected group hides nothing beyond what this chart already shows: the whole group is assembled from arbitrarily small pieces near  $e$ .

#### Corollary 2.3.5: Connected Groups Are Generated Near the Identity

Let  $G$  be a connected Lie group. Then  $G$  is generated by the image  $\exp(\mathfrak{g})$ ; equivalently,  $G$  is generated by any neighborhood of the identity. Explicitly, every  $g \in G$  is a finite product  $g = \exp(X_1)\exp(X_2)\cdots\exp(X_k)$  with  $X_1, \dots, X_k \in \mathfrak{g}$ .

**Proof:**

Let  $V$  be the neighborhood of  $e$  on which  $\exp$  is a diffeomorphism from a neighborhood  $U$  of  $0$ , as in theorem 2.3.4; then  $V \subseteq \exp(\mathfrak{g})$ , so it suffices to show that any neighborhood  $W$  of  $e$  generates  $G$ . Let  $H = \bigcup_{k \geq 1} W_{\text{sym}}^k$  be the subgroup generated by  $W_{\text{sym}} = W \cap W^{-1}$ , which is a symmetric neighborhood of  $e$  (the inversion map is a homeomorphism by proposition 2.1.2, so  $W^{-1}$  is open and  $W_{\text{sym}}$  is an open neighborhood of  $e$ ) and  $W_{\text{sym}}^k$  denotes the set of  $k$ -fold products. Then  $H$  is a subgroup of  $G$ .

The subgroup  $H$  is open: for  $h \in H$ , the translate  $hW_{\text{sym}} = L_h(W_{\text{sym}})$  is an open neighborhood of  $h$  contained in  $H$ , since  $L_h$  is a diffeomorphism (proposition 2.1.2) and  $hW_{\text{sym}} \subseteq H$  as a product of elements of  $H$ . An open subgroup is also closed: its complement  $G \setminus H = \bigsqcup_{gH \neq H} gH$  is a union of cosets, each open (being a translate of the open set  $H$ ), hence  $G \setminus H$  is open and  $H$  is closed. Thus  $H$  is a nonempty subset of the connected space  $G$  that is both open and closed, so  $H = G$ . Every  $g \in G$  therefore lies in some  $W_{\text{sym}}^k \subseteq \exp(\mathfrak{g})^k$ , giving the product representation with each factor in  $\exp(\mathfrak{g})$ , as we sought to show.

The corollary explains why the Lie algebra is so powerful a tool for connected groups: knowing  $\mathfrak{g}$  determines the local structure of  $G$  near  $e$  through the exponential chart, and connectedness then propagates that local structure to all of  $G$  by taking products. A homomorphism of connected groups is thereby determined by its behavior near the identity, and ultimately by its differential at  $e$ , a theme the next section develops in full. The corollary does not assert that  $\exp$  is surjective; the elements of  $G$  are products of exponentials, not necessarily single exponentials, and the gap between the two is real, as the exercises make concrete.

Everything so far has been intrinsic, defined through flows of left-invariant fields with no reference to matrices. To close the loop with chapter 1, the intrinsic exponential must be shown to agree with the matrix exponential on a matrix group. The verification is a matter of identifying the left-invariant field on  $\text{GL}_n(\mathbb{F})$  and recognizing its flow.

**Proposition 2.3.6: The Intrinsic Exponential is the Matrix Exponential**

Let  $G = \text{GL}_n(\mathbb{F})$ , whose Lie algebra is  $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$  identified with  $T_I \text{GL}_n(\mathbb{F})$ . For every  $X \in \mathfrak{gl}_n(\mathbb{F})$  the intrinsic exponential of definition 2.3.2 agrees with the matrix exponential of definition 1.3.2:

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!}$$

The same holds on every matrix group  $G \subseteq \text{GL}_n(\mathbb{F})$ : the intrinsic exponential of  $G$  is the restriction of the matrix exponential to  $\mathfrak{g}$ .

**Proof:**

*Step 1: the left-invariant field on  $\text{GL}_n(\mathbb{F})$ .* The group  $\text{GL}_n(\mathbb{F})$  is an open subset of the vector space  $M_n(\mathbb{F})$  (proposition 1.1.2), so its tangent space at any point  $A$  is canonically  $M_n(\mathbb{F})$ , and left translation  $L_A(B) = AB$  is the restriction of a linear map  $M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$ . The differential of a linear map is the map itself, so  $(dL_A)_I(X) = AX$  for every  $X \in M_n(\mathbb{F})$ . Hence the left-invariant field  $X^L$  generated by  $X \in T_I \text{GL}_n(\mathbb{F}) = M_n(\mathbb{F})$  is

$$X_A^L = (dL_A)_I(X) = AX \tag{2.8}$$

the vector field  $A \mapsto AX$  on  $\text{GL}_n(\mathbb{F})$ .

*Step 2: the integral curve is the matrix exponential.* By (2.8) the integral curve  $\gamma_X$  of  $X^L$  through  $I$  solves the initial value problem

$$\gamma'_X(t) = X^L_{\gamma_X(t)} = \gamma_X(t) X, \quad \gamma_X(0) = I$$

which is exactly the linear matrix differential equation (1.2) governing one-parameter subgroups of  $\mathrm{GL}_n(\mathbb{F})$ , written on the right rather than the left. Its unique solution is the matrix exponential  $t \mapsto \sum_{k \geq 0} t^k X^k / k!$ : by proposition 1.3.4(1) the matrix curve  $\eta(t) = \sum_k t^k X^k / k!$  satisfies  $\eta'(t) = \eta(t)X$  and  $\eta(0) = I$ ; uniqueness for this right-multiplication equation follows by the identical argument that theorem 1.3.10 runs for the left form  $\gamma' = X\gamma$  (if  $\eta_1, \eta_2$  both solve it, then  $(\eta_1 \eta_2^{-1})' = 0$ , so  $\eta_1 \eta_2^{-1}$  is constant, equal to  $I$  at  $t = 0$ ). Therefore  $\gamma_X(t) = \sum_k t^k X^k / k!$ , and evaluating at  $t = 1$ ,

$$\exp(X) = \gamma_X(1) = \sum_{k=0}^{\infty} \frac{X^k}{k!}$$

the matrix exponential of definition 1.3.2.

*Step 3: passage to a subgroup.* Let  $G \subseteq \mathrm{GL}_n(\mathbb{F})$  be a matrix group with Lie algebra  $\mathfrak{g} \subseteq \mathfrak{gl}_n(\mathbb{F})$ , and let  $X \in \mathfrak{g}$ . The one-parameter subgroup of  $G$  generated by  $X$  is a one-parameter subgroup of the ambient  $\mathrm{GL}_n(\mathbb{F})$  generated by the same  $X$ , because the inclusion  $G \hookrightarrow \mathrm{GL}_n(\mathbb{F})$  is a homomorphism of Lie groups carrying the left-invariant field of  $X$  on  $G$  to that on  $\mathrm{GL}_n(\mathbb{F})$ : concretely, an integral curve of  $X^L$  in  $G$  through  $I$  is also an integral curve of the ambient field  $A \mapsto AX$  through  $I$ , so by uniqueness the two one-parameter subgroups coincide as curves in  $\mathrm{GL}_n(\mathbb{F})$ . Evaluating at  $t = 1$ , the intrinsic exponential of  $G$  at  $X$  equals the intrinsic exponential of  $\mathrm{GL}_n(\mathbb{F})$  at  $X$ , which by Steps 1 and 2 is the matrix exponential. Hence the intrinsic exponential of  $G$  is the restriction of the matrix exponential to  $\mathfrak{g}$ , completing the proof.

The proposition closes the circle opened in chapter 1: the exponential built there by summing a series and the exponential built here by flowing a left-invariant field are one and the same. The two constructions look unrelated, one a computation in a normed algebra and the other the integration of a vector field on a manifold, but both solve the identical differential equation, and the uniqueness of its solution forces them to agree. From now on the notation  $\exp$  carries no ambiguity: on a matrix group it is the matrix exponential, and the abstract properties proved above specialize to the concrete properties of chapter 1. The running example makes the agreement fully explicit.

### Example 2.3.7: The Exponential on $\mathfrak{su}(2)$

For  $G = \mathrm{SU}(2)$  the intrinsic Lie algebra is  $\mathfrak{su}(2) = T_I \mathrm{SU}(2)$ , the traceless skew-Hermitian  $2 \times 2$  matrices (example 1.4.7). By proposition 2.3.6 the intrinsic exponential is the matrix exponential restricted to  $\mathfrak{su}(2)$ , so the closed form already computed in chapter 1 is the intrinsic answer. Writing  $X \in \mathfrak{su}(2)$  with  $\Phi(X) = v \in \mathbb{R}^3$  under the isomorphism  $\mathfrak{su}(2) \cong (\mathbb{R}^3, \times)$  of example 1.4.7, the computation  $X^2 = -\frac{1}{4}|v|^2 I$  splits the exponential series into its even and odd parts exactly as  $\cos$  and  $\sin$  split, giving, for  $v \neq 0$ ,

$$\exp(X) = \cos\left(\frac{|v|}{2}\right) I + \frac{\sin\left(\frac{|v|}{2}\right)}{|v|/2} X \quad (2.9)$$

This is the map of proposition 1.5.3, which established that  $\exp$  is *onto*  $\mathrm{SU}(2)$ : every element of  $\mathrm{SU}(2)$  has the form (2.9) for a suitable  $X$ , so  $\exp: \mathfrak{su}(2) \rightarrow \mathrm{SU}(2)$  is surjective. The surjectivity is a feature of  $\mathrm{SU}(2)$  as a compact connected group, not a general fact; it sharpens the product

representation of corollary 2.3.5 from a finite product of exponentials down to a single one. The intrinsic theorem 2.3.1 recovers, as its content on  $SU(2)$ , the concrete one-parameter subgroups computed in example 1.5.4: the curve  $t \mapsto \exp(tX)$  with  $\Phi(X) = v$  is a great circle on the sphere  $SU(2) \cong S^3$ , closing up with period  $4\pi/|v|$  in  $t$ .

The closed form (2.9) shows the abstract theory returning the concrete chapter 1 formula with nothing lost. The period  $4\pi/|v|$ , twice the  $SO(3)$ -side period  $2\pi/|v|$ , is the same factor of 2 that separates  $SU(2)$  from  $SO(3)$  in the double cover theorem 1.5.5: a one-parameter subgroup of  $SU(2)$  must wind twice before it closes, because a rotation by  $2\pi$  lifts to  $-I$  rather than  $I$  upstairs. The exponential map is where this doubling first becomes visible in coordinates, and it foreshadows the role  $\exp$  will play in Chapter 3 in translating between a group and its covers through their common Lie algebra.

### Exercise 2.3.1

1. Show that  $\exp: \mathfrak{g} \rightarrow G$  is a group homomorphism from  $(\mathfrak{g}, +)$  to  $G$  if and only if  $\exp(X)\exp(Y) = \exp(Y)\exp(X)$  for all  $X, Y \in \mathfrak{g}$ . Using the matrices  $X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$  in  $\mathfrak{gl}_2(\mathbb{R})$ , verify that  $\exp(X)\exp(Y) \neq \exp(X+Y)$ , so that  $\exp$  is not a homomorphism on  $\mathfrak{gl}_2(\mathbb{R})$ .
2. Prove that if  $G$  is a connected abelian Lie group then  $\exp: \mathfrak{g} \rightarrow G$  is a surjective group homomorphism. (You may use corollary 2.3.5 for surjectivity after establishing the homomorphism property; for the homomorphism property, show first that for commuting  $X, Y$  the curve  $t \mapsto \exp(tX)\exp(tY)$  is a one-parameter subgroup with velocity  $X+Y$ .)
3. Exhibit a connected Lie group  $G$  on which  $\exp$  is *not* surjective, by showing that no  $X \in \mathfrak{sl}_2(\mathbb{R})$  satisfies  $\exp(X) = \begin{pmatrix} -2 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$ , even though this matrix lies in the connected group  $SL_2(\mathbb{R})$ . (Hint: analyze the eigenvalues of  $\exp(X)$  for a traceless real  $X$  according to the sign of  $\det X$ .)
4. Show that  $\exp: \mathfrak{g} \rightarrow G$  need not be injective by computing its fibers for  $G = S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$  with  $\mathfrak{g} = i\mathbb{R}$ . Determine the kernel of  $\exp$  as a homomorphism  $(\mathfrak{g}, +) \rightarrow S^1$  and identify it with a lattice in  $\mathfrak{g}$ .
5. Prove the reparametrization identity (2.7), that  $\exp(tX) = \gamma_X(t)$ , directly from the uniqueness in theorem 2.3.1, by checking that  $s \mapsto \gamma_X(ts)$  is the one-parameter subgroup generated by  $tX$ .
6. Combining example 2.3.7 with corollary 2.3.5, explain why every element of  $SU(2)$  is a *single* exponential  $\exp(X)$ , whereas corollary 2.3.5 guarantees only a finite product of exponentials in general. State the property of  $SU(2)$  responsible, and contrast with  $SL_2(\mathbb{R})$  from exercise 2.3.1 (3).

## 2.4 The Differential of a Homomorphism and the Naturality of the Exponential

The previous three sections built, for a single Lie group  $G$ , the dictionary that assigns to  $G$  its Lie algebra  $\mathfrak{g} = \text{Lie}(G)$  and the exponential map  $\exp: \mathfrak{g} \rightarrow G$ . This section makes the dictionary functorial. A homomorphism  $\varphi: G \rightarrow H$  of Lie groups is a map of the group structure, and being smooth it also carries tangent data; its differential at the identity is a linear map  $d\varphi_e: \mathfrak{g} \rightarrow \mathfrak{h}$ , and

the central fact is that this linear map is compatible with everything the previous sections built. It respects the bracket, so it is a homomorphism of Lie algebras, and it commutes with the exponential, so the diagram relating  $\exp_G$  and  $\exp_H$  closes. These two compatibilities together say that passing from a group to its Lie algebra is a functor, and that  $\exp$  is a natural transformation.

The payoff is immediate and structural. Because  $\exp$  generates a connected group from a neighborhood of the identity, a homomorphism out of a connected group is pinned down completely by the single linear map  $d\varphi_e$ : the infinitesimal data determines the global map. This is the first half of the correspondence the book exists to develop, and it reduces questions about maps of groups to questions about maps of vector spaces equipped with a bracket. The section closes by reading the classical determinant homomorphism through this lens, recovering the identity  $\det \exp X = e^{\text{tr} X}$  of proposition 1.3.4 as an instance of naturality.

#### Definition 2.4.1: The Differential of a Homomorphism

Let  $\varphi: G \rightarrow H$  be a homomorphism of Lie groups, with Lie algebras  $\mathfrak{g} = T_e G$  and  $\mathfrak{h} = T_e H$ . The *differential* of  $\varphi$  is the linear map

$$d\varphi_e: \mathfrak{g} \longrightarrow \mathfrak{h}$$

given by the tangent map of  $\varphi$  at the identity, sending a tangent vector  $X \in T_e G$ , represented by a curve  $t \mapsto c(t)$  with  $c(0) = e$  and  $c'(0) = X$ , to the velocity  $d\varphi_e(X) = \left. \frac{d}{dt} \right|_0 \varphi(c(t)) \in T_e H$  of its image.

The differential is well defined precisely because  $\varphi$  preserves the identity:  $\varphi(e_G) = e_H$ , so  $\varphi$  maps a curve through  $e_G$  to a curve through  $e_H$ , and the tangent map at that point lands in  $\mathfrak{h} = T_{e_H} H$  as claimed. It is a linear map by the chain rule of [2], being the restriction of the smooth map  $\varphi$ 's tangent map  $d\varphi: TG \rightarrow TH$  to the fibre over the identity. So far  $d\varphi_e$  is only a linear map between vector spaces; the content of the section is that it interacts correctly with the two extra structures those vector spaces carry, the bracket and the exponential. For a homomorphism of matrix groups the differential is computed by differentiating  $\varphi$  along  $t \mapsto \exp(tX)$ , and for the classical inclusion homomorphisms it is nothing but the corresponding inclusion of matrix Lie algebras from theorem 1.4.3. A concrete instance appears within the worked example below, where  $\varphi = \det$  has differential the trace.

The mechanism that makes  $d\varphi_e$  respect the bracket is the notion of  $\varphi$ -relatedness of vector fields. Recall from [2] that two vector fields  $\xi$  on  $G$  and  $\eta$  on  $H$  are  $\varphi$ -related when  $d\varphi_g(\xi_g) = \eta_{\varphi(g)}$  for every  $g \in G$ , and that the bracket of vector fields is natural under this relation: if  $\xi_1$  is  $\varphi$ -related to  $\eta_1$  and  $\xi_2$  to  $\eta_2$ , then  $[\xi_1, \xi_2]$  is  $\varphi$ -related to  $[\eta_1, \eta_2]$ . The Lie algebra  $\mathfrak{g}$  was built in section 2.2 as the left-invariant vector fields, and the key observation is that the left-invariant field with value  $X$  at  $e$  is  $\varphi$ -related to the left-invariant field with value  $d\varphi_e X$ .

**Proposition 2.4.2: The Differential is a Lie Algebra Homomorphism**

Let  $\varphi: G \rightarrow H$  be a homomorphism of Lie groups. Then the differential  $d\varphi_e: \mathfrak{g} \rightarrow \mathfrak{h}$  is a homomorphism of Lie algebras:

$$d\varphi_e[X, Y] = [d\varphi_e X, d\varphi_e Y]$$

for all  $X, Y \in \mathfrak{g}$ , where the bracket on the left is that of  $\mathfrak{g}$  and the bracket on the right is that of  $\mathfrak{h}$ .

**Proof:**

Write  $L_g$  for left translation by  $g$ , and for a vector  $X \in \mathfrak{g} = T_e G$  let  $\tilde{X}$  denote the left-invariant vector field on  $G$  with  $\tilde{X}_e = X$ , characterized by  $\tilde{X}_g = d(L_g)_e X$  as in definition 2.2.1. By theorem 2.2.3, the bracket on  $\mathfrak{g}$  is defined by transporting the vector-field bracket, so  $[X, Y] = [\tilde{X}, \tilde{Y}]$ ; the same holds on  $H$  with fields denoted  $\widehat{(\cdot)}$ .

*Step 1: the invariant fields are  $\varphi$ -related.* Fix  $X \in \mathfrak{g}$  and set  $X' = d\varphi_e X \in \mathfrak{h}$ . The claim is that  $\tilde{X}$  on  $G$  is  $\varphi$ -related to  $\widehat{X'}$  on  $H$ , that is

$$d\varphi_g(\tilde{X}_g) = \widehat{X'}_{\varphi(g)} \quad \text{for all } g \in G \quad (2.10)$$

Because  $\varphi$  is a homomorphism,  $\varphi \circ L_g = L_{\varphi(g)} \circ \varphi$ : both sides send  $h \mapsto \varphi(gh) = \varphi(g)\varphi(h)$ . Taking differentials at  $e$  and applying the chain rule of [2],

$$d\varphi_g \circ d(L_g)_e = d(L_{\varphi(g)})_e \circ d\varphi_e$$

as maps  $T_e G \rightarrow T_{\varphi(g)} H$ . Evaluating both sides on  $X$  and using  $\tilde{X}_g = d(L_g)_e X$  on the left and  $\widehat{X'}_{\varphi(g)} = d(L_{\varphi(g)})_e X'$  with  $X' = d\varphi_e X$  on the right gives exactly (2.10). Thus  $\tilde{X}$  is  $\varphi$ -related to  $\widehat{d\varphi_e X}$  for every  $X$ .

*Step 2: apply naturality of the bracket.* Let  $X, Y \in \mathfrak{g}$ . By Step 1,  $\tilde{X}$  is  $\varphi$ -related to  $\widehat{d\varphi_e X}$  and  $\tilde{Y}$  is  $\varphi$ -related to  $\widehat{d\varphi_e Y}$ . The naturality of the vector-field bracket under  $\varphi$ -relatedness ([2]) then gives that  $[\tilde{X}, \tilde{Y}]$  is  $\varphi$ -related to  $[\widehat{d\varphi_e X}, \widehat{d\varphi_e Y}]$ . Evaluating the  $\varphi$ -relatedness at  $g = e$ , where  $\varphi(e) = e$ , yields

$$d\varphi_e([\tilde{X}, \tilde{Y}]_e) = [\widehat{d\varphi_e X}, \widehat{d\varphi_e Y}]_e$$

The left side is  $d\varphi_e[X, Y]$ , since  $[\tilde{X}, \tilde{Y}]_e = \widehat{[X, Y]}_e = [X, Y]$  by the definition of the bracket on  $\mathfrak{g}$ . The right side is  $[d\varphi_e X, d\varphi_e Y]$  by the same definition on  $\mathfrak{h}$ , since  $\widehat{Z}_e = Z$  for any  $Z \in \mathfrak{h}$ . Therefore  $d\varphi_e[X, Y] = [d\varphi_e X, d\varphi_e Y]$ , as we sought to show.

The proof isolates the one place where the group structure enters: (2.10) holds because  $\varphi$  intertwines left translations, which is a direct consequence of  $\varphi$  being a homomorphism. Everything after that is the general naturality of the bracket for smooth maps, imported from [2]. This is why the differential of a homomorphism respects the bracket while the differential of an arbitrary smooth map need not: only a homomorphism carries left-invariant fields to left-invariant fields. The result says that the assignment  $\varphi \mapsto d\varphi_e$  converts the analytic datum of a smooth group homomorphism into the purely algebraic datum of a Lie algebra homomorphism, and it is the reason the structure theory of Parts II

and III can be conducted at the level of Lie algebras and then transported back to groups.

The compatibility with the exponential is the second half, and it is the statement most often invoked in computation. The exponential of  $G$  is built from one-parameter subgroups, and  $\varphi$  carries one-parameter subgroups of  $G$  to one-parameter subgroups of  $H$ ; tracking the velocity through this transport is exactly what the differential records. The result is a square that commutes: applying  $\varphi$  after  $\exp_G$  equals applying  $\exp_H$  after  $d\varphi_e$ .

### Theorem 2.4.3: Naturality of the Exponential Map

Let  $\varphi: G \rightarrow H$  be a homomorphism of Lie groups with differential  $d\varphi_e: \mathfrak{g} \rightarrow \mathfrak{h}$ . Then for every  $X \in \mathfrak{g}$ ,

$$\varphi(\exp_G X) = \exp_H(d\varphi_e X)$$

equivalently, the square

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{d\varphi_e} & \mathfrak{h} \\ \exp_G \downarrow & & \downarrow \exp_H \\ G & \xrightarrow{\varphi} & H \end{array}$$

commutes.

#### Proof:

Fix  $X \in \mathfrak{g}$  and consider the curve

$$\sigma: \mathbb{R} \rightarrow H, \quad \sigma(t) = \varphi(\exp_G(tX))$$

*Step 1:  $\sigma$  is a one-parameter subgroup of  $H$ .* By proposition 2.3.3 the map  $t \mapsto \exp_G(tX)$  is the one-parameter subgroup of  $G$  with velocity  $X$ , so  $\exp_G((s+t)X) = \exp_G(sX) \exp_G(tX)$  and  $\exp_G(0) = e$ . Since  $\varphi$  is a homomorphism,

$$\sigma(s+t) = \varphi(\exp_G(sX) \exp_G(tX)) = \varphi(\exp_G(sX)) \varphi(\exp_G(tX)) = \sigma(s) \sigma(t)$$

and  $\sigma(0) = \varphi(e_G) = e_H$ . Being a composite of smooth maps,  $\sigma$  is smooth. Hence  $\sigma$  is a one-parameter subgroup of  $H$ .

*Step 2: identify its velocity.* Differentiating  $\sigma$  at  $t = 0$  by the chain rule of [2],

$$\sigma'(0) = \left. \frac{d}{dt} \right|_0 \varphi(\exp_G(tX)) = d\varphi_e \left( \left. \frac{d}{dt} \right|_0 \exp_G(tX) \right) = d\varphi_e(X)$$

where the last equality uses that  $t \mapsto \exp_G(tX)$  has velocity  $X$  at  $t = 0$ , again by proposition 2.3.3.

*Step 3: conclude by uniqueness.* By theorem 2.3.1, a one-parameter subgroup of  $H$  is determined by its velocity at  $t = 0$ , and the one with velocity  $d\varphi_e X$  is precisely  $t \mapsto \exp_H(t d\varphi_e X)$  by proposition 2.3.3. Steps 1 and 2 show  $\sigma$  is a one-parameter subgroup with this same velocity, so

$$\varphi(\exp_G(tX)) = \sigma(t) = \exp_H(t d\varphi_e X) \quad \text{for all } t \in \mathbb{R}$$

Setting  $t = 1$  gives  $\varphi(\exp_G X) = \exp_H(d\varphi_e X)$ , completing the proof.

The theorem is the computational engine of the correspondence. It converts the exponential of  $H$  into the exponential of  $G$  across any homomorphism, so that to evaluate  $\varphi$  on an element of the form  $\exp_G X$ , which is to say on any element near the identity, one need only know the single linear map  $d\varphi_e$  and the exponential of the target. For matrix groups, where by proposition 2.3.6 the intrinsic exponential is the matrix exponential, naturality reads:  $\varphi(\exp X) = \exp(d\varphi_e X)$  with both exponentials the matrix power series, and the classical identities of Chapter 1 fall out as special cases, as the worked example demonstrates. The commuting square also explains the terminology of  $\exp$  as a natural transformation: it is a family of maps  $\exp_G: \mathfrak{g} \rightarrow G$ , one per group, that commutes with every morphism, which is exactly the condition for a natural transformation. To be precise, both endpoints are functors into smooth manifolds: the source is  $\mathbf{Lie}$  followed by the forgetful functor  $\mathbf{LieAlg}_{\mathbb{R}} \rightarrow \mathbf{Man}$  sending  $\mathfrak{g}$  to its underlying vector space, written informally  $G \mapsto \mathfrak{g}$ , and the target is the forgetful functor  $\mathbf{LieGrp} \rightarrow \mathbf{Man}$ , written  $G \mapsto G$ ; the square above is the naturality condition relating them.

The most consequential corollary follows by combining naturality with the fact that  $\exp$  generates a connected group. Because a connected Lie group is covered by products of exponentials, and naturality determines  $\varphi$  on each exponential from  $d\varphi_e$ , the entire homomorphism is fixed by its differential.

#### Corollary 2.4.4: A Homomorphism is Determined by its Differential

Let  $G$  be a *connected* Lie group and  $H$  any Lie group. If  $\varphi, \psi: G \rightarrow H$  are homomorphisms of Lie groups with  $d\varphi_e = d\psi_e$ , then  $\varphi = \psi$ . Consequently the map  $\varphi \mapsto d\varphi_e$  from homomorphisms  $G \rightarrow H$  to Lie algebra homomorphisms  $\mathfrak{g} \rightarrow \mathfrak{h}$  is injective when  $G$  is connected.

#### Proof:

Suppose  $d\varphi_e = d\psi_e$ . For every  $X \in \mathfrak{g}$ , naturality (theorem 2.4.3) applied to each homomorphism gives

$$\varphi(\exp_G X) = \exp_H(d\varphi_e X) = \exp_H(d\psi_e X) = \psi(\exp_G X)$$

so  $\varphi$  and  $\psi$  agree on the image  $\exp_G(\mathfrak{g})$ . By theorem 2.3.4,  $\exp_G$  is a local diffeomorphism at 0, so its image contains an open neighborhood  $U$  of  $e$  in  $G$ , which we may take symmetric ( $U = U^{-1}$ ) by replacing it with  $U \cap U^{-1}$ ; and by corollary 2.3.5 a connected group  $G$  is generated by any neighborhood of  $e$ , so  $G = \langle U \rangle = \bigcup_{k \geq 1} U^k$  where  $U^k = \{u_1 \cdots u_k : u_i \in U\}$ .

It remains to propagate agreement on  $U$  to all of  $G$ . Let  $W = \{g \in G \mid \varphi(g) = \psi(g)\}$  be the set on which the two homomorphisms agree. Since  $\varphi, \psi$  are homomorphisms,  $W$  is a subgroup of  $G$ : if  $g, g' \in W$  then  $\varphi(gg'^{-1}) = \varphi(g)\varphi(g')^{-1} = \psi(g)\psi(g')^{-1} = \psi(gg'^{-1})$ . The subgroup  $W$  contains  $U$ , hence contains every product  $u_1 \cdots u_k$  of elements of  $U$ , hence contains  $\bigcup_{k \geq 1} U^k = G$ . Therefore  $W = G$ , that is  $\varphi = \psi$ . The final claim restates this: two homomorphisms with equal image under  $\varphi \mapsto d\varphi_e$ , that is with  $d\varphi_e = d\psi_e$ , are themselves equal, so the map is injective, completing the proof.

The corollary is the reason the Lie algebra deserves to be called the infinitesimal group. For a connected group there is at most one homomorphism with a prescribed differential, so the passage  $\varphi \mapsto d\varphi_e$  loses no information about  $\varphi$ ; the only question left open is existence, whether a given Lie algebra homomorphism  $\mathfrak{g} \rightarrow \mathfrak{h}$  arises from some group homomorphism. Existence requires more, and it fails in general: the differential can be an isomorphism of Lie algebras without  $\varphi$  being an isomorphism of groups, because global topology intervenes. The exercises exhibit the standard

witness, the double cover  $SU(2) \rightarrow SO(3)$ , whose differential is the isomorphism  $\mathfrak{su}(2) \cong \mathfrak{so}(3)$  of example 1.4.7; and the sharp existence statement, that when  $G$  is simply connected every Lie algebra homomorphism does integrate, is Lie's second theorem, proved in Chapter 3. Injectivity here plus that existence theorem there is the full equivalence.

Both compatibilities can now be packaged categorically. The assignment  $G \mapsto \mathfrak{g}$  on objects and  $\varphi \mapsto d\varphi_e$  on morphisms is a functor, and verifying functoriality amounts to the chain rule together with the bracket-preservation just proved.

#### Proposition 2.4.5: The Lie Functor

Let **LieGrp** be the category of Lie groups and Lie group homomorphisms, and **LieAlg<sub>ℝ</sub>** the category of finite-dimensional real Lie algebras and Lie algebra homomorphisms. The assignment

$$\text{Lie}: \mathbf{LieGrp} \rightarrow \mathbf{LieAlg}_{\mathbb{R}}, \quad G \mapsto \mathfrak{g} = T_e G, \quad (\varphi: G \rightarrow H) \mapsto (d\varphi_e: \mathfrak{g} \rightarrow \mathfrak{h})$$

is a functor. That is,  $d(\text{id}_G)_e = \text{id}_{\mathfrak{g}}$ , and for composable homomorphisms  $\varphi: G \rightarrow H$  and  $\psi: H \rightarrow K$ ,

$$d(\psi \circ \varphi)_e = d\psi_e \circ d\varphi_e$$

#### Proof:

Each object  $\mathfrak{g} = T_e G$  is a finite-dimensional real Lie algebra by theorem 2.2.3 and proposition 2.2.5, and each morphism  $d\varphi_e$  is a Lie algebra homomorphism by proposition 2.4.2, so Lie lands in **LieAlg<sub>ℝ</sub>** on both objects and morphisms. For the identity, the differential of the identity map  $\text{id}_G: G \rightarrow G$  at  $e$  is the identity linear map on  $T_e G$ , since  $\text{id}_G$  sends every curve to itself and hence every velocity to itself; thus  $d(\text{id}_G)_e = \text{id}_{\mathfrak{g}}$ . For composition, the chain rule of [2] applied to the composite  $\psi \circ \varphi$  at the identity, using  $\varphi(e_G) = e_H$ , gives

$$d(\psi \circ \varphi)_e = d\psi_{\varphi(e_G)} \circ d\varphi_e = d\psi_{e_H} \circ d\varphi_e$$

which is  $d\psi_e \circ d\varphi_e$ . Both functor axioms hold, completing the proof.

The functor Lie is the formal container for the entire correspondence. corollary 2.4.4 says Lie is faithful on the full subcategory of connected Lie groups, since it is injective on each morphism set out of a connected group. The reverse direction, reconstructing a group homomorphism, and even a group, from Lie algebra data, is the substance of Lie's theorems. When  $G$  is simply connected, every Lie algebra homomorphism  $\mathfrak{g} \rightarrow \mathfrak{h}$  is  $d\varphi_e$  for a unique  $\varphi$ , so Lie becomes fully faithful there; and every finite-dimensional real Lie algebra is  $\text{Lie}(G)$  for some simply connected  $G$ , so Lie is essentially surjective onto **LieAlg<sub>ℝ</sub>**. These two facts upgrade Lie to an equivalence between the category of simply connected Lie groups and the category of finite-dimensional real Lie algebras, which is Lie's third theorem. The equivalence, together with the integration of Lie algebra homomorphisms, is proved in Chapter 3; the present section supplies the functor itself and its faithfulness on connected groups, which is one half of the equivalence.

The determinant homomorphism is the cleanest place to watch naturality recover a Chapter 1 formula. Its differential is the trace, and feeding this through the naturality square with the multiplicative group of the field as target reproduces  $\det \exp X = e^{\text{tr} X}$  with the scalar exponential doing the work on the target side.

**Example 2.4.6: The Determinant and the Trace**

Take  $G = \mathrm{GL}_n(\mathbb{F})$  and  $H = \mathbb{F}^\times$ , the multiplicative group of nonzero scalars, a one-dimensional Lie group with Lie algebra  $\mathfrak{h} = T_1\mathbb{F}^\times = \mathbb{F}$  and exponential map the ordinary scalar exponential  $\exp_H = \exp: \mathbb{F} \rightarrow \mathbb{F}^\times, z \mapsto e^z$ . The determinant

$$\det: \mathrm{GL}_n(\mathbb{F}) \longrightarrow \mathbb{F}^\times$$

is a smooth homomorphism, since  $\det(AB) = \det A \det B$  and the determinant is polynomial in the entries.

The differential  $d(\det)_I: \mathfrak{gl}_n(\mathbb{F}) \rightarrow \mathbb{F}$  is the trace. To see this, differentiate  $\det$  along the curve  $t \mapsto \exp(tX) = I + tX + O(t^2)$  through  $I$ . Expand the determinant of  $I + tX$  as the sum over permutations  $\sigma$  of  $\pm \prod_i (I + tX)_{i\sigma(i)}$ . Any  $\sigma \neq \mathrm{id}$  moves at least two indices, so its term has at least two off-diagonal factors  $tX_{i\sigma(i)}$ , each carrying a factor  $t$ , and hence contributes only at order  $t^2$ ; only the identity permutation contributes a linear term, namely  $\prod_i (1 + tX_{ii}) = 1 + t \sum_i X_{ii} + O(t^2)$ , so

$$\det(I + tX + O(t^2)) = 1 + t \operatorname{tr} X + O(t^2)$$

whence

$$d(\det)_I(X) = \left. \frac{d}{dt} \right|_0 \det \exp(tX) = \operatorname{tr} X$$

Thus  $d(\det)_I = \operatorname{tr}: \mathfrak{gl}_n(\mathbb{F}) \rightarrow \mathbb{F}$ , which is indeed a Lie algebra homomorphism to the abelian  $\mathbb{F}$ :  $\operatorname{tr}[X, Y] = \operatorname{tr}(XY) - \operatorname{tr}(YX) = 0 = [\operatorname{tr} X, \operatorname{tr} Y]$ , consistent with proposition 2.4.2.

Now apply naturality (theorem 2.4.3) to  $\varphi = \det$ . The square reads  $\det(\exp_G X) = \exp_H(d(\det)_I X)$ , and with  $\exp_G$  the matrix exponential (proposition 2.3.6),  $\exp_H$  the scalar exponential, and  $d(\det)_I = \operatorname{tr}$ , this is

$$\det \exp X = e^{\operatorname{tr} X}$$

the identity of proposition 1.3.4. The Chapter 1 computation, which proved  $\det \exp X = e^{\operatorname{tr} X}$  by triangularizing  $X$  and multiplying eigenvalue exponentials, is here revealed as a single instance of naturality: the determinant is a homomorphism, so it must intertwine the two exponentials, and its differential can only be the trace.

The example is the template for reading any classical homomorphism through its differential. The inclusion  $\mathrm{SU}(2) \hookrightarrow \mathrm{SU}(3)$  has differential the inclusion  $\mathfrak{su}(2) \hookrightarrow \mathfrak{su}(3)$ ; the projection of a product  $G \times H \rightarrow G$  has differential the projection  $\mathfrak{g} \oplus \mathfrak{h} \rightarrow \mathfrak{g}$ ; and, as the exercises develop, conjugation  $C_g: x \mapsto gxg^{-1}$  has a differential that is a homomorphism of  $\mathfrak{g}$  to itself, the value at  $g$  of the adjoint representation constructed in section 2.5. In every case the differential is easier to compute than the map, and naturality recovers the map on exponentials from it.

**Exercise 2.4.1**

1. For a fixed  $g$  in a Lie group  $G$ , let  $C_g: G \rightarrow G$  be conjugation,  $C_g(x) = gxg^{-1}$ . Verify that  $C_g$  is a Lie group homomorphism (indeed an automorphism), and show that its differential  $d(C_g)_e: \mathfrak{g} \rightarrow \mathfrak{g}$  is a Lie algebra automorphism. For a matrix group  $G \subseteq \mathrm{GL}_n(\mathbb{F})$ , compute  $d(C_g)_e(X)$  explicitly in terms of matrix multiplication, and use theorem 2.4.3 to write  $g \exp(X) g^{-1}$  as a single exponential. (This differential, as a function of  $g$ , is the adjoint representation  $\operatorname{Ad}(g)$  constructed in section 2.5.)
2. The double cover  $p: \mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$  of theorem 1.5.5 is a homomorphism of Lie groups.

Explain why its differential  $dp_e: \mathfrak{su}(2) \rightarrow \mathfrak{so}(3)$  is an isomorphism of Lie algebras, even though  $p$  is not injective as a map of groups. Reconcile this with corollary 2.4.4, and state what additional global datum distinguishes  $SU(2)$  from  $SO(3)$ .

3. Classify the connected Lie groups of dimension 1 up to isomorphism: show there are exactly two, namely  $\mathbb{R}$  (under addition) and the circle  $S^1$ . Use that a one-dimensional Lie algebra is abelian, that  $\exp$  is a homomorphism onto the identity component for an abelian group, and corollary 2.4.4.
4. Determine all Lie group homomorphisms  $S^1 \rightarrow S^1$ , where  $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ . Identify  $\text{Lie}(S^1)$  with  $\mathbb{R}$  so that  $\exp(t) = e^{2\pi it}$ , compute the differential of the candidate map  $z \mapsto z^n$ , and use corollary 2.4.4 to show these are all of them.
5. Let  $\det: U(n) \rightarrow S^1$  be the determinant, a homomorphism onto the circle. Compute its differential  $d(\det)_I: \mathfrak{u}(n) \rightarrow \text{Lie}(S^1)$ , and verify theorem 2.4.3 directly on a skew-Hermitian  $X \in \mathfrak{u}(n)$  by comparing  $\det \exp X$  with the scalar exponential of  $d(\det)_I(X)$ .
6. Let  $\varphi: G \rightarrow H$  be a homomorphism of Lie groups with  $G$  connected. Show that if  $d\varphi_e = 0$  then  $\varphi$  is the constant map  $g \mapsto e_H$ . Deduce that if  $d\varphi_e$  is injective then  $\varphi$  is a local diffeomorphism onto its image near  $e$ , and give an example (from exercise 2.4.1 (2)) where  $d\varphi_e$  is injective but  $\varphi$  is not globally injective.

## 2.5 The Adjoint Representations Ad and ad; the Bracket Intrinsically

The intrinsic Lie algebra  $\mathfrak{g} = \text{Lie}(G)$  of definition 2.2.4 was built by transporting the bracket of vector fields from left-invariant fields to the tangent space  $T_e G$ , an operation with no visible connection to the commutator  $XY - YX$  that governed the matrix Lie algebras of chapter 1. This section supplies the missing link. The group acts on itself by conjugation, and differentiating that action twice produces two representations, Ad on the group level and ad on the algebra level, whose interplay expresses the bracket. The central result identifies the left-invariant-field bracket of theorem 2.2.3 with ad, and hence, for a matrix group, with the commutator of proposition 1.4.4. When the dust settles the bracket has three faces that turn out to be one: the commutator from Chapter 1, the left-invariant-field bracket from section 2.2, and the operator ad introduced here.

The starting point is the conjugation automorphism. For each  $g \in G$  the map  $C_g: G \rightarrow G$ ,  $C_g(h) = ghg^{-1}$ , is a Lie group automorphism fixing the identity, so its differential at  $e$  is a linear automorphism of  $T_e G = \mathfrak{g}$ . Collecting these differentials as  $g$  varies gives a representation of  $G$  on its own Lie algebra.

**Definition 2.5.1: The Adjoint Representation  $\text{Ad}$** 

Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g} = T_e G$ . For  $g \in G$  let  $C_g: G \rightarrow G$  be the conjugation automorphism  $C_g(h) = ghg^{-1}$ , a diffeomorphism fixing  $e$  (proposition 2.1.2). Its differential at  $e$  is a linear isomorphism of  $\mathfrak{g}$ , denoted

$$\text{Ad}(g) = d(C_g)_e: \mathfrak{g} \rightarrow \mathfrak{g}$$

The resulting map  $\text{Ad}: G \rightarrow \text{GL}(\mathfrak{g})$  is the *adjoint representation* of  $G$ .

That  $\text{Ad}$  deserves the name representation is the assertion that it is a Lie group homomorphism into  $\text{GL}(\mathfrak{g})$ , the group of linear automorphisms of the finite-dimensional vector space  $\mathfrak{g}$ . This is where the definition earns its keep, and the proof is a differentiation of the identity  $C_{gh} = C_g \circ C_h$ .

**Proposition 2.5.2:  $\text{Ad}$  is a Lie Group Homomorphism**

The map  $\text{Ad}: G \rightarrow \text{GL}(\mathfrak{g})$  of definition 2.5.1 is a homomorphism of Lie groups: it is smooth, and

$$\text{Ad}(gh) = \text{Ad}(g) \text{Ad}(h), \quad \text{Ad}(e) = \text{id}_{\mathfrak{g}}$$

for all  $g, h \in G$ .

**Proof:**

The conjugation maps compose by  $C_{gh} = C_g \circ C_h$ , since  $C_{gh}(x) = (gh)x(gh)^{-1} = g(hxh^{-1})g^{-1} = C_g(C_h(x))$ , and  $C_e = \text{id}_G$ . Both  $C_g$  and  $C_h$  fix  $e$ , so the chain rule for differentials applies at  $e$ :

$$\text{Ad}(gh) = d(C_{gh})_e = d(C_g \circ C_h)_e = d(C_g)_e \circ d(C_h)_e = \text{Ad}(g) \text{Ad}(h)$$

and  $\text{Ad}(e) = d(\text{id}_G)_e = \text{id}_{\mathfrak{g}}$ . In particular  $\text{Ad}(g) \text{Ad}(g^{-1}) = \text{Ad}(e) = \text{id}$ , so each  $\text{Ad}(g)$  is invertible and  $\text{Ad}$  lands in  $\text{GL}(\mathfrak{g})$ . Smoothness is a consequence of the smoothness of the group operations: the conjugation map  $(g, h) \mapsto ghg^{-1}$  is smooth on  $G \times G$  as a composite of multiplication and inversion, and  $\text{Ad}(g)$  is the partial differential in the  $h$ -variable at  $h = e$ , which depends smoothly on  $g$  because differentiation of a smooth family of maps produces a smooth family of differentials ([2]). Thus  $\text{Ad}$  is smooth, completing the proof.

The homomorphism  $\text{Ad}$  packages the conjugation action of  $G$  on itself into its infinitesimal effect on the tangent space at  $e$ . It sees only the germ of conjugation at the identity, which is exactly the data a Lie algebra is built to remember. Because  $\text{GL}(\mathfrak{g})$  is itself a Lie group, with Lie algebra the endomorphism algebra  $\mathfrak{gl}(\mathfrak{g}) = \text{End}(\mathfrak{g})$ , the machinery of section 2.4 applies to  $\text{Ad}$ : it has a differential at  $e$ , and that differential is the second of the two representations.

**Definition 2.5.3: The Adjoint Representation ad**

Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g}$ . The *adjoint representation* of  $\mathfrak{g}$  is the differential of Ad at the identity,

$$\mathrm{ad} = d(\mathrm{Ad})_e: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g}) = \mathrm{End}(\mathfrak{g})$$

Explicitly, for  $X \in \mathfrak{g}$  the endomorphism  $\mathrm{ad}(X)$  of  $\mathfrak{g}$  is

$$\mathrm{ad}(X) = \left. \frac{d}{dt} \right|_{t=0} \mathrm{Ad}(\exp tX)$$

the velocity at  $t = 0$  of the curve  $t \mapsto \mathrm{Ad}(\exp tX)$  in  $\mathrm{GL}(\mathfrak{g})$ .

The explicit formula is the naturality of exp from theorem 2.4.3 read for the homomorphism Ad: the one-parameter subgroup  $t \mapsto \exp(tX)$  of  $G$  maps under Ad to the one-parameter subgroup  $t \mapsto \mathrm{Ad}(\exp tX)$  of  $\mathrm{GL}(\mathfrak{g})$ , whose generator is  $d(\mathrm{Ad})_e(X) = \mathrm{ad}(X)$ . Since  $\mathrm{GL}(\mathfrak{g})$  is a matrix group, its exponential is the matrix exponential (proposition 2.3.6), so the naturality square specializes to

$$\mathrm{Ad}(\exp tX) = \exp(t \mathrm{ad}(X)) \quad (2.11)$$

an identity of endomorphisms of  $\mathfrak{g}$  that will reappear in the exercises as the statement  $\mathrm{Ad}(\exp X) = e^{\mathrm{ad}X}$ . Differentiating (2.11) at  $t = 0$  recovers the definition of  $\mathrm{ad}(X)$ , and integrating it recovers Ad from ad on the identity component.

By proposition 2.4.2 the differential of a Lie group homomorphism is a Lie algebra homomorphism, so ad preserves brackets:  $\mathrm{ad}([X, Y]) = [\mathrm{ad}(X), \mathrm{ad}(Y)]$ , where the bracket on the right is the commutator of endomorphisms in  $\mathrm{End}(\mathfrak{g})$ . This is the Jacobi identity in disguise, a point developed in the exercises. What remains, and it is the conceptual hinge of the chapter, is to compute ad in terms of the bracket already available on  $\mathfrak{g}$ , and to show the two agree.

The bracket on  $\mathfrak{g}$  of definition 2.2.4 was defined by carrying the bracket of vector fields across the isomorphism  $X \mapsto \tilde{X}$  of theorem 2.2.3 between  $T_e G$  and the left-invariant fields. The bracket induced by ad is a second candidate  $\mathrm{ad}(X)Y$ . The next theorem shows these coincide, which is what licenses writing  $[X, Y] = \mathrm{ad}(X)Y$  without ambiguity and, through the matrix case below, ties the whole apparatus back to the commutator.

**Theorem 2.5.4: The Bracket is the Adjoint Action**

Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g} = T_e G$ , and let  $[-, -]$  denote the bracket of definition 2.2.4, transported from the Lie bracket of left-invariant vector fields via theorem 2.2.3. Then for all  $X, Y \in \mathfrak{g}$ ,

$$[X, Y] = \mathrm{ad}(X)Y = \left. \frac{d}{dt} \right|_{t=0} \mathrm{Ad}(\exp tX) Y$$

The abstract bracket satisfying the axioms of definition 1.4.5 is therefore the infinitesimal conjugation action ad, and the two brackets on  $\mathfrak{g}$  coincide.

**Proof:**

Write  $\tilde{X}$  for the left-invariant vector field with  $\tilde{X}_e = X$ , so that by definition of the bracket on  $\mathfrak{g}$ ,

$$[X, Y] = [\tilde{X}, \tilde{Y}]_e \quad (2.12)$$

the right side being the value at  $e$  of the Lie bracket of the two left-invariant fields. The proof computes  $[\tilde{X}, \tilde{Y}]_e$  by the flow characterization of the vector-field bracket and matches it with  $\text{ad}(X)Y$ .

*Step 1: the flow of a left-invariant field is right translation.* Let  $\gamma_X(t) = \exp(tX)$  be the integral curve of  $\tilde{X}$  through  $e$ , which is the one-parameter subgroup with velocity  $X$  (proposition 2.3.3). Because  $\tilde{X}$  is left-invariant, its integral curve through an arbitrary point  $g$  is  $t \mapsto g \exp(tX) = L_g(\exp tX)$ : applying the diffeomorphism  $L_g$  carries the integral curve through  $e$  to one through  $g$ , and  $L_g$  sends the field  $\tilde{X}$  to itself. Hence the flow of  $\tilde{X}$  at time  $t$  is right translation by  $\exp(tX)$ ,

$$\Phi_t^{\tilde{X}} = R_{\exp(tX)}, \quad g \mapsto g \exp(tX) \quad (2.13)$$

These flows exist for all  $t \in \mathbb{R}$  because left-invariant fields are complete (proposition 2.2.5).

*Step 2: the bracket as a Lie derivative.* The Lie bracket of two vector fields is the Lie derivative of one along the flow of the other: for any smooth vector fields  $\tilde{X}, \tilde{Y}$ ,

$$[\tilde{X}, \tilde{Y}]_g = \left. \frac{d}{dt} \right|_{t=0} (d\Phi_{-t}^{\tilde{X}})_{\Phi_t^{\tilde{X}}(g)} (\tilde{Y}_{\Phi_t^{\tilde{X}}(g)})$$

the standard formula  $[\tilde{X}, \tilde{Y}] = \mathcal{L}_{\tilde{X}}\tilde{Y}$  for the bracket in terms of the pushforward of  $\tilde{Y}$  by the backward flow of  $\tilde{X}$  ([2]). Evaluate at  $g = e$  and substitute the flow (2.13), so  $\Phi_t^{\tilde{X}}(e) = \exp(tX)$  and  $\Phi_{-t}^{\tilde{X}} = R_{\exp(-tX)}$ :

$$[\tilde{X}, \tilde{Y}]_e = \left. \frac{d}{dt} \right|_{t=0} (dR_{\exp(-tX)})_{\exp(tX)} (\tilde{Y}_{\exp(tX)}) \quad (2.14)$$

*Step 3: rewrite the inner expression through conjugation.* The field  $\tilde{Y}$  is left-invariant, so its value at  $\exp(tX)$  is the left translate of  $Y$ ,

$$\tilde{Y}_{\exp(tX)} = (dL_{\exp(tX)})_e Y$$

Insert this into (2.14). The differential  $(dR_{\exp(-tX)})_{\exp(tX)}$  acts on the result, and since the maps  $L_{\exp(tX)}$  and  $R_{\exp(-tX)}$  commute as diffeomorphisms of  $G$  (left and right translations always commute), their composite is the conjugation map:

$$R_{\exp(-tX)} \circ L_{\exp(tX)} = (h \mapsto \exp(tX) h \exp(-tX)) = C_{\exp(tX)}$$

Both  $L_{\exp(tX)}$  and  $R_{\exp(-tX)}$  are diffeomorphisms, so the chain rule gives, for the composite evaluated on  $Y \in T_e G$ ,

$$(dR_{\exp(-tX)})_{\exp(tX)} (dL_{\exp(tX)})_e Y = d(C_{\exp(tX)})_e Y = \text{Ad}(\exp tX) Y$$

the last equality by the definition of  $\text{Ad}$  in definition 2.5.1. Substituting into (2.14),

$$[\tilde{X}, \tilde{Y}]_e = \left. \frac{d}{dt} \right|_{t=0} \text{Ad}(\exp tX) Y \quad (2.15)$$

*Step 4: identify with ad.* By definition 2.5.3 the right side of (2.15) is precisely  $\text{ad}(X)Y$ , the velocity at  $t = 0$  of the curve  $t \mapsto \text{Ad}(\exp tX)Y$  in  $\mathfrak{g}$ . Combining with (2.12) and (2.15),

$$[X, Y] = [\tilde{X}, \tilde{Y}]_e = \left. \frac{d}{dt} \right|_{t=0} \text{Ad}(\exp tX) Y = \text{ad}(X)Y$$

as we sought to show.

The theorem is the reconciliation the chapter was built toward. The bracket on  $\mathfrak{g}$  was first defined abstractly, as whatever operation the vector-field bracket induces on  $T_e G$ , with the alternating and Jacobi axioms of definition 1.4.5 guaranteed but its meaning opaque. theorem 2.5.4 gives it a concrete face:  $[X, Y]$  is the rate at which conjugation by  $\exp(tX)$  moves  $Y$ , measured at  $t = 0$ . The bracket is thus the infinitesimal record of the failure of the two one-parameter subgroups  $\exp(tX)$  and  $\exp(sY)$  to commute, exactly the interpretation the matrix commutator carried in proposition 1.4.4, now established with no matrices in sight. Every structural fact about the bracket, the Jacobi identity included, is now a fact about the conjugation action of the group on itself.

The identification also explains the notation. The operator  $\text{ad}(X)$  acts on  $\mathfrak{g}$  by  $Y \mapsto [X, Y]$ , so  $\text{ad}$  is the map sending  $X$  to left-bracketing by  $X$ . That  $\text{ad}$  is a Lie algebra homomorphism (proposition 2.4.2) then reads  $\text{ad}([X, Y]) = [\text{ad}(X), \text{ad}(Y)]$ , which unwinds to  $[[X, Y], Z] = [X, [Y, Z]] - [Y, [X, Z]]$  for all  $Z$ , a rearrangement of the Jacobi identity. The single homomorphism  $\text{ad}$  therefore encodes the entire bracket and its one nontrivial axiom at once.

Everything reduces to the commutator in the matrix case, closing the loop with Chapter 1. For a group  $G \subseteq \text{GL}_n(\mathbb{F})$  the intrinsic constructions must return the formulas computed by hand there, and the next result verifies this for  $\text{Ad}$  and  $\text{ad}$  simultaneously.

#### Proposition 2.5.5: Ad and ad for a Matrix Group

Let  $G \subseteq \text{GL}_n(\mathbb{F})$  be a matrix group with Lie algebra  $\mathfrak{g} \subseteq M_n(\mathbb{F})$ , identified with  $T_I G$  as in corollary 1.4.6. Then for all  $g \in G$  and  $X, Y \in \mathfrak{g}$ ,

$$\text{Ad}(g)X = gXg^{-1}, \quad \text{ad}(X)Y = XY - YX = [X, Y]$$

The intrinsic adjoint action is conjugation of matrices, and the intrinsic bracket  $\text{ad}(X)Y$  is the commutator of proposition 1.4.4. The intrinsic bracket on  $\mathfrak{gl}_n(\mathbb{F})$  is therefore the matrix commutator.

#### Proof:

*Step 1: the formula for Ad.* Fix  $g \in G$ . The conjugation map  $C_g(h) = ghg^{-1}$  is the restriction to  $G$  of the ambient linear map  $M \mapsto gMg^{-1}$  on  $M_n(\mathbb{F})$ , which is its own differential at every point since it is linear. For  $X \in \mathfrak{g} = T_I G$ , represent  $X$  as the velocity of the curve  $t \mapsto \exp(tX)$  through  $I$  (proposition 2.3.6 identifies the intrinsic  $\exp$  with the matrix exponential). The conjugation-equivariance of the exponential (proposition 1.3.4) gives

$$C_g(\exp tX) = g \exp(tX) g^{-1} = \exp(t gXg^{-1}) \quad (2.16)$$

the last equality because  $g \exp(tX) g^{-1} = \exp(g(tX)g^{-1})$  by the series computation  $g(\sum_k \frac{(tX)^k}{k!})g^{-1} =$

$\sum_k \frac{(g t X g^{-1})^k}{k!}$ . Differentiating (2.16) at  $t = 0$ ,

$$\text{Ad}(g)X = d(C_g)_I X = \left. \frac{d}{dt} \right|_{t=0} \exp(t g X g^{-1}) = g X g^{-1}$$

using  $\left. \frac{d}{dt} \right|_0 \exp(tZ) = Z$  with  $Z = g X g^{-1}$ . The result  $g X g^{-1}$  lies in  $\mathfrak{g}$ , consistent with  $\text{Ad}(g)$  being an endomorphism of  $\mathfrak{g}$ .

*Step 2: the formula for ad.* Apply Step 1 with  $g = \exp(sX)$ , so  $\text{Ad}(\exp sX)Y = \exp(sX) Y \exp(-sX)$  for  $Y \in \mathfrak{g}$ . Differentiating in  $s$  at  $s = 0$ , with  $\left. \frac{d}{ds} \right|_0 \exp(sX) = X$  and  $\left. \frac{d}{ds} \right|_0 \exp(-sX) = -X$ , the product rule gives

$$\text{ad}(X)Y = \left. \frac{d}{ds} \right|_{s=0} \exp(sX) Y \exp(-sX) = XY - YX = [X, Y]$$

which is the commutator. By theorem 2.5.4 the left side is the intrinsic bracket, so the intrinsic bracket equals the matrix commutator, recovering proposition 1.4.4. For  $G = \text{GL}_n(\mathbb{F})$  this says the intrinsic Lie algebra  $\mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$  carries the commutator bracket, completing the proof.

The proposition is the payoff of the whole chapter in one line: the bracket that section 2.2 manufactured from left-invariant vector fields, with no reference to matrices, becomes the ordinary commutator  $XY - YX$  the moment  $G$  is realized inside some  $\text{GL}_n(\mathbb{F})$ . Two constructions that looked unrelated, the vector-field bracket of theorem 2.2.3 and the commutator of proposition 1.4.4, are the same operation seen from two sides. The intrinsic definition is the more general one, applying to any Lie group, and proposition 2.5.5 certifies that it specializes correctly on the matrix groups where an independent computation exists.

The running example makes the abstract statement palpable. For  $\text{SU}(2)$  the adjoint representation is exactly the conjugation action that produced the double cover in Chapter 1, and ad is the cross product.

### Example 2.5.6: Ad and ad for $\text{SU}(2)$ and the Rotation Action

Take  $G = \text{SU}(2)$ , with Lie algebra  $\mathfrak{su}(2)$  the traceless skew-Hermitian  $2 \times 2$  matrices of example 1.4.7, and its basis  $X_1, X_2, X_3$  with  $[X_j, X_k] = \sum_\ell \varepsilon_{j k \ell} X_\ell$ . By proposition 2.5.5 the adjoint representation is conjugation,

$$\text{Ad}(g)X = g X g^{-1}, \quad g \in \text{SU}(2), \quad X \in \mathfrak{su}(2)$$

which is precisely the map  $\varphi(g): X \mapsto g X g^{-1}$  of theorem 1.5.5. That theorem showed  $\varphi(g)$  preserves the Euclidean structure on  $\mathfrak{su}(2) \cong \mathbb{R}^3$  given by the determinant form and lands in  $\text{SO}(3)$ ; in the present language this says

$$\text{Ad}: \text{SU}(2) \rightarrow \text{SO}(3) \subset \text{GL}(\mathfrak{su}(2))$$

is the adjoint representation, and the double cover  $\varphi = \text{Ad}$  is nothing but Ad regarded as a homomorphism onto the rotation group of the three-dimensional space  $\mathfrak{su}(2)$ . The kernel of Ad is the center  $\{\pm I\}$ , the two-to-one fibers of the cover.

On the algebra level, proposition 2.5.5 gives  $\text{ad}(X)Y = [X, Y]$ , so on the basis

$$\text{ad}(X_1)X_2 = [X_1, X_2] = X_3, \quad \text{ad}(X_1)X_3 = [X_1, X_3] = -X_2, \quad \text{ad}(X_1)X_1 = 0$$

In the ordered basis  $(X_1, X_2, X_3)$  the operator  $\text{ad}(X_1)$  has matrix

$$\text{ad}(X_1) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

the generator of rotation about the first axis, and  $\text{ad}(X_2), \text{ad}(X_3)$  are the analogous generators about the other two axes. Under the isomorphism  $\mathfrak{su}(2) \cong (\mathbb{R}^3, \times)$  of example 1.4.7 the operator  $\text{ad}(X) = [X, -]$  becomes  $v \mapsto v \times (-)$ , cross product with the vector  $v = \Phi(X)$ , which is exactly the infinitesimal rotation about the axis  $v$ . The identity  $\text{Ad}(\exp tX) = \exp(t \text{ad}(X))$  of (2.11) then states that conjugating by  $\exp(tX)$  is the finite rotation whose generator is cross product with  $v$ , recovering the computation in Step 4 of theorem 1.5.5 that  $\exp(\frac{\theta}{2}u)$  maps under Ad to the rotation by  $\theta$  about  $u$ . The factor of two between the group parameter and the rotation angle, the arithmetic heart of the double cover, is precisely the passage through  $\exp(t \text{ad}(X))$  with  $\text{ad}(X)$  the generator normalized by the bracket relations (1.5).

The example closes the circle the book opened with. In Chapter 1 the conjugation action of  $\text{SU}(2)$  on  $\mathfrak{su}(2)$  was constructed by hand to build the double cover  $\text{SU}(2) \rightarrow \text{SO}(3)$ , before the words “adjoint representation” were available; that action is Ad, and the double cover is Ad itself, the homomorphism onto the orthogonal group of the invariant form on  $\mathfrak{su}(2)$ . What Chapter 1 saw as a geometric accident, that a unit-quaternion group double-covers the rotation group, is the adjoint representation of  $\text{SU}(2)$  read through the isomorphism  $\mathfrak{su}(2) \cong \mathbb{R}^3$ .

The bracket now stands identified in three forms. It is the matrix commutator  $XY - YX$  of proposition 1.4.4, the operation Chapter 1 extracted by differentiating a conjugation curve. It is the left-invariant-field bracket  $[\tilde{X}, \tilde{Y}]_e$  of theorem 2.2.3, the intrinsic definition that requires no embedding. And it is the adjoint action  $\text{ad}(X)Y = \frac{d}{dt}\big|_0 \text{Ad}(\exp tX)Y$  of theorem 2.5.4, the infinitesimal conjugation of the group on its own algebra. These are one operation, and the equivalence of the three descriptions is what makes the Lie algebra a faithful and computable shadow of the group. The bilinear form  $\kappa(X, Y) = \text{tr}(\text{ad}X \text{ad}Y)$  built from ad, invariant under the adjoint action of  $G$ , is the *Killing form* (defined and developed in exercise 2.5.1 (4) below); it measures the semisimplicity of  $\mathfrak{g}$  and is the tool that drives the structure theory of Part II and the classification of Chapter 6 onward.

### Exercise 2.5.1

1. Prove the identity  $\text{Ad}(\exp X) = e^{\text{ad}X}$  for all  $X \in \mathfrak{g}$ , where the right side is the exponential in  $\text{GL}(\mathfrak{g})$  of the endomorphism  $\text{ad}X \in \text{End}(\mathfrak{g})$ . Deduce that for a matrix group  $G \subseteq \text{GL}_n(\mathbb{F})$ , conjugation  $e^X Y e^{-X}$  is given by the convergent series  $Y + [X, Y] + \frac{1}{2!}[X, [X, Y]] + \cdots$ .
2. Show that the kernel of  $\text{Ad}: G \rightarrow \text{GL}(\mathfrak{g})$  contains the center  $Z(G)$ , and that for a *connected* Lie group  $G$  the kernel is exactly  $Z(G)$ . Conclude that a connected  $G$  acts trivially by conjugation on its Lie algebra if and only if  $G$  is abelian.
3. Show that for each  $X \in \mathfrak{g}$  the operator  $\text{ad}(X): \mathfrak{g} \rightarrow \mathfrak{g}$  is a *derivation* of the bracket, that is

$$\text{ad}(X)[Y, Z] = [\text{ad}(X)Y, Z] + [Y, \text{ad}(X)Z]$$

for all  $Y, Z \in \mathfrak{g}$ , and prove that this identity is equivalent to the Jacobi identity of definition 1.4.5.

4. Define the *Killing form*  $\kappa(X, Y) = \text{tr}(\text{ad}X \text{ad}Y)$  on  $\mathfrak{g}$ . Show that  $\kappa$  is a symmetric bilinear form, and that it is *Ad-invariant*:  $\kappa(\text{Ad}(g)X, \text{Ad}(g)Y) = \kappa(X, Y)$  for all  $g \in G$ . Deduce the infinitesimal invariance  $\kappa([X, Y], Z) + \kappa(Y, [X, Z]) = 0$ , the associativity (or invariance) property of the Killing form.
5. For  $\mathfrak{su}(2)$  with basis  $X_1, X_2, X_3$  as in example 1.4.7, write out the three matrices  $\text{ad}(X_1), \text{ad}(X_2), \text{ad}(X_3)$  in the ordered basis  $(X_1, X_2, X_3)$ , and compute the Killing form  $\kappa$  explicitly. Verify that  $\kappa$  is negative definite, and identify it as a negative multiple of the standard inner product on  $\mathbb{R}^3$ .
6. Give an example of a Lie group  $G$  for which  $\text{ad}: \mathfrak{g} \rightarrow \text{End}(\mathfrak{g})$  is not injective, and describe its kernel. Contrast with the case  $G = \text{SU}(2)$ , where  $\text{ad}$  is injective. What feature of the Lie algebra distinguishes the two cases?

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## *The Lie Group–Lie Algebra Correspondence*

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Chapter 2 attached to every Lie group its Lie algebra and to every homomorphism its differential, assembling the assignment into a functor from Lie groups to Lie algebras. The question this chapter settles is how much of the curved object  $G$  survives in the linear object  $\mathfrak{g}$ . The answer, the content of Lie’s fundamental theorems, is that almost everything does: subgroups correspond to subalgebras, homomorphisms to their differentials, and, up to a single global subtlety carried by the fundamental group, a connected Lie group can be reconstructed in full from its Lie algebra. The correspondence is what makes Lie theory effective, for it converts questions about a nonlinear manifold with a group law into questions of linear algebra over  $\mathfrak{g}$ .

Three theorems carry the chapter. Cartan’s closed-subgroup theorem removes smoothness from the hypotheses entirely: a subgroup that is merely closed as a subset is automatically an embedded submanifold, hence a Lie subgroup, so that topology alone certifies the geometry. Lie’s integration theorems run the functor backwards: every subalgebra of  $\mathfrak{g}$  integrates to a connected subgroup of  $G$ , by way of the Frobenius theorem cited to [2], and every homomorphism of Lie algebras out of a simply-connected group integrates to a homomorphism of groups. Together they yield Lie’s third theorem, the equivalence between the category of simply-connected Lie groups and the category of finite-dimensional real Lie algebras. The chapter ends with the Baker–Campbell–Hausdorff formula, which makes the reconstruction explicit by expressing  $\exp X \exp Y$  as the exponential of a universal series in  $X$ ,  $Y$ , and their iterated brackets, so that the group law near the identity is literally written in the bracket. The one datum the Lie algebra cannot recover, the fundamental group, is what motivates the covering-group theory of chapter 4.

### 3.1 Lie Subgroups and Lie Subalgebras

Chapter 2 completed the passage from a Lie group  $G$  to its Lie algebra  $\mathfrak{g} = \text{Lie}(G)$ , and showed the passage is functorial: a homomorphism  $\varphi: G \rightarrow H$  has a differential  $d\varphi_e: \mathfrak{g} \rightarrow \mathfrak{h}$  that respects the bracket (proposition 2.4.5). This chapter turns the dictionary into a tool for recognizing and building groups. The guiding question is how the subobjects match up: which subgroups of  $G$  carry a Lie algebra sitting inside  $\mathfrak{g}$ , and conversely which subalgebras of  $\mathfrak{g}$  arise from a subgroup. The present section sets up the two notions of subobject and establishes the easy direction of their correspondence. A subgroup of  $G$  that is also a submanifold has a Lie algebra, a subalgebra of  $\mathfrak{g}$ ; the hard converse, that every subalgebra integrates to a subgroup, is the business of section 3.3.

The subtlety that organizes the section is topological. A submanifold of a manifold comes in two strengths: an *embedded* submanifold, carrying the subspace topology, and an *immersed* one, carrying its own possibly finer topology under an injective immersion. For subgroups both strengths occur, and the distinction is not a technicality. The immersed subgroups can be topologically wild, winding densely through the ambient group, and it is exactly this wildness that the closed-subgroup theorem of section 3.2 tames. The canonical example separating the two, an irrational line on the torus, is worked out at the end of the section and threads through the rest of the chapter.

#### Definition 3.1.1: Lie Subgroup

Let  $G$  be a Lie group. A *Lie subgroup* of  $G$  is a subgroup  $H \subseteq G$  equipped with a smooth manifold structure making  $H$  a Lie group, such that the inclusion

$$\iota: H \hookrightarrow G$$

is a smooth homomorphism that is an *immersion*, that is,  $d\iota_h: T_h H \rightarrow T_{\iota(h)} G$  is injective at every  $h \in H$ . The Lie subgroup is *embedded* if in addition  $\iota$  is a topological embedding, meaning the manifold topology on  $H$  agrees with the subspace topology it inherits from  $G$ ; otherwise it is (properly) *immersed*.

The definition carries two independent pieces of data that must not be conflated. The set  $H$  is a subgroup of  $G$  in the algebraic sense, and it is simultaneously the underlying set of a Lie group whose smooth structure need not be induced from  $G$ . What ties the two together is that the inclusion is required to be a smooth immersion, so the abstract Lie group  $H$  maps injectively into  $G$  with injective differential, and the Lie algebra  $\mathfrak{h} = T_e H$  maps injectively into  $\mathfrak{g} = T_e G$ . In the embedded case the topology on  $H$  is forced to be the subspace topology and nothing is lost by regarding  $H$  as a subset with its inherited geometry. In the immersed but non-embedded case the topology on  $H$  is strictly finer than the subspace topology, and  $H$  as a subset of  $G$  can fail to be locally closed: its points are visited by the immersion, but a subset open in  $H$  need not be the trace of an open subset of  $G$ ; the irrational line of example 3.1.5 exhibits exactly this failure. Every subgroup that is a regular level set of a submersion, and in particular every classical matrix group inside  $\text{GL}_n(\mathbb{F})$ , is embedded, so the immersed-only phenomenon is invisible until one meets a subgroup whose image folds back on itself.

The dual notion on the algebra side is purely linear-algebraic and requires no topology at all. A Lie algebra is a vector space with a bracket, and its subobjects are the subspaces that the bracket preserves.

**Definition 3.1.2: Lie Subalgebra**

Let  $\mathfrak{g}$  be a Lie algebra over a field  $k$ . A *Lie subalgebra* of  $\mathfrak{g}$  is a  $k$ -linear subspace  $\mathfrak{h} \subseteq \mathfrak{g}$  that is closed under the bracket:  $[X, Y] \in \mathfrak{h}$  for all  $X, Y \in \mathfrak{h}$ . With the restricted bracket  $\mathfrak{h}$  is a Lie algebra in its own right, and the inclusion  $\mathfrak{h} \hookrightarrow \mathfrak{g}$  is a homomorphism of Lie algebras.

A subalgebra is thus the linearization of the subgroup notion: where a Lie subgroup is a subgroup closed under the smooth group operations, a Lie subalgebra is a subspace closed under the one operation the algebra carries. The bracket-closure condition is the only requirement, and it is easy to verify in practice by computing brackets on a spanning set. Two matrix examples make the definition concrete before the general correspondence is proved.

**Example 3.1.3: Subalgebras of  $\mathfrak{gl}_n$** 

Inside  $\mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$  with the commutator bracket  $[X, Y] = XY - YX$ , the following are Lie subalgebras.

1. The traceless matrices  $\mathfrak{sl}_n(\mathbb{F}) = \{X \in M_n(\mathbb{F}) \mid \text{tr} X = 0\}$ . This is a subspace of codimension one, and it is bracket-closed because  $\text{tr}(XY - YX) = 0$  for all  $X, Y$ , so every commutator is traceless. This is the Lie algebra of  $\text{SL}(n, \mathbb{F})$  recorded in theorem 1.4.3.
2. The skew-symmetric matrices  $\mathfrak{so}(n) = \{X \in M_n(\mathbb{R}) \mid X^\top = -X\}$ . If  $X^\top = -X$  and  $Y^\top = -Y$  then

$$[X, Y]^\top = (XY - YX)^\top = Y^\top X^\top - X^\top Y^\top = YX - XY = -[X, Y]$$

so  $[X, Y]$  is again skew-symmetric. This is the Lie algebra of  $\text{SO}(n)$  and of  $\text{O}(n)$  from theorem 1.4.3.

3. The upper-triangular matrices, and separately the strictly upper-triangular ones. If  $X, Y$  have zero entries below the diagonal, so does  $XY$ , hence so does  $[X, Y]$ ; the strictly upper-triangular case is identical with zeros on and below the diagonal. These two subalgebras will reappear in Part II as the models of solvable and nilpotent Lie algebras.

By contrast, the symmetric matrices are a subspace but not a subalgebra: the commutator of two symmetric matrices is skew-symmetric, not symmetric, so bracket-closure fails.

The examples exhibit the two directions the section connects. Each of  $\mathfrak{sl}_n$  and  $\mathfrak{so}(n)$  is not only a subalgebra of  $\mathfrak{gl}_n$  but the Lie algebra of an actual subgroup,  $\text{SL}(n, \mathbb{F})$  and  $\text{SO}(n)$  respectively, and these subgroups are embedded, being regular level sets inside  $\text{GL}_n(\mathbb{F})$ . That the subalgebra inclusion  $\mathfrak{so}(n) \hookrightarrow \mathfrak{gl}_n$  is exactly the differential of the group inclusion  $\text{SO}(n) \hookrightarrow \text{GL}_n(\mathbb{R})$  is the content of the next proposition, in general form.

The proposition is the easy half of the Lie group–Lie algebra correspondence for subobjects. A Lie subgroup, being a Lie group with a homomorphic inclusion into  $G$ , feeds into the functor Lie of proposition 2.4.5, and the functor turns the inclusion of groups into an inclusion of algebras. The only points to check are that the differential of the inclusion is injective, so that  $\mathfrak{h}$  sits inside  $\mathfrak{g}$  as a true copy, and that the exponential maps are compatible.

**Proposition 3.1.4: The Lie Algebra of a Lie Subgroup**

Let  $H$  be a Lie subgroup of a Lie group  $G$ , with inclusion  $\iota: H \hookrightarrow G$ . Then:

1. The differential  $d\iota_e: \mathfrak{h} \rightarrow \mathfrak{g}$  is an injective homomorphism of Lie algebras, so its image  $d\iota_e(\text{Lie}(H))$  is a Lie subalgebra of  $\mathfrak{g}$ . Identifying  $\text{Lie}(H)$  with this image, one writes  $\mathfrak{h} = \text{Lie}(H) \subseteq \mathfrak{g}$ .
2. The exponential of  $G$  restricts to that of  $H$ : for all  $X \in \mathfrak{h}$ ,

$$\exp_G(X) = \iota(\exp_H(X))$$

so the one-parameter subgroups of  $G$  tangent to  $\mathfrak{h}$  are exactly those of  $H$ , and  $\exp_G(\mathfrak{h}) \subseteq H$ .

**Proof:**

The two statements are proved in turn.

1. *The differential is an injective algebra homomorphism.* By definition 3.1.1 the inclusion  $\iota$  is a smooth homomorphism of Lie groups and an immersion, so its differential  $d\iota_e: T_e H \rightarrow T_e G$  (definition 2.4.1) is injective by the immersion hypothesis. It is a homomorphism of Lie algebras by proposition 2.4.2 applied to the homomorphism  $\iota$ : for  $X, Y \in \mathfrak{h}$ ,

$$d\iota_e[X, Y]_{\mathfrak{h}} = [d\iota_e X, d\iota_e Y]_{\mathfrak{g}}$$

where the left bracket is that of  $\text{Lie}(H)$  and the right that of  $\mathfrak{g}$ . An injective linear map is an isomorphism onto its image, so the image  $d\iota_e(\mathfrak{h})$  is a linear subspace of  $\mathfrak{g}$ ; and the displayed identity, read on the image, says that this subspace is closed under the bracket of  $\mathfrak{g}$ , since  $[d\iota_e X, d\iota_e Y] = d\iota_e[X, Y]$  lies in the image. Thus  $d\iota_e(\mathfrak{h})$  is a Lie subalgebra of  $\mathfrak{g}$  in the sense of definition 3.1.2, and  $d\iota_e$  is a Lie algebra isomorphism of  $\text{Lie}(H)$  onto it. Identifying  $\text{Lie}(H)$  with its image is then harmless, which is the first assertion.

2. *Compatibility of the exponentials.* The exponential map is natural under homomorphisms: for a homomorphism  $\varphi: G_1 \rightarrow G_2$  one has  $\varphi \circ \exp_{G_1} = \exp_{G_2} \circ d\varphi_e$  (theorem 2.4.3). Apply this to  $\varphi = \iota: H \rightarrow G$ , whose differential is the inclusion  $d\iota_e$  of part (1). For  $X \in \mathfrak{h}$ ,

$$\iota(\exp_H(X)) = \exp_G(d\iota_e X) = \exp_G(X)$$

the last equality under the identification of  $X \in \mathfrak{h}$  with  $d\iota_e X \in \mathfrak{g}$ . Hence  $\exp_G$  restricted to  $\mathfrak{h}$  takes values in  $\iota(H) = H$  and coincides with  $\exp_H$  followed by the inclusion. In particular the one-parameter subgroup  $t \mapsto \exp_G(tX)$  of  $G$ , for  $X \in \mathfrak{h}$ , is the image under  $\iota$  of the one-parameter subgroup  $t \mapsto \exp_H(tX)$  of  $H$ , as we sought to show.

The proposition records that the functor  $\text{Lie}$  sends a Lie subgroup to a Lie subalgebra, with the subgroup's own exponential nothing but the restriction of the ambient one. The immersion hypothesis in definition 3.1.1 is doing real work here: it is precisely what forces  $d\iota_e$  to be injective, so that  $\mathfrak{h}$  is a genuine copy of  $\text{Lie}(H)$  inside  $\mathfrak{g}$  rather than a quotient. A reader who has met the matrix case will recognize the content: for a closed matrix group the intrinsic subalgebra is the space of matrices  $X$  with  $\exp(tX)$  in the group for all  $t$ , which is exactly the tangent-space description of definition 1.4.1,

and part (2) is the statement that this description is coordinate-free. This identification is made precise in section 3.2, where the closed-subgroup theorem supplies the missing embedded structure.

The converse question is the one this chapter exists to answer, and it is not easy. Given a Lie subalgebra  $\mathfrak{h} \subseteq \mathfrak{g}$ , is there a Lie subgroup  $H \leq G$  with  $\text{Lie}(H) = \mathfrak{h}$ ? The answer is yes, and it is the integration theorem of Lie proved in section 3.3: every Lie subalgebra integrates to a unique connected Lie subgroup, constructed as the maximal integral manifold through the identity of the left-invariant distribution spanned by  $\mathfrak{h}$ . The subgroup so produced is in general only immersed, not embedded, and the example below shows this outcome is unavoidable: there are subalgebras whose integral subgroup is a dense wind.

The distinction between embedded and immersed is not idle. The prototype that forces the distinction, and the running counterexample for the rest of the chapter, is a line of irrational slope on the two-dimensional torus. It is a one-parameter subgroup, hence a connected immersed Lie subgroup, but its image is dense, so it is as far from embedded as a subgroup can be.

### Example 3.1.5: The Irrational Line on the Torus

Let  $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$  be the two-dimensional torus, a compact abelian Lie group of dimension two, with quotient map  $\pi: \mathbb{R}^2 \rightarrow \mathbb{T}^2$ . Its Lie algebra is  $\mathbb{R}^2$  with the zero bracket, and the exponential map is the quotient projection  $\exp = \pi$ , since  $\mathbb{T}^2$  is abelian and  $\pi$  is a homomorphism whose differential at 0 is the identity: by theorem 2.3.1 a one-parameter subgroup of  $\mathbb{T}^2$  is  $t \mapsto \pi(tv)$  for a velocity vector  $v \in \mathbb{R}^2$ . Fix a slope  $\alpha \in \mathbb{R}$  and consider the one-dimensional subalgebra  $\mathfrak{h}_\alpha = \mathbb{R} \cdot (1, \alpha) \subseteq \mathbb{R}^2$  spanned by the vector  $v = (1, \alpha)$ . It is trivially a subalgebra, the bracket being zero. The corresponding one-parameter subgroup is

$$\gamma_\alpha: \mathbb{R} \rightarrow \mathbb{T}^2, \quad \gamma_\alpha(t) = \pi(t, t\alpha) \quad (3.1)$$

the image of the line of slope  $\alpha$  through the origin of  $\mathbb{R}^2$ . Whether the resulting Lie subgroup  $H_\alpha = \gamma_\alpha(\mathbb{R})$  is embedded or only immersed depends entirely on the arithmetic of  $\alpha$ .

*Rational slope: an embedded circle.* Suppose  $\alpha = p/q$  in lowest terms with  $q > 0$ . Then  $\gamma_\alpha(q) = \pi(q, p) = \pi(0, 0) = e$ , since  $(q, p) \in \mathbb{Z}^2$ , and  $q$  is the least positive such time. Thus  $\gamma_\alpha$  is periodic with period  $q$ , and it factors through an injective immersion of the circle  $\mathbb{R}/q\mathbb{Z}$  onto  $H_\alpha$ . The image is a compact embedded circle in  $\mathbb{T}^2$ , a closed one-dimensional submanifold winding  $p$  times around one generator while going once around the other. Being compact, its manifold topology and its subspace topology agree, so  $H_\alpha$  is an embedded Lie subgroup, isomorphic to the circle  $S^1$ .

*Irrational slope: a dense immersed line.* Suppose instead that  $\alpha$  is irrational. Then  $\gamma_\alpha$  is injective: if  $\gamma_\alpha(s) = \gamma_\alpha(t)$  then  $(s-t, (s-t)\alpha) \in \mathbb{Z}^2$ , so  $s-t = m$  and  $(s-t)\alpha = n$  for integers  $m, n$ , forcing  $m\alpha = n$ ; with  $\alpha$  irrational this gives  $m = n = 0$ , hence  $s = t$ . So  $\gamma_\alpha$  is an injective immersion of the line  $\mathbb{R}$  into  $\mathbb{T}^2$ , and  $H_\alpha = \gamma_\alpha(\mathbb{R})$  is a connected immersed Lie subgroup carried diffeomorphically by  $\gamma_\alpha$  from  $\mathbb{R}$ . The image is *dense*. To see this, examine where the wind meets the vertical circle  $\{\pi(0, y) \mid y \in \mathbb{R}\}$ : the point  $\gamma_\alpha(t) = \pi(t, t\alpha)$  lies on it precisely when  $t \in \mathbb{Z}$ , that is at the return times  $t = n$  for  $n \in \mathbb{Z}$ , and the second coordinates of these return points are  $\{n\alpha \bmod 1 \mid n \in \mathbb{Z}\}$ . This set is dense in  $\mathbb{R}/\mathbb{Z}$  because  $\alpha$  is irrational: the rotation  $y \mapsto y + \alpha$  of the circle has no finite orbit, so by pigeonhole two of its iterates lie within any prescribed distance, producing an arbitrarily small nonzero return, whose integer multiples then fall within that distance of every point. Density of the return points on one circle, together with the flow filling the transverse direction, makes  $H_\alpha$  dense in all of  $\mathbb{T}^2$ .

Consequently  $H_\alpha$  for irrational  $\alpha$  is an immersed but *not* embedded Lie subgroup. Its manifold

topology is that of  $\mathbb{R}$  under  $\gamma_\alpha$ , in which a bounded open interval is open, whereas in the subspace topology from  $\mathbb{T}^2$  every neighborhood of a point of  $H_\alpha$  meets infinitely many far-apart pieces of the wind, since the image is dense. The two topologies disagree, and  $H_\alpha$  is not locally closed in  $\mathbb{T}^2$ : its closure is all of  $\mathbb{T}^2$ , so  $H_\alpha$  is a proper dense subgroup and cannot be a submanifold in the subspace topology.

The example is the fault line of the whole subject. The same one-dimensional subalgebra  $\mathfrak{h}_\alpha = \mathbb{R} \cdot (1, \alpha)$  integrates to a circle for rational  $\alpha$  and to a dense line for irrational  $\alpha$ , and no purely infinitesimal datum can tell the two apart: the subalgebra is the same one-dimensional space in both cases, and the integrating subgroup exists in both cases by the same construction. What differs is a global, arithmetic fact about the slope, invisible to the Lie algebra. This is why the integration theorem of section 3.3 can promise only an immersed subgroup, and why the immersed-versus-embedded distinction of definition 3.1.1 had to be built in from the start.

The example also motivates the theorem that dominates the next section. Immersed subgroups can be topologically wild, dense in the ambient group, neither closed nor locally closed. The remedy is to impose a topological hypothesis and watch it force geometric regularity. The closed-subgroup theorem of section 3.2 does exactly this: a subgroup that is *closed* as a subset of  $G$ , with no smoothness assumed at all, is automatically an embedded Lie subgroup. The dense line  $H_\alpha$  is not closed, so it is not a counterexample; and the theorem explains why the classical groups, defined as closed subsets of  $\mathrm{GL}_n(\mathbb{F})$  cut out by equations, were embedded submanifolds without any winding pathology. The usefulness of the closed-subgroup theorem is measured precisely against how badly immersed subgroups can behave, and  $H_\alpha$  is the measure.

### Exercise 3.1.1

1. Let  $\varphi: G \rightarrow H$  be a homomorphism of Lie groups. Show that the image  $\varphi(G)$  is a subgroup of  $H$ , and that it carries the structure of an immersed Lie subgroup of  $H$  with Lie algebra  $d\varphi_e(\mathfrak{g}) \subseteq \mathfrak{h}$ . (Realize  $\varphi(G)$  as the image of the induced injective homomorphism  $G/\ker \varphi \rightarrow H$ ; you may cite the closed-subgroup theorem of section 3.2 to know that  $\ker \varphi$  is an embedded Lie subgroup and  $G/\ker \varphi$  a Lie group.) When is the image embedded?
2. Show that a subgroup  $H \subseteq G$  that is also an embedded submanifold of  $G$  is a Lie subgroup, that is, that the subspace smooth structure makes  $H$  a Lie group and the inclusion a smooth immersion. (The one thing to verify is that multiplication and inversion on  $H$  are smooth for the subspace structure; use that a smooth map into  $G$  whose image lies in an embedded submanifold is smooth as a map into that submanifold.)
3. Verify directly that  $\mathfrak{sl}_n(\mathbb{F})$  and  $\mathfrak{so}(n)$  are Lie subalgebras of  $\mathfrak{gl}_n(\mathbb{F})$ , recovering the computations of example 3.1.3, and confirm that these are the Lie algebras of the embedded Lie subgroups  $\mathrm{SL}(n, \mathbb{F})$  and  $\mathrm{SO}(n)$  named in theorem 1.4.3. Then show over  $\mathbb{R}$  that  $\mathfrak{so}(n) \subseteq \mathfrak{sl}_n(\mathbb{R})$ , and interpret this inclusion of subalgebras as the inclusion of subgroups  $\mathrm{SO}(n) \subseteq \mathrm{SL}(n, \mathbb{R})$ .
4. For the irrational line  $H_\alpha$  of example 3.1.5, show that  $H_\alpha$  is not a closed subset of  $\mathbb{T}^2$  by exhibiting a point of  $\mathbb{T}^2$  in the closure of  $H_\alpha$  but not in  $H_\alpha$ . Conclude that the closure  $\overline{H_\alpha}$  is a larger subgroup than  $H_\alpha$ , and identify it.
5. Inside  $G = \mathbb{T}^2 \times \mathbb{T}^2$ , describe the subalgebra  $\mathfrak{h} = \{(v, v) \mid v \in \mathbb{R}^2\}$  of the diagonal, and the subgroup it integrates to. Is the diagonal subgroup embedded? Contrast with the subalgebra  $\{(v, Av) \mid v \in \mathbb{R}^2\}$ , where  $A = \mathrm{diag}(1, \lambda)$  scales the second coordinate by an irrational factor  $\lambda$ , and explain which feature produces a dense immersed subgroup.

## 3.2 The Closed Subgroup Theorem

Section 3.1 drew the distinction between an immersed Lie subgroup and an embedded one, and example 3.1.5 showed the gap is real: the irrational-slope winding of a line on the torus is an immersed subgroup that is not embedded, its image dense rather than a submanifold. The example carries a warning, that a subgroup of a Lie group need be no manifold at all in any natural way, and it isolates the culprit, which is that the image of the winding is not closed. Cartan’s theorem, the subject of this section, says that closedness is the whole story. A subgroup of a Lie group that happens to be a closed subset is automatically an embedded submanifold, hence a Lie group in its own right, with no further hypothesis. This is a statement about topology forcing smooth structure, and it has no counterpart for manifolds in general.

The force of the theorem is best measured against how much it gives for how little. To exhibit a subgroup as an embedded Lie subgroup by hand, one would produce submanifold charts, verify the group operations are smooth in those charts, and match the smooth structure to the subspace topology. Cartan’s theorem replaces all of that with a single point-set condition. Every classical group of section 1.2 is closed in  $\mathrm{GL}_n(\mathbb{F})$  by inspection, being the zero set of continuous matrix equations, so the theorem subsumes the entire manifold apparatus of Chapter 1 under one hypothesis, and its companion proposition 3.2.2 recovers the concrete definition of the matrix Lie algebra intrinsically. The proof organizes itself around a single object, the set of infinitesimal directions along which  $H$  can be approached from the identity.

The candidate for the Lie algebra of  $H$  is forced by the requirement that  $\exp$  carry it into  $H$ . Mirroring the matrix definition definition 1.4.1, one collects exactly those tangent vectors whose one-parameter subgroup never leaves  $H$ .

### Theorem 3.2.1: The Closed Subgroup Theorem (Cartan)

Let  $G$  be a Lie group and let  $H \subseteq G$  be a subgroup that is closed as a subset of  $G$ . Then  $H$  is an embedded Lie subgroup of  $G$ : it is an embedded smooth submanifold, and with the induced smooth structure it is a Lie group whose inclusion into  $G$  is a smooth embedding. Its tangent space at the identity is the linear subspace

$$\mathfrak{h} = \{X \in \mathfrak{g} \mid \exp(tX) \in H \text{ for all } t \in \mathbb{R}\}$$

of  $\mathfrak{g}$ , and  $\exp$  carries a neighborhood of 0 in  $\mathfrak{h}$  diffeomorphically onto a neighborhood of  $e$  in  $H$ .

Before the proof, the shape of the argument. The set  $\mathfrak{h}$  is defined by a condition on all of  $\mathbb{R}$ , so its linearity is not visible from the definition; the first task is to show  $\mathfrak{h}$  is a linear subspace of  $\mathfrak{g}$ , and the mechanism is the same Lie product formula that carried the linear structure onto the matrix Lie algebra in proposition 1.4.2, now with the closedness of  $H$  supplying the step that closedness of the matrix group supplied there. The heart of the matter is a lemma that identifies  $\mathfrak{h}$  with the directions along which  $H$  accumulates at  $e$ : any sequence in  $H$  approaching the identity approaches it along a direction lying in  $\mathfrak{h}$ . The second task is to promote this infinitesimal information into a chart. Choosing a vector-space complement  $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ , the map  $(X, Y) \mapsto \exp X \exp Y$  is a local diffeomorphism at the origin, and the lemma forces its  $H$ -part to be exactly the  $\mathfrak{h}$ -slice near  $e$ , which is the submanifold chart. Left translation then spreads the chart over all of  $H$ .

**Proof:**

Fix an inner product on  $\mathfrak{g}$  and write  $\|\cdot\|$  for the associated norm; because  $\mathfrak{g}$  is finite-dimensional the choice is immaterial, and continuity or convergence statements below are independent of it. By theorem 2.3.4 there are open neighborhoods  $U_0$  of 0 in  $\mathfrak{g}$  and  $V_0$  of  $e$  in  $G$  with  $\exp: U_0 \rightarrow V_0$  a diffeomorphism; write  $\log: V_0 \rightarrow U_0$  for its inverse. Shrink  $U_0$  to an open ball about 0 so that  $\exp$  restricted to it is a diffeomorphism onto its image.

*Step 1: the key lemma on accumulation directions.* The claim is the following. Suppose  $(X_n)_{n \geq 1}$  is a sequence in  $\mathfrak{g}$  with  $X_n \rightarrow 0$ ,  $X_n \neq 0$ , and  $\exp(X_n) \in H$  for every  $n$ , and suppose the normalized directions converge,

$$\frac{X_n}{\|X_n\|} \rightarrow X \in \mathfrak{g}$$

Then  $X \in \mathfrak{h}$ , that is  $\exp(tX) \in H$  for every  $t \in \mathbb{R}$ .

Fix  $t \in \mathbb{R}$ . For each  $n$  set  $k_n = \lfloor t/\|X_n\| \rfloor$ , the greatest integer not exceeding  $t/\|X_n\|$ , so that  $k_n\|X_n\|$  differs from  $t$  by less than  $\|X_n\|$ . Then

$$k_n X_n = (k_n \|X_n\|) \frac{X_n}{\|X_n\|}$$

Since  $k_n\|X_n\| \rightarrow t$  and  $X_n/\|X_n\| \rightarrow X$ , the product converges,  $k_n X_n \rightarrow tX$ . The exponential is continuous, so  $\exp(k_n X_n) \rightarrow \exp(tX)$ . But  $\exp(k_n X_n) = \exp(X_n)^{k_n}$  by the one-parameter subgroup law (proposition 2.3.3), and  $\exp(X_n) \in H$ , so  $\exp(X_n)^{k_n} \in H$  because  $H$  is a subgroup. Thus  $\exp(tX)$  is a limit of a sequence in  $H$ , and  $H$  is closed, so  $\exp(tX) \in H$ . As  $t$  was arbitrary,  $X \in \mathfrak{h}$ , proving the lemma.

The lemma is where closedness earns its keep: it converts a sequence of discrete group elements shrinking toward  $e$  into a genuine one-parameter subgroup lying in  $H$ , by riding the integer multiples  $k_n X_n$  up to the target  $tX$  and closing the limit inside  $H$ . Every subsequent step consumes it.

*Step 2:  $\mathfrak{h}$  is a linear subspace of  $\mathfrak{g}$ .* That  $0 \in \mathfrak{h}$  is immediate, since  $\exp(t \cdot 0) = e \in H$  for all  $t$ . For closure under real scalars, let  $X \in \mathfrak{h}$  and  $\lambda \in \mathbb{R}$ ; then  $\exp(t\lambda X) = \exp((t\lambda)X) \in H$  for every  $t$  because  $t\lambda \in \mathbb{R}$  and  $X \in \mathfrak{h}$ , so  $\lambda X \in \mathfrak{h}$ . Closure under addition is the point where the nonlinearity of  $\exp$  must be handled, and the tool is the Lie product formula, the intrinsic analogue of the matrix formula that carried the linear structure onto the matrix Lie algebra in proposition 1.4.2. It holds in the form

$$\exp(X + Y) = \lim_{m \rightarrow \infty} \left( \exp\left(\frac{X}{m}\right) \exp\left(\frac{Y}{m}\right) \right)^m \quad (3.2)$$

for all  $X, Y \in \mathfrak{g}$ , and is proved in any Lie group directly from the exponential chart of theorem 2.3.4. Fix  $X, Y \in \mathfrak{g}$  and set  $a_m = \exp(X/m) \exp(Y/m) \in G$ . Since  $X/m, Y/m \rightarrow 0$  and  $\exp$  is continuous,  $a_m \rightarrow e$ , so  $a_m \in V_0$  for  $m$  large and  $Z_m := \log(a_m) \in U_0$  is defined with  $Z_m \rightarrow \log(e) = 0$ . The map  $m \mapsto \log(\exp(X/m) \exp(Y/m))$  extends to a smooth function of the real parameter  $s = 1/m$  near  $s = 0$ , namely  $s \mapsto \log(\exp(sX) \exp(sY))$ , whose value at  $s = 0$  is 0; its derivative there is computed by the chain rule from  $d(\exp)_0 = \text{id}$  (theorem 2.3.4), the product rule at  $e$ , and  $d(\log)_e = (d(\exp)_0)^{-1} = \text{id}$ , giving

$$\left. \frac{d}{ds} \right|_{s=0} \log(\exp(sX) \exp(sY)) = X + Y$$

Hence  $Z_m = \log(a_m) = \frac{1}{m}(X + Y) + o(1/m)$  as  $m \rightarrow \infty$ , so  $mZ_m \rightarrow X + Y$ . Now  $a_m = \exp(Z_m)$ , and by the one-parameter subgroup law (proposition 2.3.3)  $a_m^m = \exp(Z_m)^m = \exp(mZ_m)$ ; continuity of  $\exp$  then gives  $a_m^m = \exp(mZ_m) \rightarrow \exp(X + Y)$ , which is (3.2).

With the formula in hand, closure under addition follows exactly as in the matrix case. Let  $X, Y \in \mathfrak{h}$  and fix  $t \in \mathbb{R}$ . Applying (3.2) to  $tX$  and  $tY$ ,

$$\exp(t(X + Y)) = \lim_{m \rightarrow \infty} \left( \exp\left(\frac{tX}{m}\right) \exp\left(\frac{tY}{m}\right) \right)^m$$

For each fixed  $m$  the elements  $\exp(tX/m)$  and  $\exp(tY/m)$  lie in  $H$ , since  $tX, tY \in \mathfrak{h}$  and  $t/m \in \mathbb{R}$ ; their product raised to the  $m$ th power is a product of elements of the subgroup  $H$ , hence lies in  $H$ . The left side is therefore a limit of elements of  $H$ , and since  $H$  is closed,  $\exp(t(X + Y)) \in H$ . As  $t$  was arbitrary,  $X + Y \in \mathfrak{h}$ . Thus  $\mathfrak{h}$  is a linear subspace of  $\mathfrak{g}$ ; in particular it has a finite dimension, say  $k = \dim \mathfrak{h}$ , and admits a vector-space complement  $\mathfrak{m}$  with  $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ .

*Step 3:  $\mathfrak{h}$  captures every accumulation direction of  $H$  at  $e$ .* The claim, which the chart construction will use in contrapositive form, is that

$$\text{if } g_n \in H, g_n \rightarrow e, g_n \neq e, \text{ then } \log(g_n) \text{ has all its accumulation directions in } \mathfrak{h} \quad (3.3)$$

Precisely: for  $g_n \in H$  with  $g_n \rightarrow e$  and  $g_n \neq e$ , write  $Z_n = \log(g_n) \in \mathfrak{g}$  for  $n$  large (so that  $g_n \in V_0$ ); then  $Z_n \rightarrow 0$ ,  $Z_n \neq 0$ , and any subsequential limit of the unit vectors  $Z_n/\|Z_n\|$  lies in  $\mathfrak{h}$ . Indeed  $\exp(Z_n) = g_n \in H$ , and  $Z_n = \log(g_n) \rightarrow \log(e) = 0$  by continuity of  $\log$  on  $V_0$ ; moreover  $Z_n \neq 0$  because  $\exp$  is injective on  $U_0$  and  $g_n \neq e = \exp(0)$ . The unit vectors  $Z_n/\|Z_n\|$  lie on the compact unit sphere of  $\mathfrak{g}$ , so any subsequence has a further subsequence converging to some unit vector  $Z$ ; along that subsequence the hypotheses of Step 1 hold, whence  $Z \in \mathfrak{h}$ . This proves (3.3).

*Step 4: the exponential of  $\mathfrak{m}$  meets  $H$  only at  $e$  near the origin.* The claim is that there is an open neighborhood  $W$  of 0 in  $\mathfrak{m}$  such that

$$\exp(Y) \in H \text{ and } Y \in W \implies Y = 0 \quad (3.4)$$

Suppose not. Then there is a sequence  $Y_n \in \mathfrak{m}$  with  $Y_n \neq 0$ ,  $Y_n \rightarrow 0$ , and  $\exp(Y_n) \in H$ . Applying Step 1 to the sequence  $X_n = Y_n$  (its exponentials lie in  $H$ , it tends to 0, and it is nonzero), pass to a subsequence along which  $Y_n/\|Y_n\|$  converges to a unit vector  $Y \in \mathfrak{g}$ ; Step 1 gives  $Y \in \mathfrak{h}$ . But each  $Y_n \in \mathfrak{m}$ , so  $Y_n/\|Y_n\| \in \mathfrak{m}$ , and  $\mathfrak{m}$  is closed as a linear subspace, so  $Y \in \mathfrak{m}$ . Then  $Y \in \mathfrak{h} \cap \mathfrak{m} = \{0\}$ , contradicting  $\|Y\| = 1$ . Hence no such sequence exists, which is (3.4).

*Step 5: the submanifold chart at  $e$ .* Consider the map

$$\Psi: \mathfrak{h} \oplus \mathfrak{m} \rightarrow G, \quad \Psi(X, Y) = \exp(X) \exp(Y)$$

It is smooth, and its differential at  $(0, 0)$  sends  $(X, Y)$  to  $X + Y$ : differentiating  $\exp(sX) \exp(0) = \exp(sX)$  in  $s$  gives  $X$  in the first slot, differentiating  $\exp(0) \exp(sY) = \exp(sY)$  gives  $Y$  in the second, and the product rule at the identity adds them, using  $d(\exp)_0 = \text{id}$  from theorem 2.3.4. Under the identification  $\mathfrak{h} \oplus \mathfrak{m} = \mathfrak{g}$  this differential is the identity, so by the inverse function theorem ([2]) there are open balls  $B_{\mathfrak{h}} \subseteq \mathfrak{h}$  and  $B_{\mathfrak{m}} \subseteq \mathfrak{m}$  about the origins, with  $B_{\mathfrak{m}} \subseteq W$  (the set of Step 4), such that  $\Psi$  restricts to a diffeomorphism of  $B_{\mathfrak{h}} \times B_{\mathfrak{m}}$  onto an open neighborhood  $V$  of  $e$  in  $G$ . Shrinking further if needed, arrange also that  $\exp(B_{\mathfrak{h}}) \subseteq H$ , which is possible because  $B_{\mathfrak{h}} \subseteq \mathfrak{h}$  and hence  $\exp(X) \in H$  for every  $X \in B_{\mathfrak{h}}$  by definition of  $\mathfrak{h}$ .

The chart is  $\Psi^{-1}: V \rightarrow B_{\mathfrak{h}} \times B_{\mathfrak{m}}$ , and the claim is that it carries  $H \cap V$  onto the slice  $B_{\mathfrak{h}} \times \{0\}$ . One inclusion is immediate: if  $X \in B_{\mathfrak{h}}$  then  $\Psi(X, 0) = \exp(X) \in H$ , so  $\Psi(B_{\mathfrak{h}} \times \{0\}) \subseteq H \cap V$ . For the reverse, let  $h \in H \cap V$  and write  $h = \Psi(X, Y) = \exp(X) \exp(Y)$  with  $(X, Y) \in B_{\mathfrak{h}} \times B_{\mathfrak{m}}$ . Then  $\exp(X) \in H$  (as  $X \in B_{\mathfrak{h}} \subseteq \mathfrak{h}$ ), and  $h \in H$ , so

$$\exp(Y) = \exp(X)^{-1} h \in H$$

because  $H$  is a subgroup. Now  $Y \in B_{\mathfrak{m}} \subseteq W$  and  $\exp(Y) \in H$ , so (3.4) forces  $Y = 0$ . Hence  $h = \exp(X) = \Psi(X, 0)$  lies in the slice. Therefore

$$\Psi^{-1}(H \cap V) = B_{\mathfrak{h}} \times \{0\} \quad (3.5)$$

which is exactly a submanifold chart for  $H$  at  $e$ : in the coordinates  $\Psi^{-1}$ , the subset  $H$  is cut out by the vanishing of the  $\mathfrak{m}$ -coordinate. This shows  $H$  is an embedded  $k$ -dimensional submanifold near  $e$ , with tangent space  $\mathfrak{h}$  at  $e$ , and that  $\exp$  maps  $B_{\mathfrak{h}}$  diffeomorphically onto the neighborhood  $\exp(B_{\mathfrak{h}}) = H \cap V$  of  $e$  in  $H$ .

*Step 6: spreading the chart by left translation, and the Lie group structure.* The construction of Step 5 produced a submanifold chart only at  $e$ , but the homogeneity of a group upgrades it to a chart at every point. Let  $h_0 \in H$  be arbitrary. Left translation  $L_{h_0}: G \rightarrow G$  is a diffeomorphism (proposition 2.1.2) carrying  $e$  to  $h_0$ , and it carries  $H$  to  $H$  because  $H$  is a subgroup, so  $L_{h_0}(H \cap V) = (h_0 H) \cap (h_0 V) = H \cap (h_0 V)$ . Thus  $L_{h_0} \circ \Psi^{-1}$ , restricted appropriately, is a submanifold chart for  $H$  on the neighborhood  $h_0 V$  of  $h_0$ , with the same slice description (3.5) transported by the diffeomorphism  $L_{h_0}$ . Every point of  $H$  therefore has a submanifold chart, so  $H$  is an embedded smooth submanifold of  $G$  of dimension  $k = \dim \mathfrak{h}$ , and the subspace topology on  $H$  agrees with its manifold topology.

It remains to see that  $H$  with this smooth structure is a Lie group. The multiplication  $m: G \times G \rightarrow G$  and inversion  $i: G \rightarrow G$  are smooth, and they restrict to maps  $H \times H \rightarrow H$  and  $H \rightarrow H$  because  $H$  is a subgroup. Since  $H$  is an embedded submanifold, a smooth map from a manifold into  $G$  whose image lies in  $H$  is smooth as a map into  $H$  ([2]); applying this to  $m|_{H \times H}$  and  $i|_H$ , which are smooth into  $G$  with image in  $H$ , shows the group operations of  $H$  are smooth for its own manifold structure. Hence  $H$  is a Lie group, and the inclusion  $H \hookrightarrow G$  is a smooth embedding by construction, completing the proof.

The theorem is the mechanism that makes Lie theory a theory of groups rather than of specially prepared manifolds. It says the category of Lie groups is closed under the most basic algebraic operation, passing to a closed subgroup, with no analytic bookkeeping required of the user. The contrast with general manifold theory is sharp: a closed subset of a manifold that happens to be an image of some group need not be a submanifold, and it is exactly the compatibility of the algebraic structure (subgroup) with the point-set structure (closed) that forces smoothness. Nothing like this holds for an arbitrary closed subset. The engine is the accumulation lemma of Step 1, which turns the failure of  $H$  to be open, its elements crowding toward  $e$ , into positive information about the tangent directions  $H$  occupies.

The subalgebra  $\mathfrak{h}$  that emerged in the proof is more than a tangent space; it is a Lie subalgebra of  $\mathfrak{g}$ , and it deserves to be recorded as the Lie algebra of the closed subgroup  $H$ . Its defining formula is identical in shape to the matrix definition of Chapter 1, and identifying the two is what ties the intrinsic theory back to the concrete computations that opened the book.

**Proposition 3.2.2: The Lie Algebra of a Closed Subgroup**

Let  $H$  be a closed subgroup of a Lie group  $G$ , an embedded Lie subgroup by theorem 3.2.1. Its Lie algebra, the tangent space  $T_e H$  with the bracket inherited from  $\mathfrak{g}$ , is

$$\mathfrak{h} = \{X \in \mathfrak{g} \mid \exp(tX) \in H \text{ for all } t \in \mathbb{R}\}$$

and this is a Lie subalgebra of  $\mathfrak{g}$ . When  $G = \mathrm{GL}_n(\mathbb{F})$  and  $H \subseteq \mathrm{GL}_n(\mathbb{F})$  is a closed matrix group,  $\mathfrak{h}$  coincides with the matrix Lie algebra of definition 1.4.1.

**Proof:**

The equality  $T_e H = \mathfrak{h}$  was established in Step 5 of theorem 3.2.1: the chart  $\Psi^{-1}$  identifies  $T_e H$  with the slice direction  $\mathfrak{h}$ . That  $\mathfrak{h}$  is a Lie subalgebra, that is closed under the bracket of  $\mathfrak{g}$ , is the general fact (proposition 3.1.4) that the Lie algebra of an embedded Lie subgroup is a subalgebra: the inclusion  $H \hookrightarrow G$  is a Lie group homomorphism, and its differential at  $e$  is the injective Lie algebra homomorphism  $\mathfrak{h} \hookrightarrow \mathfrak{g}$ , whose image is therefore closed under the bracket.

For the last claim, let  $G = \mathrm{GL}_n(\mathbb{F})$ , so that  $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$  and the intrinsic exponential of  $G$  is the matrix exponential (proposition 2.3.6). For a closed matrix group  $H \subseteq \mathrm{GL}_n(\mathbb{F})$ , the set  $\mathfrak{h}$  above reads

$$\mathfrak{h} = \{X \in M_n(\mathbb{F}) \mid \exp(tX) \in H \text{ for all } t \in \mathbb{R}\}$$

which is verbatim the definition of the matrix Lie algebra in definition 1.4.1. The two constructions therefore agree, as we sought to show.

The proposition closes the loop opened in Chapter 1. There the Lie algebra of a matrix group was defined by fiat as the set of matrices whose one-parameter subgroup stays in  $G$ , a definition justified only by its usefulness. Cartan's theorem now shows that this ad hoc formula is the intrinsic Lie algebra  $T_e H$  of  $H$  as a manifold, computed with no reference to matrices, and that the matrix formula is its specialization to  $G = \mathrm{GL}_n(\mathbb{F})$ . The classical groups of section 1.2 are all closed in  $\mathrm{GL}_n(\mathbb{F})$ , so each of them is covered twice over, once by the explicit regular-value construction of Chapter 1 and once by the single hypothesis of theorem 3.2.1, and the two computations of the Lie algebra necessarily agree.

The running example makes this concrete. The group  $\mathrm{SU}(2)$  was built in Chapter 1 as a matrix group and shown to be diffeomorphic to  $S^3$ ; it is a closed subgroup of  $\mathrm{GL}_2(\mathbb{C})$ , so Cartan's theorem applies and returns its Lie algebra intrinsically.

**Example 3.2.3:  $\mathrm{SU}(2)$  and  $\mathrm{SO}(3)$  as Closed Subgroups**

The unitary group  $\mathrm{U}(2)$  is closed in  $\mathrm{GL}_2(\mathbb{C})$ , being the set of matrices  $g$  with  $g^*g = I$ , a system of continuous equations in the entries; and  $\mathrm{SU}(2) = \{g \in \mathrm{U}(2) : \det g = 1\}$  is closed in  $\mathrm{U}(2)$ , hence in  $\mathrm{GL}_2(\mathbb{C})$ , as the further zero set of the continuous function  $\det - 1$ . By theorem 3.2.1,  $\mathrm{SU}(2)$  is an embedded Lie subgroup of  $\mathrm{GL}_2(\mathbb{C})$ , and by proposition 3.2.2 its Lie algebra is

$$\mathfrak{su}(2) = \{X \in M_2(\mathbb{C}) \mid \exp(tX) \in \mathrm{SU}(2) \text{ for all } t\}$$

Differentiating the two defining conditions  $\exp(tX)^* \exp(tX) = I$  and  $\det \exp(tX) = 1$  at  $t = 0$  returns  $X^* + X = 0$  and  $\mathrm{tr} X = 0$ , the traceless skew-Hermitian matrices already found in theorem 1.4.3, a three-dimensional real space matching  $\dim \mathrm{SU}(2) = \dim S^3 = 3$ . The same argument applies to  $\mathrm{SO}(3)$ , closed in  $\mathrm{GL}_3(\mathbb{R})$  as the set of  $g$  with  $g^T g = I$  and  $\det g = 1$ , whose

Lie algebra  $\mathfrak{so}(3)$  is the skew-symmetric  $3 \times 3$  matrices. The double cover  $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$  of theorem 1.5.5 is then a homomorphism between two closed subgroups, and it induces the isomorphism  $\mathfrak{su}(2) \cong \mathfrak{so}(3)$  of their Lie algebras, both being the cross-product algebra on  $\mathbb{R}^3$ ; the groups differ only by the central kernel  $\{\pm I\}$ , invisible to the Lie algebra.

The example exhibits the theorem doing exactly the work Chapter 1 did by hand, now for free. What required a regular-value computation there, verifying that  $\mathrm{SU}(2)$  is a submanifold and extracting its tangent space, is here the single observation that  $\mathrm{SU}(2)$  is a closed subset of a Lie group. The Lie algebra comes out identical because proposition 3.2.2 guarantees it.

A second consequence of Cartan’s theorem is about maps rather than subgroups, and it is of the same character: a purely topological hypothesis forces smoothness. A homomorphism between Lie groups that is merely continuous, with no assumption of differentiability, is automatically smooth. The proof is a device of the same flavor as the theorem itself, realizing the map as a closed subgroup.

#### Corollary 3.2.4: Continuous Homomorphisms Are Smooth

Let  $G$  and  $H$  be Lie groups and let  $\varphi: G \rightarrow H$  be a continuous group homomorphism. Then  $\varphi$  is smooth, and hence a homomorphism of Lie groups in the sense of definition 2.1.4.

#### Proof:

Consider the graph of  $\varphi$ ,

$$\Gamma = \{(g, \varphi(g)) \mid g \in G\} \subseteq G \times H$$

inside the product Lie group  $G \times H$ . It is a subgroup: since  $\varphi$  is a homomorphism,  $(g, \varphi(g))(g', \varphi(g')) = (gg', \varphi(g)\varphi(g')) = (gg', \varphi(gg')) \in \Gamma$ , and  $(e, e) \in \Gamma$ , and  $(g, \varphi(g))^{-1} = (g^{-1}, \varphi(g)^{-1}) = (g^{-1}, \varphi(g^{-1})) \in \Gamma$ . It is closed:  $\Gamma$  is the preimage of the diagonal  $\Delta_H = \{(h, h) : h \in H\} \subseteq H \times H$  under the continuous map  $G \times H \rightarrow H \times H$ ,  $(g, h) \mapsto (\varphi(g), h)$ , and  $\Delta_H$  is closed because  $H$  is Hausdorff. Being a closed subgroup of  $G \times H$ , the graph  $\Gamma$  is an embedded Lie subgroup by theorem 3.2.1.

Now consider the two coordinate projections restricted to  $\Gamma$ ,

$$\pi_G: \Gamma \rightarrow G, \quad (g, \varphi(g)) \mapsto g, \quad \pi_H: \Gamma \rightarrow H, \quad (g, \varphi(g)) \mapsto \varphi(g)$$

Both are smooth, as restrictions of the smooth projections of the product to the embedded submanifold  $\Gamma$ . The first,  $\pi_G$ , is a smooth bijective homomorphism of Lie groups: it is injective because the first coordinate determines the pair  $(g, \varphi(g))$ , and surjective because every  $g$  has the image  $\varphi(g)$ . Its set-theoretic inverse is  $g \mapsto (g, \varphi(g))$ .

First,  $\dim \Gamma = \dim G$ . Indeed  $\pi_G$  is a continuous bijection whose set-theoretic inverse  $g \mapsto (g, \varphi(g))$  is continuous, by continuity of  $\varphi$ ; so  $\pi_G$  is a homeomorphism, and an embedded submanifold  $\Gamma$  of  $G \times H$  homeomorphic to  $G$  has  $G$ ’s dimension by invariance of dimension.

Second,  $d(\pi_G)_{(e,e)}$  is injective. This is the Lie-group fact that an injective homomorphism of Lie groups has injective differential at  $e$ : the kernel of  $d(\pi_G)_{(e,e)}$  is the Lie algebra of the closed subgroup  $\ker \pi_G$  ([2]), and  $\ker \pi_G = \{(e, e)\}$  because  $\pi_G$  is injective, so this Lie algebra is  $\{0\}$  and  $d(\pi_G)_{(e,e)}$  has trivial kernel. An injective linear map between spaces of equal dimension is an isomorphism, so by  $\dim \Gamma = \dim G$  the differential  $d(\pi_G)_{(e,e)}$  is a linear isomorphism, and left-translating shows the differential is an isomorphism at every point, so  $\pi_G$  is a diffeomorphism by

the inverse function theorem ([2]). Therefore  $\pi_G^{-1}: G \rightarrow \Gamma$ ,  $g \mapsto (g, \varphi(g))$ , is smooth. Composing,

$$\varphi = \pi_H \circ \pi_G^{-1}$$

is a composite of smooth maps, hence smooth, completing the proof.

The corollary removes smoothness from the definition of a homomorphism of Lie groups. Whatever category one starts in, continuous or smooth homomorphisms of Lie groups are the same maps, so the passage from topology to differential geometry is automatic at the level of morphisms as well as objects. The device is the same one that powers Cartan’s theorem: an object one wants to control, here a bare continuous map, is repackaged as a closed subgroup, whose smooth structure is then handed over by theorem 3.2.1. The graph device will recur in the next section, where the graph of a Lie algebra homomorphism inside  $\mathfrak{g} \oplus \mathfrak{h}$  integrates to the graph of the group homomorphism it induces.

Taken together, the closed subgroup theorem and its corollary express a rigidity peculiar to Lie groups. Neither subgroups nor homomorphisms carry a smoothness hypothesis: closedness of a subset and continuity of a map, both purely topological, suffice to produce the full differential-geometric structure. There is no analogue for manifolds in general, where a closed subset or a continuous map is under no obligation to be smooth. The reason is that a Lie group’s smooth structure is rigidly determined near the identity by its algebraic structure, through the exponential chart, and this rigidity propagates: the algebra of the group leaves the topology no freedom.

### Exercise 3.2.1

1. Show that  $O(n)$ ,  $U(n)$ , and  $Sp(2n, \mathbb{R})$  are closed subsets of the appropriate  $GL_N(\mathbb{F})$ , and conclude from theorem 3.2.1 that each is an embedded Lie subgroup. Recover in each case the Lie algebra of theorem 1.4.3 from proposition 3.2.2 by differentiating the defining equations along  $\exp(tX)$  at  $t = 0$ .
2. Let  $S$  be a subgroup of a Lie group  $G$  (not assumed closed). Prove that its topological closure  $\overline{S}$  is again a subgroup of  $G$ , and hence, by theorem 3.2.1, an embedded Lie subgroup. Deduce that the closure of the irrational-slope one-parameter subgroup of the torus  $\mathbb{T}^2$  (example 3.1.5) is all of  $\mathbb{T}^2$ .
3. Let  $H$  be a closed subgroup of a connected Lie group  $G$ , and suppose  $H$  has countable index in  $G$  (that is,  $G/H$  is countable). Prove that  $H = G$ . (Hint:  $G$  is a countable union of the cosets of  $H$ ; use theorem 3.2.1 to compare dimensions, and a Baire-category argument to rule out  $\dim H < \dim G$ .)
4. Use corollary 3.2.4 to show that the only continuous automorphisms of the additive group  $(\mathbb{R}, +)$  are the maps  $x \mapsto cx$  for  $c \in \mathbb{R}^\times$ . Contrast with the existence of discontinuous additive maps  $\mathbb{R} \rightarrow \mathbb{R}$ , and explain why the continuity hypothesis cannot be dropped even though the conclusion of smoothness is automatic once continuity is present.
5. Prove that the center  $Z(G)$  of a Lie group  $G$  is a closed subgroup, hence an embedded Lie subgroup, and identify its Lie algebra as  $\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} \mid [X, Y] = 0 \text{ for all } Y \in \mathfrak{g}\}$  when  $G$  is connected. (Use the characterization  $\ker \text{Ad} = Z(G)$  for connected  $G$  from exercise 2.5.1 (2).)
6. Explain precisely which hypothesis of theorem 3.2.1 the irrational-slope one-parameter subgroup of  $\mathbb{T}^2$  (example 3.1.5) violates, and identify the step of the proof that consequently fails.

Why do the accumulation directions of this subgroup at the identity fill out all of  $\mathfrak{g}$  rather than a one-dimensional line?

### 3.3 Homomorphisms and Lie's Integration Theorems

The closed subgroup theorem of section 3.2 answered a question about *closed* subsets: a closed subgroup of a Lie group is automatically a submanifold, hence carries a Lie algebra  $\mathfrak{h} = \{X \in \mathfrak{g} \mid \exp(tX) \in H \forall t\}$  (proposition 3.2.2). This section runs the correspondence in the reverse direction and drops the closedness. Starting from a Lie subalgebra  $\mathfrak{h} \subseteq \mathfrak{g}$ , an algebraic object with no a priori geometric content, the goal is to manufacture a subgroup  $H \leq G$  whose Lie algebra is exactly  $\mathfrak{h}$ . The subgroup produced need not be closed, as the irrational line of example 3.1.5 already warned, so the natural category is that of immersed Lie subgroups (definition 3.1.1). The mechanism is the Frobenius theorem: a subalgebra  $\mathfrak{h}$  determines a left-invariant distribution on  $G$ , and the bracket-closure of  $\mathfrak{h}$  is precisely the involutivity that Frobenius needs to integrate that distribution to a foliation, one of whose leaves is the sought subgroup.

Once subalgebras integrate to subgroups, the same device integrates *homomorphisms*. A Lie algebra homomorphism  $\psi: \mathfrak{g} \rightarrow \mathfrak{h}$  has a graph, a subalgebra of  $\mathfrak{g} \oplus \mathfrak{h}$ ; the subgroup integrating that graph is the graph of the group homomorphism sought, provided the graph subgroup projects isomorphically onto  $G$ . That projection is a covering map, and simple connectivity of  $G$  is exactly what forces a covering of  $G$  to be trivial. This is Lie's second theorem, and it upgrades the faithfulness of the Lie functor (corollary 2.4.4) to a bijection  $\text{Hom}(G, H) \xrightarrow{\sim} \text{Hom}(\mathfrak{g}, \mathfrak{h})$  for  $G$  simply connected. The object-level companion, that every Lie algebra is  $\text{Lie}(G)$  for some  $G$ , is Lie's third theorem and is deferred to section 3.4, where the same integration is applied one last time to the adjoint representation.

The starting observation is that a subspace of  $\mathfrak{g}$  spreads out over all of  $G$  by left translation, producing a field of tangent subspaces, that is a distribution. This is the geometric shadow of the subspace, and everything below is organized around it.

#### Definition 3.3.1: The Left-Invariant Distribution of a Subspace

Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g} = T_e G$ , and let  $\mathfrak{h} \subseteq \mathfrak{g}$  be a linear subspace of dimension  $k$ . The *left-invariant distribution*  $D^{\mathfrak{h}}$  determined by  $\mathfrak{h}$  assigns to each  $g \in G$  the  $k$ -dimensional subspace of  $T_g G$

$$D_g^{\mathfrak{h}} = (dL_g)_e \mathfrak{h} \subseteq T_g G$$

the image of  $\mathfrak{h}$  under the differential of left translation  $L_g: x \mapsto gx$ . Equivalently, if  $X_1, \dots, X_k$  is a basis of  $\mathfrak{h}$  with left-invariant extensions  $X_1^L, \dots, X_k^L$  (definition 2.2.1), then  $D_g^{\mathfrak{h}} = \text{span}\{(X_1^L)_g, \dots, (X_k^L)_g\}$ .

The distribution  $D^{\mathfrak{h}}$  is smooth because it is globally framed by the left-invariant fields  $X_1^L, \dots, X_k^L$ , which are smooth and, being the translates of a basis of  $\mathfrak{h}$ , linearly independent at every point (left translation is a diffeomorphism, so  $(dL_g)_e$  is a linear isomorphism). A smooth  $k$ -plane field is exactly the data the Frobenius theorem operates on. The one hypothesis Frobenius requires is involutivity: the bracket of two vector fields tangent to the distribution must again be tangent to it. For a left-invariant distribution this hypothesis translates directly into an algebraic condition on  $\mathfrak{h}$ .

**Proposition 3.3.2: Involutivity Equals Bracket-Closure**

Let  $\mathfrak{h} \subseteq \mathfrak{g}$  be a linear subspace and  $D^{\mathfrak{h}}$  its left-invariant distribution (definition 3.3.1). Then  $D^{\mathfrak{h}}$  is involutive, that is  $[V, W]$  is a section of  $D^{\mathfrak{h}}$  whenever  $V, W$  are, if and only if  $\mathfrak{h}$  is a Lie subalgebra of  $\mathfrak{g}$  (definition 3.1.2).

**Proof:**

Fix a basis  $X_1, \dots, X_k$  of  $\mathfrak{h}$  and let  $X_i^L$  be the left-invariant extensions, which frame  $D^{\mathfrak{h}}$  by definition 3.3.1.

*Step 1: bracket-closure of  $\mathfrak{h}$  implies involutivity.* Suppose  $\mathfrak{h}$  is closed under the bracket, so  $[X_i, X_j] \in \mathfrak{h}$  for all  $i, j$ . Under the identification of  $\mathfrak{g}$  with left-invariant fields (theorem 2.2.3) the bracket on  $\mathfrak{g}$  is computed by the bracket of the fields,  $[X_i, X_j]^L = [X_i^L, X_j^L]$ , and  $[X_i, X_j] \in \mathfrak{h}$  means this field is a constant-coefficient combination of the frame,  $[X_i^L, X_j^L] = \sum_{\ell} c_{ij}^{\ell} X_{\ell}^L$  with  $c_{ij}^{\ell} \in \mathbb{R}$  the structure constants. Any two sections of  $D^{\mathfrak{h}}$  have the form  $V = \sum_i f_i X_i^L$  and  $W = \sum_j g_j X_j^L$  with  $f_i, g_j$  smooth functions on  $G$ . Expanding the bracket by the Leibniz rule,

$$[V, W] = \sum_{i,j} f_i g_j [X_i^L, X_j^L] + \sum_{i,j} f_i (X_i^L g_j) X_j^L - \sum_{i,j} g_j (X_j^L f_i) X_i^L$$

The second and third sums are already sections of  $D^{\mathfrak{h}}$ , being combinations of the frame  $X_{\ell}^L$  with smooth coefficients; and the first sum equals  $\sum_{i,j,\ell} f_i g_j c_{ij}^{\ell} X_{\ell}^L$ , again a section of  $D^{\mathfrak{h}}$ . Hence  $[V, W]$  is a section of  $D^{\mathfrak{h}}$ , so  $D^{\mathfrak{h}}$  is involutive.

*Step 2: involutivity implies bracket-closure.* Conversely suppose  $D^{\mathfrak{h}}$  is involutive. The fields  $X_i^L$  are themselves sections of  $D^{\mathfrak{h}}$ , so  $[X_i^L, X_j^L]$  is a section, that is  $[X_i^L, X_j^L]_g \in D_g^{\mathfrak{h}} = (dL_g)_e \mathfrak{h}$  for every  $g$ . Evaluating at  $g = e$  gives  $[X_i^L, X_j^L]_e \in \mathfrak{h}$ . But  $[X_i^L, X_j^L]_e = [X_i, X_j]$  is the value at  $e$  of the bracket field, which is the bracket on  $\mathfrak{g}$  by theorem 2.2.3. Thus  $[X_i, X_j] \in \mathfrak{h}$  for all basis vectors, and by bilinearity  $[\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h}$ , so  $\mathfrak{h}$  is a subalgebra, completing the proof.

The proposition is the pivot on which the entire integration turns. It says that the algebraic condition defining a subalgebra, closure under the bracket, is not a formal accident but the exact geometric condition that lets the associated plane field be integrated into a foliation. The bracket, which Chapter 2 built as the infinitesimal record of noncommutativity, reappears here as the obstruction to fitting the subspaces  $D_g^{\mathfrak{h}}$  together into submanifolds. When the obstruction vanishes, which is precisely when  $\mathfrak{h}$  absorbs its own brackets, the subspaces knit into leaves, and the leaf through  $e$  will be the subgroup. The Frobenius theorem supplies the knitting; the burden of proof here is entirely the group theory that follows, since the analytic integration is imported.

**3.3.3 Note: The Frobenius theorem as a cited input** The Frobenius theorem states that a smooth involutive distribution  $D$  of rank  $k$  on a manifold  $M$  is integrable: through each point  $p \in M$  there is a unique maximal connected *integral manifold*, an immersed  $k$ -dimensional submanifold  $L$  everywhere tangent to  $D$  ( $T_q L = D_q$  for all  $q \in L$ ), and these leaves partition  $M$  into a foliation. Its proof, via the simultaneous straightening of commuting flows, belongs to the theory of smooth manifolds and is taken as given here; see [2]. Everything below uses Frobenius as a black box and proves the group-theoretic content in full.

With involutivity secured by proposition 3.3.2, the Frobenius theorem hands back the leaf through the identity. The work that remains is to show this leaf is a subgroup and that its intrinsic Lie algebra is the  $\mathfrak{h}$  that was started with. Left-invariance of the distribution is what makes the leaf a subgroup: translating a leaf carries it to a leaf, and the translate through  $e$  is again the leaf through  $e$ .

#### Theorem 3.3.4: Integration of Subalgebras

Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g}$ , and let  $\mathfrak{h} \subseteq \mathfrak{g}$  be a Lie subalgebra (definition 3.1.2). Then there is a unique connected immersed Lie subgroup  $H \leq G$  (definition 3.1.1) whose Lie algebra is  $\mathfrak{h}$ , that is  $\text{Lie}(H) = \mathfrak{h} \subseteq \mathfrak{g}$  under the inclusion of proposition 3.1.4. Concretely,  $H$  is the maximal connected integral manifold through  $e$  of the left-invariant distribution  $D^{\mathfrak{h}}$  of definition 3.3.1.

#### Proof:

By proposition 3.3.2 the distribution  $D^{\mathfrak{h}}$  is smooth and involutive, so the Frobenius theorem (box 3.3.3, [2]) applies. Let  $H$  be the maximal connected integral manifold of  $D^{\mathfrak{h}}$  through  $e$ ; it is an immersed connected  $k$ -dimensional submanifold with  $T_g H = D_g^{\mathfrak{h}}$  for all  $g \in H$ . The proof shows  $H$  is a subgroup, computes its Lie algebra, and establishes uniqueness.

*Step 1: left translation permutes the leaves.* The distribution  $D^{\mathfrak{h}}$  is left-invariant: for any  $a \in G$  the diffeomorphism  $L_a$  carries  $D^{\mathfrak{h}}$  to itself, because

$$(dL_a)_g D_g^{\mathfrak{h}} = (dL_a)_g (dL_g)_e \mathfrak{h} = (dL_{ag})_e \mathfrak{h} = D_{ag}^{\mathfrak{h}}$$

using  $L_a \circ L_g = L_{ag}$  and the chain rule. A diffeomorphism carrying a distribution to itself carries integral manifolds to integral manifolds and maximal connected ones to maximal connected ones. Hence  $L_a(H)$  is the maximal connected integral manifold through  $L_a(e) = a$ ; that is, the leaf through  $a$  is  $aH$ .

*Step 2:  $H$  is a subgroup.* Let  $g \in H$ . By Step 1 the leaf through  $g$  is  $gH$ , and it also equals  $H$  itself, because  $g \in H$  and the leaves partition  $G$  into disjoint sets, so the unique leaf containing  $g$  is  $H$ ; thus  $gH = H$ . Consequently for any  $h \in H$  the product  $gh \in gH = H$ , so  $H$  is closed under multiplication. For inverses,  $gH = H$  gives  $e = g^{-1}g \in g^{-1}H$ , so the leaf  $g^{-1}H$  through  $g^{-1}$  contains  $e$ ; the leaf through  $e$  is  $H$ , and distinct leaves are disjoint, so  $g^{-1}H = H$ , whence  $g^{-1} = g^{-1}e \in H$ . Thus  $H$  is a subgroup of  $G$ . The multiplication and inversion of  $G$  restrict to smooth maps on the immersed submanifold  $H$ , because  $H \hookrightarrow G$  is an immersion whose image is a union of leaves and the group operations preserve the foliation by Step 1; this is the standard fact that a subgroup which is an integral manifold of a left-invariant distribution is a Lie subgroup ([2]). Hence  $H$  is an immersed Lie subgroup in the sense of definition 3.1.1.

*Step 3: the Lie algebra of  $H$  is  $\mathfrak{h}$ .* By construction  $H$  is an integral manifold of  $D^{\mathfrak{h}}$ , so its tangent space at the identity is

$$T_e H = D_e^{\mathfrak{h}} = (dL_e)_e \mathfrak{h} = \mathfrak{h}$$

since  $L_e = \text{id}_G$ . Under the inclusion of proposition 3.1.4,  $\text{Lie}(H) = T_e H$  is identified with a subalgebra of  $\mathfrak{g}$ , and the identification is compatible with the bracket because the inclusion  $\iota: H \hookrightarrow G$  is a Lie group homomorphism whose differential  $d\iota_e$  is the inclusion  $T_e H = \mathfrak{h} \hookrightarrow \mathfrak{g}$ ; by proposition 2.4.2 this differential preserves brackets. Therefore  $\text{Lie}(H) = \mathfrak{h}$  as a subalgebra of  $\mathfrak{g}$ .

*Step 4: uniqueness.* Suppose  $H' \leq G$  is another connected immersed Lie subgroup with  $\text{Lie}(H') = \mathfrak{h}$ . Then  $T_e H' = \mathfrak{h}$ , and by left-invariance of the tangent bundle of a subgroup, for

each  $g \in H'$  the tangent space  $T_g H' = (dL_g)_e T_e H' = (dL_g)_e \mathfrak{h} = D_g^{\mathfrak{h}}$ . So  $H'$  is a connected integral manifold of  $D^{\mathfrak{h}}$  containing  $e$ , hence contained in the maximal one,  $H' \subseteq H$ . Both are connected immersed subgroups of the same dimension  $k$  with the same tangent space at  $e$ ; the inclusion  $H' \hookrightarrow H$  is therefore an immersion between manifolds of equal dimension, an open map, so  $H'$  is an open subgroup of the connected group  $H$ . An open subgroup of a connected group is the whole group, since its cosets are open and partition  $H$ , forcing a single coset. Hence  $H' = H$ , establishing uniqueness and completing the proof.

The theorem completes the local dictionary between Lie groups and Lie algebras at the level of subobjects: subalgebras of  $\mathfrak{g}$  correspond exactly to connected immersed subgroups of  $G$ . The correspondence is not perfect in the category of embedded subgroups, precisely because integration can produce a subgroup that fails to be closed, but in the category of immersed subgroups it is a clean bijection. The subgroup  $H$  is intrinsically determined by  $\mathfrak{h}$  and is often called the *integral subgroup* or *analytic subgroup* generated by  $\mathfrak{h}$ ; it may be described concretely as the subgroup generated by  $\exp(\mathfrak{h})$ , the smallest subgroup containing all one-parameter subgroups whose generators lie in  $\mathfrak{h}$ .

The irrational line makes the necessity of the immersed category concrete. Its Lie algebra is a one-dimensional subalgebra of the abelian  $\mathfrak{g} = \mathbb{R}^2$  of the torus, and integration returns exactly that dense, non-embedded winding line.

**Example 3.3.5: The Irrational Line as an Integral Subgroup**

Take  $G = \mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ , whose Lie algebra  $\mathfrak{g} = \mathbb{R}^2$  is abelian, so every linear subspace is a subalgebra. Fix an irrational  $\alpha$  and let  $\mathfrak{h} = \mathbb{R} \cdot (1, \alpha) \subseteq \mathbb{R}^2$ , a one-dimensional subalgebra. The left-invariant distribution  $D^{\mathfrak{h}}$  is translation-invariant, since the group is abelian:  $D_g^{\mathfrak{h}}$  is the  $(1, \alpha)$ -direction at every  $g$ , the constant line field of slope  $\alpha$ . Its integral manifold through 0 is the image of

$$\mathbb{R} \rightarrow \mathbb{T}^2, \quad t \mapsto (t, \alpha t) \bmod \mathbb{Z}^2$$

the one-parameter subgroup  $H$  with generator  $(1, \alpha)$ . This is exactly the immersed, non-embedded subgroup of example 3.1.5: it is a subgroup, an injectively immersed copy of  $\mathbb{R}$ , and dense in  $\mathbb{T}^2$  because  $\alpha$  is irrational, so its subspace topology from  $\mathbb{T}^2$  is strictly coarser than its manifold topology as  $\mathbb{R}$ . The integration theorem produces  $H$  with  $\text{Lie}(H) = \mathfrak{h}$ , and the failure of  $H$  to be embedded is not a defect of the theorem but a feature of the subalgebra: no embedded subgroup of  $\mathbb{T}^2$  has Lie algebra the irrational-slope line, because the only embedded connected subgroups are  $\{0\}$ , the rational circles, and  $\mathbb{T}^2$  itself. The immersed category is forced.

For a matrix group the integral subgroup of a matrix subalgebra is generated by exponentials, matching the concrete picture of Chapter 1. If  $G \subseteq \text{GL}_n(\mathbb{F})$  and  $\mathfrak{h} \subseteq \mathfrak{g} \subseteq M_n(\mathbb{F})$  is a subalgebra of matrices closed under the commutator, the integral subgroup  $H$  is the subgroup generated by  $\{\exp X : X \in \mathfrak{h}\}$ , since these one-parameter subgroups are tangent to  $\mathfrak{h}$  at  $e$  and generate a connected subgroup whose Lie algebra is  $\mathfrak{h}$  by the same argument. When  $\mathfrak{h} = \mathfrak{g}$  this recovers the standard fact that the identity component of a matrix group is generated by exponentials.

With subalgebras integrating to subgroups, the integration of homomorphisms follows by a device that recurs throughout Lie theory: encode a map as its graph, and integrate the graph. A Lie algebra homomorphism  $\psi: \mathfrak{g} \rightarrow \mathfrak{h}$  is not itself a subalgebra, but its graph is a subalgebra of the direct sum, and the subgroup integrating that graph is, up to the projection isomorphism, the graph of the homomorphism sought. This is Lie’s second theorem. The role of simple connectivity is isolated in a single step, where a covering map with connected simply connected base is shown to be a diffeomorphism.

Before the theorem, one preparatory fact about the direct product. The Lie algebra of  $G \times H$  is  $\mathfrak{g} \oplus \mathfrak{h}$  with the componentwise bracket, and a homomorphism  $\psi: \mathfrak{g} \rightarrow \mathfrak{h}$  produces a subalgebra of this direct sum.

**Lemma 3.3.6: The Graph of a Lie Algebra Homomorphism**

Let  $\mathfrak{g}, \mathfrak{h}$  be Lie algebras and  $\psi: \mathfrak{g} \rightarrow \mathfrak{h}$  a Lie algebra homomorphism. The graph

$$\mathfrak{k} = \{(X, \psi X) \mid X \in \mathfrak{g}\} \subseteq \mathfrak{g} \oplus \mathfrak{h}$$

is a Lie subalgebra of  $\mathfrak{g} \oplus \mathfrak{h} = \text{Lie}(G \times H)$ , isomorphic to  $\mathfrak{g}$  via the projection to the first factor.

**Proof:**

The set  $\mathfrak{k}$  is a linear subspace, the graph of the linear map  $\psi$ . To see it is closed under the componentwise bracket of  $\mathfrak{g} \oplus \mathfrak{h}$ , compute for  $X, X' \in \mathfrak{g}$

$$[(X, \psi X), (X', \psi X')] = ([X, X'], [\psi X, \psi X']) = ([X, X'], \psi[X, X'])$$

the second equality because  $\psi$  is a Lie algebra homomorphism,  $[\psi X, \psi X'] = \psi[X, X']$ . The result has the form  $(Z, \psi Z)$  with  $Z = [X, X']$ , so it lies in  $\mathfrak{k}$ ; hence  $\mathfrak{k}$  is a subalgebra. The projection  $\text{pr}_{\mathfrak{g}}: \mathfrak{k} \rightarrow \mathfrak{g}$ ,  $(X, \psi X) \mapsto X$ , is a linear bijection with inverse  $X \mapsto (X, \psi X)$ , and it preserves brackets by the computation just made, so it is a Lie algebra isomorphism  $\mathfrak{k} \cong \mathfrak{g}$ , completing the proof.

The lemma is the algebraic half of the graph trick; the theorem is the geometric half. The identification  $\text{Lie}(G \times H) = \mathfrak{g} \oplus \mathfrak{h}$  used above is the componentwise statement that the exponential and bracket of a product are computed factorwise, a direct consequence of the naturality of  $\exp$  (theorem 2.4.3) applied to the two projections  $G \times H \rightarrow G$  and  $G \times H \rightarrow H$ . Now the graph subalgebra integrates to a subgroup of  $G \times H$ , and reading off its two projections produces the homomorphism.

**Theorem 3.3.7: Lie's Second Theorem: Integration of Homomorphisms**

Let  $G$  be a *simply connected* Lie group,  $H$  any Lie group, and  $\psi: \mathfrak{g} \rightarrow \mathfrak{h}$  a homomorphism of their Lie algebras. Then there is a unique Lie group homomorphism  $\varphi: G \rightarrow H$  with  $d\varphi_e = \psi$ .

**Proof:**

Uniqueness is immediate from the connectivity of  $G$ : if  $\varphi_1, \varphi_2: G \rightarrow H$  both have differential  $\psi$  at  $e$ , then  $d(\varphi_1)_e = d(\varphi_2)_e$ , and by corollary 2.4.4 a homomorphism out of a connected group is determined by its differential, so  $\varphi_1 = \varphi_2$ . It remains to construct  $\varphi$ .

*Step 1: integrate the graph.* Form the product Lie group  $G \times H$ , with Lie algebra  $\mathfrak{g} \oplus \mathfrak{h}$ . By lemma 3.3.6 the graph  $\mathfrak{k} = \{(X, \psi X) \mid X \in \mathfrak{g}\}$  is a Lie subalgebra of  $\mathfrak{g} \oplus \mathfrak{h}$ . By the integration theorem (theorem 3.3.4) there is a unique connected immersed Lie subgroup  $K \leq G \times H$  with  $\text{Lie}(K) = \mathfrak{k}$ .

*Step 2: the first projection restricts to a homomorphism with invertible differential.* Let  $p = \text{pr}_G: G \times H \rightarrow G$  be the projection, a Lie group homomorphism with differential the projection  $\mathfrak{g} \oplus \mathfrak{h} \rightarrow \mathfrak{g}$ . Restrict  $p$  to  $K$  and call it  $q = p|_K: K \rightarrow G$ , a homomorphism of Lie groups. Its

differential at  $e$  is the composite

$$\mathfrak{k} \hookrightarrow \mathfrak{g} \oplus \mathfrak{h} \xrightarrow{\text{pr}_{\mathfrak{g}}} \mathfrak{g}, \quad (X, \psi X) \mapsto X$$

which is the isomorphism  $\mathfrak{k} \xrightarrow{\sim} \mathfrak{g}$  of lemma 3.3.6. Thus  $dq_e$  is a linear isomorphism.

*Step 3:  $q$  is a covering map.* Since  $dq_e: \mathfrak{k} \rightarrow \mathfrak{g}$  is an isomorphism and  $q$  is a homomorphism, the differential at an arbitrary  $g \in K$  factors as  $dq_g = (dL_{q(g)})_e \circ dq_e \circ (dL_{g^{-1}})_g$  (translation-equivariance of a homomorphism), a composite of isomorphisms, so  $dq_g$  is an isomorphism at every point. Hence  $q$  is a local diffeomorphism, and its image is an open subgroup of  $G$ ; as  $G$  is connected, an open subgroup is all of  $G$ , so  $q$  is surjective. The kernel  $\Gamma = \ker q$  is discrete:  $q$  is a local diffeomorphism near  $e$ , so  $e$  is isolated in  $\Gamma = q^{-1}(e)$ , and left translation by elements of  $\Gamma$  spreads this to isolation everywhere. A surjective homomorphism of Lie groups that is a local diffeomorphism with discrete kernel realizes  $G$  as the quotient  $K/\Gamma$ , with  $\Gamma$  acting freely and properly by the deck transformations  $k \mapsto k\gamma$ ; the covering-space criterion of [5] identifies  $q: K \rightarrow K/\Gamma = G$  as a covering map.

*Step 4: simple connectivity trivializes the cover.* The base  $G$  is simply connected. A connected covering of a connected, simply connected, locally path connected space is a homeomorphism, by the lifting theory of [5]: the number of sheets of  $q$  equals the index of the image of  $\pi_1(K)$  in  $\pi_1(G) = 1$ , hence equals one, so the discrete fiber  $\Gamma$  is trivial and  $q$  is bijective. A bijective local diffeomorphism is a diffeomorphism, so  $q: K \rightarrow G$  is an isomorphism of Lie groups, with inverse  $q^{-1}: G \rightarrow K$  a Lie group homomorphism.

*Step 5: read off  $\varphi$ .* Define

$$\varphi = \text{pr}_H \circ q^{-1}: G \rightarrow H$$

the composite of the isomorphism  $q^{-1}: G \xrightarrow{\sim} K$  with the projection  $\text{pr}_H: G \times H \rightarrow H$  (restricted to  $K$ ). As a composite of Lie group homomorphisms,  $\varphi$  is a Lie group homomorphism. Its differential at  $e$  is

$$d\varphi_e = d(\text{pr}_H)_e \circ (dq_e)^{-1}$$

Now  $(dq_e)^{-1}: \mathfrak{g} \rightarrow \mathfrak{k}$  is  $X \mapsto (X, \psi X)$  by Step 2, and  $d(\text{pr}_H)_e: \mathfrak{g} \oplus \mathfrak{h} \rightarrow \mathfrak{h}$  is the projection  $(A, B) \mapsto B$ ; composing,

$$d\varphi_e(X) = d(\text{pr}_H)_e(X, \psi X) = \psi X$$

so  $d\varphi_e = \psi$ . This is the homomorphism sought, completing the proof.

The theorem is the existence half of the morphism-level correspondence, and its hypothesis pinpoints the exact obstruction. The differential  $d\varphi_e = \psi$  can always be reconstructed on a neighborhood of  $e$ , where  $\exp$  is a diffeomorphism and one sets  $\varphi(\exp X) = \exp(\psi X)$ ; the difficulty is globalizing this local homomorphism to all of  $G$ , and the obstruction is exactly  $\pi_1(G)$ . Running the local homomorphism around a noncontractible loop in  $G$  can return an inconsistent value, and the covering  $K \rightarrow G$  of the proof is precisely the space on which the local homomorphism unwinds consistently. Simple connectivity says every loop is contractible, so no inconsistency arises and the local homomorphism extends. When  $G$  is not simply connected the extension can fail, and the failure is measured by which homomorphisms  $\pi_1(G) \rightarrow H$  the deck group forces.

Packaging both halves, faithfulness from corollary 2.4.4 and now existence, gives the morphism-level equivalence for simply connected groups.

**Corollary 3.3.8: The Correspondence on Homomorphisms**

Let  $G$  be a simply connected Lie group and  $H$  any Lie group. The differential map

$$\mathrm{Hom}(G, H) \longrightarrow \mathrm{Hom}(\mathfrak{g}, \mathfrak{h}), \quad \varphi \mapsto d\varphi_e$$

from Lie group homomorphisms to Lie algebra homomorphisms is a bijection.

**Proof:**

The map is well defined, sending a Lie group homomorphism to a Lie algebra homomorphism by proposition 2.4.2. It is injective by corollary 2.4.4, since  $G$  is connected (being simply connected). It is surjective by theorem 3.3.7: every Lie algebra homomorphism  $\psi: \mathfrak{g} \rightarrow \mathfrak{h}$  equals  $d\varphi_e$  for some  $\varphi \in \mathrm{Hom}(G, H)$ . An injective and surjective map is a bijection, completing the proof.

The corollary is the sharp form of the statement that the Lie functor is fully faithful on simply connected groups. For a simply connected  $G$ , the group homomorphisms out of  $G$  are in bijection with the algebra homomorphisms out of  $\mathfrak{g}$ : nothing is lost and nothing is gained by passing to differentials. This is an equivalence at the level of *morphisms only*. The complementary object-level statement, that every finite-dimensional real Lie algebra is  $\mathfrak{g} = \mathrm{Lie}(G)$  for some simply connected  $G$ , is Lie's third theorem, which requires one further integration, of the adjoint representation, together with Ado's embedding theorem as an input; that argument is carried out in section 3.4. Together the morphism bijection here and the object statement there assemble into an equivalence of categories between simply connected Lie groups and finite-dimensional real Lie algebras.

The running example is the cleanest illustration of both the power and the limit of the corollary. The isomorphism of Lie algebras  $\mathfrak{su}(2) \cong \mathfrak{so}(3)$  integrates in one direction but not the other, and the asymmetry is exactly simple connectivity.

**Example 3.3.9: Integrating and Failing to Integrate for  $\mathrm{SU}(2)$  and  $\mathrm{SO}(3)$** 

The Lie algebras  $\mathfrak{su}(2)$  and  $\mathfrak{so}(3)$  are isomorphic (example 1.4.7), both being  $(\mathbb{R}^3, \times)$ ; fix the isomorphism  $\psi: \mathfrak{su}(2) \xrightarrow{\sim} \mathfrak{so}(3)$ . Consider its two directions as candidate inputs to theorem 3.3.7.

1. Take  $G = \mathrm{SU}(2)$ , which is diffeomorphic to  $S^3$  (proposition 1.5.1) and hence simply connected. Then theorem 3.3.7 applies: the isomorphism  $\psi: \mathfrak{su}(2) \rightarrow \mathfrak{so}(3)$  integrates to a unique Lie group homomorphism  $\varphi: \mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$  with  $d\varphi_e = \psi$ . This  $\varphi$  is exactly the double cover of theorem 1.5.5, recovered here from purely algebraic data. The integration theorem produces it without any of the explicit quaternion computation of Chapter 1; the algebra isomorphism alone forces the covering homomorphism into existence.
2. Take instead  $G = \mathrm{SO}(3)$ , which is *not* simply connected:  $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}/2\mathbb{Z}$  (corollary 1.5.6). Now theorem 3.3.7 does not apply, and indeed the inverse isomorphism  $\psi^{-1}: \mathfrak{so}(3) \rightarrow \mathfrak{su}(2)$  does *not* integrate to a group homomorphism  $\mathrm{SO}(3) \rightarrow \mathrm{SU}(2)$ . If it did, that homomorphism  $\rho$  would satisfy  $d\rho_e = \psi^{-1} = (d\varphi_e)^{-1}$ , so  $d(\varphi \circ \rho)_e = \mathrm{id}$ ; by corollary 2.4.4 on the connected  $\mathrm{SO}(3)$ ,  $\varphi \circ \rho = \mathrm{id}_{\mathrm{SO}(3)}$ , making  $\rho$  a continuous section of the double cover  $\varphi: \mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$ . A connected double cover with nontrivial  $\pi_1$  downstairs admits no continuous section, since a section would exhibit  $\mathrm{SO}(3)$  as a retract of  $\mathrm{SU}(2)$  and force  $\pi_1(\mathrm{SO}(3))$  to inject into  $\pi_1(\mathrm{SU}(2)) = 1$ , contradicting  $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}/2\mathbb{Z}$ . The obstruction is precisely this nontrivial fundamental group that the double cover measures.

The asymmetry is the entire content of the simple connectivity hypothesis: the algebra isomorphism is symmetric, but integration is possible only from the simply connected side.  $SU(2)$  is the universal cover, and morphisms out of it realize every algebra morphism;  $SO(3)$  realizes only those algebra morphisms compatible with its fundamental group.

The example shows that corollary 3.3.8 is not a restatement of the algebra data. For  $SU(2)$  the set  $\text{Hom}(SU(2), SO(3))$  contains the double cover, and its bijection with  $\text{Hom}(\mathfrak{su}(2), \mathfrak{so}(3))$  accounts for it; for  $SO(3)$  the map  $\text{Hom}(SO(3), SU(2)) \rightarrow \text{Hom}(\mathfrak{so}(3), \mathfrak{su}(2))$  is not surjective, missing the algebra isomorphism  $\psi^{-1}$ . Simple connectivity is what removes the topological obstruction and turns the differential map into a bijection.

### Exercise 3.3.1

1. Show that for any Lie group  $G$ , the Lie group homomorphisms  $\mathbb{R} \rightarrow G$  are exactly the one-parameter subgroups of  $G$ , and deduce from corollary 3.3.8 that  $\text{Hom}(\mathbb{R}, G) \rightarrow \text{Hom}(\mathbb{R}, \mathfrak{g}) = \mathfrak{g}$ ,  $\varphi \mapsto d\varphi_e(1)$ , is a bijection. Verify that this bijection is the one of theorem 2.3.1, and check that the hypothesis of corollary 3.3.8 holds for the source group  $\mathbb{R}$ .
2. Let  $\mathfrak{n} \subseteq \mathfrak{gl}_n(\mathbb{R})$  be the subalgebra of strictly upper-triangular matrices, closed under the commutator. Show that its integral subgroup (theorem 3.3.4) inside  $GL_n(\mathbb{R})$  is the group  $N$  of upper-triangular matrices with 1’s on the diagonal (the unipotent group), that  $\exp: \mathfrak{n} \rightarrow N$  is a diffeomorphism, and that  $N$  is embedded and closed. Contrast this with the irrational line of example 3.3.5: what makes the unipotent case embedded?
3. Let  $G$  be a connected Lie group with center  $Z(G)$ , and let  $\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} \mid [X, Y] = 0 \ \forall Y\}$  be the center of  $\mathfrak{g}$ . Show that  $\mathfrak{z}(\mathfrak{g})$  is a subalgebra and that its integral subgroup is the identity component of  $Z(G)$ . (Use that  $X \in \mathfrak{z}(\mathfrak{g})$  iff  $\text{ad}X = 0$  iff  $\text{Ad}(\exp tX) = \text{id}$  for all  $t$ , via (2.11) of section 2.5.)
4. Let  $G$  be simply connected with Lie algebra a direct sum of ideals  $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$ . Using theorem 3.3.7 and theorem 3.3.4, show that  $G \cong G_1 \times G_2$  for simply connected  $G_1, G_2$  with  $\text{Lie}(G_i) = \mathfrak{g}_i$ . (Integrate the two projections  $\mathfrak{g} \rightarrow \mathfrak{g}_i$ .)
5. Let  $H = S^1 = \mathbb{R}/\mathbb{Z}$  and consider a one-parameter subgroup  $\gamma_c: \mathbb{R} \rightarrow S^1$ ,  $t \mapsto e^{2\pi i c t}$ , generated by  $c \in \mathbb{R} = \mathfrak{h}$ . Determine for which  $c$  the map  $\gamma_c$  descends to a Lie group homomorphism  $S^1 = \mathbb{R}/\mathbb{Z} \rightarrow S^1$ , and explain how the answer is consistent with corollary 3.3.8 failing when the source  $S^1$  is not simply connected. Identify the resulting image of the differential map  $\text{Hom}(S^1, S^1) \rightarrow \text{Hom}(\mathbb{R}, \mathbb{R}) = \mathbb{R}$ .
6. Prove that the integral subgroup  $H$  of theorem 3.3.4 is embedded (a closed submanifold) if and only if  $H$  is closed in  $G$ , and that when  $H$  is closed the integration theorem and the closed subgroup theorem (theorem 3.2.1) produce the same subgroup. Use  $\mathbb{T}^2$  to exhibit both an embedded and a non-embedded integral subgroup.

## 3.4 Simply-Connected Groups and Lie’s Third Theorem

The correspondence built so far runs in one direction with a caveat. Passing from a Lie group to its Lie algebra is a functor  $\text{Lie}$  (proposition 2.4.5), and theorem 3.3.7 showed that on a *simply connected* domain this functor is faithful and full at the level of homomorphisms: a Lie-algebra map integrates

to a unique group map. Two questions remain, and they are the two halves of Lie's third theorem. First, does every finite-dimensional real Lie algebra arise as  $\text{Lie}(G)$  for some Lie group  $G$ , so that the functor is essentially surjective? Second, what is the relationship among the possibly many groups sharing a fixed Lie algebra, once simple-connectivity is dropped? This section answers both. The mechanism is the universal cover. Every connected Lie group  $G$  has a universal covering space  $\tilde{G}$ , and the group structure lifts to make  $\tilde{G}$  a simply connected Lie group with the same Lie algebra; the group  $G$  is recovered from  $\tilde{G}$  by quotienting by a discrete central subgroup isomorphic to  $\pi_1(G)$ . The upshot is an equivalence of categories between simply connected Lie groups and finite-dimensional real Lie algebras, and a complete classification of all connected groups with a given algebra as the quotients of one universal object. The Lie algebra determines the group up to exactly one piece of data the linear theory cannot see: the discrete central subgroup that is the fundamental group.

The starting point is a topological fact upgraded to a group-theoretic one. A connected, locally path-connected, semilocally simply connected space has a universal covering space ([5]); a Lie group, being a connected manifold, satisfies these hypotheses, so  $\tilde{G}$  exists as a space. The content of the next proposition is that the group structure of  $G$  transports to  $\tilde{G}$  through the covering, and does so uniquely once a basepoint over the identity is fixed.

**Proposition 3.4.1: The Universal Covering Group**

Let  $G$  be a connected Lie group and let  $\pi: \tilde{G} \rightarrow G$  be the universal covering space, with a chosen point  $\tilde{e} \in \tilde{G}$  over the identity  $e \in G$ . Then  $\tilde{G}$  carries a unique Lie group structure with identity  $\tilde{e}$  for which  $\pi$  is a homomorphism of Lie groups. Moreover:

1. the kernel  $\ker \pi$  is a discrete subgroup of the center  $Z(\tilde{G})$ , and  $\ker \pi \cong \pi_1(G)$  as groups;
2. the differential  $d\pi_{\tilde{e}}: \text{Lie}(\tilde{G}) \rightarrow \text{Lie}(G)$  is an isomorphism of Lie algebras.

The Lie group  $\tilde{G}$  is called the *universal covering group* of  $G$ , and it is simply connected.

**Proof:**

The strategy is to lift the multiplication and inversion maps of  $G$  through the covering, using the lifting criterion for maps into a covering space. Recall ([5]) that if  $Y$  is connected and locally path connected and  $f: Y \rightarrow G$  is a continuous map, then  $f$  lifts to a map  $\tilde{f}: Y \rightarrow \tilde{G}$  with  $\pi \circ \tilde{f} = f$  and a prescribed value at one point precisely when  $f_*\pi_1(Y) \subseteq \pi_*\pi_1(\tilde{G})$ ; since  $\tilde{G}$  is simply connected the right side is trivial, so the criterion is met exactly when  $f_*\pi_1(Y) = 0$ . When  $Y$  is itself simply connected this holds automatically, and the lift is unique once its value at one point is fixed. Every lift produced below is unique for this reason.

*Step 1: lifting the multiplication.* Consider the composite

$$\mu \circ (\pi \times \pi): \tilde{G} \times \tilde{G} \rightarrow G, \quad (\tilde{a}, \tilde{b}) \mapsto \pi(\tilde{a})\pi(\tilde{b})$$

where  $\mu: G \times G \rightarrow G$  is the multiplication of  $G$ . The domain  $\tilde{G} \times \tilde{G}$  is simply connected: the universal cover of a connected manifold is simply connected, and a product of simply connected spaces is simply connected ([5]). By the lifting criterion there is a unique smooth map

$$\tilde{\mu}: \tilde{G} \times \tilde{G} \rightarrow \tilde{G}$$

with  $\pi \circ \tilde{\mu} = \mu \circ (\pi \times \pi)$  and  $\tilde{\mu}(\tilde{e}, \tilde{e}) = \tilde{e}$ . That  $\tilde{\mu}$  is smooth, not only continuous, is automatic: a continuous lift of a smooth map through a covering by a smooth manifold is smooth, since  $\pi$

is a local diffeomorphism and the lift is locally  $\pi^{-1} \circ f$  in charts. Define inversion similarly by lifting  $\iota \circ \pi: \tilde{G} \rightarrow G$ ,  $\tilde{a} \mapsto \pi(\tilde{a})^{-1}$ , to the unique smooth  $\tilde{\iota}: \tilde{G} \rightarrow \tilde{G}$  with  $\tilde{\iota}(\tilde{e}) = \tilde{e}$ .

*Step 2: the group axioms.* The axioms hold because their two sides are lifts of the same map and agree at  $\tilde{e}$ , so they coincide by uniqueness of lifts. For associativity, both  $(\tilde{a}, \tilde{b}, \tilde{c}) \mapsto \tilde{\mu}(\tilde{\mu}(\tilde{a}, \tilde{b}), \tilde{c})$  and  $(\tilde{a}, \tilde{b}, \tilde{c}) \mapsto \tilde{\mu}(\tilde{a}, \tilde{\mu}(\tilde{b}, \tilde{c}))$  are maps  $\tilde{G}^3 \rightarrow \tilde{G}$  that project under  $\pi$  to the single product  $(\tilde{a}, \tilde{b}, \tilde{c}) \mapsto \pi(\tilde{a})\pi(\tilde{b})\pi(\tilde{c})$  on  $G$ , associative in  $G$ , and both send  $(\tilde{e}, \tilde{e}, \tilde{e})$  to  $\tilde{e}$ ; since  $\tilde{G}^3$  is simply connected, the two lifts agree. The identity and inverse axioms follow the same way:  $\tilde{a} \mapsto \tilde{\mu}(\tilde{e}, \tilde{a})$  and the identity map  $\tilde{a} \mapsto \tilde{a}$  both lift  $\tilde{a} \mapsto \pi(\tilde{a})$  and agree at  $\tilde{e}$ , hence are equal, so  $\tilde{e}$  is a left identity; and  $\tilde{a} \mapsto \tilde{\mu}(\tilde{a}\tilde{e}, \tilde{a})$  and the constant map at  $\tilde{e}$  both lift the constant map  $e$  and agree at  $\tilde{e}$ , so  $\tilde{a}\tilde{e}$  is a left inverse. Right identity and right inverse are identical. Thus  $\tilde{G}$  is a Lie group with  $\tilde{e}$  its identity, and  $\pi$  is a homomorphism by construction:  $\pi(\tilde{\mu}(\tilde{a}, \tilde{b})) = \pi(\tilde{a})\pi(\tilde{b})$ . Uniqueness of the group structure is uniqueness of the lifted multiplication.

*Step 3: the kernel is discrete, central, and  $\cong \pi_1(G)$ .* As the kernel of a homomorphism,  $\ker \pi$  is a subgroup of  $\tilde{G}$ ; as the fiber  $\pi^{-1}(e)$  of a covering map, it is a discrete subset. So  $\ker \pi$  is a discrete normal subgroup of the connected group  $\tilde{G}$ . A discrete normal subgroup  $N$  of a connected topological group is central: for fixed  $n \in N$  the map  $\tilde{G} \rightarrow N$ ,  $\tilde{a} \mapsto \tilde{a}n\tilde{a}^{-1}$ , is continuous with connected domain  $\tilde{G}$  and image inside the discrete set  $N$  (normality gives  $\tilde{a}n\tilde{a}^{-1} \in N$ ), hence constant; its value at  $\tilde{a} = \tilde{e}$  is  $n$ , so  $\tilde{a}n\tilde{a}^{-1} = n$  for all  $\tilde{a}$ , that is  $n \in Z(\tilde{G})$ . Thus  $\ker \pi \subseteq Z(\tilde{G})$ . Finally,  $\pi$  is the universal cover of the connected group  $G$ ; the deck transformation group of the universal cover acts simply transitively on the fiber  $\pi^{-1}(e) = \ker \pi$  and is isomorphic to  $\pi_1(G, e)$ , and for a covering group these deck transformations are exactly left translations by the elements of  $\ker \pi$ , so the deck group is  $\ker \pi$  itself and  $\ker \pi \cong \pi_1(G)$  as groups ([5]).

*Step 4: the differential is an isomorphism.* A covering map is a local diffeomorphism, so  $d\pi_{\tilde{e}}: T_{\tilde{e}}\tilde{G} \rightarrow T_eG$  is a linear isomorphism of tangent spaces. It is a Lie-algebra homomorphism because  $\pi$  is a group homomorphism, so its differential preserves the bracket (proposition 2.4.2, via the differential of definition 2.4.1). A bijective Lie-algebra homomorphism is an isomorphism of Lie algebras, so  $d\pi_{\tilde{e}}$  identifies  $\text{Lie}(\tilde{G})$  with  $\text{Lie}(G)$ , completing the proof.

The proposition is the technical heart of the section, and its consequences organize everything that follows. It says that among the connected Lie groups with a given Lie algebra there is a distinguished one, the simply connected cover, from which all the others descend by a discrete central quotient. The covering map is not just a topological device: it is a homomorphism, so  $\pi_1(G)$  enters the algebra of the situation as a central subgroup of  $\tilde{G}$ . The differential being an isomorphism records the invisibility of the covering to the linear theory. The groups  $G$  and  $\tilde{G}$  have literally the same Lie algebra, and no infinitesimal computation can tell them apart; the distinction lives entirely in the discrete data of the kernel.

The centrality argument in Step 3 deserves a second look, because it is the recurring reason discrete kernels behave so rigidly. A discrete normal subgroup of a connected group cannot be twisted by conjugation: conjugation is a continuous motion, the subgroup is a scatter of isolated points, and a continuous path cannot slide one isolated point to another without leaving the subgroup. So conjugation fixes each element, which is centrality. This is why the fundamental group of a Lie group is always abelian, a fact that follows the moment  $\pi_1(G) \cong \ker \pi \subseteq Z(\tilde{G})$ .

The concrete instance to keep in mind is the circle and its cover, which exhibits every feature of the proposition in the smallest possible setting.

**Example 3.4.2: The Real Line Covering the Circle**

Let  $G = S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$  with multiplication of complex numbers, a connected abelian Lie group of dimension one with Lie algebra  $\text{Lie}(S^1) = \mathbb{R}$ , the tangent line at 1 with trivial bracket. The universal cover is

$$\pi: \mathbb{R} \rightarrow S^1, \quad \pi(t) = e^{2\pi i t}$$

with  $\tilde{e} = 0$ . The lifted multiplication on  $\mathbb{R}$  is ordinary addition: the map  $(s, t) \mapsto s + t$  satisfies  $\pi(s + t) = e^{2\pi i(s+t)} = e^{2\pi i s} e^{2\pi i t} = \pi(s)\pi(t)$  and sends  $(0, 0)$  to 0, so by uniqueness of lifts it is  $\tilde{\mu}$ . Thus  $\tilde{G} = (\mathbb{R}, +)$ , the simply connected group with Lie algebra  $\mathbb{R}$ . The kernel is  $\ker \pi = \mathbb{Z} \subset \mathbb{R}$ , a discrete central subgroup (central because  $\mathbb{R}$  is abelian), and  $\pi_1(S^1) \cong \mathbb{Z}$  matches  $\ker \pi = \mathbb{Z}$  exactly. The differential  $d\pi_0: \mathbb{R} = T_0\mathbb{R} \rightarrow T_1S^1 = i\mathbb{R}$  sends the generator 1 to the velocity  $\frac{d}{dt}|_{t=0} e^{2\pi i t} = 2\pi i$ , so it is multiplication by  $2\pi i$  into the tangent line  $i\mathbb{R}$ , a linear isomorphism, as the proposition demands. The circle is recovered as the quotient  $S^1 = \mathbb{R}/\mathbb{Z}$ , the archetype of a connected group presented as  $\tilde{G}/\Gamma$  for a discrete central  $\Gamma$ .

With the universal cover in hand, the essential-surjectivity half of Lie's third theorem is within reach. The remaining input is purely algebraic: every finite-dimensional real Lie algebra can be realized inside a matrix algebra. This is Ado's theorem, whose proof is a substantial piece of Lie-algebra structure theory carried out in the algebraic companion volume; here it is cited and used as a black box.

**Theorem 3.4.3: Ado's Theorem**

Every finite-dimensional Lie algebra  $\mathfrak{g}$  over a field of characteristic zero admits a faithful finite-dimensional representation: there is an injective Lie-algebra homomorphism  $\mathfrak{g} \hookrightarrow \mathfrak{gl}_n(\mathbb{R})$  for some  $n$ . In particular every finite-dimensional real Lie algebra is isomorphic to a Lie subalgebra of  $\mathfrak{gl}_n(\mathbb{R}) = \text{Lie}(\text{GL}_n(\mathbb{R}))$ .

**Proof Key ideas:**

The proof uses the structure theory of Lie algebras, in particular the Levi decomposition into a solvable radical and a semisimple part together with a delicate extension of representations across that decomposition; the tools it requires are developed in Part II of this book, and the full argument is carried out in [3] and is not repeated here, as those tools lie downstream of the present chapter. The result is stated here because it is the one external input to the existence half of Lie's third theorem, entering only as the assertion that a faithful finite-dimensional representation exists. That single assertion, a faithful finite-dimensional representation for every  $\mathfrak{g}$ , is all the integration machinery below will draw on.

The interplay of Ado's theorem with the integration theorem of the previous section is the crux of the existence argument. Ado places  $\mathfrak{g}$  inside  $\mathfrak{gl}_n(\mathbb{R})$ , the Lie algebra of the concrete group  $\text{GL}_n(\mathbb{R})$ ; the integration theorem (theorem 3.3.4) turns that subalgebra into an actual subgroup; and the universal cover (proposition 3.4.1) upgrades that subgroup to a simply connected group with the same algebra. Chaining these produces a group for *every* algebra, which is essential surjectivity. Combined with the full faithfulness already established, this is the equivalence of categories.

**Theorem 3.4.4: Lie’s Third Theorem**

The functor  $\text{Lie}$  sending a simply connected Lie group  $G$  to its Lie algebra  $\text{Lie}(G)$ , and a homomorphism  $\varphi: G \rightarrow H$  to its differential  $d\varphi_e: \text{Lie}(G) \rightarrow \text{Lie}(H)$ , is an equivalence of categories

$$\text{Lie}: \{\text{simply connected Lie groups}\} \xrightarrow{\sim} \{\text{finite-dimensional real Lie algebras}\}$$

Concretely: every finite-dimensional real Lie algebra  $\mathfrak{g}$  is  $\text{Lie}(G)$  for a simply connected Lie group  $G$ , unique up to isomorphism, and for simply connected  $G, H$  the map  $\varphi \mapsto d\varphi_e$  is a bijection  $\text{Hom}(G, H) \xrightarrow{\sim} \text{Hom}(\text{Lie}(G), \text{Lie}(H))$ .

**Proof:**

A functor is an equivalence of categories precisely when it is fully faithful and essentially surjective. Both properties are established by assembling results already in hand.

*Step 1: full faithfulness.* For simply connected  $G$  and  $H$ , the map  $\varphi \mapsto d\varphi_e$  from  $\text{Hom}(G, H)$  to  $\text{Hom}(\text{Lie}(G), \text{Lie}(H))$  is a bijection: this is exactly corollary 3.3.8, the corollary of Lie’s second theorem. Its injectivity is that a homomorphism from a connected group is determined by its differential (corollary 2.4.4), and its surjectivity is that a Lie-algebra map integrates to a group map when the domain is simply connected (theorem 3.3.7). Full faithfulness is precisely this bijection on each hom-set.

*Step 2: essential surjectivity.* Let  $\mathfrak{g}$  be a finite-dimensional real Lie algebra. By Ado’s theorem (theorem 3.4.3) there is an injective Lie-algebra homomorphism  $\mathfrak{g} \hookrightarrow \mathfrak{gl}_n(\mathbb{R}) = \text{Lie}(\text{GL}_n(\mathbb{R}))$  for some  $n$ , identifying  $\mathfrak{g}$  with a Lie subalgebra  $\mathfrak{h} \subseteq \mathfrak{gl}_n(\mathbb{R})$ . By the integration theorem (theorem 3.3.4) there is a connected immersed Lie subgroup  $H \leq \text{GL}_n(\mathbb{R})$  with  $\text{Lie}(H) = \mathfrak{h} \cong \mathfrak{g}$ . The group  $H$  is connected but need not be simply connected; pass to its universal covering group  $\tilde{H}$  (proposition 3.4.1). Then  $\tilde{H}$  is simply connected, and the covering map  $\pi: \tilde{H} \rightarrow H$  has differential an isomorphism, so

$$\text{Lie}(\tilde{H}) \cong \text{Lie}(H) = \mathfrak{h} \cong \mathfrak{g}$$

Thus  $\mathfrak{g} \cong \text{Lie}(\tilde{H})$  with  $\tilde{H}$  simply connected, and  $\text{Lie}$  hits every algebra up to isomorphism.

*Step 3: uniqueness of the realizing group.* If  $G_1$  and  $G_2$  are simply connected with  $\text{Lie}(G_1) \cong \text{Lie}(G_2)$ , then the isomorphism  $\psi: \text{Lie}(G_1) \rightarrow \text{Lie}(G_2)$  and its inverse integrate by Step 1 to homomorphisms  $\varphi: G_1 \rightarrow G_2$  and  $\varphi': G_2 \rightarrow G_1$  with  $d\varphi_e = \psi$  and  $d\varphi'_e = \psi^{-1}$ . Then  $\varphi' \circ \varphi$  and  $\varphi \circ \varphi'$  have differentials  $\psi^{-1}\psi = \text{id}$  and  $\psi\psi^{-1} = \text{id}$ , so by corollary 2.4.4 they equal the identity maps of  $G_1$  and  $G_2$ ; hence  $\varphi$  is an isomorphism and  $G_1 \cong G_2$ . A fully faithful, essentially surjective functor is an equivalence, completing the proof.

Lie’s third theorem is the summit of the correspondence and closes Part I’s central question. The passage from a group to its Lie algebra, which began in Chapter 1 as the extraction of a tangent space of matrices, is now known to lose no information at all on the simply connected side: the categories of simply connected Lie groups and of finite-dimensional real Lie algebras are the same category wearing two descriptions. A theorem about groups can be proved by computing with brackets, and a finite-dimensional Lie algebra, an object of finite linear data, is guaranteed to be the infinitesimal shadow of an actual manifold-with-multiplication. The one hypothesis that cannot be dropped is simple connectivity; without it the functor forgets the fundamental group, and it is precisely that

forgotten data the next corollary restores.

Dropping simple connectivity, the theorem still controls the situation completely, because every connected group is a quotient of its simply connected cover. The classification of connected groups with a fixed Lie algebra therefore reduces to the classification of discrete central subgroups of a single object.

**Corollary 3.4.5: Connected Groups with a Given Lie Algebra**

Let  $\mathfrak{g}$  be a finite-dimensional real Lie algebra and let  $\tilde{G}$  be the simply connected Lie group with  $\text{Lie}(\tilde{G}) \cong \mathfrak{g}$  of theorem 3.4.4. Then the connected Lie groups  $G$  with  $\text{Lie}(G) \cong \mathfrak{g}$  are, up to isomorphism, exactly the quotients

$$G \cong \tilde{G}/\Gamma, \quad \Gamma \subseteq Z(\tilde{G}) \text{ a discrete subgroup}$$

and  $\pi_1(G) \cong \Gamma$ .

**Proof:**

Let  $G$  be a connected Lie group with  $\text{Lie}(G) \cong \mathfrak{g}$ . By proposition 3.4.1 its universal covering group  $\tilde{G}'$  is simply connected with  $\text{Lie}(\tilde{G}') \cong \text{Lie}(G) \cong \mathfrak{g}$ , so by the uniqueness in theorem 3.4.4 there is an isomorphism  $\tilde{G}' \cong \tilde{G}$ ; identify the two. The covering map  $\pi: \tilde{G} \rightarrow G$  is then a surjective homomorphism with discrete central kernel  $\Gamma = \ker \pi \subseteq Z(\tilde{G})$ , and  $\Gamma \cong \pi_1(G)$  by proposition 3.4.1. The first isomorphism theorem ([1] for the algebraic content) gives  $G \cong \tilde{G}/\Gamma$ , and the quotient is a Lie group with  $\pi$  its covering map because  $\Gamma$  is discrete.

Conversely, for any discrete subgroup  $\Gamma \subseteq Z(\tilde{G})$  the quotient  $\tilde{G}/\Gamma$  is a Lie group, and the quotient map  $\tilde{G} \rightarrow \tilde{G}/\Gamma$  is a covering: a discrete central subgroup acts on  $\tilde{G}$  by translations, and this action is free and properly discontinuous, so the orbit map is a covering ([5]). A covering map is a local diffeomorphism, so  $\text{Lie}(\tilde{G}/\Gamma) \cong \text{Lie}(\tilde{G}) \cong \mathfrak{g}$ , and  $\tilde{G}$  being simply connected makes the quotient map the universal cover, whence  $\pi_1(\tilde{G}/\Gamma) \cong \Gamma$  by proposition 3.4.1. Thus the connected groups with algebra  $\mathfrak{g}$  are exactly the  $\tilde{G}/\Gamma$ , and their fundamental groups are the corresponding  $\Gamma$ , as we sought to show.

The corollary reduces a differential-geometric classification to a discrete computation inside one group. To list every connected Lie group with a given Lie algebra, one finds the simply connected model  $\tilde{G}$ , computes its center  $Z(\tilde{G})$ , and enumerates the discrete subgroups of that center. Each such subgroup  $\Gamma$  produces a group  $\tilde{G}/\Gamma$  with fundamental group  $\Gamma$ ; the choice  $\Gamma = \{e\}$  recovers  $\tilde{G}$  itself, and the largest possible  $\Gamma = Z(\tilde{G})$  produces the smallest group in the family, the adjoint group  $\tilde{G}/Z(\tilde{G})$  when  $Z(\tilde{G})$  is finite, whose fundamental group  $\pi_1 \cong Z(\tilde{G})$  is correspondingly the largest. Everything hinges on the center of the simply connected group, a finite-dimensional and often finite piece of arithmetic.

The running example makes the enumeration explicit and recovers the double cover of Chapter 1 as an instance of the corollary.

**Example 3.4.6: The Two Connected Groups with Algebra  $\mathfrak{su}(2)$**

Take  $\mathfrak{g} = \mathfrak{su}(2)$ , the Lie algebra of example 1.4.7, isomorphic to  $(\mathbb{R}^3, \times)$ . Its simply connected model is  $\tilde{G} = \text{SU}(2)$ : the group  $\text{SU}(2)$  is diffeomorphic to the 3-sphere  $S^3$  and hence simply connected (proposition 1.5.1), and its Lie algebra is  $\mathfrak{su}(2)$ , so by the uniqueness in theorem 3.4.4 it

is the simply connected group with this algebra. To classify all connected groups with algebra  $\mathfrak{su}(2)$ , compute the center. The center of  $SU(2)$  is

$$Z(SU(2)) = \{I, -I\} \cong \mathbb{Z}/2$$

as a direct computation shows: a matrix  $g \in SU(2)$  commuting with the diagonal  $\text{diag}(i, -i) \in SU(2)$  must itself be diagonal, say  $g = \text{diag}(a, d)$ , since  $\text{diag}(i, -i)$  has distinct eigenvalues; and commuting in addition with  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in SU(2)$  forces  $a = d$ , so  $g = aI$  with  $a^2 = 1$  and  $|a| = 1$ , whence  $g = \pm I$ . The discrete subgroups of  $Z(SU(2)) = \mathbb{Z}/2$  are only two, the trivial group  $\{I\}$  and the whole center  $\{\pm I\}$ . By corollary 3.4.5 there are therefore exactly two connected Lie groups with Lie algebra  $\mathfrak{su}(2)$ :

1.  $\Gamma = \{I\}$  gives  $SU(2)/\{I\} = SU(2)$  itself, simply connected,  $\pi_1 = 0$ ;
2.  $\Gamma = \{\pm I\}$  gives  $SU(2)/\{\pm I\} \cong SO(3)$ , with  $\pi_1(SO(3)) \cong \mathbb{Z}/2$ .

The second quotient is the double cover  $SU(2) \rightarrow SO(3)$  of theorem 1.5.5: that map is a surjective homomorphism with kernel exactly  $\{\pm I\}$ , so  $SO(3) \cong SU(2)/\{\pm I\}$ , and its fundamental group  $\mathbb{Z}/2$  (corollary 1.5.6) is the kernel  $\Gamma$ , as the corollary predicts. There is no third connected group with this algebra, because  $\mathbb{Z}/2$  has no other subgroups. The pair  $SU(2), SO(3)$  is the complete list, and the double cover is not an accident of the classical groups but the only nontrivial central quotient the algebra  $\mathfrak{su}(2)$  permits.

The example is the sharpest illustration of the section’s thesis. The two groups  $SU(2)$  and  $SO(3)$  are indistinguishable by their Lie algebra: both have  $\text{Lie} = \mathfrak{su}(2)$ , the same exponential map on a neighborhood of the identity, the same brackets, the same local structure. What separates them is a single bit of global topology,  $\pi_1 = 0$  against  $\pi_1 = \mathbb{Z}/2$ , encoded as the central subgroup  $\{\pm I\}$  one quotients by. Chapter 1 constructed the double cover by an explicit computation with quaternions and rotations; the present corollary explains why that computation had to produce exactly a  $\mathbb{Z}/2$  and no more, and why  $SU(2)$  and  $SO(3)$  exhaust the connected groups with this algebra. The linear theory sees  $\mathfrak{su}(2)$ ; the topology sees the choice of  $\Gamma$ .

This is the exact boundary of the Lie group–Lie algebra dictionary. The Lie algebra determines a connected group completely except for the discrete central data of the fundamental group, the one invariant the infinitesimal, linear theory cannot detect. Two connected groups with isomorphic Lie algebras are locally isomorphic and differ only in which discrete central subgroup of the common simply connected cover has been divided out. Reading the correspondence in this light turns the fundamental group from a topological afterthought into the governing invariant of the classification, and it is the covering-group theory of the next chapter that develops the tools to compute it: group actions, quotient manifolds, and the systematic study of covering groups and their centers. The center of the simply connected group, whose subgroups index the connected forms, becomes an object of study in its own right, and the fundamental group  $\pi_1(G) \cong \Gamma$  becomes a piece of the group’s structure rather than a fact about its underlying space.

### Exercise 3.4.1

1. Verify in detail that  $\pi: \mathbb{R} \rightarrow S^1$ ,  $\pi(t) = e^{2\pi it}$ , is the universal covering group of the circle: show it is a Lie group homomorphism, that its kernel is  $\mathbb{Z}$ , and that  $\pi_1(S^1) \cong \mathbb{Z}$ . Identify explicitly the discrete central subgroup  $\Gamma \subseteq \mathbb{R}$  with  $S^1 \cong \mathbb{R}/\Gamma$ .

2. Prove that every connected abelian Lie group is isomorphic to  $\mathbb{R}^k \times \mathbb{T}^m$  for some  $k, m \geq 0$ , where  $\mathbb{T}^m = (S^1)^m$ . Use that the simply connected abelian group with Lie algebra  $\mathbb{R}^n$  is  $(\mathbb{R}^n, +)$ , obtained by integrating the trivial bracket via theorem 3.4.4, then apply corollary 3.4.5 to a discrete central subgroup.
3. Deduce from the previous exercise that a *simply connected* connected abelian Lie group of dimension  $n$  is isomorphic to  $(\mathbb{R}^n, +)$ . Which value of  $m$  in the description  $\mathbb{R}^k \times \mathbb{T}^m$  corresponds to simple connectivity, and why?
4. State the fact that  $\mathrm{SU}(n)$  is simply connected for every  $n \geq 1$  (proved topologically in [5]). Using corollary 3.4.5 and the center  $Z(\mathrm{SU}(n)) \cong \mathbb{Z}/n$ , describe how many connected Lie groups have Lie algebra  $\mathfrak{su}(n)$ , and identify  $\mathrm{SU}(2)$  and  $\mathrm{SO}(3)$  as the two groups in the case  $n = 2$ .
5. Let  $\mathfrak{g}$  be a finite-dimensional real Lie algebra whose simply connected group  $\tilde{G}$  has *trivial* center  $Z(\tilde{G}) = \{e\}$ . Prove that  $\tilde{G}$  is the only connected Lie group with Lie algebra  $\mathfrak{g}$ . Give an example of such a  $\mathfrak{g}$  together with its unique connected group.
6. With  $G = \mathrm{SO}(3)$ ,  $H = \mathrm{SU}(2)$ , and  $\psi = \mathrm{id}$  the identity on  $\mathfrak{su}(2) = \mathrm{Lie}(\mathrm{SO}(3)) = \mathrm{Lie}(\mathrm{SU}(2))$ , show that  $\psi$  does *not* arise as  $d\varphi_e$  for any Lie group homomorphism  $\varphi: \mathrm{SO}(3) \rightarrow \mathrm{SU}(2)$ . Explain how this is consistent with corollary 3.3.8.

## 3.5 The Baker-Campbell-Hausdorff Formula

Lie's third theorem (theorem 3.4.4) asserts that a simply-connected Lie group is recovered from its Lie algebra, an equivalence of categories that trades an analytic object for an algebraic one. The proof runs through Ado's theorem and the integration of subalgebras, and it is in a sense non-constructive: it produces the group but does not exhibit its multiplication. This section supplies what is missing near the identity. The exponential map is a local diffeomorphism (theorem 2.3.4), so the product of two group elements  $\exp X$  and  $\exp Y$  close to  $e$  is encoded by a single element  $Z$  of the Lie algebra with  $\exp Z = \exp X \exp Y$ ; the content of this section is a closed formula for  $Z = \log(\exp X \exp Y)$  purely in terms of  $X$ ,  $Y$ , and their iterated brackets. The formula is universal, the same combination of brackets for every Lie group, and its existence makes precise the claim that the bracket determines the group law near the identity.

The formula was found in incomplete form by Campbell, Poincare, Baker, and Hausdorff around the turn of the twentieth century, with the explicit bracket-series expression due to Dynkin in 1947. Its structural message is sharper than any single coefficient: the product of two exponentials is again an exponential, and the exponent lies in the Lie subalgebra generated by  $X$  and  $Y$  under the bracket alone. Everything the group multiplication does infinitesimally is bracketing.

The starting point is a direct expansion in the matrix setting, where  $\exp$  and  $\log$  are the convergent series of definition 1.3.2 and proposition 1.3.6. Multiplying the two exponential series and feeding the product into the logarithm series produces, order by order in  $X$  and  $Y$ , a polynomial in  $X$  and  $Y$ ; the content of the theorem is that after collecting terms this polynomial is expressible entirely through commutators. Before stating the general formula the low-order terms are worth isolating, since they carry the whole conceptual load and are what one uses in practice.

**Theorem 3.5.1: The Baker–Campbell–Hausdorff Formula**

Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g}$ . There is a neighborhood  $V$  of 0 in  $\mathfrak{g}$  such that for all  $X, Y \in V$  the product  $\exp X \exp Y$  lies in the image of the exponential chart, and

$$\log(\exp X \exp Y) = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}([X, [X, Y]] + [Y, [Y, X]]) - \frac{1}{24}[Y, [X, [X, Y]]] + \cdots$$

where every term after  $X + Y$  is a rational multiple of an iterated bracket of  $X$  and  $Y$ , and the omitted terms have bracket-degree  $\geq 5$ . The right-hand side is a single *universal Lie series*: the same rational coefficients occur for every Lie group, and the series depends on  $G$  only through its bracket. In Dynkin’s closed form,

$$\log(\exp X \exp Y) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \sum_{\substack{p_i + q_i \geq 1 \\ 1 \leq i \leq n}} \frac{[X^{p_1} Y^{q_1} \cdots X^{p_n} Y^{q_n}]}{(\sum_{i=1}^n (p_i + q_i)) \prod_{i=1}^n p_i! q_i!} \quad (3.6)$$

where  $[X^{p_1} Y^{q_1} \cdots X^{p_n} Y^{q_n}]$  denotes the right-nested iterated bracket  $[Z_1, [Z_2, [\dots, [Z_{N-1}, Z_N] \cdots]]]$  whose arguments  $Z_1, \dots, Z_N$  are, read left to right, the letters  $X$  occurring  $p_i$  times and  $Y$  occurring  $q_i$  times in the  $i$ th block; a bracket whose innermost two arguments coincide is understood to be zero.

The three lowest-order terms are the ones that appear in nearly every application, and they already display the structure. To first order the group is abelian:  $\log(\exp X \exp Y) = X + Y + O(2)$ , so infinitesimally  $\exp$  turns addition in  $\mathfrak{g}$  into multiplication in  $G$ . The first correction is  $\frac{1}{2}[X, Y]$ , the leading measure of noncommutativity, and the third-order terms  $\frac{1}{12}([X, [X, Y]] + [Y, [Y, X]])$  are its refinement. That no term ever involves anything but iterated brackets is the substance of the theorem: a product of exponentials returns to the Lie subalgebra generated by  $X$  and  $Y$ , never leaving it for the ambient associative algebra. This is why the Lie algebra, a purely bracket-theoretic object, suffices to reconstruct the local group.

The Dynkin expression (3.6) is explicit but combinatorially dense; in practice one computes the first few terms directly from the  $\exp/\log$  series and invokes the closed form only to certify that all higher terms are brackets. The convergence of the series is a separate matter, taken up next, and the recursion that generates the coefficients is developed in the proof.

Before the general proof, the convergence claim deserves its own statement, since it is what makes the formula an identity between elements of  $\mathfrak{g}$  rather than a formal manipulation of symbols.

**Proposition 3.5.2: Convergence of the BCH Series**

Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g}$ , and fix any norm  $\|\cdot\|$  on the finite-dimensional space  $\mathfrak{g}$ . There is a constant  $\rho > 0$ , depending only on the bracket of  $\mathfrak{g}$  and the chosen norm, such that the Dynkin series (3.6) converges absolutely whenever  $\|X\| + \|Y\| < \rho$ , and its sum is  $\log(\exp X \exp Y)$ . In particular the series defines a real-analytic  $\mathfrak{g}$ -valued function of  $(X, Y)$  on the ball  $\|X\| + \|Y\| < \rho$ .

**Proof:**

*Step 1: reduction to the matrix case.* By theorem 3.4.3 (cited to [3]), the Lie algebra  $\mathfrak{g}$  admits a faithful finite-dimensional representation  $\iota: \mathfrak{g} \hookrightarrow \mathfrak{gl}_m(\mathbb{R}) = M_m(\mathbb{R})$  for some  $m$ , a Lie-algebra embedding into a matrix algebra. Both sides of (3.6) are built from  $X, Y$ , and the bracket, and  $\iota$  preserves the bracket, so the series for  $\mathfrak{g}$  converges if and only if the corresponding series for the matrices  $\iota(X), \iota(Y) \in M_m(\mathbb{R})$  converges, and to the image under  $\iota$  of the same element. It therefore suffices to prove absolute convergence and the identity in  $M_m(\mathbb{R})$ , where  $\exp$  and  $\log$  are the concrete series of Chapter 1. Fix the operator norm  $\|\cdot\|$  on  $M_m(\mathbb{R})$ , which is submultiplicative, so that  $\|[A, B]\| \leq 2\|A\|\|B\|$ .

*Step 2: absolute convergence of the term-by-term series.* Set  $W = \exp X \exp Y - I$ . The bound  $\|\exp X\| \leq e^{\|X\|}$  of proposition 1.3.3 gives

$$\|W\| = \|\exp X \exp Y - I\| \leq e^{\|X\|} e^{\|Y\|} - 1 = e^{\|X\| + \|Y\|} - 1$$

so if  $\|X\| + \|Y\| < \log 2$  then  $\|W\| < 1$  and the logarithm series  $\log(I+W) = \sum_{k \geq 1} (-1)^{k+1} W^k / k$  of proposition 1.3.6 converges absolutely to a well-defined matrix  $Z = \log(\exp X \exp Y)$ . Expanding  $W = \sum_{p+q \geq 1} X^p Y^q / (p! q!)$  from the two exponential series and substituting into the logarithm series produces, after grouping monomials by total degree, exactly the summands of (3.6) before the passage from associative monomials to brackets; the associated series of absolute values is dominated term by term by the scalar series obtained by replacing  $X, Y$  with  $\|X\|, \|Y\|$ , namely  $-\log(2 - e^{s+t})$  with  $s = \|X\|, t = \|Y\|$ , which is finite precisely when  $e^{s+t} < 2$ . Hence the associative form of the series converges absolutely for  $\|X\| + \|Y\| < \log 2$ .

*Step 3: the bracket form and its radius.* The passage from the associative monomials to the iterated brackets of (3.6) is a finite regrouping within each fixed total degree  $n$ : the Dynkin coefficient of a bracket of degree  $n$  is a fixed rational number, and there are finitely many brackets of each degree, so the bracket series is the absolutely convergent associative series collected into finite blocks. Each iterated bracket of degree  $n$  is bounded by  $\|[X^{p_1} \cdots Y^{q_n}]\| \leq 2^{n-1} \|X\|^{\sum p_i} \|Y\|^{\sum q_i}$ , by iterating  $\|[A, B]\| \leq 2\|A\|\|B\|$ . Summing these bounds against the Dynkin coefficients gives a scalar majorant convergent on a ball  $\|X\| + \|Y\| < \rho$  for some  $\rho \in (0, \log 2)$  determined by the coefficient growth and the factor 2 from the bracket estimate. On that ball the bracket series converges absolutely to the same sum  $Z = \log(\exp X \exp Y)$  as the associative series, since the two are absolute rearrangements of one another. The transferred constant  $\rho$  for the original  $\mathfrak{g}$  absorbs the distortion of the norm under the fixed linear isomorphism  $\iota$  onto its image. A power series converging absolutely to a matrix-valued (hence  $\mathfrak{g}$ -valued) function is real-analytic there, completing the proof.

The proposition upgrades the formula from a formal identity to an equation between elements of  $\mathfrak{g}$ , valid on an explicit neighborhood. The radius  $\rho$  is not sharp and depends on the algebra: for a nilpotent Lie algebra the series terminates and  $\rho = \infty$ , while for  $\mathfrak{sl}_2(\mathbb{R})$  the exponential fails to be globally surjective and no global formula can hold (exercise 3.5.1 (6)). What matters for the structure theory is only that some positive radius exists, so that the group law is analytic in the canonical coordinates  $\exp$  provides.

The proof of the theorem now proceeds in two parts: the explicit low-order computation, done from the series, and the recursion that produces the higher coefficients and certifies they are brackets. The low-order part is the one the reader should be able to reproduce; the recursion is where Dynkin's closed form comes from, and it is stated and its key step verified, with the closed evaluation of the coefficients cited.

**Proof Proof of theorem 3.5.1:**

By proposition 3.5.2 it suffices to work in  $M_m(\mathbb{R})$  after the faithful embedding of its Step 1, so throughout  $X, Y \in M_m(\mathbb{R})$  and  $\exp, \log$  are the series of definition 1.3.2 and proposition 1.3.6.

*Step 1: the low-order terms by direct expansion.* Write the two exponential series to third order,

$$\exp X = I + X + \frac{1}{2}X^2 + \frac{1}{6}X^3 + \cdots, \quad \exp Y = I + Y + \frac{1}{2}Y^2 + \frac{1}{6}Y^3 + \cdots$$

and multiply, collecting by total degree,

$$\exp X \exp Y = I + (X + Y) + \frac{1}{2}(X^2 + 2XY + Y^2) + \frac{1}{6}(X^3 + 3X^2Y + 3XY^2 + Y^3) + \cdots$$

where in each degree the cross terms carry  $X$ 's to the left of  $Y$ 's, since  $\exp X$  is the left factor. Denote by  $W = \exp X \exp Y - I$  the sum of the terms of degree  $\geq 1$  displayed above, and apply the logarithm series  $\log(I + W) = W - \frac{1}{2}W^2 + \frac{1}{3}W^3 - \cdots$ , retaining only terms of total degree  $\leq 3$ .

The degree-one part of  $\log(I + W)$  is the degree-one part of  $W$ , namely  $X + Y$ . For degree two,  $W$  contributes  $\frac{1}{2}(X^2 + 2XY + Y^2)$  and  $-\frac{1}{2}W^2$  contributes  $-\frac{1}{2}(X + Y)^2 = -\frac{1}{2}(X^2 + XY + YX + Y^2)$ ; adding,

$$\frac{1}{2}(X^2 + 2XY + Y^2) - \frac{1}{2}(X^2 + XY + YX + Y^2) = \frac{1}{2}(XY - YX) = \frac{1}{2}[X, Y]$$

The squares  $X^2$  and  $Y^2$  cancel exactly and only the commutator survives, the second-order term of the theorem.

*Step 2: the third-order term.* Collect the degree-three contributions from  $W$ ,  $-\frac{1}{2}W^2$ , and  $\frac{1}{3}W^3$ . Write  $W = W_1 + W_2 + W_3 + \cdots$  by degree, with  $W_1 = X + Y$  and  $W_2 = \frac{1}{2}(X^2 + 2XY + Y^2)$ . The degree-three part of  $W$  is  $W_3 = \frac{1}{6}(X^3 + 3X^2Y + 3XY^2 + Y^3)$ ; the degree-three part of  $-\frac{1}{2}W^2$  is  $-\frac{1}{2}(W_1W_2 + W_2W_1)$ ; the degree-three part of  $\frac{1}{3}W^3$  is  $\frac{1}{3}W_1^3 = \frac{1}{3}(X + Y)^3$ . Summing these and tallying every associative monomial of degree three, the pure powers  $X^3, Y^3$  cancel and the mixed monomials assemble into

$$[\log(\exp X \exp Y)]_{\deg 3} = \frac{1}{12}([X, [X, Y]] + [Y, [Y, X]])$$

with the stated coefficient  $\frac{1}{12}$ . The same bookkeeping at degree four yields the term  $-\frac{1}{24}[Y, [X, [X, Y]]]$  recorded in the theorem.

*Step 3: the recursion and universality.* That the phenomenon of Steps 1 and 2 persists to all orders, so that every term is a bracket, follows from a differential recursion. Fix  $X$  and set  $Z(t) = \log(\exp X \exp tY)$ , a curve in  $M_m(\mathbb{R})$  with  $Z(0) = X$ . Differentiating  $\exp Z(t) = \exp X \exp tY$  in  $t$  and using the standard formula for the differential of the exponential,

$$\frac{d}{dt} \exp Z(t) = \left( \int_0^1 \exp(s \operatorname{ad} Z(t)) ds \right) (\dot{Z}(t)) \exp Z(t)$$

together with  $\frac{d}{dt}(\exp X \exp tY) = \exp X \exp tY \cdot Y = \exp Z(t) Y$ , one uses the conjugation identity  $\exp Z(t) Y = (e^{\operatorname{ad} Z(t)} Y) \exp Z(t)$  to move the right factor across, and then, cancelling  $\exp Z(t)$  on the right, obtains the identity

$$\left( \int_0^1 \exp(s \operatorname{ad} Z(t)) ds \right) (\dot{Z}(t)) = e^{\operatorname{ad} Z(t)} Y$$

The operator  $\int_0^1 \exp(s \operatorname{ad} Z) ds$  equals  $g(\operatorname{ad} Z)$  for the analytic function  $g(u) = (e^u - 1)/u$ , which is invertible near  $u = 0$ ; composing its inverse  $u/(e^u - 1)$  with the factor  $e^{\operatorname{ad} Z}$  on the right produces the analytic function  $u/(1 - e^{-u})$ , the generating function of the Bernoulli numbers in the convention  $B_1 = +\frac{1}{2}$ ,

$$\frac{u}{1 - e^{-u}} = \sum_{k=0}^{\infty} \frac{B_k}{k!} u^k = 1 + \frac{1}{2}u + \frac{1}{12}u^2 - \frac{1}{720}u^4 + \cdots \quad (3.7)$$

Inverting gives the differential equation, whose operator ratio is read off from its power series on the right, so that the kernel of  $\operatorname{ad} Z$ , forced by  $[Z, Z] = 0$ , poses no problem of dividing by a noninvertible  $I - e^{-\operatorname{ad} Z}$ ,

$$\dot{Z}(t) = \frac{\operatorname{ad} Z(t)}{I - \exp(-\operatorname{ad} Z(t))} (Y) = \sum_{k=0}^{\infty} \frac{B_k}{k!} (\operatorname{ad} Z(t))^k (Y) \quad (3.8)$$

with initial condition  $Z(0) = X$ . Every term on the right of (3.8) is an iterated bracket of  $Z(t)$  and  $Y$ , and  $Z(0) = X$  is a single letter; solving by Picard iteration therefore produces  $Z(1) = \log(\exp X \exp Y)$  as a series each of whose terms is an iterated bracket of  $X$  and  $Y$ , with rational coefficients built from the  $B_k$ . Because (3.8) refers only to the bracket and the universal constants  $B_k$ , the resulting series is the same for every Lie algebra, which is the asserted universality. Integrating (3.8) order by order reproduces  $X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}([X, [X, Y]] + [Y, [Y, X]]) + \cdots$  and, carried out in closed form, yields precisely Dynkin's expression (3.6); the combinatorial evaluation of the coefficients is a bookkeeping computation whose details are in [3]. This establishes the formula and that all its terms are brackets, completing the proof.

The proof exhibits two faces of the formula. The direct expansion of Steps 1 and 2 is elementary and self-checking: multiply the series, take the logarithm, watch the pure powers cancel. The recursion of Step 3 is the engine, encoding the entire series in a single ordinary differential equation (3.8) whose only ingredients are the bracket and the Bernoulli numbers. The appearance of the Bernoulli generating function (3.7) is not accidental: it is the same series that governs the differential of the exponential map, and it is why the coefficients  $\frac{1}{2}, \frac{1}{12}, 0, -\frac{1}{720}, \dots$  of BCH are exactly the Bernoulli numbers divided by factorials. That  $B_3 = 0$ , and more generally  $B_k = 0$  for odd  $k \geq 3$ , is the reason the series thins out at odd orders.

The formula answers the question left open by Lie's third theorem: not only does a Lie algebra determine a simply-connected group, it determines the group multiplication explicitly near the identity. This is the local reconstruction, recorded as the following corollary.

#### Corollary 3.5.3: The Local Group Law from the Bracket

Let  $G$  and  $G'$  be Lie groups with Lie algebras  $\mathfrak{g}$  and  $\mathfrak{g}'$ , and suppose  $\varphi: \mathfrak{g} \rightarrow \mathfrak{g}'$  is an isomorphism of Lie algebras. Then there are neighborhoods  $U$  of  $e$  in  $G$  and  $U'$  of  $e'$  in  $G'$  and a diffeomorphism  $\Phi: U \rightarrow U'$  that is a *local isomorphism*:  $\Phi(e) = e'$ , and  $\Phi(gh) = \Phi(g)\Phi(h)$  whenever  $g, h$ , and  $gh$  all lie in  $U$ . Consequently the multiplication of  $G$  on a neighborhood of  $e$  is determined by the bracket of  $\mathfrak{g}$  alone, and two Lie groups with isomorphic Lie algebras are locally isomorphic.

**Proof:**

Choose the exponential charts  $\exp: V \rightarrow U$  and  $\exp': V' \rightarrow U'$  of theorem 2.3.4, shrinking  $V, V'$  so that the BCH series of proposition 3.5.2 converges on  $V$  and on  $\varphi(V)$  and so that  $\varphi(V) \subseteq V'$ . Define  $\Phi = \exp' \circ \varphi \circ \exp^{-1}: U \rightarrow U'$ , a diffeomorphism carrying  $e = \exp 0$  to  $e' = \exp' 0$ . For  $g = \exp X$  and  $h = \exp Y$  with  $X, Y \in V$  small enough that  $\log(gh) \in V$ , the formula (3.6) gives  $\log(gh) = B(X, Y)$ , where  $B(X, Y) = X + Y + \frac{1}{2}[X, Y] + \cdots$  is the universal bracket series. The map  $\varphi$  is a Lie-algebra isomorphism and so commutes with every bracket, and  $B$  is a universal series in the bracket with the same coefficients on both sides, so  $\varphi(B(X, Y)) = B(\varphi X, \varphi Y)$ . Therefore

$$\Phi(g)\Phi(h) = \exp'(\varphi X) \exp'(\varphi Y) = \exp'(B(\varphi X, \varphi Y)) = \exp'(\varphi B(X, Y)) = \exp'(\varphi(\log(gh))) = \Phi(gh)$$

using theorem 3.5.1 in  $G'$  for the second equality and universality for the third. Thus  $\Phi$  is a local homomorphism, hence a local isomorphism, since  $\varphi^{-1}$  produces the inverse the same way. The multiplication on  $U$  is then  $gh = \exp(B(\log g, \log h))$ , a formula in the bracket alone, as we sought to show.

This is the local half of Lie's third theorem made explicit. Lie's third theorem (theorem 3.4.4) says an isomorphism of Lie algebras integrates to an isomorphism of the simply-connected groups; the corollary extracts the part that needs no global topology, namely that the algebras being isomorphic already forces the groups to look identical near the identity. Global obstructions, differences in the fundamental group or the center, are invisible to the bracket and appear only when the local isomorphism is continued around noncontractible loops. The pair  $SU(2)$  and  $SO(3)$  is the standing example: their Lie algebras are isomorphic (both isomorphic to  $(\mathbb{R}^3, \times)$ , with  $\mathfrak{so}(3) \cong \mathfrak{su}(2) \cong (\mathbb{R}^3, \times)$ ), so BCH gives the same local multiplication, and the double cover  $SU(2) \rightarrow SO(3)$  of theorem 1.5.5 is a local isomorphism that fails to be global precisely because  $\pi_1(SO(3)) = \mathbb{Z}/2$  is nontrivial (corollary 1.5.6). The classification of the connected groups sharing one algebra as quotients  $\tilde{G}/\Gamma$  (corollary 3.4.5) is the global completion of this local picture.

The formula also isolates exactly when the group is abelian near the identity. If  $[X, Y] = 0$  then every bracket in the BCH series vanishes and the formula collapses to its first term.

**Corollary 3.5.4: Commuting Exponentials**

Let  $G$  be a Lie group with Lie algebra  $\mathfrak{g}$ , and let  $X, Y \in \mathfrak{g}$  with  $[X, Y] = 0$ . Then, for  $X, Y$  in the domain of convergence of proposition 3.5.2,

$$\exp X \exp Y = \exp(X + Y) = \exp Y \exp X$$

Conversely, if  $\exp(sX)$  and  $\exp(tY)$  commute for all small  $s, t$ , then  $[X, Y] = 0$ .

**Proof:**

If  $[X, Y] = 0$  then every iterated bracket in (3.6) of degree  $\geq 2$  contains a factor  $[X, Y]$  or  $[Y, X]$  as an innermost commutator, hence vanishes; the series reduces to  $X + Y$ , so  $\log(\exp X \exp Y) = X + Y$  and  $\exp X \exp Y = \exp(X + Y)$ . Exchanging the roles of  $X$  and  $Y$  and using  $\exp(X + Y) = \exp(Y + X)$  gives  $\exp X \exp Y = \exp Y \exp X$ . For the converse, the group commutator  $\exp(sX) \exp(tY) \exp(-sX) \exp(-tY)$  has BCH expansion  $st[X, Y] + O(3)$  in  $s, t$ ; if it is the identity for all small  $s, t$  then its second-order term  $st[X, Y]$  vanishes identically, forcing  $[X, Y] = 0$ .

| 0, completing the proof.

In the matrix case this recovers the additivity of the exponential on commuting matrices proved directly in proposition 1.3.5: there the identity  $\exp X \exp Y = \exp(X + Y)$  for  $XY = YX$  was obtained by rearranging the product of two absolutely convergent series, and BCH now explains it structurally, as the statement that all correction terms are brackets and therefore disappear the instant the bracket does. The bracket is the entire obstruction to the naive exponential law, measured to leading order by  $\frac{1}{2}[X, Y]$  and to all orders by the BCH series.

### Example 3.5.5: BCH on the Heisenberg Algebra

Let  $\mathfrak{h}_3$  be the three-dimensional *Heisenberg algebra*: the Lie algebra with basis  $P, Q, C$  and single nontrivial bracket  $[P, Q] = C$ , all other brackets among basis elements zero. It is realized concretely as the strictly upper-triangular  $3 \times 3$  real matrices, with

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

so that  $[P, Q] = PQ - QP = C$  and  $C$  is central. Every triple bracket vanishes:  $[Z, [Z', Z'']]$  for basis elements is a bracket involving  $C$ , which brackets to zero with everything. Hence in (3.6) every term of bracket-degree  $\geq 3$  is zero, and the series terminates after the  $\frac{1}{2}[X, Y]$  term:

$$\log(\exp X \exp Y) = X + Y + \frac{1}{2}[X, Y] \quad \text{for all } X, Y \in \mathfrak{h}_3$$

an exact identity with no domain restriction, since a terminating series converges everywhere. Writing  $X = aP + bQ + cC$  and  $X' = a'P + b'Q + c'C$ , the bracket is  $[X, X'] = (ab' - a'b)C$ , so

$$\exp X \exp X' = \exp\left((a + a')P + (b + b')Q + \left(c + c' + \frac{1}{2}(ab' - a'b)\right)C\right)$$

The corresponding Heisenberg group  $H_3$  of unipotent upper-triangular matrices therefore has the multiplication law  $(a, b, c) \cdot (a', b', c') = (a + a', b + b', c + c' + \frac{1}{2}(ab' - a'b))$  in exponential coordinates, the standard cocycle presentation. This is the smallest nonabelian example in which BCH is visible in full: the group is nonabelian, yet the failure of commutativity is captured entirely by the single quadratic term  $\frac{1}{2}(ab' - a'b)$ .

The Heisenberg example is the cleanest illustration of the formula because the series stops after two terms. More generally, on any *nilpotent* Lie algebra the iterated brackets eventually vanish, so the BCH series is a polynomial and converges everywhere; the exponential map is then a global diffeomorphism onto a simply-connected nilpotent group, and BCH presents its multiplication as a polynomial map, a point developed with the nilpotent Lie algebras of Part II. At the opposite extreme, on an abelian Lie algebra every bracket is zero, BCH reduces to  $X + Y$ , and  $\exp$  is a homomorphism from  $(\mathfrak{g}, +)$  to  $G$ , recovering that a connected abelian Lie group is a quotient of its own Lie algebra. Between these poles the BCH series measures exactly how far a group departs from being the additive group of its Lie algebra.

### Exercise 3.5.1

1. Verify the third-order term of the BCH formula directly. Using  $\log(\exp X \exp Y) = W - \frac{1}{2}W^2 + \frac{1}{3}W^3 - \dots$  with  $W = \exp X \exp Y - I$ , collect all degree-three associative monomials

in  $X$  and  $Y$  and confirm they sum to  $\frac{1}{12}([X, [X, Y]] + [Y, [Y, X]])$ .

2. Show from  $\exp X \exp Y = \exp Z$  that BCH satisfies the symmetry  $\log(\exp X \exp Y) = -\log(\exp(-Y) \exp(-X))$ . Deduce that the degree- $n$  term  $Z_n(X, Y)$  obeys  $Z_n(X, Y) = (-1)^{n+1} Z_n(Y, X)$ , and check this against the terms  $X + Y$ ,  $\frac{1}{2}[X, Y]$ , and  $\frac{1}{12}([X, [X, Y]] + [Y, [Y, X]])$ .
3. Prove the conjugation identity  $\exp X \exp Y \exp(-X) = \exp(\text{Ad}(\exp X) Y) = \exp(e^{\text{ad} X} Y)$  for a matrix group, and explain why, unlike BCH, it holds globally and needs no logarithm. Use proposition 2.5.5 and the identity  $\text{Ad}(\exp X) = e^{\text{ad} X}$ .
4. On the Heisenberg algebra  $\mathfrak{h}_3$  of example 3.5.5, compute  $\exp(P)$ ,  $\exp(Q)$ , and  $\exp(P) \exp(Q)$  explicitly as  $3 \times 3$  matrices, and confirm by direct multiplication that  $\exp(P) \exp(Q) = \exp(P + Q + \frac{1}{2}C)$ . Then verify  $\exp(P) \exp(Q) \exp(-P) \exp(-Q) = \exp(C)$ , the group commutator, and interpret it in light of corollary 3.5.4.
5. Let  $\mathfrak{g}$  be an abelian Lie algebra and  $G$  a connected Lie group with  $\text{Lie}(G) = \mathfrak{g}$ . Show  $\exp: (\mathfrak{g}, +) \rightarrow G$  is a surjective Lie group homomorphism with discrete kernel, and conclude  $G \cong \mathfrak{g}/\Gamma$  for a discrete subgroup  $\Gamma \subseteq \mathfrak{g}$ . Identify  $\Gamma$  for  $G = \mathbb{T} = \mathbb{R}/\mathbb{Z}$  and for  $G = \mathbb{R}_{>0}$ .
6. Explain why the BCH series cannot converge on all of  $\mathfrak{sl}_2(\mathbb{R}) \times \mathfrak{sl}_2(\mathbb{R})$ . Show that the matrix  $g = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \in \text{SL}(2, \mathbb{R})$  is not  $\exp Z$  for any  $Z \in \mathfrak{sl}_2(\mathbb{R})$ , then exhibit  $X, Y \in \mathfrak{sl}_2(\mathbb{R})$  with  $\exp X \exp Y = g$ , so that no  $Z = \log(\exp X \exp Y)$  exists for these  $X, Y$ . Relate this to the failure of  $\exp$  to be surjective on  $\text{SL}(2, \mathbb{R})$ .



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# *Homogeneous Spaces, Actions, and Covering Groups*

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The correspondence of chapter 3 turned the Lie algebra into a tool for building and recognizing groups. This chapter turns the same machinery outward, onto the spaces a Lie group acts on. The organizing observation is that a great many of the manifolds of geometry, the spheres, the projective spaces, the Grassmannians and flag manifolds, are not merely acted on by a Lie group but are *homogeneous*: the group acts transitively, so that every point looks like every other, and the manifold is recovered as a quotient  $G/H$  by the stabilizer of a single point. The quotient construction, and the theorem that  $G/H$  is again a smooth manifold whenever  $H$  is closed, converts group theory into a factory for manifolds with a large symmetry group already built in.

Two threads run through the chapter. The first is the quotient itself: for a closed subgroup  $H \leq G$  the coset space  $G/H$  carries a unique smooth structure making  $G \rightarrow G/H$  a submersion, indeed a principal  $H$ -bundle, and the orbit-stabilizer theorem identifies every transitive action with such a quotient. The isotropy representation of  $H$  on the tangent space  $\mathfrak{g}/\mathfrak{h}$  then linearizes the geometry at a point, the first step toward the curvature computations of the symmetric-space theory in Part IV. The second thread returns to the loose end of chapter 3: the fundamental group. Covering maps of a connected Lie group are themselves homomorphisms, with central discrete kernel, so the connected groups sharing a fixed Lie algebra are organized by the subgroups of  $\pi_1$ , and the universal cover sits atop the tower. The running example gathers both threads at once, for the Hopf fibration  $SU(2) \rightarrow SU(2)/U(1) = S^2$  and the double cover  $SU(2) \rightarrow SO(3)$  exhibit the same group as a bundle over a sphere and as a two-sheeted cover of a rotation group.

## 4.1 Group Actions and the Quotient Manifold $G/H$

Chapter 3 turned the algebraic structure of a Lie group into differential-geometric structure: a closed subgroup is automatically an embedded submanifold (theorem 3.2.1), and a continuous homomorphism is automatically smooth (corollary 3.2.4). This chapter puts that machinery to work building manifolds. The construction that does the building is the coset space  $G/H$  of a closed subgroup, and the principle behind it is that a Lie group is a producer of symmetric spaces: whenever a group acts smoothly and transitively on a set, the set is a coset space of the group, and if the stabilizer is closed, that coset space is a manifold. Spheres, projective spaces, Grassmannians, and flag manifolds all arise this way, each as  $G/H$  for a suitable classical group  $G$  and a suitable closed subgroup  $H$ . This section builds the factory; the next two run the examples through it.

The starting point is the notion of a group acting on a manifold. Actions formalize symmetry, and the two invariants attached to a point, its orbit and its stabilizer, are the raw material from which coset spaces are assembled. The first task is to set down the vocabulary of smooth actions carefully, because the quotient construction depends on knowing that stabilizers are closed and that orbits are the fibers of a smooth map.

### Definition 4.1.1: Smooth Action, Orbit, and Stabilizer

Let  $G$  be a Lie group and  $M$  a smooth manifold. A *smooth (left) action* of  $G$  on  $M$  is a smooth map

$$\theta: G \times M \rightarrow M, \quad \theta(g, x) = g \cdot x$$

satisfying  $e \cdot x = x$  and  $g \cdot (h \cdot x) = (gh) \cdot x$  for all  $g, h \in G$  and  $x \in M$ . For each fixed  $g$  the map  $\theta_g: M \rightarrow M, x \mapsto g \cdot x$ , is a diffeomorphism with inverse  $\theta_{g^{-1}}$ . Given  $x \in M$ :

- the *orbit* of  $x$  is  $G \cdot x = \{g \cdot x \mid g \in G\} \subseteq M$ ;
- the *stabilizer* (or isotropy subgroup) of  $x$  is  $G_x = \{g \in G \mid g \cdot x = x\} \subseteq G$ .

If  $G$  also acts smoothly on a manifold  $N$ , a smooth map  $F: M \rightarrow N$  is  *$G$ -equivariant* if  $F(g \cdot x) = g \cdot F(x)$  for all  $g \in G$  and  $x \in M$ . The action is

- *effective* (or faithful) if  $g \cdot x = x$  for all  $x$  forces  $g = e$ , that is, only the identity acts trivially;
- *free* if  $g \cdot x = x$  for some  $x$  forces  $g = e$ , that is, every stabilizer is trivial;
- *transitive* if  $G \cdot x = M$  for some (equivalently every)  $x \in M$ , that is, there is a single orbit.

The two invariants pull in opposite directions. The orbit  $G \cdot x$  measures how far  $G$  can move  $x$ ; a transitive action has a single orbit and so identifies  $M$  with the shape of  $G$  as seen from  $x$ . The stabilizer  $G_x$  measures the redundancy in this identification, the group elements that fix  $x$  and are therefore invisible to it. The relation between them is the orbit-stabilizer correspondence: the elements of the orbit are in bijection with the cosets  $G/G_x$ , since  $g \cdot x = g' \cdot x$  exactly when  $g^{-1}g' \in G_x$ . This is the set-theoretic skeleton of every homogeneous space (section 4.2 upgrades it to a diffeomorphism). A free action is one with no redundancy, so the orbit is a faithful copy of  $G$ ; an effective action is the weaker condition that no single group element is invisible at every point at once. The conjugation action below is effective whenever the center is trivial but rarely free, and this

contrast is exactly why the center of a group is worth naming.

The definition is abstract, so it is worth pinning it to the actions this chapter actually uses. Four are basic, and each will reappear.

### Example 4.1.2: The Four Basic Actions

1.  $\mathrm{GL}_n(\mathbb{R})$  on  $\mathbb{R}^n$ . The map  $(A, v) \mapsto Av$  is smooth, bilinear in fact, and satisfies  $Iv = v$  and  $A(Bv) = (AB)v$ ; it is the defining representation. The orbit of 0 is  $\{0\}$ , and the orbit of any  $v \neq 0$  is all of  $\mathbb{R}^n \setminus \{0\}$ , since an invertible matrix carries  $v$  to any other nonzero vector (extend  $v$  and the target each to a basis and take the change-of-basis matrix). The action is effective: if  $Av = v$  for every  $v$  then  $A = I$ . It is not transitive on  $\mathbb{R}^n$  (the origin is its own orbit) and not free (the stabilizer of  $e_1$  contains every  $A$  fixing  $e_1$ , a copy of the affine group of  $\mathbb{R}^{n-1}$ ).
2.  $\mathrm{O}(n)$  on  $S^{n-1}$ . An orthogonal matrix preserves the Euclidean norm, so it maps the unit sphere  $S^{n-1} = \{v \in \mathbb{R}^n \mid |v| = 1\}$  to itself; the restriction  $(A, v) \mapsto Av$  is a smooth action on  $S^{n-1}$ . It is transitive: given unit vectors  $v, w$ , extend each to an orthonormal basis and let  $A$  be the change-of-basis matrix, which is orthogonal and carries  $v$  to  $w$ . The stabilizer of  $e_n \in S^{n-1}$  is the copy of  $\mathrm{O}(n-1)$  acting on the orthogonal complement  $e_n^\perp$ , so this action realizes  $S^{n-1}$  as the coset space  $\mathrm{O}(n)/\mathrm{O}(n-1)$ , the theme of section 4.3.
3.  $G$  on itself by left translation. The map  $(g, x) \mapsto gx$  is smooth because multiplication is smooth, and  $\theta_g = L_g$  is the left translation, a diffeomorphism by proposition 2.1.2. This action is transitive (any  $x$  is carried to any  $y$  by  $g = yx^{-1}$ ) and free (if  $gx = x$  then  $g = e$ ). Homogeneity of a Lie group, the fact that every point looks like every other, is precisely the transitivity of this action, and it is the reason a chart at  $e$  can be spread over the whole group.
4.  $G$  on itself by conjugation. The map  $(g, x) \mapsto gxg^{-1}$  is smooth, and  $\theta_g = C_g$  is the inner automorphism, a diffeomorphism again by proposition 2.1.2. The orbits are the conjugacy classes of  $G$ , and the stabilizer of  $x$  is its centralizer  $\{g \mid gx = xg\}$ . The action is effective on a group with trivial center but never free unless  $G = \{e\}$ , since every element fixes  $e$  and, more generally, every element of the center acts trivially. Its differential at  $e$  is the adjoint representation  $\mathrm{Ad}$  of definition 2.5.1, the infinitesimal shadow of conjugation.

The left-translation action is the quiet workhorse of the entire theory. It is transitive and free, so  $G$  is a single orbit with no redundancy, which is why the quotient by a subgroup below is so well behaved: left translation by  $G$  descends to a transitive action on  $G/H$ , and it is this residual symmetry that spreads a chart near the identity coset over the whole quotient. The conjugation action, by contrast, is the source of the adjoint representation and, further downstream, of the entire structure theory of Part II; its orbits are the conjugacy classes, and its infinitesimal version  $\mathrm{Ad}$  already appeared in Chapter 2.

With actions in hand, the central construction of the section is the quotient. Fix a closed subgroup  $H$  of  $G$  and let

$$G/H = \{gH \mid g \in G\}$$

be the set of left cosets, with the quotient topology induced by the projection  $\pi: G \rightarrow G/H$ ,  $g \mapsto gH$ . The group  $G$  acts on  $G/H$  on the left by  $a \cdot (gH) = (ag)H$ , and this action is transitive by construction. The question is whether  $G/H$  carries a smooth structure making this a smooth action and  $\pi$  a

smooth map. It does, and the closedness of  $H$  is exactly what makes it work:  $H$  is an embedded submanifold by theorem 3.2.1, and this is what allows the cosets to be separated by a smooth slice.

**Theorem 4.1.3: The Quotient Manifold Theorem**

Let  $G$  be a Lie group and  $H \subseteq G$  a closed subgroup, with Lie algebras  $\mathfrak{g}$  and  $\mathfrak{h}$ . Then the coset space  $G/H$  carries a unique smooth manifold structure for which the projection  $\pi: G \rightarrow G/H$  is a smooth submersion. With this structure:

1.  $\dim(G/H) = \dim G - \dim H$ ;
2. the left action  $G \times G/H \rightarrow G/H$ ,  $(a, gH) \mapsto (ag)H$ , is smooth and transitive;
3. the quotient topology on  $G/H$  is Hausdorff and second countable, so  $G/H$  is a smooth manifold in the usual sense.

The idea of the proof is to build a chart on  $G/H$  near the identity coset  $eH$  by choosing a slice through  $e$  transverse to  $H$ . Since  $H$  is an embedded submanifold with tangent space  $\mathfrak{h}$  at  $e$ , a complementary subspace  $\mathfrak{m}$  with  $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$  gives, via the exponential, a submanifold  $S = \exp(\text{ball in } \mathfrak{m})$  that crosses  $H$  transversally at  $e$ . The point is that  $S$  meets each coset near  $eH$  exactly once, so  $S$  is a local model for  $G/H$  and the map  $\pi|_S$  is the chart. Left translation by  $G$  then spreads this chart over the whole quotient, and the smoothness of the transition maps is where the group structure and the exponential chart do their work. The submersion and constant-rank facts about smooth maps are cited to [2]; the coset charts are built here.

**Proof:**

Fix an inner product on  $\mathfrak{g}$ , giving norms and open balls, and choose a vector-space complement  $\mathfrak{m}$  to  $\mathfrak{h}$ , so that

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$$

The strategy is to use  $\mathfrak{m}$  as the coordinate directions transverse to  $H$  and  $\pi$  as the chart map. There are five steps: build the slice, show it meets nearby cosets once, define the identity chart, spread it by translation, and verify the transitions and the submersion property.

*Step 1: the slice through  $e$ .* Consider the map

$$\Phi: \mathfrak{m} \oplus \mathfrak{h} \rightarrow G, \quad \Phi(Y, X) = \exp(Y) \exp(X)$$

Its differential at  $(0, 0)$  sends  $(Y, X)$  to  $Y + X$ , by the same product-rule computation as in theorem 3.2.1: differentiating  $\exp(sY) \exp(0)$  in  $s$  gives  $Y$ , differentiating  $\exp(0) \exp(sX)$  gives  $X$ , and the product rule at  $e$  adds them, using  $d(\exp)_0 = \text{id}$  from theorem 2.3.4. Under  $\mathfrak{m} \oplus \mathfrak{h} = \mathfrak{g}$  this differential is the identity, so the inverse function theorem ([2]) furnishes open balls  $B_{\mathfrak{m}} \subseteq \mathfrak{m}$  and  $B_{\mathfrak{h}} \subseteq \mathfrak{h}$  about the origins on which  $\Phi$  restricts to a diffeomorphism onto an open neighborhood  $\Omega$  of  $e$  in  $G$ . Set

$$S = \exp(B_{\mathfrak{m}}) \subseteq G$$

an embedded submanifold of  $G$  through  $e$  with tangent space  $\mathfrak{m}$  at  $e$ ; it is the *slice*. Because  $\Phi$  is a diffeomorphism on  $B_{\mathfrak{m}} \times B_{\mathfrak{h}}$ , every element  $g \in \Omega$  factors *uniquely* as  $g = \exp(Y) \exp(X)$  with  $Y \in B_{\mathfrak{m}}$  and  $X \in B_{\mathfrak{h}}$ ; since  $\exp(X) \in H$  for  $X \in B_{\mathfrak{h}} \subseteq \mathfrak{h}$ , this says  $g$  lies in the coset  $\exp(Y)H$  with  $Y \in B_{\mathfrak{m}}$  determined by  $g$ .

*Step 2: the slice meets nearby cosets exactly once.* The claim is that, after shrinking  $B_{\mathfrak{m}}$  if necessary, the exponentials of distinct points of  $B_{\mathfrak{m}}$  lie in distinct cosets:

$$Y_1, Y_2 \in B_{\mathfrak{m}}, \quad \exp(Y_1)H = \exp(Y_2)H \implies Y_1 = Y_2 \quad (4.1)$$

The equality of cosets means  $h := \exp(Y_2)^{-1} \exp(Y_1) \in H$ . Suppose no shrinking made (4.1) hold. Then there are sequences  $Y_1^{(n)}, Y_2^{(n)} \in \mathfrak{m}$  both tending to 0, with  $Y_1^{(n)} \neq Y_2^{(n)}$ , such that  $h_n := \exp(Y_2^{(n)})^{-1} \exp(Y_1^{(n)}) \in H$  for all  $n$ . Now  $h_n \rightarrow e$ , so for large  $n$  the element  $h_n$  lies in  $\Omega$  and factors as  $h_n = \exp(Y_n) \exp(X_n)$  with  $Y_n \in B_{\mathfrak{m}}$ ,  $X_n \in B_{\mathfrak{h}}$ , both tending to 0. But  $h_n \in H$  forces  $\exp(Y_n) = h_n \exp(X_n)^{-1} \in H$  with  $Y_n \in \mathfrak{m}$ ; the transversality established in theorem 3.2.1 (its Step 4, that  $\exp$  of a small nonzero element of  $\mathfrak{m}$  never lands in  $H$ ) then gives  $Y_n = 0$  once  $n$  is large. Hence  $h_n = \exp(X_n) \in \exp(B_{\mathfrak{h}})$  for large  $n$ . Feeding this back,  $\exp(Y_1^{(n)}) = \exp(Y_2^{(n)}) \exp(X_n)$  expresses one element of  $\Omega$  in two ways as  $\Phi(\cdot, \cdot)$ ; the uniqueness of the factorization in Step 1 forces  $Y_1^{(n)} = Y_2^{(n)}$  and  $X_n = 0$ , contradicting  $Y_1^{(n)} \neq Y_2^{(n)}$ . Therefore (4.1) holds on a small enough ball, which we fix and continue to call  $B_{\mathfrak{m}}$ .

Combining Steps 1 and 2, the restriction  $\pi|_S: S \rightarrow G/H$  is injective, and its image  $\pi(S)$  is the set of cosets  $\exp(Y)H$  with  $Y \in B_{\mathfrak{m}}$ . Moreover  $\pi(S)$  is open in  $G/H$ . By definition of the quotient topology it suffices that the saturation  $\pi^{-1}(\pi(S)) = S \cdot H = \bigcup_{h \in H} Sh$  be open in  $G$ , and it is: each  $Sh = R_h(S)$  is the image of the open-in-its-slice set under a diffeomorphism, and the union  $S \cdot H \supseteq \Omega$  contains the open neighborhood  $\Omega$  of  $e$  translated across  $H$ , so  $S \cdot H = \bigcup_{h \in H} \Omega h$  is a union of open sets.

*Step 3: the identity chart.* Define

$$\varphi_e: \pi(S) \rightarrow B_{\mathfrak{m}}, \quad \varphi_e(\exp(Y)H) = Y$$

This is well defined and injective by (4.1), and surjective onto  $B_{\mathfrak{m}}$  by construction, hence a bijection from the open set  $\pi(S)$  onto the open ball  $B_{\mathfrak{m}} \subseteq \mathfrak{m} \cong \mathbb{R}^{\dim G - \dim H}$ . It is a homeomorphism. Its inverse  $Y \mapsto \exp(Y)H = \pi(\exp Y)$  is continuous, being a composite of the continuous  $\exp$  and  $\pi$ . For continuity of  $\varphi_e$  itself, note that  $\varphi_e = \log_{\mathfrak{m}} \circ \sigma$ , where  $\sigma: \pi(S) \rightarrow S$  is the section  $\exp(Y)H \mapsto \exp(Y)$  and  $\log_{\mathfrak{m}} = (\exp|_{B_{\mathfrak{m}}})^{-1}$  is a diffeomorphism  $S \rightarrow B_{\mathfrak{m}}$ ;  $\sigma$  is continuous because  $\pi|_S$  is a continuous open bijection onto  $\pi(S)$ , hence a homeomorphism, and  $\sigma$  is its inverse followed by the inclusion  $S \hookrightarrow G$ . Thus  $\varphi_e$  is a chart on  $G/H$  around the coset  $eH$ , valued in  $\mathbb{R}^{\dim G - \dim H}$ .

*Step 4: spreading the chart by left translation.* For each  $a \in G$  let  $\tau_a: G/H \rightarrow G/H$  be the map  $gH \mapsto (ag)H$ ; it is a homeomorphism, a bijection with continuous inverse  $\tau_{a^{-1}}$ , both induced by the continuous open  $\pi$  and left translation  $L_a$  on  $G$  via  $\pi \circ L_a = \tau_a \circ \pi$ . Define charts around an arbitrary coset  $aH$  by transporting the identity chart:

$$\varphi_a: \tau_a(\pi(S)) \rightarrow B_{\mathfrak{m}}, \quad \varphi_a = \varphi_e \circ \tau_a^{-1}$$

The domains  $\tau_a(\pi(S))$  are open and cover  $G/H$ : given a coset  $bH$ , take  $a = b$ , and then  $bH = \tau_b(eH) \in \tau_b(\pi(S))$  since  $eH \in \pi(S)$ . So  $\{(\tau_a(\pi(S)), \varphi_a)\}_{a \in G}$  is an atlas into  $\mathbb{R}^{\dim G - \dim H}$ .

*Step 5: smooth transitions and the submersion property.* Two charts  $\varphi_a, \varphi_b$  with overlapping domains have transition map

$$\varphi_b \circ \varphi_a^{-1} = \varphi_e \circ (\tau_b^{-1} \circ \tau_a) \circ \varphi_e^{-1} = \varphi_e \circ \tau_c \circ \varphi_e^{-1}, \quad c = b^{-1}a$$

on its domain of definition. To see this is smooth, unwind it through  $G$ . On its domain,  $\varphi_e^{-1}(Y) = \exp(Y)H$ ; applying  $\tau_c$  gives  $(c \exp(Y))H$ ; and applying  $\varphi_e$  requires factoring  $c \exp(Y)$

into slice-times- $H$  form. Where the overlap is nonempty,  $c \exp(Y)$  lies in the region  $\Omega$  where  $\Phi$  is a diffeomorphism, so its slice component is  $P_{\mathfrak{m}}(\Phi^{-1}(c \exp(Y)))$ , the  $\mathfrak{m}$ -part of  $\Phi^{-1}$ , where  $P_{\mathfrak{m}}: \mathfrak{m} \oplus \mathfrak{h} \rightarrow \mathfrak{m}$  is the linear projection. This is a composite of the smooth maps  $Y \mapsto c \exp(Y)$  (smooth, as multiplication and  $\exp$  are smooth), the smooth  $\Phi^{-1}$ , and the linear  $P_{\mathfrak{m}}$ . Hence  $\varphi_b \circ \varphi_a^{-1}$  is smooth, and being invertible with a symmetrically smooth inverse, it is a diffeomorphism between open subsets of  $\mathbb{R}^{\dim G - \dim H}$ . The atlas is therefore smooth, and it defines a smooth manifold structure on  $G/H$  of dimension  $\dim G - \dim H$ , giving part (1).

In these charts the projection is a submersion. Read  $\pi$  near a point of  $G$  in the source chart  $\Phi$  (coordinates  $(Y, X) \in B_{\mathfrak{m}} \times B_{\mathfrak{h}}$ , after left-translating so the point is  $e$ ) and the target chart  $\varphi_e$ . The composite  $\varphi_e \circ \pi \circ \Phi$  sends  $(Y, X) \mapsto \exp(Y) \exp(X)H \mapsto Y$ , that is, it is the linear projection  $(Y, X) \mapsto Y$ . This is a smooth submersion, its differential the surjection  $\mathfrak{m} \oplus \mathfrak{h} \rightarrow \mathfrak{m}$ ; left translation carries the same local form to every point of  $G$ , so  $\pi$  is a smooth submersion everywhere, and its differential at  $e$  is the quotient map  $\mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{h} \cong \mathfrak{m}$  with kernel  $\mathfrak{h}$ .

The left action  $G \times G/H \rightarrow G/H$  is smooth: in the charts above,  $(a, gH) \mapsto (ag)H$  reads, after the same local factoring, as a composite of the smooth multiplication on  $G$  and the smooth chart maps, exactly as the transition maps were shown smooth. It is transitive by construction, since  $(gx^{-1}) \cdot (xH) = gH$  for any  $x, g$ . This is part (2).

For part (3), the quotient is Hausdorff and second countable. The projection  $\pi$  is an open continuous surjection: for  $U \subseteq G$  open,  $\pi^{-1}(\pi(U)) = UH = \bigcup_{h \in H} Uh$  is a union of open sets, hence open, so  $\pi(U)$  is open. Second countability then passes from  $G$  to its continuous open image  $G/H$ . For Hausdorffness the closedness of  $H$  is used. The set

$$R = \{(g, g') \in G \times G \mid g^{-1}g' \in H\}$$

is the preimage of the closed set  $H$  under the continuous map  $(g, g') \mapsto g^{-1}g'$ , hence closed in  $G \times G$ ;  $R$  is the graph of the coset equivalence relation, and the quotient of a topological group by a closed subgroup, with its open projection, is Hausdorff precisely when this relation is closed ([5]). Concretely, if  $gH \neq g'H$  then  $g^{-1}g' \notin H$ ; as  $H$  is closed there is a symmetric neighborhood  $W$  of  $e$  with  $Wg^{-1}g'W \cap H = \emptyset$ , and then  $\pi(gW)$  and  $\pi(g'W)$  are disjoint open neighborhoods separating  $gH$  and  $g'H$ . Thus  $G/H$  is a Hausdorff, second-countable smooth manifold.

*Uniqueness.* Suppose  $G/H$  carries a second smooth structure for which  $\pi$  is a smooth submersion. A submersion admits local smooth sections ([2]); using such a section  $s$  for the first structure, the identity map from the first structure to the second equals  $(\pi$ -descent of)  $\pi \circ s$  locally and is smooth, and symmetrically its inverse is smooth, so the identity is a diffeomorphism between the two structures, which therefore coincide. (This uniqueness is restated cleanly through the universal property in proposition 4.1.5.) This completes the proof.

The theorem is the manifold factory promised at the start. It converts a purely algebraic datum, a closed subgroup  $H \leq G$ , into a manifold  $G/H$  that comes with a transitive  $G$ -action already installed. The dimension count  $\dim G - \dim H$  is the tangible payoff: to build a manifold of a prescribed dimension, one finds a group  $G$  and a closed subgroup  $H$  whose dimensions differ by the target, arranges transitivity, and reads off the space. The slice  $S = \exp(B_{\mathfrak{m}})$  is the geometric heart of the construction, a submanifold transverse to  $H$  that meets each nearby coset once, so that the  $\mathfrak{m}$ -directions become coordinates on the quotient while the  $\mathfrak{h}$ -directions are quotiented away. Closedness of  $H$  enters in two places, both indispensable: it makes  $H$  an embedded submanifold, which is what gives the transversality of Step 2, and it makes the relation  $R$  closed, which is what makes the quotient Hausdorff. Drop closedness, as in the irrational winding line on the torus

(example 3.1.5), and the quotient is non-Hausdorff with no manifold structure.

A concrete instance is worth carrying through immediately, before the systematic examples of the next two sections. The sphere is the first coset manifold, and it exhibits every ingredient of the theorem.

**Example 4.1.4: The Sphere  $S^{n-1} = O(n)/O(n-1)$**

Take  $G = O(n)$  acting on  $S^{n-1} \subseteq \mathbb{R}^n$  by  $(A, v) \mapsto Av$ , as in example 4.1.2. The action is transitive, and the stabilizer of the north pole  $e_n = (0, \dots, 0, 1)$  is

$$O(n)_{e_n} = \{A \in O(n) \mid Ae_n = e_n\} = \left\{ \begin{pmatrix} B & 0 \\ 0 & 1 \end{pmatrix} \mid B \in O(n-1) \right\} \cong O(n-1)$$

since an orthogonal matrix fixing  $e_n$  must preserve its orthogonal complement  $e_n^\perp = \mathbb{R}^{n-1}$ , on which it acts by an arbitrary element of  $O(n-1)$ . This stabilizer is closed in  $O(n)$ , being the intersection of  $O(n)$  with the closed condition  $Ae_n = e_n$ , so theorem 4.1.3 makes  $O(n)/O(n-1)$  a smooth manifold with

$$\dim(O(n)/O(n-1)) = \binom{n}{2} - \binom{n-1}{2} = n-1$$

using  $\dim O(k) = \binom{k}{2}$  from section 1.2. The dimension matches  $\dim S^{n-1} = n-1$ , exactly as it must: the orbit map  $A \mapsto Ae_n$  is the  $G$ -equivariant diffeomorphism  $O(n)/O(n-1) \xrightarrow{\sim} S^{n-1}$  that section 4.2 will establish in general. The projection  $\pi: O(n) \rightarrow S^{n-1}$  is  $A \mapsto Ae_n$ , the map reading off the last column, and it is a submersion: its fibers are the cosets  $A O(n-1)$ , and its slice at  $e_n$  is a small patch of directions transverse to the stabilizer.

The example is the pattern every homogeneous space will follow: a transitive action, a closed stabilizer, and a dimension count that confirms the coset space is the space one started with. Here  $S^{n-1}$  was already a manifold, so the coset presentation is a new description rather than a new object, but the same machine, applied to abstract cosets, manufactures Grassmannians and flag manifolds that have no prior definition.

What makes the smooth structure on  $G/H$  canonical, rather than one choice among many, is a universal property. The projection  $\pi$  is a submersion, and submersions are characterized by how smooth maps factor through them: a map out of  $G/H$  is smooth exactly when its pullback along  $\pi$  is smooth. This is the working criterion for smoothness on a quotient, and it is used constantly, since a map defined on cosets is almost always specified by a formula in  $g$  that happens to be constant on  $H$ -cosets.

**Proposition 4.1.5: Universal Property of the Quotient**

Let  $H \subseteq G$  be a closed subgroup and  $\pi: G \rightarrow G/H$  the quotient projection of theorem 4.1.3. Then:

1. for any smooth manifold  $N$ , a map  $f: G/H \rightarrow N$  is smooth if and only if  $f \circ \pi: G \rightarrow N$  is smooth;
2. consequently the smooth structure of theorem 4.1.3 is the unique one making  $\pi$  a submersion, and  $G/H$  is determined up to a unique  $G$ -equivariant diffeomorphism by this property;
3.  $\pi$  admits local smooth sections: every point of  $G/H$  has an open neighborhood  $V$  and a smooth map  $s: V \rightarrow G$  with  $\pi \circ s = \text{id}_V$ .

**Proof:**

1. If  $f$  is smooth then  $f \circ \pi$  is smooth, a composite of smooth maps. Conversely suppose  $f \circ \pi$  is smooth. Smoothness of  $f$  is local, so fix a coset  $gH$ . By part (3), proved independently below, there is an open neighborhood  $V$  of  $gH$  and a smooth section  $s: V \rightarrow G$  with  $\pi \circ s = \text{id}_V$ . Then on  $V$ ,

$$f = f \circ \text{id}_V = f \circ (\pi \circ s) = (f \circ \pi) \circ s$$

is a composite of the smooth map  $f \circ \pi$  with the smooth section  $s$ , hence smooth on  $V$ . As  $gH$  was arbitrary,  $f$  is smooth.

2. Any smooth structure on  $G/H$  making  $\pi$  a submersion satisfies the factoring property of part (1), by the same argument, which used only that  $\pi$  is a submersion (to obtain local sections). Given two such structures, apply part (1) to  $f = \text{id}$  viewed from the first structure to the second:  $\text{id} \circ \pi = \pi$  is smooth into the target structure, so  $\text{id}$  is smooth from the first structure to the second, and symmetrically its inverse is smooth, so  $\text{id}$  is a diffeomorphism and the structures coincide. The diffeomorphism intertwining any two realizations of  $G/H$  is  $G$ -equivariant because  $\pi$  is  $G$ -equivariant, and it is unique because a  $G$ -equivariant map is determined by its value at the base point  $eH$ , whose orbit is all of  $G/H$ . Hence  $G/H$  is canonical up to a unique  $G$ -equivariant diffeomorphism.
3. The section is read off the slice of theorem 4.1.3. At the identity coset, the chart  $\varphi_e: \pi(S) \rightarrow B_m$  has continuous inverse  $\varphi_e^{-1}(Y) = \exp(Y)H$ , and the map

$$s_e: \pi(S) \rightarrow G, \quad s_e(\exp(Y)H) = \exp(Y)$$

is smooth (it equals  $\exp \circ \varphi_e$ , a composite of smooth maps) and satisfies  $\pi \circ s_e = \text{id}_{\pi(S)}$ , since  $\pi(\exp(Y)) = \exp(Y)H$ . This is a smooth section over the neighborhood  $\pi(S)$  of  $eH$ . For an arbitrary coset  $aH$ , transport by left translation:  $s_a = L_a \circ s_e \circ \tau_{a^{-1}}$  is a smooth section over the neighborhood  $\tau_a(\pi(S))$  of  $aH$ , using that  $L_a: G \rightarrow G$  is a diffeomorphism (proposition 2.1.2) and  $\pi \circ L_a = \tau_a \circ \pi$ , so  $\pi \circ s_a = \tau_a \circ \pi \circ s_e \circ \tau_{a^{-1}} = \tau_a \circ \tau_{a^{-1}} = \text{id}$ . Every point of  $G/H$  thus has a local smooth section, as we sought to show.

The universal property is the practical tool. It says  $G/H$  is the best quotient of  $G$  by the equivalence relation of  $H$ -cosets: smoothness downward is free, and smoothness upward is the criterion one checks. In practice a map on a coset space is defined by an  $H$ -invariant formula on  $G$ , a function of  $g$  that does not change when  $g$  is replaced by  $gh$ , and the proposition certifies that such a formula defines a smooth map on  $G/H$  exactly when it is smooth on  $G$ . Uniqueness up to canonical diffeomorphism means the notation  $G/H$  is unambiguous: any two constructions of a manifold with a transitive  $G$ -action and stabilizer  $H$  are the same, so one may speak of the homogeneous space  $G/H$ . Local sections are the recurring technical device, the smooth choice of coset representative that trivializes  $\pi$  over a small open set. They are precisely what can fail to exist globally, and that failure is the twisting that makes  $G \rightarrow G/H$  a nontrivial bundle in section 4.5.

The construction has converted a closed subgroup into a manifold carrying a built-in transitive  $G$ -action, and this is the mechanism behind every example of section 4.3. Two structural facts round out the picture, both consequences of the theorem worth isolating. When  $H$  is normal, the quotient is not just a manifold but a Lie group, and  $\pi$  is a homomorphism: the group operations on  $G/H$  are smooth because they pull back to the smooth operations on  $G$ , and the universal property upgrades this to smoothness on the quotient. And the tangent space of  $G/H$  at the identity coset is the quotient vector space  $\mathfrak{g}/\mathfrak{h}$ , since  $d\pi_e$  is the surjection  $\mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{h}$  with kernel  $\mathfrak{h}$ ; this identification is the seed of the isotropy representation in section 4.5, where  $H$  acts on  $\mathfrak{g}/\mathfrak{h}$  and the geometry of  $G/H$  near  $eH$  is read from that action. Both facts appear as exercises below.

### Exercise 4.1.1

1. Let  $G$  be a Lie group and  $G^0$  its identity component (definition 2.1.6), a closed normal subgroup. Show that  $G/G^0$  is a 0-dimensional manifold, that is, a discrete group, and identify it with the component group of  $G$ . More generally, prove that if a closed subgroup  $H \leq G$  is open, then  $G/H$  is discrete.
2. Let  $H$  be a *closed normal* subgroup of a Lie group  $G$ . Prove that the quotient group  $G/H$ , with the smooth structure of theorem 4.1.3, is a Lie group and that  $\pi: G \rightarrow G/H$  is a homomorphism of Lie groups. (Use proposition 4.1.5 to check that multiplication and inversion on  $G/H$  are smooth, and corollary 3.2.4 to reduce smoothness of  $\pi$  to its continuity.)
3. With  $G$ ,  $H$ ,  $\mathfrak{g}$ ,  $\mathfrak{h}$  as in theorem 4.1.3, prove that the differential  $d\pi_e: \mathfrak{g} \rightarrow T_{eH}(G/H)$  is surjective with kernel exactly  $\mathfrak{h}$ , and conclude that there is a canonical linear isomorphism  $T_{eH}(G/H) \cong \mathfrak{g}/\mathfrak{h}$ . Verify the dimension count of theorem 4.1.3 against this.
4. Show that  $\mathrm{GL}_n(\mathbb{R})$  acts smoothly and transitively on the set  $\mathcal{B}$  of ordered bases of  $\mathbb{R}^n$  by  $A \cdot (v_1, \dots, v_n) = (Av_1, \dots, Av_n)$ , and compute the stabilizer of the standard basis. Conclude that  $\mathcal{B}$ , given the coset-space structure, is diffeomorphic to  $\mathrm{GL}_n(\mathbb{R})$  itself, and explain why the stabilizer being trivial forces this.
5. Let  $H = \{(e^{2\pi i t}, e^{2\pi i \alpha t}) \mid t \in \mathbb{R}\} \subseteq \mathbb{T}^2$  be the image of the irrational-slope one-parameter subgroup with  $\alpha$  irrational (example 3.1.5). Explain why the quotient  $\mathbb{T}^2/H$  is not a Hausdorff manifold, identifying the step of theorem 4.1.3 that fails and the hypothesis responsible.
6. Give an example of a coset space  $G/H$  for which  $\pi: G \rightarrow G/H$  admits no global smooth section  $s: G/H \rightarrow G$  with  $\pi \circ s = \mathrm{id}$ , and explain how this is consistent with proposition 4.1.5, which guarantees only *local* sections. (Consider  $\mathrm{SO}(3) \rightarrow S^2$  and what a global section would say about the tangent bundle of  $S^2$ .)

## 4.2 Transitive Actions and Homogeneous Spaces

The quotient construction of section 4.1 started from a subgroup and produced a manifold  $G/H$ . This section reverses the arrow. Many of the manifolds one meets in geometry, the spheres and projective spaces and Grassmannians of the next section, arrive already carrying an action of a Lie group, and the action has a feature that makes the manifold rigid: it is transitive. Every point can be moved to every other by the group, so there is no intrinsic way to tell two points apart, and the manifold looks the same seen from anywhere. A space with this degree of symmetry is called homogeneous, and the central result of the section is that a homogeneous space is nothing other than a coset space  $G/H$  in disguise. The disguise is stripped by a single computation: fix a point, take its stabilizer  $H$ , and the orbit map  $gH \mapsto g \cdot x$  is a diffeomorphism  $G/H \rightarrow M$ .

This is the observation that turns the abstract quotient of the previous section into a practical tool. To recognize a given manifold  $M$  as a coset space, one does not construct charts or verify a submersion by hand. It suffices to exhibit a Lie group acting transitively on  $M$  and to compute the stabilizer of one point. The orbit-stabilizer diffeomorphism then supplies the identification  $M \cong G/H$  for free, together with the smooth structure and the group action. The section develops this identification, first for transitive actions, where it is a clean diffeomorphism, and then for general actions, where each individual orbit is a homogeneous space for  $G$  and sits inside  $M$  as an immersed submanifold.

### Definition 4.2.1: Homogeneous Space

A smooth action of a Lie group  $G$  on a smooth manifold  $M$  (definition 4.1.1) is *transitive* if for every pair of points  $x, y \in M$  there is an element  $g \in G$  with  $g \cdot x = y$ ; equivalently,  $M$  consists of a single orbit. A smooth manifold  $M$  equipped with a transitive smooth action of  $G$  is a *homogeneous  $G$ -space*, or simply a *homogeneous space* when the group is understood.

Transitivity is a strong hypothesis with an immediate consequence for the shape of  $M$ : because  $G$  moves any point to any other by a diffeomorphism (the action of a fixed  $g$  is a diffeomorphism of  $M$  with inverse the action of  $g^{-1}$ ), every local or infinitesimal feature at one point is carried to every other. A homogeneous space has no distinguished point, no special locus, no place where it is more curved or more singular than elsewhere. This uniformity is what makes such spaces the natural setting for the geometry of symmetry, and it is the source of the eventual identification with a coset space, where the distinguished coset  $eH$  is a choice, not an intrinsic feature.

The prototypical homogeneous space is the coset space itself. Before the general theorem, it is worth seeing that  $G/H$  is homogeneous under the action that defines it, since this is the target of the identification to come.

### Example 4.2.2: The Coset Space $G/H$

Let  $H$  be a closed subgroup of a Lie group  $G$ , so that  $G/H$  is a smooth manifold by theorem 4.1.3. The group  $G$  acts on  $G/H$  by left translation,

$$g \cdot (g'H) = (gg')H$$

which is well defined because left multiplication by  $g$  carries the coset  $g'H$  to  $(gg')H$  regardless of the representative  $g'$ , and which is smooth because it is induced from the smooth left multiplication  $G \times G \rightarrow G$  through the submersion  $\pi: G \rightarrow G/H$  (proposition 4.1.5). The action is transitive: given two cosets  $g'H$  and  $g''H$ , the element  $g = g''(g')^{-1}$  satisfies  $g \cdot (g'H) = g''H$ . Thus  $G/H$  is a homogeneous  $G$ -space. The stabilizer of the base coset  $eH$  is exactly  $H$ , since  $g \cdot eH = eH$

means  $gH = H$ , that is  $g \in H$ . This example is the model to which the next theorem reduces every homogeneous space.

The theorem is the converse of the example. Any homogeneous space, however it arises, is diffeomorphic to a coset space of the acting group, and the diffeomorphism respects the action. The mechanism is the orbit map through a chosen point: it factors through the coset space of the stabilizer and lands in the manifold as an equivariant bijection, which the manifold structure then forces to be a diffeomorphism.

### Theorem 4.2.3: Smooth Orbit-Stabilizer

Let  $G$  be a Lie group acting smoothly and transitively on a smooth manifold  $M$ , and fix a point  $x \in M$ . Let  $H = G_x = \{g \in G \mid g \cdot x = x\}$  be the stabilizer of  $x$ . Then  $H$  is a closed subgroup of  $G$ , so that  $G/H$  is a smooth manifold, and the orbit map

$$\Phi: G/H \longrightarrow M, \quad \Phi(gH) = g \cdot x$$

is a well-defined  $G$ -equivariant diffeomorphism. In particular  $\dim M = \dim G - \dim H$ .

#### Proof:

*Step 1: the stabilizer is closed, so  $G/H$  is a manifold.* The orbit map of the point  $x$ , namely  $\theta_x: G \rightarrow M$  with  $\theta_x(g) = g \cdot x$ , is smooth, being the composite of  $g \mapsto (g, x)$  with the smooth action  $G \times M \rightarrow M$  (definition 4.1.1); in particular it is continuous. The stabilizer  $H = \theta_x^{-1}(\{x\})$  is the preimage of the one-point set  $\{x\}$ , which is closed because  $M$  is Hausdorff, so  $H$  is a closed subset of  $G$ . It is a subgroup:  $e \cdot x = x$ , and if  $g \cdot x = x$  and  $g' \cdot x = x$  then  $(gg') \cdot x = g \cdot (g' \cdot x) = g \cdot x = x$  and  $g^{-1} \cdot x = g^{-1} \cdot (g \cdot x) = (g^{-1}g) \cdot x = x$ . Being a closed subgroup,  $H$  is an embedded Lie subgroup by theorem 3.2.1, and theorem 4.1.3 then endows  $G/H$  with its smooth manifold structure, with  $\pi: G \rightarrow G/H$  a smooth submersion.

*Step 2: the orbit map descends and is a smooth equivariant bijection.* The map  $\theta_x: G \rightarrow M$  is constant on the cosets of  $H$ : if  $g' = gh$  with  $h \in H$ , then  $\theta_x(gh) = (gh) \cdot x = g \cdot (h \cdot x) = g \cdot x = \theta_x(g)$ . Hence  $\theta_x$  factors through the quotient, inducing a well-defined map  $\Phi: G/H \rightarrow M$  with  $\Phi(gH) = g \cdot x$  and  $\Phi \circ \pi = \theta_x$ . Because  $\theta_x$  is smooth and  $\pi$  is a surjective submersion,  $\Phi$  is smooth by the universal property of the quotient (proposition 4.1.5). It is  $G$ -equivariant: for  $g_0 \in G$ ,

$$\Phi(g_0 \cdot (gH)) = \Phi((g_0g)H) = (g_0g) \cdot x = g_0 \cdot (g \cdot x) = g_0 \cdot \Phi(gH)$$

It is surjective because the action is transitive: every  $y \in M$  equals  $g \cdot x = \Phi(gH)$  for some  $g$ . It is injective because  $\Phi(gH) = \Phi(g'H)$  means  $g \cdot x = g' \cdot x$ , whence  $(g^{-1}g') \cdot x = x$ , so  $g^{-1}g' \in H$  and  $gH = g'H$ . Thus  $\Phi$  is a smooth equivariant bijection.

*Step 3: a smooth equivariant bijection of homogeneous spaces has constant rank.* It remains to promote the bijection  $\Phi$  to a diffeomorphism, and the obstruction is that a smooth bijection need not have smooth inverse (the map  $t \mapsto t^3$  on  $\mathbb{R}$  is the standard warning). Equivariance removes the obstruction by making the rank of  $\Phi$  the same at every point. Write  $\ell_g: G/H \rightarrow G/H$  for the diffeomorphism  $g'H \mapsto (gg')H$  given by the action of  $g$  on  $G/H$ , and  $L_g: M \rightarrow M$  for the diffeomorphism  $y \mapsto g \cdot y$  given by the action of  $g$  on  $M$ . Equivariance is the identity  $\Phi \circ \ell_g = L_g \circ \Phi$ . Differentiating at an arbitrary point  $g'H$  and using the chain rule,

$$d\Phi_{(gg')H} \circ (d\ell_g)_{g'H} = (dL_g)_{\Phi(g'H)} \circ d\Phi_{g'H}$$

Both  $(d\ell_g)_{g'H}$  and  $(dL_g)_{\Phi(g'H)}$  are linear isomorphisms, being differentials of diffeomorphisms, so  $d\Phi_{(gg')H}$  and  $d\Phi_{g'H}$  have the same rank. Since  $G$  acts transitively on  $G/H$  (every coset is  $\ell_g(eH)$  for a suitable  $g$ ), the rank of  $d\Phi$  is the same at every point of  $G/H$ . Thus  $\Phi$  is a smooth map of constant rank.

*Step 4: constant rank plus bijectivity forces a diffeomorphism.* A smooth map of constant rank  $r$  that is injective is an immersion, and one that is surjective is a submersion; this is the global rank theorem ([2]). The bijection  $\Phi$  is both injective and surjective, so it is simultaneously an immersion and a submersion, which forces

$$r = \dim(G/H) = \dim M$$

so that  $d\Phi$  is an isomorphism at every point. A bijective local diffeomorphism is a diffeomorphism, so  $\Phi$  is a diffeomorphism. The dimension count follows from  $\dim(G/H) = \dim G - \dim H$  (theorem 4.1.3), giving  $\dim M = \dim G - \dim H$ , completing the proof.

The theorem is the master tool for the examples of the next section, and its practical content is worth isolating. To recognize a manifold  $M$  as a coset space  $G/H$ , one need only produce a Lie group  $G$  acting smoothly and transitively on  $M$  and compute the stabilizer  $H$  of a single point. The theorem then delivers the diffeomorphism  $M \cong G/H$  with no further work, and in particular the dimension identity  $\dim M = \dim G - \dim H$  becomes a consistency check: if the arithmetic fails, the proposed action cannot be transitive. The choice of base point  $x$  is immaterial up to isomorphism, since different base points give conjugate stabilizers, a point the exercises make precise. What the theorem does not require is any preexisting knowledge of the smooth structure of  $M$  beyond the manifold axioms; the structure is matched automatically to that of  $G/H$ .

The archetype is the action of the rotation group on the sphere, which the next section develops in full. A first instance is the running example, where the identification of  $\mathrm{SU}(2)$  with the three-sphere lets one read off a homogeneous-space presentation of the two-sphere.

#### Example 4.2.4: $\mathrm{SU}(2)$ Acting on $S^2$

The group  $\mathrm{SU}(2)$  acts on the two-sphere  $S^2 \subseteq \mathbb{R}^3$  through the double cover  $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$  of theorem 1.5.5: an element  $g \in \mathrm{SU}(2)$  maps to a rotation  $R(g) \in \mathrm{SO}(3)$ , which rotates  $S^2$ . The action is transitive because  $\mathrm{SO}(3)$  acts transitively on  $S^2$  (any unit vector can be rotated to any other), and  $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$  is surjective. Fix the north pole  $x = (0, 0, 1)$ . Its stabilizer in  $\mathrm{SO}(3)$  is the group of rotations about the vertical axis, a circle  $\mathrm{SO}(2)$ ; its stabilizer in  $\mathrm{SU}(2)$  is the preimage of this circle, which is the diagonal torus

$$\mathrm{U}(1) = \left\{ \begin{pmatrix} e^{i\vartheta} & 0 \\ 0 & e^{-i\vartheta} \end{pmatrix} \mid \vartheta \in \mathbb{R} \right\}$$

a closed subgroup isomorphic to the circle. By theorem 4.2.3,

$$S^2 \cong \mathrm{SU}(2)/\mathrm{U}(1)$$

as homogeneous spaces, and the dimension arithmetic checks:  $\dim \mathrm{SU}(2) - \dim \mathrm{U}(1) = 3 - 1 = 2 = \dim S^2$ . Since  $\mathrm{SU}(2)$  is diffeomorphic to  $S^3$  (proposition 1.5.1), the quotient map  $\mathrm{SU}(2) \rightarrow \mathrm{SU}(2)/\mathrm{U}(1)$  is a smooth map  $S^3 \rightarrow S^2$  with circle fibers, the Hopf fibration; the next section returns to it as example 4.3.3.

The example is a template. The manifold  $S^2$  was presented as a quotient without ever writing down a chart on it, purely by exhibiting the transitive action of  $SU(2)$  and identifying one stabilizer. Every homogeneous-space identification in the next section follows this pattern, and the dimension count  $\dim G - \dim H = \dim M$  is the routine sanity check that the action is transitive.

The orbit-stabilizer theorem assumed transitivity. Dropping that hypothesis, a smooth action need not sweep out all of  $M$  from a single point; instead  $M$  decomposes into orbits, and each orbit is a homogeneous space in its own right. The question is how an individual orbit sits inside  $M$ . The theorem's argument applies verbatim to the action restricted to one orbit, identifying the orbit with a coset space, but the orbit carries the subspace topology from  $M$ , which may or may not agree with its intrinsic manifold topology. This is exactly the immersed-versus-embedded distinction of section 3.1, and the outcome is that orbits are always immersed submanifolds and are embedded under a mild closedness condition.

#### Proposition 4.2.5: Orbits as Homogeneous Submanifolds

Let  $G$  be a Lie group acting smoothly on a smooth manifold  $M$ , and let  $x \in M$ . Then the stabilizer  $G_x$  is a closed subgroup of  $G$ , and the orbit map

$$\Phi_x: G/G_x \longrightarrow M, \quad gG_x \mapsto g \cdot x$$

is an injective smooth immersion whose image is the orbit  $G \cdot x$ . Thus  $G \cdot x$  is an injectively immersed submanifold of  $M$ , diffeomorphic as a homogeneous  $G$ -space to  $G/G_x$ , and its dimension is  $\dim G - \dim G_x$ . The orbit is an embedded submanifold whenever it is locally closed in  $M$ ; in particular, every orbit of a compact Lie group is embedded.

#### Proof:

*Step 1: the orbit is a homogeneous space and  $\Phi_x$  is an injective immersion.* That  $G_x$  is a closed subgroup is the argument of Step 1 of theorem 4.2.3, which used only smoothness of the action, not transitivity:  $G_x = \theta_x^{-1}(\{x\})$  is closed, and it is a subgroup, so it is an embedded Lie subgroup by theorem 3.2.1 and  $G/G_x$  is a manifold. The map  $\Phi_x: G/G_x \rightarrow M$  is smooth, well defined, injective, and  $G$ -equivariant by Steps 1 and 2 of theorem 4.2.3, whose verifications used only smoothness of the action, never transitivity of  $G$  on all of  $M$ ; the same descent through the submersion  $\pi: G \rightarrow G/G_x$  and the same injectivity computation apply. By the constant-rank argument of Step 3 there, equivariance makes  $d\Phi_x$  of constant rank, and injectivity of  $\Phi_x$  forces that constant rank to equal  $\dim(G/G_x)$ , so  $\Phi_x$  is an injective immersion. Its image is the set of all  $g \cdot x$ , which is precisely the orbit  $G \cdot x$ . Thus  $G \cdot x$  is an injectively immersed submanifold, carrying via  $\Phi_x$  the manifold structure of  $G/G_x$ , and  $\dim(G \cdot x) = \dim(G/G_x) = \dim G - \dim G_x$ .

*Step 2: local closedness upgrades the immersion to an embedding.* An injective immersion is an embedding precisely when it is a homeomorphism onto its image with the subspace topology, and for an orbit this holds exactly when the orbit is locally closed, that is open in its closure. Suppose  $G \cdot x$  is locally closed. The homogeneity of the orbit makes the embedding condition need checking at only one point: the transitive  $G$ -action carries a neighborhood of  $x$  in  $G \cdot x$  to a neighborhood of any other point, so if the immersion is a homeomorphism onto its image near  $x$  it is so everywhere. Local closedness at  $x$  provides an open set  $U \subseteq M$  with  $G \cdot x \cap U$  closed in  $U$ ; a standard Baire-category argument on the second-countable group  $G$  ([2]) then shows the injective immersion  $\Phi_x$  is a homeomorphism onto  $G \cdot x$  with the subspace topology, hence an embedding.

*Step 3: orbits of a compact group are embedded.* Suppose  $G$  is compact. The orbit map  $\theta_x: G \rightarrow M$  is continuous and  $G$  is compact, so the orbit  $G \cdot x = \theta_x(G)$  is a compact subset of the Hausdorff space  $M$ , hence closed; a closed set is in particular locally closed, so by Step 2 the orbit is embedded. The conclusion is also visible directly:  $\Phi_x: G/G_x \rightarrow M$  is a continuous injection from the compact space  $G/G_x$  (a continuous image of the compact  $G$ ) into the Hausdorff space  $M$ , and a continuous injection from a compact space to a Hausdorff space is a homeomorphism onto its image; combined with the immersion property of Step 1,  $\Phi_x$  is an embedding. Therefore every orbit of a compact Lie group is an embedded submanifold, completing the proof.

The proposition records the general shape of an action: the manifold  $M$  is partitioned into orbits, each a homogeneous space  $G/G_x$  immersed inside  $M$ , with the stabilizer varying from orbit to orbit. The immersion is always injective, so each orbit is an image of a coset space, but the way it sits in  $M$  can be complicated, exactly as an immersed Lie subgroup can wind densely rather than embed. The irrational winding of example 3.1.5 is the paradigm: the additive group  $\mathbb{R}$  acting on the torus  $\mathbb{T}^2$  by an irrational-slope flow has a single orbit that is dense, an injectively immersed but non-embedded copy of  $\mathbb{R}$ , and the orbit fails to be locally closed. The compact case is clean because a continuous image of a compact group is compact, hence closed, hence locally closed, and the pathology cannot occur.

The two results together say that transitivity is what turns an immersed orbit into a diffeomorphism onto the whole manifold. When  $G$  acts transitively there is one orbit, it is all of  $M$ , and the immersion of proposition 4.2.5 becomes the diffeomorphism of theorem 4.2.3. The orbit-stabilizer diffeomorphism is thus the transitive special case where the immersed orbit fills the manifold, and it is this special case that the next section exploits: each of the spheres, projective spaces, Grassmannians, and flag manifolds is exhibited as a single orbit of a classical group acting transitively, with the manifold recovered as  $G/H$  by computing one stabilizer.

### Exercise 4.2.1

1. Verify directly, without invoking theorem 4.2.3, that the left-translation action of  $G$  on the coset space  $G/H$  (example 4.2.2) is transitive and that the stabilizer of the base coset  $eH$  is exactly  $H$ . Conclude that under the identification of theorem 4.2.3 a homogeneous space is recovered as the coset space of the stabilizer of any chosen point.
2. Let  $G$  act transitively and smoothly on  $M$ , and let  $x, y \in M$ . Prove that the stabilizers  $G_x$  and  $G_y$  are conjugate subgroups of  $G$ : if  $y = g \cdot x$  then  $G_y = g G_x g^{-1}$ . Deduce that the diffeomorphism type of the presentation  $M \cong G/G_x$  does not depend on the base point  $x$ , and conversely that if  $H, H' \leq G$  are closed subgroups then  $G/H$  and  $G/H'$  are  $G$ -equivariantly diffeomorphic if and only if  $H$  and  $H'$  are conjugate in  $G$ .
3. Show that the standard action of  $O(n)$  on  $\mathbb{R}^n$  restricts to a transitive action on the unit sphere  $S^{n-1}$ , compute the stabilizer of the point  $e_n = (0, \dots, 0, 1)$ , and conclude from theorem 4.2.3 that  $S^{n-1} \cong O(n)/O(n-1)$ . Check the dimension count.
4. Let the additive group  $\mathbb{R}$  act on the torus  $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$  by  $t \cdot (\bar{u}, \bar{v}) = (\overline{u+t}, \overline{v+\alpha t})$ , with  $\alpha$  irrational. Show that every orbit is an injectively immersed copy of  $\mathbb{R}$  that is dense in  $\mathbb{T}^2$ , hence not embedded, and identify precisely which hypothesis of the embeddedness conclusion of proposition 4.2.5 fails. Explain why replacing  $\mathbb{R}$  by the compact group  $S^1$  (available when  $\alpha$  is rational) makes every orbit embedded.
5. Let  $G$  be a connected Lie group acting transitively on a manifold  $M$ . Prove that  $M$  is

connected. Then, using  $\dim M = \dim G - \dim G_x$ , show that if a compact connected Lie group  $G$  acts transitively on  $M$  with  $\dim M = \dim G$  and trivial stabilizer  $G_x = \{e\}$ , then the action is free and  $G \rightarrow M$  is a diffeomorphism, so  $M \cong G$  as a homogeneous space.

6. Suppose a Lie group  $G$  acts smoothly and transitively on  $M$ . Prove that the orbit map  $\theta_x: G \rightarrow M$  is a surjective submersion, and use this to give a second proof that  $\dim M \leq \dim G$ , with equality if and only if the stabilizer  $G_x$  is discrete.

### 4.3 Examples: Spheres, Projective Spaces, Grassmannians, and Flag Manifolds

The two preceding sections built the machine. Theorem 4.1.3 manufactures a smooth manifold  $G/H$  from a Lie group  $G$  and a closed subgroup  $H$ , and theorem 4.2.3 identifies any manifold  $M$  carrying a transitive  $G$ -action with such a quotient, through the orbit map  $gH \mapsto g \cdot x$ . This section runs the machine on the manifolds a geometer meets first. Each of the classical spaces, the spheres, the projective spaces, the Grassmannians of  $k$ -planes, and the flag manifolds, turns out to be a single orbit of a classical group, and reading off the stabilizer of a base point presents it as a coset space  $G/H$  with the symmetry group displayed in the notation itself.

The gain is not just uniform notation. A presentation  $M = G/H$  makes the full group  $G$  act on  $M$  by isometries of whatever  $G$ -invariant geometry  $M$  carries, and it reduces questions about  $M$ , its dimension, its tangent space at a point, its cohomology, its invariant metrics, to questions about the pair  $(\mathfrak{g}, \mathfrak{h})$  of Lie algebras. Every computation below follows the same three steps demanded by theorem 4.2.3: exhibit a transitive action, compute the stabilizer of a base point, and confirm the dimension arithmetic  $\dim M = \dim G - \dim H$  against the manifold dimensions of Chapter 1. The running example  $SU(2)$  reappears at the end as the smallest nontrivial instance, where the coset presentation is exactly the Hopf fibration.

The most basic homogeneous space is the sphere. The orthogonal group acts on  $\mathbb{R}^n$  preserving lengths, so it permutes the vectors of any fixed length; the transitivity of this action on the unit sphere is the statement that any unit vector can be rotated to any other, which is the content of the Gram-Schmidt process. Fixing the north pole and reading off which orthogonal transformations leave it alone presents the sphere as a coset space of  $O(n)$ , and its determinant-one subgroup gives a second presentation.

#### Example 4.3.1: Spheres as Quotients of Orthogonal and Unitary Groups

The unit sphere  $S^{n-1} = \{x \in \mathbb{R}^n \mid \|x\| = 1\}$  is a homogeneous space for both the orthogonal group and the rotation group,

$$S^{n-1} \cong O(n)/O(n-1) \cong SO(n)/SO(n-1)$$

and the unit sphere  $S^{2n-1} = \{z \in \mathbb{C}^n \mid \|z\| = 1\}$  in  $\mathbb{C}^n \cong \mathbb{R}^{2n}$  is a homogeneous space for both the unitary and the special unitary group,

$$S^{2n-1} \cong U(n)/U(n-1) \cong SU(n)/SU(n-1)$$

all four presentations being  $G$ -equivariant diffeomorphisms in the sense of theorem 4.2.3.

The verification of the real case runs through the three required checks. *Transitivity.* Let  $x, y \in S^{n-1}$  be unit vectors. Extend  $x$  to an orthonormal basis  $x = u_1, u_2, \dots, u_n$  and  $y$  to an orthonormal basis  $y = v_1, v_2, \dots, v_n$  of  $\mathbb{R}^n$ , possible by the Gram-Schmidt process. The matrix  $g$  carrying  $u_i \mapsto v_i$  is orthogonal, since it sends one orthonormal basis to another, and it satisfies  $gx = y$ . Thus  $O(n)$  acts transitively on  $S^{n-1}$ ; the action is smooth, being the restriction of the linear action  $O(n) \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ .

*Stabilizer.* Take the base point  $e_n = (0, \dots, 0, 1)$ , the north pole. A matrix  $g \in O(n)$  fixes  $e_n$  if and only if its last column is  $e_n$ ; orthogonality then forces the last row to be  $e_n^T$  as well, since the columns are orthonormal and the  $n$ th column is  $e_n$ , so the top-left  $(n-1) \times (n-1)$  block is itself orthogonal and the off-diagonal blocks vanish. Hence the stabilizer is

$$\text{Stab}_{O(n)}(e_n) = \left\{ \begin{pmatrix} h & 0 \\ 0 & 1 \end{pmatrix} \mid h \in O(n-1) \right\} \cong O(n-1)$$

embedded as the upper-left block. This is a closed subgroup, as proposition 4.2.5 guarantees and as one checks directly, and theorem 4.2.3 delivers the diffeomorphism  $O(n)/O(n-1) \cong S^{n-1}$ .

*Dimension.* By proposition 1.2.4,  $\dim O(n) = \binom{n}{2}$ , and the dimension count reads

$$\dim O(n) - \dim O(n-1) = \binom{n}{2} - \binom{n-1}{2} = \frac{n(n-1)}{2} - \frac{(n-1)(n-2)}{2} = n-1$$

which is the dimension of  $S^{n-1}$ . The same base point  $e_n$  works for the rotation group  $SO(n)$ : the action is still transitive, because for  $n \geq 2$  the matrix  $g$  above can be corrected to have determinant 1 by negating one of the auxiliary basis vectors  $v_2, \dots, v_n$ , none of which is the target  $y = v_1$ ; and the stabilizer of  $e_n$  in  $SO(n)$  consists of the block matrices above whose block  $h$  has determinant 1, that is  $SO(n-1)$ . The dimension count is identical,  $\binom{n}{2} - \binom{n-1}{2} = n-1$ , since  $SO(m)$  and  $O(m)$  have the same dimension by proposition 1.2.4.

The complex case is the exact analogue with the Hermitian inner product in place of the Euclidean one. The unitary group  $U(n)$  preserves  $\|z\|^2 = \sum |z_i|^2$  on  $\mathbb{C}^n$ , hence acts on the unit sphere  $S^{2n-1} \subset \mathbb{C}^n$ , and the action is transitive by the Hermitian Gram-Schmidt process: any unit vector extends to a unitary basis, so any unit vector is carried to any other by a unitary matrix. Fixing the base point  $e_n \in \mathbb{C}^n$ , a unitary matrix fixes it if and only if it has the block form  $\begin{pmatrix} h & 0 \\ 0 & 1 \end{pmatrix}$  with  $h \in U(n-1)$ , by the same column-orthonormality argument, so the stabilizer is  $U(n-1)$ . The dimension check uses  $\dim U(n) = n^2$  from proposition 1.2.6:

$$\dim U(n) - \dim U(n-1) = n^2 - (n-1)^2 = 2n-1$$

the dimension of  $S^{2n-1}$ . For  $SU(n)$  the action on  $S^{2n-1}$  is again transitive for  $n \geq 2$ : build the unitary  $g$  carrying  $z$  to  $w$  so that  $w$  is its first column, exactly as above, and then correct its determinant to 1 by rescaling one of the auxiliary columns  $2, \dots, n$  (none of which is the target column) by a unit scalar, which adjusts  $\det g$  freely while leaving the first column, hence  $gz = w$ , untouched. Its stabilizer of  $e_n$  is the block group with  $h \in U(n-1)$  and  $\det h = 1$ , that is  $SU(n-1)$ . By proposition 1.2.6,  $\dim SU(n) = n^2 - 1$ , and

$$\dim SU(n) - \dim SU(n-1) = (n^2 - 1) - ((n-1)^2 - 1) = 2n-1$$

again the dimension of  $S^{2n-1}$ .

The four presentations of the sphere are not idle repetition; they record which symmetries survive as one shrinks the group. That  $S^{n-1} = SO(n)/SO(n-1)$  already exhibits the round metric as

$\mathrm{SO}(n)$ -invariant, so the isometry group of the sphere contains  $\mathrm{SO}(n)$ , and the coset picture will let Part IV read off the sphere as a symmetric space. That the same sphere is  $\mathrm{O}(n)/\mathrm{O}(n-1)$  records the extra reflection symmetry. The complex presentations  $S^{2n-1} = \mathrm{U}(n)/\mathrm{U}(n-1)$  and  $\mathrm{SU}(n)/\mathrm{SU}(n-1)$  see the odd sphere as a unitary orbit, and this is the presentation that will produce the Hopf fibration below, once the scalar circle inside  $\mathrm{U}(n)$  is quotiented to pass from  $S^{2n-1}$  to complex projective space.

Passing to that quotient is the subject of the next family of examples. Projective space is the space of lines through the origin, and a Grassmannian is the space of  $k$ -dimensional subspaces. A classical group acts transitively on each because any subspace of a given dimension can be moved to any other by an orthogonal (or unitary) transformation, and the stabilizer of a fixed subspace splits into the symmetries of the subspace and the symmetries of its orthogonal complement. This block structure is the source of the product subgroups appearing below.

### Example 4.3.2: Projective Spaces, Grassmannians, and Flag Manifolds

The real projective space, complex projective space, real Grassmannian of  $k$ -planes, and full complex flag manifold are the homogeneous spaces

$$\mathbb{RP}^n \cong \mathrm{O}(n+1)/(\mathrm{O}(1) \times \mathrm{O}(n)) \quad (4.2)$$

$$\mathbb{CP}^n \cong \mathrm{SU}(n+1)/\mathrm{S}(\mathrm{U}(1) \times \mathrm{U}(n)) \quad (4.3)$$

$$\mathrm{Gr}_k(\mathbb{R}^n) \cong \mathrm{O}(n)/(\mathrm{O}(k) \times \mathrm{O}(n-k)) \quad (4.4)$$

$$\mathrm{Fl}(n) \cong \mathrm{U}(n)/\mathbb{T}^n \quad (4.5)$$

of real dimensions  $n$ ,  $2n$ ,  $k(n-k)$ , and  $n(n-1)$  respectively, where  $\mathrm{Gr}_k(\mathbb{R}^n)$  is the set of  $k$ -dimensional linear subspaces of  $\mathbb{R}^n$ ,  $\mathrm{Fl}(n)$  is the set of complete flags  $0 \subset L_1 \subset \cdots \subset L_n = \mathbb{C}^n$  with  $\dim_{\mathbb{C}} L_j = j$ , and  $\mathbb{T}^n \subset \mathrm{U}(n)$  is the diagonal maximal torus of unitary diagonal matrices.

Real projective space  $\mathbb{RP}^n$  is the set of lines through the origin in  $\mathbb{R}^{n+1}$ , equivalently the Grassmannian  $\mathrm{Gr}_1(\mathbb{R}^{n+1})$ , so it is the  $k=1$  case of (4.4), and it is worth treating first to fix the pattern. The orthogonal group  $\mathrm{O}(n+1)$  acts on lines by  $g \cdot \mathbb{R}x = \mathbb{R}(gx)$ . *Transitivity*: any line  $\mathbb{R}x$  with  $\|x\|=1$  is carried to any other line  $\mathbb{R}y$  by an orthogonal  $g$  with  $gx=y$ , which exists by the sphere case above. *Stabilizer*: fix the line  $\ell = \mathbb{R}e_1$ . A matrix  $g \in \mathrm{O}(n+1)$  maps  $\ell$  to itself if and only if  $ge_1 = \pm e_1$ ; orthogonality then forces  $g$  to preserve the orthogonal complement  $e_1^\perp = \mathbb{R}^n$  and to act orthogonally on it, so

$$\mathrm{Stab}_{\mathrm{O}(n+1)}(\ell) = \left\{ \begin{pmatrix} \varepsilon & 0 \\ 0 & h \end{pmatrix} \mid \varepsilon \in \{\pm 1\}, h \in \mathrm{O}(n) \right\} \cong \mathrm{O}(1) \times \mathrm{O}(n)$$

recording the sign  $\varepsilon = \pm 1$  on the line ( $\mathrm{O}(1) = \{\pm 1\}$ ) and the orthogonal symmetry  $h$  of the complementary hyperplane. *Dimension*: with  $\dim \mathrm{O}(1) = 0$  and  $\dim \mathrm{O}(n) = \binom{n}{2}$ ,

$$\dim \mathrm{O}(n+1) - \dim(\mathrm{O}(1) \times \mathrm{O}(n)) = \binom{n+1}{2} - \binom{n}{2} = n$$

the dimension of  $\mathbb{RP}^n$ , and theorem 4.2.3 gives (4.2).

The general real Grassmannian  $\mathrm{Gr}_k(\mathbb{R}^n)$  repeats this argument with a  $k$ -plane in place of a line. The group  $\mathrm{O}(n)$  acts on  $k$ -dimensional subspaces by  $g \cdot W = gW$ . Transitivity is again Gram-Schmidt: given two  $k$ -planes  $W, W'$ , extend orthonormal bases of  $W$  and  $W'$  to orthonormal bases of  $\mathbb{R}^n$ ; the orthogonal matrix carrying the first basis to the second carries  $W$  to  $W'$ . Fix the base point  $W_0 = \mathbb{R}^k = \mathrm{span}(e_1, \dots, e_k)$ . A matrix  $g \in \mathrm{O}(n)$  preserves  $W_0$  if and only if it also preserves

its orthogonal complement  $W_0^\perp = \text{span}(e_{k+1}, \dots, e_n)$ , because  $g$  orthogonal and  $gW_0 = W_0$  forces  $gW_0^\perp = (gW_0)^\perp = W_0^\perp$ ; on each summand  $g$  acts orthogonally, so

$$\text{Stab}_{\text{O}(n)}(W_0) = \left\{ \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \mid a \in \text{O}(k), d \in \text{O}(n-k) \right\} \cong \text{O}(k) \times \text{O}(n-k)$$

This is a closed subgroup, being the intersection of  $\text{O}(n)$  with the closed subgroup of block-diagonal matrices, so theorem 4.2.3 yields (4.4). The dimension count is

$$\dim \text{O}(n) - \dim(\text{O}(k) \times \text{O}(n-k)) = \binom{n}{2} - \binom{k}{2} - \binom{n-k}{2} = k(n-k)$$

where the last equality is the identity  $\binom{n}{2} = \binom{k}{2} + \binom{n-k}{2} + k(n-k)$ , which counts the pairs  $\{i, j\}$  from  $\{1, \dots, n\}$  according to whether both indices lie in the first block, both in the second, or one in each. Setting  $k = 1$  recovers  $\dim \mathbb{R}P^n = 1 \cdot n = n$ .

Complex projective space  $\mathbb{C}P^n$  is the set of complex lines through the origin in  $\mathbb{C}^{n+1}$ , on which  $\text{U}(n+1)$  acts transitively by the Hermitian version of the same argument. The presentation (4.3) uses the special unitary group, so the stabilizer requires slightly more care. Fix the base line  $\ell_0 = \mathbb{C}e_1$ . A matrix  $g \in \text{SU}(n+1)$  preserves  $\ell_0$  if and only if  $ge_1 = \lambda e_1$  for some unit scalar  $\lambda \in \text{U}(1)$ ; unitarity forces  $g$  to preserve  $e_1^\perp = \mathbb{C}^n$  and act unitarily there, so  $g$  has the block form  $\begin{pmatrix} \lambda & 0 \\ 0 & B \end{pmatrix}$  with  $\lambda \in \text{U}(1)$  and  $B \in \text{U}(n)$ , and the determinant constraint  $\det g = \lambda \det B = 1$  ties the two blocks together. The stabilizer is therefore

$$\text{Stab}_{\text{SU}(n+1)}(\ell_0) = \left\{ \begin{pmatrix} \lambda & 0 \\ 0 & B \end{pmatrix} \mid \lambda \in \text{U}(1), B \in \text{U}(n), \lambda \det B = 1 \right\} =: \text{S}(\text{U}(1) \times \text{U}(n))$$

the group  $\text{S}(\text{U}(1) \times \text{U}(n))$  of block-diagonal unitary matrices of total determinant 1. Its dimension is  $\dim(\text{U}(1) \times \text{U}(n)) - 1 = (n^2 + 1) - 1 = n^2$ , the determinant condition cutting one real dimension. The dimension count is

$$\dim \text{SU}(n+1) - \dim \text{S}(\text{U}(1) \times \text{U}(n)) = ((n+1)^2 - 1) - n^2 = 2n$$

the real dimension of  $\mathbb{C}P^n$ , which as a complex  $n$ -manifold has real dimension  $2n$ . This gives (4.3).

The full flag manifold  $\text{Fl}(n)$  carries the pattern to its extreme, refining a single subspace into a complete tower. A complete flag in  $\mathbb{C}^n$  is a nested chain  $0 \subset L_1 \subset L_2 \subset \dots \subset L_n = \mathbb{C}^n$  of subspaces with  $\dim_{\mathbb{C}} L_j = j$ . The unitary group acts by  $g \cdot (L_j) = (gL_j)$ , and transitivity follows because any orthonormal basis  $v_1, \dots, v_n$  adapted to a flag (with  $L_j = \text{span}(v_1, \dots, v_j)$ ) can be carried to any other adapted orthonormal basis by a unitary matrix. Fix the standard flag  $L_j^0 = \text{span}(e_1, \dots, e_j)$ . A unitary  $g$  fixes this flag if and only if it maps each  $L_j^0$  into itself, which forces  $g$  to be upper triangular; a unitary upper-triangular matrix is diagonal, and its diagonal entries are unit scalars, so

$$\text{Stab}_{\text{U}(n)}((L_j^0)) = \{\text{diag}(t_1, \dots, t_n) \mid t_j \in \text{U}(1)\} = \mathbb{T}^n$$

the diagonal maximal torus. The dimension count, with  $\dim \text{U}(n) = n^2$  and  $\dim \mathbb{T}^n = n$ , is

$$\dim \text{U}(n) - \dim \mathbb{T}^n = n^2 - n = n(n-1)$$

so (4.5) holds and  $\text{Fl}(n)$  has real dimension  $n(n-1)$ , an even number, as befits a complex manifold (of complex dimension  $\binom{n}{2}$ ).

The four presentations of example 4.3.2 share one grammar: a base point is a distinguished piece of linear-algebraic data, a line, a  $k$ -plane, or a flag, and its stabilizer is the group of symmetries preserving that data, which decomposes along the pieces the data cuts out. The invariant that varies is how finely the data slices  $\mathbb{R}^n$  or  $\mathbb{C}^n$ : a Grassmannian cuts it into two blocks and gets a product stabilizer  $O(k) \times O(n-k)$ , while the full flag cuts it into  $n$  ordered lines and gets the finest stabilizer available, a maximal torus. This last presentation,  $\text{Fl}(n) = \text{U}(n)/\mathbb{T}^n$ , is the geometric object at the center of the highest-weight theory of Part III: the coset space of a compact group by its maximal torus is where the Weyl character formula and the Borel-Weil construction of irreducible representations live.

The running example  $\text{SU}(2)$  is the smallest case where the sphere presentation and the projective presentation collide, and their collision is the Hopf fibration. The special unitary group  $\text{SU}(2)$  acts on the complex projective line  $\mathbb{CP}^1$ , which is a 2-sphere, and the stabilizer of a point is a circle; the resulting coset presentation, read as a map of total spaces, is the fibration  $S^3 \rightarrow S^2$  with circle fibers. This is the same group  $\text{SU}(2)$  whose double cover of  $\text{SO}(3)$  organized Chapter 1, now seen from the side of its action on the sphere rather than its action on its own Lie algebra.

### Example 4.3.3: The Hopf Fibration $S^3 \rightarrow S^2$

Complex projective space  $\mathbb{CP}^1$  is diffeomorphic to the 2-sphere  $S^2$ , and the action of  $\text{SU}(2)$  on  $\mathbb{CP}^1$  is transitive with stabilizer the diagonal maximal torus  $\text{U}(1) \subset \text{SU}(2)$ , giving

$$S^2 \cong \mathbb{CP}^1 \cong \text{SU}(2)/\text{U}(1)$$

Combined with the diffeomorphism  $\text{SU}(2) \cong S^3$  of proposition 1.5.1, the quotient map  $\pi: \text{SU}(2) \rightarrow \text{SU}(2)/\text{U}(1)$  is the *Hopf fibration*

$$S^1 \hookrightarrow S^3 \xrightarrow{\pi} S^2$$

a smooth surjection with every fiber a circle.

The presentation is the  $n = 1$  case of the complex projective example, specialized to the special unitary group. Complex projective space  $\mathbb{CP}^1$  is the set of complex lines in  $\mathbb{C}^2$ ; specializing (4.3) to  $n = 1$  gives  $\mathbb{CP}^1 \cong \text{SU}(2)/\text{S}(\text{U}(1) \times \text{U}(1))$ , and the stabilizer  $\text{S}(\text{U}(1) \times \text{U}(1)) = \{\text{diag}(\lambda, \mu) \mid \lambda, \mu \in \text{U}(1), \lambda\mu = 1\}$  is exactly the diagonal subgroup  $\{\text{diag}(\lambda, \lambda^{-1}) \mid \lambda \in \text{U}(1)\}$ , a circle, which is the diagonal maximal torus of  $\text{SU}(2)$ . The dimension count  $\dim \text{SU}(2) - \dim \text{U}(1) = 3 - 1 = 2$  confirms that the quotient is a surface, and it is the 2-sphere:  $\mathbb{CP}^1$  is diffeomorphic to  $S^2$  by stereographic coordinates, sending the line through  $(z_0, z_1) \neq 0$  to the point  $z_1/z_0 \in \mathbb{C} \cup \{\infty\}$ , the Riemann sphere, a standard identification from the theory of Riemann surfaces (the series volume ‘NateComplexAnalysis’).

Reading the coset map through the identifications  $\text{SU}(2) \cong S^3$  (proposition 1.5.1) and  $\text{SU}(2)/\text{U}(1) \cong S^2$  produces a smooth surjection  $\pi: S^3 \rightarrow S^2$  whose fibers are the cosets of  $\text{U}(1)$ , each a circle  $S^1$ . In coordinates: writing a point of  $S^3 \subset \mathbb{C}^2$  as  $(\alpha, \beta)$  with  $|\alpha|^2 + |\beta|^2 = 1$ , the Hopf map sends  $(\alpha, \beta)$  to the line  $\mathbb{C}(\alpha, \beta) \in \mathbb{CP}^1$ , equivalently to the ratio  $[\alpha : \beta] \in \mathbb{CP}^1 \cong S^2$ ; two points  $(\alpha, \beta)$  and  $(\alpha', \beta')$  have the same image precisely when  $(\alpha', \beta') = \lambda(\alpha, \beta)$  for a unit scalar  $\lambda \in \text{U}(1)$ , so the fiber over each point is a circle, the  $\text{U}(1)$ -orbit. The principal-bundle structure of this map, that  $\pi$  is a principal  $\text{U}(1)$ -bundle, is the general fact recorded for every coset projection and proved in the next section.

The Hopf fibration completes the portrait of  $\text{SU}(2)$  begun in Chapter 1. There the two-to-one map  $\text{SU}(2) \rightarrow \text{SO}(3)$  of theorem 1.5.5 realized  $\text{SU}(2)$  as the double cover of the rotation group, a covering of a 3-manifold by a 3-manifold. Here the map  $\text{SU}(2) \rightarrow S^2$  realizes the same group as the total space of a circle bundle over a 2-manifold, a fibration that drops dimension. The two maps out of

$SU(2) \cong S^3$  are complementary: one is the running example seen as a covering group, the object of the next section, and the other is the running example seen as a fiber bundle, the object of the section after. That a single 3-sphere fibers nontrivially over the 2-sphere with connected circle fibers is the origin of the fact that  $\pi_3(S^2) \neq 0$ , generated by the Hopf class, a computation from homotopy theory (the series volume ‘NateAlgTop’); the Hopf fibration is the geometric witness to the nontriviality of that homotopy group.

The examples of this section make one point about the whole subject. A single classical group presents an entire family of the manifolds a geometer cares about, each as a quotient  $G/H$ , and the presentation is not a bookkeeping convenience but a statement of symmetry: the group  $G$  acts on  $G/H$ , and every  $G$ -invariant structure on the quotient, a metric, a volume form, a complex structure, is thereby forced to be homogeneous, looking the same at every point. The isotropy subgroup  $H$  controls the local geometry through its action on the tangent space at the base coset, and this is the mechanism that the next two sections and the symmetric-space theory of Part IV will exploit: an invariant Riemannian metric on  $G/H$  is exactly an  $\text{Ad}(H)$ -invariant inner product on the complementary subspace  $\mathfrak{m}$  of  $\mathfrak{h}$  in  $\mathfrak{g}$ , so the differential geometry of the quotient is encoded in the finite-dimensional representation theory of  $H$ . The sphere  $S^2 = SU(2)/U(1)$  carries its round metric  $SO(3)$ -invariantly for exactly this reason, and the Grassmannians and flag manifolds inherit their canonical metrics by the same principle.

### Exercise 4.3.1

1. Verify the identity  $\binom{n}{2} = \binom{k}{2} + \binom{n-k}{2} + k(n-k)$  used in example 4.3.2, and conclude that  $\dim \text{Gr}_k(\mathbb{R}^n) = k(n-k)$  from the presentation (4.4). Check directly that this dimension is symmetric in the exchange  $k \leftrightarrow n-k$ , and explain geometrically why  $\text{Gr}_k(\mathbb{R}^n)$  and  $\text{Gr}_{n-k}(\mathbb{R}^n)$  are diffeomorphic.
2. Show that the complex flag manifold  $SU(n)/\mathbb{T}$ , where  $\mathbb{T} = \mathbb{T}^{n-1}$  is the diagonal maximal torus of  $SU(n)$ , is even-dimensional, and compute its dimension. (Note  $SU(n)$  has dimension  $n^2 - 1$  and its maximal torus has dimension  $n - 1$ .) Reconcile your answer with the presentation  $\text{Fl}(n) = U(n)/\mathbb{T}^n$  of (4.5) by checking that the two quotients have the same dimension.
3. Using the presentation  $S^2 = SU(2)/U(1)$  of example 4.3.3, explain why the round metric on  $S^2$  is invariant under the action of  $SO(3)$ . (Hint:  $SU(2)$  acts on  $\mathbb{CP}^1 = S^2$  through its quotient  $SU(2)/\{\pm I\} = SO(3)$  of theorem 1.5.5, since the center  $\{\pm I\} \subset U(1)$  acts trivially on lines.)
4. The *Stiefel manifold*  $V_k(\mathbb{R}^n)$  is the set of orthonormal  $k$ -frames in  $\mathbb{R}^n$ , that is, ordered  $k$ -tuples  $(v_1, \dots, v_k)$  of orthonormal vectors. Show that  $O(n)$  acts transitively on  $V_k(\mathbb{R}^n)$  with stabilizer  $O(n-k)$ , giving  $V_k(\mathbb{R}^n) \cong O(n)/O(n-k)$ , and compute its dimension. Identify the special cases  $k = 1$  and  $k = n$ .
5. With notation as in exercise 4.3.1 (4), exhibit a natural surjection  $V_k(\mathbb{R}^n) \rightarrow \text{Gr}_k(\mathbb{R}^n)$  and identify its fibers with a group. Verify that the resulting dimension arithmetic is consistent:  $\dim V_k(\mathbb{R}^n) = \dim \text{Gr}_k(\mathbb{R}^n) + \dim O(k)$ .
6. Confirm the dimension count for (4.3) by computing  $\dim S(U(1) \times U(n))$  directly, and check that  $\dim \mathbb{CP}^n = 2n$  agrees with the count of complex homogeneous coordinates  $[z_0 : \dots : z_n]$  modulo scaling.

## 4.4 Covering Groups, the Fundamental Group, and Discrete Central Kernels

The correspondence of theorem 3.4.4 identifies a simply connected Lie group with its Lie algebra, and corollary 3.4.5 extended this to all connected groups by dividing the simply connected model  $\tilde{G}$  by a discrete central subgroup. That corollary answered the algebraic question, which connected groups carry a given Lie algebra, by reducing it to a computation inside  $Z(\tilde{G})$ . This section reads the same data from the topological side. The subgroup one quotients by is exactly the fundamental group  $\pi_1(G)$ , sitting as the kernel of the covering homomorphism  $\tilde{G} \rightarrow G$ , and the covers of a fixed group  $G$  are organized not by an arbitrary list but by the lattice of subgroups of  $\pi_1(G)$ . The result is a tower of Lie groups, all sharing the single Lie algebra  $\text{Lie}(G)$ , indexed by the subgroup structure of one finite or finitely generated abelian group.

Two facts drive everything. The first is that the kernel of a covering homomorphism is not just discrete but central, so that  $\pi_1(G)$  lives inside the center of  $\tilde{G}$  and, being a fiber of a covering of a topological group, is abelian. The second is the covering-space correspondence itself, imported from ‘NateTopology’, which matches connected covers of  $G$  with subgroups of  $\pi_1(G)$ ; the content of this section is that the correspondence respects Lie group structure, so every connected cover of a Lie group is again a Lie group covering homomorphically, with the same Lie algebra. The fundamental group, the one datum Lie’s third theorem could not detect, thereby becomes the deck group governing a tower of covers.

The centrality fact appeared once already, embedded in Step 3 of the construction of the universal covering group (proposition 3.4.1). It deserves to be recorded on its own, because it holds for any discrete normal subgroup of any connected group and is the reason discrete kernels behave so rigidly. The argument is a single application of the principle that a continuous map from a connected space into a discrete space is constant.

### Proposition 4.4.1: Discrete Central Kernels and the Abelian Fundamental Group

Let  $G$  be a connected Lie group.

1. Every discrete normal subgroup  $N \trianglelefteq G$  is contained in the center  $Z(G)$ .
2. If  $\pi: \tilde{G} \rightarrow G$  is a covering homomorphism of connected Lie groups, its kernel  $\ker \pi$  is a discrete subgroup of  $Z(\tilde{G})$ , and  $\ker \pi \cong \pi_1(G)/\pi_*\pi_1(\tilde{G})$ .
3. The fundamental group  $\pi_1(G)$  is abelian.

#### **Proof:**

The three parts are proved in turn; each rests on the same principle, that a continuous map from a connected space to a discrete space is constant.

1. Fix  $n \in N$  and consider the conjugation map

$$c_n: G \rightarrow G, \quad c_n(g) = g n g^{-1}$$

It is continuous, being a composition of multiplications and inversion. Because  $N$  is normal,  $g n g^{-1} \in N$  for every  $g \in G$ , so  $c_n$  restricts to a continuous map  $G \rightarrow N$  into the discrete subspace  $N$ . The domain  $G$  is connected, and a continuous map from a connected space

into a discrete space is constant, so  $c_n$  is constant on  $G$ ; its value at  $g = e$  is  $c_n(e) = n$ . Hence  $g n g^{-1} = n$  for all  $g \in G$ , that is  $n \in Z(G)$ . As  $n \in N$  was arbitrary,  $N \subseteq Z(G)$ .

2. As the kernel of a homomorphism,  $K = \ker \pi$  is a normal subgroup of  $\tilde{G}$ . As the fiber  $\pi^{-1}(e)$  of a covering map, it is a discrete subset of  $\tilde{G}$ : a covering map is a local homeomorphism, so each point of the fiber over  $e$  has a neighborhood meeting the fiber only in that point. Thus  $K$  is a discrete normal subgroup of the connected group  $\tilde{G}$ , and part (1) applied to  $\tilde{G}$  places  $K \subseteq Z(\tilde{G})$ . The isomorphism  $K \cong \pi_1(G)/\pi_*\pi_1(\tilde{G})$  is the covering-space computation of the fiber: for a covering  $\pi: \tilde{G} \rightarrow G$  of connected, locally path-connected spaces the fiber  $\pi^{-1}(e)$  is in bijection with the coset space  $\pi_1(G, e)/\pi_*\pi_1(\tilde{G}, \tilde{e})$ , and for a covering homomorphism this bijection is a group isomorphism because the deck transformations are left translations by the elements of  $K$  ([5]). When  $\tilde{G}$  is the universal cover,  $\pi_*\pi_1(\tilde{G})$  is trivial and this recovers  $K \cong \pi_1(G)$  of proposition 3.4.1.
3. Apply part (2) to the universal covering homomorphism  $\pi: \tilde{G} \rightarrow G$  of proposition 3.4.1, whose existence is guaranteed there. Its kernel  $\ker \pi$  is contained in  $Z(\tilde{G})$  and is therefore abelian, being a subgroup of an abelian group. By proposition 3.4.1 the kernel is isomorphic to  $\pi_1(G)$ , so  $\pi_1(G)$  is abelian, as we sought to show.

The proposition explains why the fundamental group of a Lie group is so much better behaved than the fundamental group of a general space. For an arbitrary path-connected space  $\pi_1$  can be any group at all, and is typically nonabelian, as the figure-eight already shows with its free group on two generators. A Lie group is a topological group, and the group structure forces its fundamental group to be abelian. The mechanism is entirely contained in part (3):  $\pi_1(G)$  is realized as a subgroup of the center of  $\tilde{G}$ , and central subgroups are abelian. This is the same phenomenon that makes the fundamental group of any topological group abelian, seen here through the covering homomorphism rather than through the Eckmann-Hilton argument on loop concatenation.

The centrality of the kernel is what makes the covering theory of Lie groups a theory of central extensions. A connected group  $G$  and its universal cover fit into a short exact sequence

$$1 \longrightarrow \pi_1(G) \longrightarrow \tilde{G} \xrightarrow{\pi} G \longrightarrow 1$$

in which the kernel  $\pi_1(G)$  is central. The infinitesimal theory sees only the middle and right terms as the same object, since  $d\pi_{\tilde{e}}$  is an isomorphism of Lie algebras (proposition 3.4.1); the entire content of the extension is the discrete central subgroup on the left. Chapter 3 read this sequence algebraically, as  $G \cong \tilde{G}/\Gamma$  with  $\Gamma$  a discrete subgroup of  $Z(\tilde{G})$ ; the covering-space language reads the same  $\Gamma$  as  $\pi_1(G)$ , the deck group of the cover.

A first instance is already at hand. The circle  $S^1$  is a connected Lie group, and its universal cover is the additive group  $\mathbb{R}$  via  $t \mapsto e^{2\pi i t}$ , with kernel the discrete central subgroup  $\mathbb{Z} \subseteq \mathbb{R}$ . By proposition 4.4.1 this kernel is central, which is automatic since  $\mathbb{R}$  is abelian, and  $\pi_1(S^1) \cong \mathbb{Z}$ , abelian as the proposition requires. The winding number of a loop, the classical invariant  $\pi_1(S^1) = \mathbb{Z}$ , is literally the element of the kernel through which the loop lifts.

With the kernel understood, the covers of a fixed group  $G$  can be classified. The covering-space correspondence of ‘NateTopology’ matches connected covers of a reasonable space  $X$  with subgroups of  $\pi_1(X)$ ; the theorem below is that for a Lie group this correspondence lands inside the category of Lie groups, so every connected cover is again a Lie group covering  $G$  by a homomorphism. This is the global, discrete refinement of the Lie group-Lie algebra dictionary, complementing corollary 3.4.5: where the corollary fixed the Lie algebra and varied the group, the theorem fixes the group  $G$  and organizes the groups lying above it.

**Theorem 4.4.2: Classification of Connected Covering Groups**

Let  $G$  be a connected Lie group with universal covering group  $\tilde{G}$  and fundamental group  $\pi_1(G) \cong \ker(\tilde{G} \rightarrow G) \subseteq Z(\tilde{G})$  (propositions 3.4.1 and 4.4.1). Assigning to a subgroup  $\Lambda \subseteq \pi_1(G)$  the quotient  $\tilde{G}/\Lambda$  gives an order-reversing bijection

$$\{\text{subgroups } \Lambda \subseteq \pi_1(G)\} \longleftrightarrow \{\text{connected covering groups of } G\} / \cong$$

under which:

1. each  $\tilde{G}/\Lambda$  is a connected Lie group and the induced map  $p_\Lambda: \tilde{G}/\Lambda \rightarrow G$  is a covering homomorphism with kernel  $\pi_1(G)/\Lambda$ , so  $\pi_1(\tilde{G}/\Lambda) \cong \Lambda$ ;
2.  $\Lambda = \{e\}$  gives the universal cover  $\tilde{G}$  (on top) and  $\Lambda = \pi_1(G)$  gives  $G$  itself (on the bottom);
3. every group in the tower has the same Lie algebra:  $\text{Lie}(\tilde{G}/\Lambda) \cong \text{Lie}(G)$ , and every covering homomorphism in the tower is a local diffeomorphism inducing this identity on Lie algebras.

**Proof:**

Write  $\Pi = \pi_1(G)$ , identified with the discrete central subgroup  $\ker(\pi: \tilde{G} \rightarrow G) \subseteq Z(\tilde{G})$  via proposition 3.4.1. Since  $\Pi$  is central, every subgroup  $\Lambda \subseteq \Pi$  is normal in  $\tilde{G}$ , so the quotients below are groups.

*Step 1: each quotient  $\tilde{G}/\Lambda$  is a Lie group and  $p_\Lambda$  is a covering homomorphism.* Let  $\Lambda \subseteq \Pi$  be a subgroup. As a subgroup of the discrete group  $\Pi$ , it is itself discrete, and it is central in  $\tilde{G}$ . A discrete central subgroup acts on  $\tilde{G}$  by left translations, freely (translation by a nonidentity element has no fixed point) and properly discontinuously (discreteness of  $\Lambda$  separates the orbit of any point), so the quotient map  $q_\Lambda: \tilde{G} \rightarrow \tilde{G}/\Lambda$  is a covering and  $\tilde{G}/\Lambda$  inherits a smooth manifold structure for which  $q_\Lambda$  is a local diffeomorphism ([5]). The group operations descend because  $\Lambda$  is normal, and they are smooth because they lift through the local diffeomorphism  $q_\Lambda$  to the smooth operations of  $\tilde{G}$ ; hence  $\tilde{G}/\Lambda$  is a Lie group. Because  $\Lambda \subseteq \Pi = \ker \pi$ , the universal covering homomorphism  $\pi$  factors through  $q_\Lambda$ : there is a unique homomorphism  $p_\Lambda: \tilde{G}/\Lambda \rightarrow G$  with  $p_\Lambda \circ q_\Lambda = \pi$ , given on cosets by  $p_\Lambda(\tilde{g}\Lambda) = \pi(\tilde{g})$ . It is well defined since  $\Lambda \subseteq \ker \pi$ , surjective since  $\pi$  is, and its kernel is  $\ker \pi / \Lambda = \Pi / \Lambda$ . That  $p_\Lambda$  is a covering follows from  $\pi = p_\Lambda \circ q_\Lambda$  being a covering with  $q_\Lambda$  a covering: a local homeomorphism  $p_\Lambda$  whose composite with the surjective covering  $q_\Lambda$  is a covering is itself a covering, with fiber the discrete group  $\Pi/\Lambda$  ([5]). Since  $\tilde{G}$  is simply connected,  $q_\Lambda$  is the universal cover of  $\tilde{G}/\Lambda$ , so  $\pi_1(\tilde{G}/\Lambda) \cong \Lambda$  by proposition 3.4.1.

*Step 2: surjectivity of the assignment.* Let  $p: H \rightarrow G$  be any connected covering homomorphism. Its universal cover is a simply connected Lie group covering  $G$ , so by uniqueness of the universal covering group (proposition 3.4.1) it is isomorphic to  $\tilde{G}$ , and the covering  $\tilde{G} \rightarrow H \xrightarrow{p} G$  realizes  $\tilde{G}$  as the universal cover of  $H$  compatibly with  $\pi$ . The kernel  $\Lambda := \ker(\tilde{G} \rightarrow H)$  is then a discrete central subgroup of  $\tilde{G}$  contained in  $\ker \pi = \Pi$  (since the composite  $\tilde{G} \rightarrow H \rightarrow G$  is  $\pi$ ), and  $H \cong \tilde{G}/\Lambda$  as covering groups of  $G$  by the first isomorphism theorem, with the isomorphism commuting with the maps to  $G$ . Thus every connected covering group of  $G$  arises as some  $\tilde{G}/\Lambda$ .

*Step 3: injectivity and order-reversal.* Suppose  $\tilde{G}/\Lambda \cong \tilde{G}/\Lambda'$  as covering groups of  $G$ , that is, there is an isomorphism  $\psi$  with  $p_{\Lambda'} \circ \psi = p_\Lambda$ . Lifting to the common universal cover  $\tilde{G}$ , the

isomorphism  $\psi$  lifts to an automorphism of  $\tilde{G}$  commuting with  $\pi$ ; the only such automorphism fixing the basepoint is the identity, since a lift of the identity map of  $G$  through the universal cover that fixes  $\tilde{e}$  is the identity ([5]). Hence  $q_{\Lambda'} = \psi \circ q_{\Lambda}$  forces  $\Lambda = \Lambda'$  as the respective kernels. Order-reversal is immediate from the construction: if  $\Lambda \subseteq \Lambda'$  then  $\tilde{G}/\Lambda'$  is a further quotient of  $\tilde{G}/\Lambda$ , so  $\tilde{G}/\Lambda$  sits above  $\tilde{G}/\Lambda'$  in the tower; larger subgroups give smaller covers. This proves the bijection is order-reversing and establishes claims (1) and (2), with  $\Lambda = \{e\}$  giving  $\tilde{G}$  on top and  $\Lambda = \Pi$  giving  $\tilde{G}/\Pi \cong G$  on the bottom.

*Step 4: the common Lie algebra.* Each covering homomorphism in the tower is a covering map, hence a local diffeomorphism, so its differential at the identity is a linear isomorphism of tangent spaces and a Lie-algebra homomorphism because the map is a group homomorphism. A bijective Lie-algebra homomorphism is an isomorphism, so  $\text{Lie}(\tilde{G}/\Lambda) \cong \text{Lie}(\tilde{G}) \cong \text{Lie}(G)$  for every  $\Lambda$ , with the identifications compatible across the tower. This is claim (3), completing the proof.

The theorem completes the picture begun in Chapter 3. There the varying object was the Lie algebra's group, listed as central quotients of  $\tilde{G}$ ; here the fixed object is  $G$ , and the theorem organizes the groups above it as a lattice indexed by the subgroups of  $\pi_1(G)$ . The two descriptions are the same short exact sequence read from opposite ends. Corollary 3.4.5 started from  $\mathfrak{g}$  and produced  $\tilde{G}/\Gamma$  for each discrete  $\Gamma \subseteq Z(\tilde{G})$ ; the present theorem starts from  $G = \tilde{G}/\pi_1(G)$  and produces the intermediate covers  $\tilde{G}/\Lambda$  for  $\Lambda \subseteq \pi_1(G)$ . Since  $\pi_1(G)$  is one particular such  $\Gamma$ , its subgroups are exactly the discrete central subgroups sitting between  $\{e\}$  and  $\pi_1(G)$ , and the tower of covers of  $G$  is the interval  $[\{e\}, \pi_1(G)]$  in the subgroup lattice of  $Z(\tilde{G})$ .

The order-reversal is the content worth remembering. A larger subgroup  $\Lambda$  of  $\pi_1(G)$  means a larger quotient  $\tilde{G}/\Lambda$ , hence a smaller cover, hence fewer sheets: passing up the subgroup lattice of  $\pi_1(G)$  passes down the tower of covers toward  $G$ . The universal cover, with  $\Lambda = \{e\}$ , is maximally unwound;  $G$  itself, with  $\Lambda = \pi_1(G)$ , is maximally wound. Everything in between is a cover of  $G$  and simultaneously a quotient of  $\tilde{G}$ , and none of these distinctions is visible to the Lie algebra, which is the same  $\text{Lie}(G)$  at every stage.

The worked examples make the tower concrete, from the one-line case of the circle to the double cover of the rotation group and the cases that reach beyond the matrix groups entirely.

#### Example 4.4.3: Covering Groups: Circle, Rotations, Spin, and $\text{SL}_2(\mathbb{R})$

The four examples run from the simplest infinite tower to a case that leaves the matrix groups behind.

1. *The circle.* The map  $\pi: \mathbb{R} \rightarrow S^1$ ,  $\pi(t) = e^{2\pi it}$ , is a homomorphism of Lie groups with kernel the discrete central subgroup  $\mathbb{Z}$ , so it is the universal covering group of  $S^1$  and  $\pi_1(S^1) \cong \mathbb{Z}$ . Since  $\mathbb{R}$  is abelian its center is all of  $\mathbb{R}$ , and the subgroups of  $\pi_1(S^1) = \mathbb{Z}$  are exactly  $n\mathbb{Z}$  for  $n \geq 0$ . By theorem 4.4.2 the connected covering groups of  $S^1$  are the quotients  $\mathbb{R}/n\mathbb{Z}$ : for  $n = 0$  this is  $\mathbb{R}$  itself (the universal cover), and for  $n \geq 1$  the quotient  $\mathbb{R}/n\mathbb{Z} \cong S^1$  is the circle covering itself by the  $n$ -fold map  $z \mapsto z^n$ , with  $\pi_1 \cong n\mathbb{Z} \cong \mathbb{Z}$  realized as the index- $n$  subgroup. The tower is the lattice of subgroups of  $\mathbb{Z}$ , an infinite chain of circles above  $S^1$  topped by the line  $\mathbb{R}$ .
2. *The rotation group.* The double cover  $\varphi: \text{SU}(2) \rightarrow \text{SO}(3)$  of theorem 1.5.5 has kernel exactly the center  $\{\pm I\}$  of  $\text{SU}(2)$ , a discrete central subgroup isomorphic to  $\mathbb{Z}/2$ . Since  $\text{SU}(2) \cong S^3$  is simply connected (proposition 1.5.1),  $\varphi$  is the universal cover of  $\text{SO}(3)$ , and  $\pi_1(\text{SO}(3)) \cong \mathbb{Z}/2$  (corollary 1.5.6). The subgroups of  $\mathbb{Z}/2$  are only  $\{e\}$  and  $\mathbb{Z}/2$ , so by theorem 4.4.2 the tower of connected covers of  $\text{SO}(3)$  has exactly two floors:  $\text{SU}(2)$  on top ( $\Lambda = \{e\}$ ) and  $\text{SO}(3)$

itself on the bottom ( $\Lambda = \mathbb{Z}/2$ ). There is no intermediate cover, because  $\mathbb{Z}/2$  has no proper nontrivial subgroup. The nontrivial loop in  $\mathrm{SO}(3)$ , a full  $2\pi$  rotation about a fixed axis, lifts to a path in  $\mathrm{SU}(2)$  from  $I$  to  $-I$ , and the element  $-I \in \ker \varphi$  is the generator of  $\pi_1(\mathrm{SO}(3))$ .

3. *The spin groups.* The pattern of  $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$  persists in every dimension. For  $n \geq 3$  the special orthogonal group  $\mathrm{SO}(n)$  has fundamental group  $\pi_1(\mathrm{SO}(n)) \cong \mathbb{Z}/2$ , and its universal cover is the spin group  $\mathrm{Spin}(n)$ , fitting into a double cover

$$1 \longrightarrow \mathbb{Z}/2 \longrightarrow \mathrm{Spin}(n) \longrightarrow \mathrm{SO}(n) \longrightarrow 1$$

with central kernel  $\mathbb{Z}/2$ . The case  $n = 3$  is  $\mathrm{Spin}(3) = \mathrm{SU}(2)$ , the double cover of this example, so the running example is the first spin group. As with  $\mathrm{SO}(3)$ , the tower over  $\mathrm{SO}(n)$  has just two floors for  $n \geq 3$ , since  $\pi_1 = \mathbb{Z}/2$ ; the case  $n = 2$  is different, with  $\mathrm{SO}(2) = S^1$  and the infinite tower of item (1). The spin groups are the arena of the spinor representations, which live on  $\mathrm{Spin}(n)$  but not on  $\mathrm{SO}(n)$  precisely because they fail to descend across the central  $\mathbb{Z}/2$ ; the construction of  $\mathrm{Spin}(n)$  inside the Clifford algebra is developed in the Riemannian and index-theoretic sequels to this book.

4. *A cover that is not a matrix group.* The examples above all have matrix groups at every floor of the tower, but this is not forced. The group  $\mathrm{SL}_2(\mathbb{R})$  is connected with fundamental group  $\pi_1(\mathrm{SL}_2(\mathbb{R})) \cong \mathbb{Z}$ , so by theorem 4.4.2 its tower of covers is the infinite chain indexed by the subgroups  $n\mathbb{Z} \subseteq \mathbb{Z}$ , topped by the universal cover  $\widetilde{\mathrm{SL}_2(\mathbb{R})}$ . This universal cover admits no faithful finite-dimensional representation, and in particular is not isomorphic to any matrix group: any continuous homomorphism  $\widetilde{\mathrm{SL}_2(\mathbb{R})} \rightarrow \mathrm{GL}_N(\mathbb{C})$  must kill an infinite subgroup of the central  $\mathbb{Z}$ , because the center of every connected matrix group with this Lie algebra is finite. Thus  $\widetilde{\mathrm{SL}_2(\mathbb{R})}$  is a Lie group that cannot be realized as a closed subgroup of any  $\mathrm{GL}_N$ , the first object in this book outside the matrix world of Chapter 1. Its representation theory belongs to the analysis of noncompact real groups in Part IV, where  $\mathrm{SL}_2(\mathbb{R})$  and its covers organize the noncompact side of the theory.

The examples span the range of behavior the classification allows. The circle and  $\mathrm{SL}_2(\mathbb{R})$  have infinite fundamental group  $\mathbb{Z}$  and so infinite towers; the rotation and spin groups have  $\pi_1 = \mathbb{Z}/2$  and so two-floor towers. In every case the Lie algebra is blind to the height in the tower, and the topology, encoded in  $\pi_1$ , is what distinguishes the floors. The last example is the one that breaks the frame of Chapter 1:  $\widetilde{\mathrm{SL}_2(\mathbb{R})}$  is a bona fide Lie group produced by the abstract correspondence, yet it is not a matrix group, which is why the correspondence had to be developed intrinsically rather than through matrices, and why Part IV must handle noncompact groups by tools beyond the classical-group computations.

The section closes the account of Part I's dictionary. Lie's third theorem (theorem 3.4.4) established that the Lie algebra determines the simply connected group, and named the single invisible invariant: the fundamental group, which the infinitesimal theory cannot see. That invariant is now organized. It is the discrete central kernel of the universal cover (proposition 4.4.1), it is abelian, and its subgroups index a tower of connected covers (theorem 4.4.2), all sharing the one Lie algebra  $\mathrm{Lie}(G)$ . A Lie group is thus a pair of data: a Lie algebra, carrying all the local structure, and a discrete central subgroup of the associated simply connected group, carrying the global topology. The rest of the book computes with the first; this section named and organized the second.

### Exercise 4.4.1

1. Show that the  $n$ -torus  $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$  is a connected Lie group with universal cover  $\mathbb{R}^n$ , and that  $\pi_1(\mathbb{T}^n) \cong \mathbb{Z}^n$ . Using theorem 4.4.2, describe the connected covering groups of  $\mathbb{T}^n$  in terms of the subgroups of  $\mathbb{Z}^n$ , and verify directly that the kernel of the covering  $\mathbb{R}^n \rightarrow \mathbb{T}^n$  is central.
2. Let  $G$  be a connected Lie group with finite center  $Z(G)$ . Prove that  $G$  has only finitely many connected covering groups (up to isomorphism) if and only if  $\pi_1(G)$  is finite. (Use theorem 4.4.2 to reduce to counting subgroups of  $\pi_1(G)$ , and note that  $\pi_1(G) \subseteq Z(\tilde{G})$ ; the finiteness of  $Z(G)$  and of  $\pi_1(G)$  together are equivalent to finiteness of  $Z(\tilde{G})$ .)
3. Prove that the unitary group  $U(n)$  is isomorphic to the quotient  $(SU(n) \times U(1))/\mu_n$ , where  $\mu_n = \{(\zeta^{-1}I, \zeta) \mid \zeta^n = 1\}$  is the subgroup of the center generated by  $(\omega^{-1}I, \omega)$  for  $\omega = e^{2\pi i/n}$ . Identify the map  $SU(n) \times U(1) \rightarrow U(n)$  realizing this quotient and compute its kernel.
4. Prove that the center of  $SU(n)$  is the cyclic group of order  $n$  consisting of the scalar matrices  $\zeta I$  with  $\zeta^n = 1$ , that is  $Z(SU(n)) = \{\zeta I \mid \zeta^n = 1\} \cong \mathbb{Z}/n$ . Deduce that the adjoint group  $SU(n)/Z(SU(n)) = PSU(n)$  has fundamental group  $\pi_1(PSU(n)) \cong \mathbb{Z}/n$ , and that  $SU(n)$  is its universal cover.
5. Using  $\pi_1(SO(4)) \cong \mathbb{Z}/2$  and the exceptional isomorphism  $\text{Spin}(4) \cong SU(2) \times SU(2)$ , describe the tower of connected covers of  $SO(4)$ . In particular, identify the subgroup of  $\pi_1(SO(4))$  corresponding to the intermediate group  $SO(4)$ 's relation to  $SU(2) \times SU(2)$ , and explain why  $SO(4)$  has exactly two connected covers despite  $\text{Spin}(4)$  splitting as a product.
6. Prove that for connected Lie groups  $G$  and  $H$  the fundamental group of the product satisfies  $\pi_1(G \times H) \cong \pi_1(G) \times \pi_1(H)$ , and that the universal cover of  $G \times H$  is  $\tilde{G} \times \tilde{H}$ . Use this and theorem 4.4.2 to compute the tower of covers of the torus  $\mathbb{T}^2 = S^1 \times S^1$  from the towers of its factors, and reconcile the answer with exercise 4.4.1(1).

## 4.5 Fiber Bundles from Subgroups and the Isotropy Representation

The quotient  $G/H$  of theorem 4.1.3 was built as a smooth manifold, and the orbit-stabilizer theorem 4.2.3 identified it with any homogeneous  $G$ -space carrying a point of stabilizer  $H$ . This section reads the projection  $\pi: G \rightarrow G/H$  not as a bare submersion but as a bundle: the group  $H$  acts on  $G$  by right translation, its orbits are exactly the fibers of  $\pi$ , and near each point of the base this action looks like the trivial action of  $H$  on a product. That is the content of being a principal  $H$ -bundle, and it is the structure that organizes the geometry attached to a homogeneous space. Every  $H$ -space  $F$  then produces an associated bundle  $G \times_H F$  over  $G/H$  by gluing copies of  $F$  along the transition data of  $\pi$ , and the associated bundle attached to the most geometrically loaded  $F$  is the tangent bundle of  $G/H$  itself.

The payoff of this reading is the isotropy representation. A homogeneous space is the same at every point, so its local geometry is encoded at the single base point  $eH$ , in the tangent space  $T_{eH}(G/H)$  together with the linear action of the stabilizer  $H$  on it. This section identifies that tangent space with the quotient  $\mathfrak{g}/\mathfrak{h}$  of Lie algebras and the action with  $\text{Ad}(H)$  descended to the quotient. When a complement to  $\mathfrak{h}$  can be chosen  $\text{Ad}(H)$ -invariantly, which happens whenever  $H$  is compact, the tangent space is realized inside  $\mathfrak{g}$  itself as a subspace  $\mathfrak{m}$  carrying an  $H$ -representation, and a  $G$ -invariant metric on  $G/H$  becomes the same datum as an  $\text{Ad}(H)$ -invariant inner product on  $\mathfrak{m}$ . This is the input to the theory of Riemannian symmetric spaces developed in Part IV.

The right action of  $H$  on  $G$  is the starting point. For  $h \in H$  the map  $R_h: g \mapsto gh$  is a diffeomorphism of  $G$ , it preserves each left coset  $gH$ , and  $\pi(gh) = ghH = gH = \pi(g)$ , so  $R_h$  acts within the fibers of  $\pi$ . The action is free, since  $gh = g$  forces  $h = e$ , and its orbits are precisely the fibers  $\pi^{-1}(gH) = gH$ . A principal bundle is the assertion that this fiberwise free action is, in suitable local coordinates, the translation action of  $H$  on a product  $U \times H$ .

### Theorem 4.5.1: $G \rightarrow G/H$ Is a Principal $H$ -Bundle

Let  $G$  be a Lie group and  $H \leq G$  a closed subgroup, and equip  $G/H$  with the smooth structure of theorem 4.1.3. Then the projection  $\pi: G \rightarrow G/H$ , together with the right translation action of  $H$  on  $G$ , is a smooth principal  $H$ -bundle. That is:

1.  $H$  acts smoothly and freely on  $G$  on the right by  $g \cdot h = gh$ , preserving each fiber of  $\pi$ , and acts simply transitively on each fiber.
2. Every point of  $G/H$  has an open neighborhood  $U$  and a diffeomorphism  $\Phi: \pi^{-1}(U) \xrightarrow{\sim} U \times H$  that is  $H$ -equivariant and covers the identity, meaning  $\Phi(g) = (\pi(g), \varphi(g))$  with  $\varphi(gh) = \varphi(g)h$  for all  $h \in H$ .

Consequently, for every smooth left  $H$ -space  $F$  the associated bundle  $G \times_H F = (G \times F)/H$ , with  $H$  acting by  $h \cdot (g, f) = (gh^{-1}, h \cdot f)$ , is a smooth fiber bundle over  $G/H$  with fiber  $F$  and projection  $[g, f] \mapsto \pi(g)$ .

#### Proof:

*Step 1: the right action and freeness.* The right action  $G \times H \rightarrow G$ ,  $(g, h) \mapsto gh$ , is the restriction of the smooth multiplication of  $G$ , hence smooth. If  $gh = g$  then  $h = e$  by cancellation, so the action is free. For  $g, g' \in G$  the equality  $\pi(g) = \pi(g')$  means  $g^{-1}g' \in H$ , so  $g' = g(g^{-1}g') = g \cdot (g^{-1}g')$  lies in the  $H$ -orbit of  $g$ ; and any  $gh$  lies in the same fiber as  $g$  since  $\pi(gh) = gH = \pi(g)$ . Thus the orbit of  $g$  is exactly  $\pi^{-1}(\pi(g))$ , and freeness makes the action simply transitive on that fiber.

*Step 2: a local section through the base point.* The construction of theorem 4.1.3 produced, from a vector-space complement  $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$  and the exponential chart of theorem 2.3.4, an open ball  $B \subseteq \mathfrak{m}$  about 0 for which  $Y \mapsto \pi(\exp Y)$  is a diffeomorphism of  $B$  onto an open neighborhood  $U_0 := \pi(\exp B)$  of the base point  $eH$ , the coset chart of theorem 4.1.3. Its inverse followed by  $\exp$  is a smooth map

$$s_0: U_0 \rightarrow G, \quad \pi(\exp Y) \mapsto \exp Y \quad (Y \in B)$$

with  $\pi \circ s_0 = \text{id}_{U_0}$ : the slice  $\exp(B)$  is transverse to  $H$  near  $e$  and meets each nearby fiber in exactly one point. A map  $s: U \rightarrow G$  on an open set  $U \subseteq G/H$  with  $\pi \circ s = \text{id}_U$  is a *local section* of  $\pi$ ; the map  $s_0$  is such a section on  $U_0$ .

*Step 3: the trivialization from a local section.* Given a smooth local section  $s: U \rightarrow G$ , define

$$\Psi: U \times H \rightarrow \pi^{-1}(U), \quad \Psi(x, h) = s(x)h$$

This is smooth, being built from  $s$  and the multiplication of  $G$ . It is a bijection onto  $\pi^{-1}(U)$ : given  $g \in \pi^{-1}(U)$ , set  $x = \pi(g) \in U$ ; then  $\pi(s(x)) = x = \pi(g)$ , so  $s(x)^{-1}g \in H$ , and  $g = s(x)(s(x)^{-1}g) = \Psi(x, s(x)^{-1}g)$ , giving surjectivity, while if  $s(x)h = s(x')h'$  then applying  $\pi$

gives  $x = x'$  and then  $h = h'$  by cancellation, giving injectivity. Its inverse is

$$\Phi: \pi^{-1}(U) \rightarrow U \times H, \quad \Phi(g) = (\pi(g), s(\pi(g))^{-1}g)$$

which is smooth because  $\pi$ ,  $s$ , inversion, and multiplication all are; here  $s(\pi(g))^{-1}g \in H$  lies in the embedded Lie subgroup  $H$ , and the corestriction to  $H$  is smooth by the embedded-submanifold property ([2]). Thus  $\Phi$  is a diffeomorphism. Writing  $\varphi(g) = s(\pi(g))^{-1}g$  for the  $H$ -component, equivariance is the computation

$$\varphi(gh) = s(\pi(gh))^{-1}(gh) = s(\pi(g))^{-1}gh = \varphi(g)h$$

using  $\pi(gh) = \pi(g)$ . This establishes the trivialization over  $U_0$  from the section  $s_0$  of Step 2.

*Step 4: trivializing over every point by translation.* Let  $gH \in G/H$  be arbitrary. Left translation  $L_g: G \rightarrow G$  is a diffeomorphism, and it descends to a diffeomorphism  $\bar{L}_g: G/H \rightarrow G/H$ ,  $\bar{L}_g(g'H) = gg'H$ , because  $L_g$  carries fibers to fibers and  $\pi \circ L_g = \bar{L}_g \circ \pi$  by proposition 4.1.5. Then  $s := L_g \circ s_0 \circ \bar{L}_g^{-1}$  is a smooth local section over  $U := \bar{L}_g(U_0)$ , an open neighborhood of  $gH$ : indeed  $\pi \circ s = \pi \circ L_g \circ s_0 \circ \bar{L}_g^{-1} = \bar{L}_g \circ \pi \circ s_0 \circ \bar{L}_g^{-1} = \bar{L}_g \circ \bar{L}_g^{-1} = \text{id}_U$ , using  $\pi \circ s_0 = \text{id}$ . Step 3 turns this section into an  $H$ -equivariant trivialization over  $U$ . Since  $gH$  was arbitrary,  $\pi$  is locally trivial, and with the free fiberwise action of Step 1 this is precisely a principal  $H$ -bundle.

*Step 5: the associated bundle.* Let  $F$  be a smooth left  $H$ -space. The group  $H$  acts on  $G \times F$  by  $h \cdot (g, f) = (gh^{-1}, h \cdot f)$ , freely, since freeness in the first factor already forces  $h = e$ . The quotient  $G \times_H F := (G \times F)/H$  carries the quotient topology, and the map  $q: G \times F \rightarrow G/H$ ,  $(g, f) \mapsto \pi(g)$ , is constant on  $H$ -orbits, hence descends to  $p: G \times_H F \rightarrow G/H$ ,  $[g, f] \mapsto \pi(g)$ . Over a trivializing set  $U$  with section  $s$ , the map

$$U \times F \rightarrow p^{-1}(U), \quad (x, f) \mapsto [s(x), f]$$

is a bijection with inverse  $[g, f] \mapsto (\pi(g), \varphi(g) \cdot f)$ , where  $\varphi(g) = s(\pi(g))^{-1}g$  as in Step 3: this is well defined because  $[gh^{-1}, h \cdot f] \mapsto (\pi(g), \varphi(gh^{-1}) \cdot (h \cdot f)) = (\pi(g), \varphi(g)h^{-1}h \cdot f) = (\pi(g), \varphi(g) \cdot f)$ , using the equivariance  $\varphi(gh^{-1}) = \varphi(g)h^{-1}$ . These charts are smoothly compatible because on overlaps the two local sections differ by a smooth  $H$ -valued function, which acts smoothly on  $F$ ; transporting the smooth structure of  $U \times F$  through them makes  $G \times_H F$  a smooth manifold and  $p$  a smooth fiber bundle with fiber  $F$ , completing the proof.

The theorem promotes the submersion  $\pi: G \rightarrow G/H$  to a bundle whose structure group is  $H$  acting on itself by translation, and it does so with no new analytic input: the single local section  $s_0$  built from the coset chart of theorem 4.1.3, spread over the base by the homogeneity of  $G$ , is the entire construction. The mechanism is the one that recurs throughout the chapter, that a group acts transitively on its own quotient and so a chart at one point suffices everywhere. The associated-bundle construction  $F \rightsquigarrow G \times_H F$  then manufactures a vector bundle, a sphere bundle, or a bundle of frames over  $G/H$  from any  $H$ -space  $F$ , all sharing the transition functions of  $\pi$ ; the geometry of a homogeneous space is thereby reduced to the representation theory of the single group  $H$ .

The archetype is the Hopf fibration, where  $G = \text{SU}(2)$ ,  $H = \text{U}(1)$ , and the base is the two-sphere.

#### Example 4.5.2: The Hopf Fibration as a Principal $\text{U}(1)$ -Bundle

Realize  $\text{U}(1) = \{\text{diag}(e^{i\theta}, e^{-i\theta}) \mid \theta \in \mathbb{R}\}$  as the diagonal maximal torus of  $\text{SU}(2)$ , a closed subgroup. By proposition 1.5.1 the group  $\text{SU}(2)$  is diffeomorphic to  $S^3$ , and by example 4.3.3 the coset space  $\text{SU}(2)/\text{U}(1)$  is diffeomorphic to  $S^2$ , with the projection realizing the Hopf map  $S^3 \rightarrow S^2$ .

Theorem 4.5.1 identifies this projection as a principal  $U(1)$ -bundle

$$S^1 \hookrightarrow S^3 \longrightarrow S^2$$

the circle  $U(1) \cong S^1$  acting freely on  $S^3$  on the right by  $g \mapsto g \operatorname{diag}(e^{i\theta}, e^{-i\theta})$ . Freeness is immediate from Step 1, and the fibers are the circles  $g U(1)$ . Local triviality is not global: were the bundle trivial,  $S^3$  would be diffeomorphic to  $S^2 \times S^1$ , but these are distinguished by their fundamental groups,  $\pi_1(S^3) = 0$  while  $\pi_1(S^2 \times S^1) = \mathbb{Z}$  ([5]). The Hopf bundle is therefore a nontrivial principal circle bundle, the smallest example in which the local product structure of theorem 4.5.1 fails to be global.

The example shows the principal-bundle language capturing a nontriviality that the plain diffeomorphism  $SU(2)/U(1) \cong S^2$  does not see: the total space  $S^3$  is assembled from circles over  $S^2$  in a way that cannot be undone globally. The obstruction is measured by the topology of the base and structure group, and it is the kind of invariant that the associated-bundle construction converts into characteristic classes in a later theory. Here the point is more elementary, that  $\pi: G \rightarrow G/H$  is a genuine bundle and its local sections need not patch to a global one.

The associated bundle attached to the geometrically decisive  $H$ -space produces the tangent bundle of  $G/H$  itself. To see which  $H$ -space that is, fix attention at the base point  $o := eH \in G/H$ . Its stabilizer under the  $G$ -action of theorem 4.2.3 is exactly  $H$ , so  $H$  acts on the tangent space  $T_o(G/H)$  by differentials of the maps  $\bar{L}_h: G/H \rightarrow G/H$  fixing  $o$ . This linear action is the datum that linearizes the homogeneous space at  $o$ .

#### Definition 4.5.3: The Linear Isotropy Representation

Let  $H \leq G$  be a closed subgroup,  $o = eH$  the base point of  $G/H$ , and for  $h \in H$  let  $\bar{L}_h: G/H \rightarrow G/H$ ,  $gH \mapsto hgH$ , be the diffeomorphism induced by left translation by  $h$ . Each  $\bar{L}_h$  fixes  $o$ , since  $hH = H$ , so its differential at  $o$  is a linear automorphism of  $T_o(G/H)$ . The *linear isotropy representation* of  $H$  is the homomorphism

$$\chi: H \rightarrow \operatorname{GL}(T_o(G/H)), \quad \chi(h) = d(\bar{L}_h)_o$$

Under the canonical isomorphism  $T_o(G/H) \cong \mathfrak{g}/\mathfrak{h}$  induced by the submersion  $d\pi_e: \mathfrak{g} \rightarrow T_o(G/H)$  (whose kernel is  $\mathfrak{h} = T_eH$ ), the isotropy representation is  $\operatorname{Ad}(h)$  descended to the quotient: writing  $\overline{\operatorname{Ad}}(h): \mathfrak{g}/\mathfrak{h} \rightarrow \mathfrak{g}/\mathfrak{h}$  for the map induced by  $\operatorname{Ad}(h)$ , well defined because  $\operatorname{Ad}(H)$  preserves  $\mathfrak{h}$  (definition 2.5.1), the relation

$$\chi(h) = d\pi_e \circ \operatorname{Ad}(h) \circ (d\pi_e)^{-1} = \overline{\operatorname{Ad}}(h)$$

holds for every  $h \in H$ .

The identification demands two verifications, and both are short enough to record at once. First,  $d\pi_e: \mathfrak{g} \rightarrow T_o(G/H)$  is surjective because  $\pi$  is a submersion (theorem 4.1.3), and its kernel is  $\mathfrak{h}$ : the fiber  $\pi^{-1}(o) = H$  has tangent space  $\mathfrak{h}$  at  $e$ , and a tangent vector at  $e$  lies in  $\ker d\pi_e$  iff it is tangent to the fiber, so  $\ker d\pi_e = T_eH = \mathfrak{h}$ . Hence  $d\pi_e$  descends to the isomorphism  $\mathfrak{g}/\mathfrak{h} \xrightarrow{\sim} T_o(G/H)$ . Second, that  $\operatorname{Ad}(H)$  preserves  $\mathfrak{h}$ : for  $h \in H$  the conjugation  $C_h$  maps  $H$  to  $H$ , so its differential  $\operatorname{Ad}(h) = d(C_h)_e$  maps  $T_eH = \mathfrak{h}$  into  $\mathfrak{h}$ , and therefore  $\operatorname{Ad}(h)$  descends to  $\overline{\operatorname{Ad}}(h)$  on  $\mathfrak{g}/\mathfrak{h}$ . The naturality square  $\pi \circ C_h = \bar{L}_h \circ \pi$ , valid because  $C_h(g)H = hgh^{-1}H = hgH = \bar{L}_h(gH)$  (the middle step using  $h^{-1} \in H$ ), differentiates at  $e$  to  $d\pi_e \circ \operatorname{Ad}(h) = \chi(h) \circ d\pi_e$ , which is exactly the claimed relation

$\chi(h) = \overline{\text{Ad}}(h)$  under the identification.

The isotropy representation is the linear shadow of a homogeneous space at a point. Because  $G$  moves any point to any other,  $T_o(G/H)$  together with the  $H$ -action  $\chi$  carries the pointwise structure of  $G/H$ : a  $G$ -invariant tensor field on  $G/H$ , evaluated at  $o$ , is an  $H$ -invariant tensor on  $T_o(G/H)$  for the isotropy action, and conversely each such  $H$ -invariant tensor at  $o$  spreads by  $G$  to a unique  $G$ -invariant field. The formula  $\chi = \overline{\text{Ad}}$  makes this computable: the isotropy representation is nothing but the adjoint representation of  $G$ , restricted to  $H$  and pushed to  $\mathfrak{g}/\mathfrak{h}$ . The linear algebra of  $G/H$  is thereby read off from the pair  $(\mathfrak{g}, \mathfrak{h})$  and the adjoint action.

The running example illustrates the identification in both of its extremes, a discrete stabilizer where the quotient does not collapse and a one-dimensional stabilizer where it does.

**Example 4.5.4: Isotropy for  $\text{SO}(3) = \text{SU}(2)/\{\pm I\}$  and for  $S^2$**

Take first  $H = \{\pm I\}$ , the center of  $G = \text{SU}(2)$ , a closed discrete subgroup with  $\mathfrak{h} = 0$ . Then  $G/H = \text{SO}(3)$  by theorem 1.5.5, and  $\mathfrak{g}/\mathfrak{h} = \mathfrak{su}(2) \cong \mathbb{R}^3$  with no collapse. The isotropy representation  $\chi: \{\pm I\} \rightarrow \text{GL}(\mathfrak{su}(2))$  is  $\overline{\text{Ad}}$ ; since  $\text{Ad}(\pm I)X = (\pm I)X(\pm I)^{-1} = X$ , the representation is trivial, as it must be because a discrete stabilizer acts trivially on the tangent space of a covering quotient. This recovers that  $\text{SO}(3)$  has the same infinitesimal structure at every point as  $\text{SU}(2)$ , the two differing only by the central kernel.

Take next  $H = \text{U}(1)$ , the diagonal torus, so  $G/H = S^2$  by example 4.3.3. Use the basis  $X_1, X_2, X_3$  of  $\mathfrak{su}(2)$  from example 1.4.7, with  $X_3 = \frac{1}{2} \text{diag}(-i, i)$  and  $[X_1, X_2] = X_3$ ,  $[X_2, X_3] = X_1$ ,  $[X_3, X_1] = X_2$ . Since  $\exp(tX_3) = \text{diag}(e^{-it/2}, e^{it/2})$ , the diagonal torus is  $\text{U}(1) = \{\exp(-2\theta X_3) : \theta \in \mathbb{R}\} = \{\text{diag}(e^{i\theta}, e^{-i\theta})\}$ , so  $\mathfrak{h} = \mathfrak{u}(1) = \mathbb{R}X_3$  is one-dimensional and  $\mathfrak{g}/\mathfrak{h}$  is two-dimensional, spanned by the images of  $X_1, X_2$ , matching  $\dim S^2 = 2$ . For  $u_\theta = \exp(-2\theta X_3) \in \text{U}(1)$  the adjoint action is  $\text{Ad}(u_\theta) = \exp(-2\theta \text{ad}X_3)$ ; on the plane  $\text{span}(X_1, X_2)$  the operator  $\text{ad}X_3$  sends  $X_1 \mapsto X_2$  and  $X_2 \mapsto -X_1$ , a rotation generator, so

$$\text{Ad}(u_\theta)X_1 = \cos(2\theta)X_1 - \sin(2\theta)X_2, \quad \text{Ad}(u_\theta)X_2 = \sin(2\theta)X_1 + \cos(2\theta)X_2$$

a rotation through angle  $2\theta$  of the  $(X_1, X_2)$ -plane, while  $\text{Ad}(u_\theta)X_3 = X_3$  fixes  $\mathfrak{h}$ . Descending to  $\mathfrak{g}/\mathfrak{h}$ , the isotropy representation  $\chi(u_\theta)$  is exactly this rotation through  $2\theta$  of the tangent plane  $T_o S^2$ . The circle  $\text{U}(1)$  thus acts on the tangent plane to the sphere at the base point by rotations, the isotropy action under which the round metric on  $S^2$  is invariant.

The example separates the two extremes: a discrete stabilizer gives a trivial isotropy representation, so a covering quotient is infinitesimally indistinguishable from its total space, whereas a positive-dimensional stabilizer acts by a nontrivial rotation, encoding the rotational symmetry of the round sphere about each point. The doubling of the angle, rotation by  $2\theta$  under  $\text{Ad}(u_\theta)$ , is the same factor of two that made  $\text{SU}(2) \rightarrow \text{SO}(3)$  a double cover, now seen infinitesimally in how the torus turns the tangent plane. In both cases the tangent space and its symmetry are computed entirely inside  $\mathfrak{g}$ , with no reference to charts on  $G/H$ .

The quotient  $\mathfrak{g}/\mathfrak{h}$  is intrinsic, but working with a quotient of vector spaces is less convenient than working with a subspace. A *reductive* homogeneous space is one where the quotient can be split off as an  $\text{Ad}(H)$ -invariant complement, realizing  $T_o(G/H)$  as a concrete subspace of  $\mathfrak{g}$  on which  $H$  acts by  $\text{Ad}$  with no descent. The splitting exists whenever  $H$  is compact, by an averaging argument that will recur when invariant integration is developed for compact groups in Part III.

**Proposition 4.5.5: Reductive Decompositions and the Isotropy Representation**

Let  $H \leq G$  be a closed subgroup with Lie algebras  $\mathfrak{h} \subseteq \mathfrak{g}$ . Suppose there is a vector-space decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m} \quad \text{with} \quad \text{Ad}(h)\mathfrak{m} \subseteq \mathfrak{m} \quad \text{for all } h \in H \quad (4.6)$$

Then  $\mathfrak{m}$  is called a *reductive complement* and  $G/H$  a *reductive homogeneous space*. In this case the composite

$$\mathfrak{m} \hookrightarrow \mathfrak{g} \xrightarrow{d\pi_e} T_o(G/H)$$

is an isomorphism of vector spaces, and it intertwines the restriction  $\text{Ad}|_H$  of the adjoint representation to  $\mathfrak{m}$  with the linear isotropy representation  $\chi$  of definition 4.5.3. That is,  $T_o(G/H) \cong \mathfrak{m}$  as  $H$ -representations, with  $H$  acting on  $\mathfrak{m}$  by  $h \mapsto \text{Ad}(h)|_{\mathfrak{m}}$ . A reductive complement exists whenever  $H$  is compact.

**Proof:**

*Step 1: the complement realizes the tangent space.* By the identification in definition 4.5.3,  $d\pi_e: \mathfrak{g} \rightarrow T_o(G/H)$  is surjective with kernel  $\mathfrak{h}$ . Restricting to  $\mathfrak{m}$ , the map  $d\pi_e|_{\mathfrak{m}}: \mathfrak{m} \rightarrow T_o(G/H)$  has kernel  $\mathfrak{m} \cap \mathfrak{h} = 0$  by the directness of the sum (4.6), so it is injective, and  $\dim \mathfrak{m} = \dim \mathfrak{g} - \dim \mathfrak{h} = \dim T_o(G/H)$  by theorem 4.1.3, so it is an isomorphism.

*Step 2: it intertwines the two actions.* For  $h \in H$  the invariance  $\text{Ad}(h)\mathfrak{m} \subseteq \mathfrak{m}$  of (4.6) makes  $\text{Ad}(h)|_{\mathfrak{m}}$  a linear automorphism of  $\mathfrak{m}$ . The naturality relation  $d\pi_e \circ \text{Ad}(h) = \chi(h) \circ d\pi_e$  established in definition 4.5.3, restricted to  $\mathfrak{m}$ , reads

$$d\pi_e(\text{Ad}(h)|_{\mathfrak{m}} Y) = \chi(h) d\pi_e(Y) \quad (Y \in \mathfrak{m})$$

which says precisely that the isomorphism  $d\pi_e|_{\mathfrak{m}}$  carries the action  $\text{Ad}|_H$  on  $\mathfrak{m}$  to the isotropy action  $\chi$  on  $T_o(G/H)$ . Hence  $\mathfrak{m} \cong T_o(G/H)$  as  $H$ -representations.

*Step 3: existence for compact  $H$ .* Suppose  $H$  is compact. Choose any inner product  $\langle -, - \rangle_0$  on  $\mathfrak{g}$  and average it over  $H$  against the bi-invariant (Haar) probability measure  $dh$  on  $H$ :

$$\langle X, Y \rangle = \int_H \langle \text{Ad}(h)X, \text{Ad}(h)Y \rangle_0 dh$$

A compact group carries such a measure, its Haar measure, whose construction is developed for compact groups in Part III and whose existence for locally compact groups is cited to [4]; the integrand is continuous in  $h$ , and the integral of a positive-definite form against a probability measure is again positive definite, so  $\langle -, - \rangle$  is an inner product on  $\mathfrak{g}$ . It is  $\text{Ad}(H)$ -invariant: for  $h_0 \in H$  the substitution  $h \mapsto hh_0$  preserves the measure and gives  $\langle \text{Ad}(h_0)X, \text{Ad}(h_0)Y \rangle = \int_H \langle \text{Ad}(hh_0)X, \text{Ad}(hh_0)Y \rangle_0 dh = \langle X, Y \rangle$ . Now  $\mathfrak{h}$  is  $\text{Ad}(H)$ -invariant (definition 4.5.3), so its orthogonal complement

$$\mathfrak{m} = \mathfrak{h}^\perp = \{Y \in \mathfrak{g} \mid \langle Y, X \rangle = 0 \text{ for all } X \in \mathfrak{h}\}$$

is  $\text{Ad}(H)$ -invariant as well, since  $\text{Ad}(h)$  preserves the inner product and maps  $\mathfrak{h}$  to itself, hence maps  $\mathfrak{h}^\perp$  to  $\mathfrak{h}^\perp$ . The orthogonal decomposition  $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$  then satisfies (4.6), completing the proof.

The proposition trades the quotient  $\mathfrak{g}/\mathfrak{h}$  for a subspace  $\mathfrak{m}$  carrying the same  $H$ -representation, and this is the form in which the isotropy data is used in practice. The isotropy representation of a reductive homogeneous space is simply  $\text{Ad}(H)$  acting on  $\mathfrak{m}$ , a representation of  $H$  on a piece of the Lie algebra of  $G$ , and every  $G$ -invariant geometric structure on  $G/H$  is now an  $H$ -invariant algebraic structure on  $\mathfrak{m}$ . The compactness of  $H$  is a sufficient condition by averaging, not a necessary one; the split (4.6) can exist for noncompact  $H$ , and the symmetric spaces of Part IV supply a canonical reductive complement, the negative eigenspace of the Cartan involution, with no compactness assumption on  $H$ .

The construction identifies exactly the data from which a  $G$ -invariant metric is assembled. A  $G$ -invariant Riemannian metric on a reductive  $G/H$  is the same thing as an  $\text{Ad}(H)$ -invariant inner product on  $\mathfrak{m}$ : the value of the metric at the base point  $o$  is an inner product on  $T_o(G/H) \cong \mathfrak{m}$ , it must be invariant under the isotropy action  $\text{Ad}(H)|_{\mathfrak{m}}$  for the metric to be  $G$ -invariant, and any such inner product spreads by the transitive  $G$ -action to a unique  $G$ -invariant metric on all of  $G/H$ . When  $H$  is compact the averaging of Step 3 produces such an inner product, so every reductive homogeneous space with compact isotropy carries a  $G$ -invariant metric. This is the entry point to the theory of Riemannian symmetric spaces: there the reductive complement  $\mathfrak{m}$  is distinguished by a further bracket condition,  $[\mathfrak{m}, \mathfrak{m}] \subseteq \mathfrak{h}$ , which turns the geodesic symmetry at  $o$  into an isometry and pins down the curvature of the invariant metric in terms of the Lie bracket alone. Part IV takes up that theory, with the Cartan decomposition supplying  $\mathfrak{m}$  and the isotropy representation governing the geometry.

### Exercise 4.5.1

1. Verify directly that the Hopf fibration is a principal  $U(1)$ -bundle in the sense of theorem 4.5.1: realize  $U(1)$  as the diagonal torus of  $SU(2)$ , describe the right action on  $S^3 \cong SU(2)$  (proposition 1.5.1) explicitly, and exhibit the fibers as circles. Explain why the bundle cannot be globally trivial by comparing the fundamental groups of  $S^3$  and  $S^2 \times S^1$ .
2. Show that the tangent bundle of  $G/H$  is the associated bundle  $T(G/H) \cong G \times_H (\mathfrak{g}/\mathfrak{h})$ , where  $H$  acts on  $\mathfrak{g}/\mathfrak{h}$  by the isotropy representation  $\overline{\text{Ad}}$  of definition 4.5.3. When  $G/H$  is reductive with complement  $\mathfrak{m}$ , rewrite this as  $T(G/H) \cong G \times_H \mathfrak{m}$  and identify the fiber over  $gH$  with  $g \cdot \mathfrak{m}$ .
3. Let  $G/H$  be reductive with  $H$  compact, and let  $\langle -, - \rangle$  be an  $\text{Ad}(H)$ -invariant inner product on the reductive complement  $\mathfrak{m}$  (which exists by proposition 4.5.5). Construct from it a  $G$ -invariant Riemannian metric on  $G/H$ , and prove it is well defined, that is, independent of the coset representative  $g$  used to identify  $T_{gH}(G/H)$  with  $\mathfrak{m}$ .
4. Let  $M$  be a smooth  $n$ -manifold and let  $\text{Fr}(M)$  be its frame bundle, whose fiber over  $p \in M$  is the set of ordered bases (frames) of  $T_p M$ . Show that the right action of  $\text{GL}_n(\mathbb{R})$  on frames makes  $\text{Fr}(M) \rightarrow M$  a principal  $\text{GL}_n(\mathbb{R})$ -bundle, and identify the tangent bundle  $TM$  as the associated bundle  $\text{Fr}(M) \times_{\text{GL}_n(\mathbb{R})} \mathbb{R}^n$  for the standard representation of  $\text{GL}_n(\mathbb{R})$  on  $\mathbb{R}^n$ .
5. For the sphere  $S^{n-1} = \text{SO}(n)/\text{SO}(n-1)$  (example 4.3.1), compute the isotropy representation of  $H = \text{SO}(n-1)$  on  $\mathfrak{g}/\mathfrak{h} = \mathfrak{so}(n)/\mathfrak{so}(n-1)$ . Identify the reductive complement  $\mathfrak{m}$  explicitly and show the isotropy representation is the standard representation of  $\text{SO}(n-1)$  on  $\mathbb{R}^{n-1}$ . Conclude that  $S^{n-1}$  carries an  $\text{SO}(n)$ -invariant metric, the round metric.
6. Give an example of a closed subgroup  $H \leq G$  for which no reductive complement exists, so that  $G/H$  is not reductive, and explain which step of the compact-case argument in

proposition 4.5.5 fails. (Hint: arrange the adjoint action of  $H$  on  $\mathfrak{g}$  so that  $\mathfrak{h}$  is an invariant subspace with no invariant complement, which is possible only when  $H$  is noncompact.)



## Part II

# The Structure of Lie Algebras



# 5

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## *Nilpotent and Solvable Lie Algebras*

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Part I built the dictionary between a Lie group and its Lie algebra. From here through the end of Part II the group recedes and the Lie algebra is studied on its own terms, as a finite-dimensional vector space equipped with a bracket. The destination of the part is the classification of the semisimple Lie algebras by their Dynkin diagrams, the combinatorial skeleton that will drive the representation theory of Part III. The route to it runs through the opposite of semisimplicity: to isolate the semisimple algebras one must first understand the solvable ones, the algebras built by successive abelian extensions, and the two classical theorems, Engel's and Lie's, that bring a solvable or nilpotent algebra of operators into triangular form.

This chapter develops solvability and nilpotency from the ground up. Two descending chains of ideals, the derived series and the lower central series, measure how far a Lie algebra stands from being abelian; the algebra is called solvable or nilpotent precisely when the respective chain reaches zero. Engel's theorem characterizes nilpotency infinitesimally, through the nilpotence of the operators  $\text{ad}X$ , while Lie's theorem shows that over an algebraically closed field of characteristic zero every solvable algebra of matrices can be put simultaneously into upper-triangular form. The chapter closes by extracting from an arbitrary Lie algebra its radical, the largest solvable ideal, whose vanishing is the definition of the semisimple algebras taken up in ???. Unless stated otherwise the ground field throughout Part II is  $\mathbb{C}$ , or any algebraically closed field of characteristic zero where only those hypotheses are used; the descent from  $\mathfrak{g} \otimes \mathbb{C}$  back to a real Lie algebra is the business of Part IV.

### 5.1 Ideals, Quotients, and the Derived Series

Part I built the dictionary between a Lie group and its Lie algebra and turned it into a tool. Part II now spends the whole of its effort inside the algebra. The reason is one of leverage: the structure theory of Lie algebras is finite-dimensional linear algebra with a bracket, and once it is complete it transfers back to groups through the correspondence theorems of Chapter 3 without further analysis.

The goal of the next four chapters is a classification of the simplest interesting Lie algebras, the semisimple ones, and the route runs through their opposite. A semisimple algebra is one with no solvable part at all; to isolate it one must first understand what solvable means, which is the business of this chapter.

The organizing idea is that a Lie algebra can be dismantled the way a group is, by quotienting out a substructure and studying the pieces. For groups the substructures that permit quotients are the normal subgroups; for Lie algebras they are the ideals. This section installs ideals, the quotient algebra, and the homomorphism theorems that make the quotient behave, then uses them to build the first structural invariant, the derived series, whose termination is the definition of solvability. Solvable algebras are the ones assembled from abelian layers by repeated extension, and they are exactly the algebras that Lie's theorem of section 5.4 will triangularize.

The subalgebra of definition 3.1.2 was the linear analogue of a subgroup, a subspace closed under the one operation the algebra carries. It is the wrong notion for forming quotients. To divide  $\mathfrak{g}$  by a subspace  $\mathfrak{a}$  and inherit a bracket on the cosets, the value  $[X + \mathfrak{a}, Y + \mathfrak{a}]$  must not depend on the coset representatives, and this forces a stronger closure condition than bracket closure within  $\mathfrak{a}$ : the bracket of anything in  $\mathfrak{g}$  with anything in  $\mathfrak{a}$  must land back in  $\mathfrak{a}$ . That condition is what an ideal records, and it is the exact linearization of normality.

#### Definition 5.1.1: Ideal and Quotient Lie Algebra

Let  $\mathfrak{g}$  be a Lie algebra over a field  $k$ . An *ideal* of  $\mathfrak{g}$  is a  $k$ -linear subspace  $\mathfrak{a} \subseteq \mathfrak{g}$  such that

$$[\mathfrak{g}, \mathfrak{a}] \subseteq \mathfrak{a}, \quad \text{that is} \quad [X, Y] \in \mathfrak{a} \text{ for all } X \in \mathfrak{g}, Y \in \mathfrak{a}$$

written  $\mathfrak{a} \trianglelefteq \mathfrak{g}$ . Because the bracket is antisymmetric, the one-sided condition  $[\mathfrak{g}, \mathfrak{a}] \subseteq \mathfrak{a}$  already gives  $[\mathfrak{a}, \mathfrak{g}] \subseteq \mathfrak{a}$ , so there is no distinction between left and right ideals. Every ideal is in particular a subalgebra.

Given an ideal  $\mathfrak{a} \trianglelefteq \mathfrak{g}$ , the *quotient Lie algebra*  $\mathfrak{g}/\mathfrak{a}$  is the quotient vector space with the bracket

$$[X + \mathfrak{a}, Y + \mathfrak{a}] := [X, Y] + \mathfrak{a}$$

This is well defined, and it makes  $\mathfrak{g}/\mathfrak{a}$  a Lie algebra for which the projection  $\pi: \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{a}$ ,  $X \mapsto X + \mathfrak{a}$ , is a homomorphism.

The one point requiring proof in the definition is that the bracket on cosets is well defined; everything else is inherited from  $\mathfrak{g}$ . Suppose  $X + \mathfrak{a} = X' + \mathfrak{a}$  and  $Y + \mathfrak{a} = Y' + \mathfrak{a}$ , so  $X' = X + A$  and  $Y' = Y + B$  with  $A, B \in \mathfrak{a}$ . Bilinearity expands the bracket of the primed representatives,

$$[X', Y'] = [X + A, Y + B] = [X, Y] + [X, B] + [A, Y] + [A, B]$$

and each of the last three terms lies in  $\mathfrak{a}$ : the terms  $[X, B]$  and  $[A, B]$  because  $B \in \mathfrak{a}$  and  $\mathfrak{a}$  is an ideal, and  $[A, Y]$  because  $A \in \mathfrak{a}$ . Hence  $[X', Y'] - [X, Y] \in \mathfrak{a}$ , so  $[X', Y'] + \mathfrak{a} = [X, Y] + \mathfrak{a}$ , and the coset bracket does not depend on the representatives. The bracket axioms of definition 1.4.5, bilinearity, the alternating law, and the Jacobi identity, then pass to the quotient because each is an identity that holds representative by representative. This is the same mechanism that produces the quotient group of a group by a normal subgroup, transported to the additive setting where the subspace  $\mathfrak{a}$  plays the role of the normal subgroup and the ideal condition plays the role of normality.

The parallel with groups is more than formal; it is the source of the homomorphism theorems, which hold for Lie algebras in exactly the form the reader knows from [1]. These are the tools that let one

recognize an ideal as a kernel and a quotient as an image, and they are used at every turn in the structure theory to follow.

**Proposition 5.1.2: Kernel, Image, and the First Isomorphism Theorem**

Let  $\varphi: \mathfrak{g} \rightarrow \mathfrak{h}$  be a homomorphism of Lie algebras.

1. The kernel  $\ker \varphi = \{X \in \mathfrak{g} \mid \varphi(X) = 0\}$  is an ideal of  $\mathfrak{g}$ .
2. The image  $\varphi(\mathfrak{g})$  is a subalgebra of  $\mathfrak{h}$ .
3. The induced map  $\bar{\varphi}: \mathfrak{g}/\ker \varphi \rightarrow \varphi(\mathfrak{g})$ ,  $X + \ker \varphi \mapsto \varphi(X)$ , is an isomorphism of Lie algebras.

**Proof:**

The three statements are proved in order, each a short verification once the ideal condition is written out.

1. *Kernel is an ideal.* The kernel is a linear subspace, being the null space of the linear map  $\varphi$ . For the ideal condition, let  $X \in \mathfrak{g}$  and  $Y \in \ker \varphi$ . Then

$$\varphi([X, Y]) = [\varphi(X), \varphi(Y)] = [\varphi(X), 0] = 0$$

using that  $\varphi$  preserves brackets and  $\varphi(Y) = 0$ , so  $[X, Y] \in \ker \varphi$ . Hence  $[\mathfrak{g}, \ker \varphi] \subseteq \ker \varphi$ , and  $\ker \varphi$  is an ideal.

2. *Image is a subalgebra.* The image is a linear subspace, being the range of a linear map. If  $\varphi(X), \varphi(Y)$  are two elements of  $\varphi(\mathfrak{g})$ , then their bracket  $[\varphi(X), \varphi(Y)] = \varphi([X, Y])$  is again in the image, so  $\varphi(\mathfrak{g})$  is closed under the bracket and is a subalgebra. It need not be an ideal of  $\mathfrak{h}$ : the inclusion of a subalgebra that is not an ideal, such as the diagonal matrices inside  $\mathfrak{gl}_2$ , is a homomorphism whose image is not an ideal.

3. *First isomorphism theorem.* By part (1) the quotient  $\mathfrak{g}/\ker \varphi$  is a Lie algebra. The linear map  $\bar{\varphi}(X + \ker \varphi) = \varphi(X)$  is well defined, since  $X - X' \in \ker \varphi$  gives  $\varphi(X) = \varphi(X')$ , and it is a linear isomorphism onto  $\varphi(\mathfrak{g})$  by the linear-algebra first isomorphism theorem of [1]. It preserves the bracket:

$$\bar{\varphi}([X + \ker \varphi, Y + \ker \varphi]) = \bar{\varphi}([X, Y] + \ker \varphi) = \varphi([X, Y]) = [\varphi(X), \varphi(Y)]$$

which equals  $[\bar{\varphi}(X + \ker \varphi), \bar{\varphi}(Y + \ker \varphi)]$ . A bracket-preserving linear isomorphism is an isomorphism of Lie algebras, as we sought to show.

The proposition says that ideals and kernels are the same objects seen from two sides: every kernel is an ideal by part (1), and conversely every ideal  $\mathfrak{a}$  is the kernel of its own projection  $\pi: \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{a}$ . So the ideals of  $\mathfrak{g}$  are precisely the substructures one can quotient by, and each quotient realizes  $\mathfrak{g}$  as an extension of  $\mathfrak{g}/\mathfrak{a}$  by  $\mathfrak{a}$ . The contrast with the subalgebra of definition 3.1.2 is the whole point of introducing the two notions separately: a subalgebra sits inside  $\mathfrak{g}$  and is a Lie algebra in its own right, but one cannot in general form  $\mathfrak{g}$  modulo a subalgebra, because the coset bracket is ill defined; an ideal is a subalgebra strong enough to divide by. The diagonal matrices  $\mathfrak{d} \subseteq \mathfrak{gl}_2$  mark the gap concretely:  $\mathfrak{d}$  is a subalgebra of  $\mathfrak{gl}_2$ , being abelian and closed under the bracket, but it is not an ideal,

since  $[E_{11}, E_{12}] = E_{12} \notin \mathfrak{d}$  places the bracket of a diagonal matrix with an off-diagonal one outside  $\mathfrak{d}$ . This same  $\mathfrak{d}$  reappears inside  $\mathfrak{b}_2$  in the computation of example 5.1.7 below as a subalgebra that is not an ideal of the solvable algebra  $\mathfrak{b}_2$ .

A ready supply of ideals comes from the bracket itself. The most important is built by bracketing  $\mathfrak{g}$  with itself and then iterating, and it measures how far  $\mathfrak{g}$  is from being abelian.

### Definition 5.1.3: The Derived Algebra and the Derived Series

Let  $\mathfrak{g}$  be a Lie algebra. The *derived algebra* of  $\mathfrak{g}$  is the linear span

$$[\mathfrak{g}, \mathfrak{g}] = \text{span}_k \{[X, Y] \mid X, Y \in \mathfrak{g}\}$$

The *derived series* of  $\mathfrak{g}$  is the descending chain of subalgebras defined by

$$\mathfrak{g}^{(0)} = \mathfrak{g}, \quad \mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]$$

so that  $\mathfrak{g}^{(1)} = [\mathfrak{g}, \mathfrak{g}]$  is the derived algebra,  $\mathfrak{g}^{(2)} = [[\mathfrak{g}, \mathfrak{g}], [\mathfrak{g}, \mathfrak{g}]]$ , and so on. The chain

$$\mathfrak{g} = \mathfrak{g}^{(0)} \supseteq \mathfrak{g}^{(1)} \supseteq \mathfrak{g}^{(2)} \supseteq \dots$$

is decreasing, and each  $\mathfrak{g}^{(k)}$  is an ideal of  $\mathfrak{g}$ .

Two claims in the definition need verification: that each  $\mathfrak{g}^{(k)}$  is an ideal of  $\mathfrak{g}$  (not only of  $\mathfrak{g}^{(k-1)}$ ), and that the chain descends. Both follow from a single fact about how the bracket construction interacts with symmetries of  $\mathfrak{g}$ .

### Lemma 5.1.4: The Derived Series Consists of Characteristic Ideals

For every Lie algebra  $\mathfrak{g}$  and every  $k \geq 0$ , the subspace  $\mathfrak{g}^{(k)}$  is an ideal of  $\mathfrak{g}$ , and it is moreover *characteristic*: every derivation  $D: \mathfrak{g} \rightarrow \mathfrak{g}$  satisfies  $D(\mathfrak{g}^{(k)}) \subseteq \mathfrak{g}^{(k)}$ . In particular  $\mathfrak{g}^{(k)}$  is invariant under  $\text{ad}X$  for every  $X \in \mathfrak{g}$ , and the chain descends,  $\mathfrak{g}^{(k+1)} \subseteq \mathfrak{g}^{(k)}$ .

#### Proof:

Recall that a *derivation* of  $\mathfrak{g}$  is a linear map  $D: \mathfrak{g} \rightarrow \mathfrak{g}$  satisfying the Leibniz rule  $D[X, Y] = [DX, Y] + [X, DY]$ ; for each  $X \in \mathfrak{g}$  the operator  $\text{ad}X$  of definition 2.5.3 is a derivation, which is precisely the Jacobi identity of definition 1.4.5 read as  $\text{ad}X[Y, Z] = [\text{ad}X Y, Z] + [Y, \text{ad}X Z]$ . The proof is by induction on  $k$ , the characteristic property being the strengthening that makes the induction close.

*Base case.* For  $k = 0$ ,  $\mathfrak{g}^{(0)} = \mathfrak{g}$  is carried into itself by any linear map  $\mathfrak{g} \rightarrow \mathfrak{g}$ , so it is characteristic.

*Inductive step.* Suppose  $\mathfrak{g}^{(k)}$  is characteristic, and let  $D$  be any derivation. A general element of  $\mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]$  is a sum of brackets  $[U, V]$  with  $U, V \in \mathfrak{g}^{(k)}$ , so by linearity of  $D$  it suffices to treat one such bracket. The Leibniz rule gives

$$D[U, V] = [DU, V] + [U, DV]$$

By the inductive hypothesis  $DU, DV \in \mathfrak{g}^{(k)}$ , so both  $[DU, V]$  and  $[U, DV]$  lie in  $[\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}] = \mathfrak{g}^{(k+1)}$ . Hence  $D(\mathfrak{g}^{(k+1)}) \subseteq \mathfrak{g}^{(k+1)}$ , and  $\mathfrak{g}^{(k+1)}$  is characteristic.

Specializing  $D = \text{ad}X$  shows  $[X, W] \in \mathfrak{g}^{(k)}$  for every  $X \in \mathfrak{g}$  and  $W \in \mathfrak{g}^{(k)}$ , which is the statement

$[\mathfrak{g}, \mathfrak{g}^{(k)}] \subseteq \mathfrak{g}^{(k)}$ , so each  $\mathfrak{g}^{(k)}$  is an ideal of  $\mathfrak{g}$ . The chain descends because  $\mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}] \subseteq \mathfrak{g}^{(k)}$ , the bracket of the subalgebra  $\mathfrak{g}^{(k)}$  with itself landing inside  $\mathfrak{g}^{(k)}$ , completing the proof.

The lemma isolates the reason the derived series is a good invariant: it is built canonically from the bracket, so it is preserved not only by inner automorphisms but by every derivation, and in particular by every automorphism of  $\mathfrak{g}$ . An invariant this rigid cannot be moved by a change of basis or by any symmetry of the algebra, which is what allows the next definition to read off a genuine property of  $\mathfrak{g}$  from where the series stops.

The derived algebra  $[\mathfrak{g}, \mathfrak{g}]$  vanishes exactly when  $\mathfrak{g}$  is abelian, so the first step of the derived series measures the failure of commutativity. Iterating asks whether that failure itself commutes, and so on down. If the process reaches zero, the algebra has been resolved into abelian layers.

#### Definition 5.1.5: Solvable Lie Algebra

A Lie algebra  $\mathfrak{g}$  is *solvable* if its derived series reaches zero: there is an integer  $n \geq 0$  with

$$\mathfrak{g}^{(n)} = 0$$

The least such  $n$  is the *derived length* of  $\mathfrak{g}$ .

An abelian Lie algebra has  $\mathfrak{g}^{(1)} = [\mathfrak{g}, \mathfrak{g}] = 0$  and so is solvable of derived length at most one; conversely a solvable algebra of derived length at most one is abelian. Solvability is thus the property of being built from abelian pieces by iterated extension. Reading the derived series from the bottom,  $\mathfrak{g}^{(n)} = 0$  says  $\mathfrak{g}^{(n-1)}$  is abelian; each quotient  $\mathfrak{g}^{(k)}/\mathfrak{g}^{(k+1)}$  is abelian by construction, since  $[\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}] = \mathfrak{g}^{(k+1)}$  is exactly what is divided out; and  $\mathfrak{g}$  is assembled from the abelian quotients  $\mathfrak{g}^{(0)}/\mathfrak{g}^{(1)}, \mathfrak{g}^{(1)}/\mathfrak{g}^{(2)}, \dots, \mathfrak{g}^{(n-1)}/\mathfrak{g}^{(n)}$  by a tower of extensions. This is the sense in which solvable algebras are the ones with no rigid, non-abelian core, and it is the exact linear counterpart of a solvable group, whose derived series of iterated commutator subgroups terminates. The name is inherited from Galois theory through that group-theoretic parallel, where solvability of a group governs solvability of a polynomial by radicals.

The closure properties of the solvable class are what make it usable. They say solvability is preserved by the two operations that build and dismantle algebras, passing to subalgebras and quotients, and that it is stable under extension and under summing ideals. The last of these is the fact that section 5.5 will use to produce a unique maximal solvable ideal, the radical, whose vanishing defines semisimplicity.

#### Proposition 5.1.6: Closure Properties of Solvable Lie Algebras

Let  $\mathfrak{g}$  be a Lie algebra.

1. Every subalgebra and every quotient of a solvable Lie algebra is solvable.
2. If  $\mathfrak{a} \trianglelefteq \mathfrak{g}$  is an ideal such that both  $\mathfrak{a}$  and  $\mathfrak{g}/\mathfrak{a}$  are solvable, then  $\mathfrak{g}$  is solvable.
3. If  $\mathfrak{a}, \mathfrak{b} \trianglelefteq \mathfrak{g}$  are solvable ideals, then their sum  $\mathfrak{a} + \mathfrak{b}$  is a solvable ideal.

**Proof:**

The proofs turn on one observation about how the derived series behaves under homomorphisms. If  $\varphi: \mathfrak{g} \rightarrow \mathfrak{h}$  is a homomorphism, then

$$\varphi(\mathfrak{g}^{(k)}) = \varphi(\mathfrak{g})^{(k)} \quad \text{for every } k \geq 0 \quad (5.1)$$

because  $\varphi$  preserves brackets:  $\varphi([\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]) = [\varphi(\mathfrak{g}^{(k)}), \varphi(\mathfrak{g}^{(k)})]$ , and induction on  $k$  propagates the equality up the series from the base case  $\varphi(\mathfrak{g}^{(0)}) = \varphi(\mathfrak{g})$ . Two consequences are used repeatedly. For a subalgebra  $\mathfrak{h} \subseteq \mathfrak{g}$ , applying the bracket inside  $\mathfrak{h}$  gives  $\mathfrak{h}^{(k)} \subseteq \mathfrak{g}^{(k)}$ . For the quotient projection  $\pi: \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{a}$ , equation (5.1) reads  $(\mathfrak{g}/\mathfrak{a})^{(k)} = \pi(\mathfrak{g}^{(k)})$ .

1. *Subalgebras and quotients.* Let  $\mathfrak{g}$  be solvable with  $\mathfrak{g}^{(n)} = 0$ . If  $\mathfrak{h} \subseteq \mathfrak{g}$  is a subalgebra, then  $\mathfrak{h}^{(n)} \subseteq \mathfrak{g}^{(n)} = 0$ , so  $\mathfrak{h}$  is solvable. If  $\pi: \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{a}$  is the quotient by an ideal, then  $(\mathfrak{g}/\mathfrak{a})^{(n)} = \pi(\mathfrak{g}^{(n)}) = \pi(0) = 0$ , so the quotient is solvable.
2. *Extensions.* Suppose  $\mathfrak{a}$  and  $\mathfrak{g}/\mathfrak{a}$  are solvable, say  $(\mathfrak{g}/\mathfrak{a})^{(m)} = 0$  and  $\mathfrak{a}^{(p)} = 0$ . Let  $\pi: \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{a}$  be the projection. From  $(\mathfrak{g}/\mathfrak{a})^{(m)} = \pi(\mathfrak{g}^{(m)}) = 0$  one gets  $\mathfrak{g}^{(m)} \subseteq \ker \pi = \mathfrak{a}$ . The derived series composes with itself, in the sense that  $(\mathfrak{g}^{(m)})^{(p)} = \mathfrak{g}^{(m+p)}$ , since applying the derived operation  $p$  times to  $\mathfrak{g}^{(m)}$  continues the same recursion. Because  $\mathfrak{g}^{(m)} \subseteq \mathfrak{a}$  is a subalgebra, the subalgebra case of the opening observation gives  $(\mathfrak{g}^{(m)})^{(p)} \subseteq \mathfrak{a}^{(p)}$ , and therefore

$$\mathfrak{g}^{(m+p)} = (\mathfrak{g}^{(m)})^{(p)} \subseteq \mathfrak{a}^{(p)} = 0$$

Hence  $\mathfrak{g}^{(m+p)} = 0$  and  $\mathfrak{g}$  is solvable.

3. *Sum of solvable ideals.* That  $\mathfrak{a} + \mathfrak{b}$  is an ideal is immediate: it is a subspace, and  $[\mathfrak{g}, \mathfrak{a} + \mathfrak{b}] \subseteq [\mathfrak{g}, \mathfrak{a}] + [\mathfrak{g}, \mathfrak{b}] \subseteq \mathfrak{a} + \mathfrak{b}$ . For solvability, apply the second isomorphism theorem, valid for Lie algebras by the same coset argument as for modules ([1]): since  $\mathfrak{a}$  and  $\mathfrak{b}$  are ideals,  $\mathfrak{b} \trianglelefteq \mathfrak{a} + \mathfrak{b}$  and

$$(\mathfrak{a} + \mathfrak{b})/\mathfrak{b} \cong \mathfrak{a}/(\mathfrak{a} \cap \mathfrak{b})$$

as Lie algebras. The right-hand side is a quotient of the solvable algebra  $\mathfrak{a}$ , hence solvable by part (1). Thus  $(\mathfrak{a} + \mathfrak{b})/\mathfrak{b}$  is solvable, and  $\mathfrak{b}$  is solvable by hypothesis, so  $\mathfrak{a} + \mathfrak{b}$  is solvable by part (2), completing the proof.

The three properties fit together into a single structural picture. Parts (1) and (2) say the solvable algebras form a class closed under sub-objects, quotients, and extensions, which is precisely the closure a class must have to support a decomposition theory. Part (3) is the one that does not follow formally from the others, and it carries the most weight downstream: among all ideals of  $\mathfrak{g}$ , the solvable ones are closed under addition, so their sum is again solvable. Since  $\mathfrak{g}$  is finite-dimensional this sum stabilizes, and section 5.5 takes the sum of *all* solvable ideals to define the radical  $\text{rad } \mathfrak{g}$ , the unique maximal solvable ideal, well defined precisely because of part (3). Semisimplicity is then the vanishing of this radical, so the entire dichotomy that organizes the classification, solvable versus semisimple, rests on the closure statement just proved.

The definitions have so far been abstract; the class of examples that makes them concrete, and that recurs through the whole chapter, is the upper-triangular matrices. This is where solvability is not just a formal property but a visible one, the derived series descending step by step through the strictly-upper-triangular layers until it reaches zero.

**Example 5.1.7: The Upper Triangular Algebra  $\mathfrak{b}_n$  is Solvable**

Let  $\mathfrak{b}_n \subseteq \mathfrak{gl}_n(\mathbb{C})$  be the subalgebra of upper-triangular matrices, those with  $X_{ij} = 0$  for  $i > j$ , and let  $\mathfrak{n}_n \subseteq \mathfrak{b}_n$  be the *strictly* upper-triangular matrices, with  $X_{ij} = 0$  for  $i \geq j$ . Both are subalgebras of  $\mathfrak{gl}_n$ : the product of two upper-triangular matrices is upper-triangular, so the commutator is too, and likewise for strictly upper-triangular matrices. Write  $E_{ij}$  for the matrix unit with a single 1 in position  $(i, j)$ ; the bracket of matrix units is

$$[E_{ij}, E_{kl}] = \delta_{jk} E_{il} - \delta_{li} E_{kj} \quad (5.2)$$

The claim is that the derived algebra of  $\mathfrak{b}_n$  is exactly  $\mathfrak{n}_n$ ,

$$[\mathfrak{b}_n, \mathfrak{b}_n] = \mathfrak{n}_n$$

One inclusion is a diagonal count. Write  $X = D_X + N_X$  with  $D_X$  diagonal and  $N_X \in \mathfrak{n}_n$ ; the diagonal entries of a product of upper-triangular matrices are the products of the diagonal entries, so the diagonal part of  $XY$  is  $D_X D_Y$ , and diagonal matrices commute. Hence the diagonal part of  $[X, Y]$  is  $D_X D_Y - D_Y D_X = 0$ , so every bracket  $[X, Y]$  has zero diagonal, giving  $[\mathfrak{b}_n, \mathfrak{b}_n] \subseteq \mathfrak{n}_n$ . For the reverse inclusion, produce each generator  $E_{ij}$  with  $i < j$  as a bracket. Using (5.2) with the diagonal unit  $E_{ii} \in \mathfrak{b}_n$ ,

$$[E_{ii}, E_{ij}] = \delta_{ii} E_{ij} - \delta_{ji} E_{ii} = E_{ij} \quad (i < j)$$

since  $\delta_{ii} = 1$  and  $\delta_{ji} = 0$  for  $i \neq j$ , and  $E_{ij} \in \mathfrak{b}_n$ . Thus every strictly-upper-triangular generator is a bracket of two elements of  $\mathfrak{b}_n$ , so  $\mathfrak{n}_n \subseteq [\mathfrak{b}_n, \mathfrak{b}_n]$ , and the two inclusions give  $[\mathfrak{b}_n, \mathfrak{b}_n] = \mathfrak{n}_n$ .

The derived series now descends through the strictly-upper-triangular layers. Bracketing raises the superdiagonal on which the nonzero entries sit: write  $\mathfrak{n}_n^{[p]}$  for the matrices supported on the  $p$ th superdiagonal and beyond (entries  $X_{ij}$  with  $j - i \geq p$ ), so  $\mathfrak{n}_n = \mathfrak{n}_n^{[1]}$ . If  $X \in \mathfrak{n}_n^{[p]}$  and  $Y \in \mathfrak{n}_n^{[q]}$ , then (5.2) applied entrywise shows every nonzero entry of  $[X, Y]$  has row-to-column gap at least  $p + q$ , so  $[\mathfrak{n}_n^{[p]}, \mathfrak{n}_n^{[q]}] \subseteq \mathfrak{n}_n^{[p+q]}$ . Consequently the derived series of  $\mathfrak{b}_n$  satisfies

$$\mathfrak{b}_n^{(0)} = \mathfrak{b}_n, \quad \mathfrak{b}_n^{(1)} = \mathfrak{n}_n = \mathfrak{n}_n^{[1]}, \quad \mathfrak{b}_n^{(2)} \subseteq \mathfrak{n}_n^{[2]}, \quad \mathfrak{b}_n^{(k)} \subseteq \mathfrak{n}_n^{[2^{k-1}]} \quad (k \geq 1)$$

the last inclusion by induction, since  $\mathfrak{b}_n^{(k+1)} = [\mathfrak{b}_n^{(k)}, \mathfrak{b}_n^{(k)}] \subseteq [\mathfrak{n}_n^{[2^{k-1}]}, \mathfrak{n}_n^{[2^{k-1}]}] \subseteq \mathfrak{n}_n^{[2^k]}$ . Once  $2^{k-1} \geq n$  the superdiagonal index exceeds the matrix size and  $\mathfrak{n}_n^{[2^{k-1}]} = 0$ , so  $\mathfrak{b}_n^{(k)} = 0$  for  $k$  large. Therefore  $\mathfrak{b}_n$  is solvable. For  $n = 2$  the computation is fully explicit:  $\mathfrak{b}_2$  is spanned by  $E_{11}, E_{22}, E_{12}$ , its derived algebra is  $\mathfrak{n}_2 = \mathbb{C}E_{12}$ , which is one-dimensional and abelian, so  $\mathfrak{b}_2^{(2)} = 0$  and  $\mathfrak{b}_2$  has derived length 2.

The example is the concrete shape of solvability. The upper-triangular algebra  $\mathfrak{b}_n$  is not abelian, yet its derived series collapses to zero, and the descent is legible: each application of the bracket pushes the nonzero entries a full block up toward the corner, and the finite size of the matrix forces termination. Two features of this example are worth recording. The subalgebra  $\mathfrak{b}_n$  is the standard model of a *Borel subalgebra*, the maximal solvable subalgebra of  $\mathfrak{gl}_n$ ; and its diagonal subalgebra  $\mathfrak{d}$ , the diagonal matrices, is a subalgebra of  $\mathfrak{b}_n$  that is not an ideal of  $\mathfrak{b}_n$ , since  $[E_{11}, E_{12}] = E_{12}$  lands outside  $\mathfrak{d}$ . This is the concrete instance of the subalgebra-not-ideal gap promised after proposition 5.1.2.

The other feature is a warning that separates this chapter's two central notions. Although  $\mathfrak{b}_n$  is solvable, it is *not* nilpotent in the sense of section 5.2: its lower central series does not reach zero,

because bracketing the off-diagonal entries against the diagonal part of  $\mathfrak{b}_n$  reproduces those same off-diagonal entries indefinitely rather than shrinking them, as the computation  $[E_{ii}, E_{ij}] = E_{ij}$  above shows. The strictly upper-triangular algebra  $\mathfrak{n}_n$ , by contrast, is nilpotent, as example 5.2.5 will verify. Solvability and nilpotency are therefore distinct properties, and the pair  $\mathfrak{n}_n \subseteq \mathfrak{b}_n$  is the minimal example exhibiting the difference: nilpotency will imply solvability in section 5.2, but the diagonal directions of  $\mathfrak{b}_n$  obstruct the reverse implication.

Solvability is the first of the two structural properties Part II uses to carve out the semisimple algebras. In the language that section 5.5 will make precise, a Lie algebra is *semisimple* when it has no nonzero solvable ideal, so semisimplicity is the exact negation of harboring a solvable part. In this sense the upper-triangular algebra  $\mathfrak{b}_n$  sits at the opposite pole from the semisimple algebras: it is solvable, hence equal to its own radical, hence as far from semisimple as an algebra can be. The whole of Part II studies how a general Lie algebra is glued from a solvable piece and a semisimple piece, a splitting (the Levi decomposition) that the Cartan criteria of ?? make available; the equivalence of the radical-zero definition of semisimplicity with the nondegeneracy of the Killing form is proved there as well. Everything downstream begins with the solvable end of the spectrum, which is why it comes first.

### Exercise 5.1.1

1. Prove that a Lie algebra  $\mathfrak{g}$  is solvable if and only if there is a chain of subalgebras

$$\mathfrak{g} = \mathfrak{g}_0 \supseteq \mathfrak{g}_1 \supseteq \cdots \supseteq \mathfrak{g}_m = 0$$

in which each  $\mathfrak{g}_{i+1}$  is an ideal of  $\mathfrak{g}_i$  and each quotient  $\mathfrak{g}_i/\mathfrak{g}_{i+1}$  is abelian. (This characterizes solvability by an abelian tower; note that the  $\mathfrak{g}_i$  need only be ideals in the preceding term, not in  $\mathfrak{g}$ .)

2. Let  $\mathfrak{g}$  be the two-dimensional Lie algebra over  $\mathbb{C}$  with basis  $X, Y$  and bracket  $[X, Y] = X$ , the remaining brackets fixed by antisymmetry. Verify that this is a Lie algebra, compute its derived series, and conclude that it is solvable but not abelian. Realize  $\mathfrak{g}$  as a subalgebra of  $\mathfrak{b}_2$ .
3. Show that the quotient  $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$  is abelian, and that it is the *maximal* abelian quotient of  $\mathfrak{g}$ : for any ideal  $\mathfrak{a} \trianglelefteq \mathfrak{g}$  with  $\mathfrak{g}/\mathfrak{a}$  abelian, one has  $[\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{a}$ , so  $\mathfrak{g}/\mathfrak{a}$  is a quotient of  $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ . (This quotient is the *abelianization* of  $\mathfrak{g}$ .)
4. Let  $\mathfrak{a} \trianglelefteq \mathfrak{g}$  be an ideal. Prove that  $\mathfrak{g}$  is solvable if and only if both  $\mathfrak{a}$  and  $\mathfrak{g}/\mathfrak{a}$  are solvable. (One direction is proposition 5.1.6; supply the other and assemble the equivalence.)
5. Give an example of two solvable subalgebras  $\mathfrak{h}_1, \mathfrak{h}_2$  of a Lie algebra  $\mathfrak{g}$  whose sum  $\mathfrak{h}_1 + \mathfrak{h}_2$  is not solvable, and explain why this does not contradict part (3) of proposition 5.1.6. (Hint: inside  $\mathfrak{sl}_2(\mathbb{C})$ , take the upper-triangular and lower-triangular traceless matrices.)
6. Let  $\mathfrak{g}$  be a finite-dimensional Lie algebra. Using proposition 5.1.6(3) and finite-dimensionality, prove that  $\mathfrak{g}$  has a unique maximal solvable ideal, that is, a solvable ideal containing every solvable ideal of  $\mathfrak{g}$ . (This is the radical of section 5.5; the point here is that finite-dimensionality plus closure under sums forces existence.)

## 5.2 The Lower Central Series and Nilpotency

The derived series of section 5.1 measures how far a Lie algebra stands from being abelian by iterating the commutator on the whole of the previous term: at each stage one brackets  $\mathfrak{g}^{(k)}$  with itself. There is a second, finer descending chain that brackets against all of  $\mathfrak{g}$  at every stage, refusing to let the terms shrink so quickly. This is the lower central series, and the algebras it drives to zero are the *nilpotent* ones. Nilpotency is a strictly stronger condition than solvability, and the gap between the two chains is exactly the gap between the two structure theorems that organize this chapter: solvability is the hypothesis of Lie's theorem, nilpotency the hypothesis of Engel's. The nilpotent algebras are the closest a nonabelian Lie algebra comes to being abelian, built by iterated *central* extensions rather than abelian ones, and they are the algebras on which the exponential map behaves best, as the Heisenberg computation of example 3.5.5 already hinted.

The definition below brackets against  $\mathfrak{g}$  at every stage. Where the derived series replaces  $\mathfrak{g}^{(k)}$  by  $[\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]$ , keeping both slots inside the shrinking term, the lower central series keeps one slot pinned to the ambient algebra. This single change is what separates the two notions.

### Definition 5.2.1: Lower Central Series

Let  $\mathfrak{g}$  be a Lie algebra. Its *lower central series* is the descending chain of subspaces  $\mathfrak{g}^1 \supseteq \mathfrak{g}^2 \supseteq \dots$  defined recursively by

$$\mathfrak{g}^1 = \mathfrak{g}, \quad \mathfrak{g}^{k+1} = [\mathfrak{g}, \mathfrak{g}^k]$$

where  $[\mathfrak{g}, \mathfrak{g}^k]$  denotes the subspace spanned by all brackets  $[X, Y]$  with  $X \in \mathfrak{g}$  and  $Y \in \mathfrak{g}^k$ . The second term  $\mathfrak{g}^2 = [\mathfrak{g}, \mathfrak{g}]$  is the derived algebra of definition 5.1.3.

Each bracket in the recursion pins its first slot to the whole of  $\mathfrak{g}$ , so the lower central series never brackets a small term against itself; it always drags all of  $\mathfrak{g}$  across. Two facts about the terms need recording before the definition of nilpotency can be stated cleanly: every  $\mathfrak{g}^k$  is an ideal, and the series descends at least as slowly as the derived series.

### Proposition 5.2.2: The Terms of the Lower Central Series

Let  $\mathfrak{g}$  be a Lie algebra with lower central series  $\mathfrak{g}^k$  and derived series  $\mathfrak{g}^{(k)}$ .

1. Each  $\mathfrak{g}^k$  is an ideal of  $\mathfrak{g}$ , and  $[\mathfrak{g}^i, \mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$  for all  $i, j \geq 1$ .
2.  $\mathfrak{g}^{(k)} \subseteq \mathfrak{g}^{k+1}$  for every  $k \geq 0$ ; in particular the derived series descends at least as fast as the lower central series.

#### *Proof:*

1. *Each term is an ideal, and the bracket inclusion.* That  $\mathfrak{g}^1 = \mathfrak{g}$  is an ideal is immediate. Proceeding by induction, suppose  $\mathfrak{g}^k$  is an ideal. Then  $\mathfrak{g}^{k+1} = [\mathfrak{g}, \mathfrak{g}^k]$  is a subspace, and for  $X \in \mathfrak{g}$  and a spanning bracket  $[Y, Z] \in \mathfrak{g}^{k+1}$  with  $Y \in \mathfrak{g}$ ,  $Z \in \mathfrak{g}^k$ , the Jacobi identity gives

$$[X, [Y, Z]] = [[X, Y], Z] + [Y, [X, Z]]$$

The first term lies in  $[\mathfrak{g}, \mathfrak{g}^k] = \mathfrak{g}^{k+1}$  because  $[X, Y] \in \mathfrak{g}$  and  $Z \in \mathfrak{g}^k$ ; the second lies in  $[\mathfrak{g}, \mathfrak{g}^k] = \mathfrak{g}^{k+1}$  because  $Y \in \mathfrak{g}$  and  $[X, Z] \in \mathfrak{g}^k$ , the latter since  $\mathfrak{g}^k$  is an ideal. Thus

$[X, [Y, Z]] \in \mathfrak{g}^{k+1}$ ; as such brackets span  $\mathfrak{g}^{k+1}$  and  $X$  was arbitrary,  $[\mathfrak{g}, \mathfrak{g}^{k+1}] \subseteq \mathfrak{g}^{k+1}$ , so  $\mathfrak{g}^{k+1}$  is an ideal.

For the inclusion  $[\mathfrak{g}^i, \mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$ , argue by induction on  $j$ , the statement being taken to hold for all  $i \geq 1$  simultaneously at each fixed  $j$ . The base case  $j = 1$  reads  $[\mathfrak{g}^i, \mathfrak{g}] \subseteq \mathfrak{g}^{i+1}$  for all  $i$ , which holds because  $\mathfrak{g}^i$  is an ideal, so  $[\mathfrak{g}^i, \mathfrak{g}] = [\mathfrak{g}, \mathfrak{g}^i] = \mathfrak{g}^{i+1}$ . Assume the statement for  $j$ , again for all  $i$ . A spanning generator of  $\mathfrak{g}^{j+1} = [\mathfrak{g}, \mathfrak{g}^j]$  has the form  $[U, V]$  with  $U \in \mathfrak{g}$ ,  $V \in \mathfrak{g}^j$ , and for  $W \in \mathfrak{g}^i$  the Jacobi identity gives

$$[W, [U, V]] = [[W, U], V] + [U, [W, V]]$$

In the first term,  $[W, U] \in [\mathfrak{g}^i, \mathfrak{g}] \subseteq \mathfrak{g}^{i+1}$  by the base case, and  $V \in \mathfrak{g}^j$ , so the inductive hypothesis at index pair  $(i+1, j)$  gives  $[[W, U], V] \in [\mathfrak{g}^{i+1}, \mathfrak{g}^j] \subseteq \mathfrak{g}^{i+1+j}$ . In the second term,  $[W, V] \in [\mathfrak{g}^i, \mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$  by the inductive hypothesis at  $(i, j)$ , so  $[U, [W, V]] \in [\mathfrak{g}, \mathfrak{g}^{i+j}] = \mathfrak{g}^{i+j+1}$ . Both terms lie in  $\mathfrak{g}^{i+j+1}$ , and since such generators span  $[\mathfrak{g}^i, \mathfrak{g}^{j+1}]$ , one concludes  $[\mathfrak{g}^i, \mathfrak{g}^{j+1}] \subseteq \mathfrak{g}^{i+j+1}$  for all  $i$ , completing the induction.

2. *The derived series descends faster.* Argue by induction on  $k$ . For  $k = 0$ ,  $\mathfrak{g}^{(0)} = \mathfrak{g} = \mathfrak{g}^1$ , so  $\mathfrak{g}^{(0)} \subseteq \mathfrak{g}^1 = \mathfrak{g}^{0+1}$ . Suppose  $\mathfrak{g}^{(k)} \subseteq \mathfrak{g}^{k+1}$ . Then

$$\mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}] \subseteq [\mathfrak{g}^{k+1}, \mathfrak{g}^{k+1}] \subseteq [\mathfrak{g}, \mathfrak{g}^{k+1}] = \mathfrak{g}^{k+2}$$

the first inclusion from the inductive hypothesis (bracket is monotone in each slot), the second because  $\mathfrak{g}^{k+1} \subseteq \mathfrak{g}$ . Thus  $\mathfrak{g}^{(k+1)} \subseteq \mathfrak{g}^{(k+1)+1}$ , as we sought to show.

The bracket inclusion  $[\mathfrak{g}^i, \mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$  is the arithmetic backbone of the whole theory: it says the lower central filtration is compatible with the bracket in the way a grading would be, the index adding under multiplication. It is exactly this additivity that will force iterated brackets to vanish once the filtration reaches zero, and it is why a nilpotent Lie algebra has all sufficiently long brackets equal to zero regardless of how the factors are associated. The containment  $\mathfrak{g}^{(k)} \subseteq \mathfrak{g}^{k+1}$  records the coarser fact that a chain bracketing against all of  $\mathfrak{g}$  cannot outrun the chain bracketing against the smaller term: reaching zero along the slower chain is the harder condition.

### Definition 5.2.3: Nilpotent Lie Algebra

A Lie algebra  $\mathfrak{g}$  is *nilpotent* if  $\mathfrak{g}^k = 0$  for some  $k \geq 1$ ; the least such  $k - 1$  is the *nilpotency class* of  $\mathfrak{g}$ . Equivalently,  $\mathfrak{g}$  is nilpotent of class  $\leq c$  if every iterated bracket of  $c + 1$  elements vanishes,

$$[X_1, [X_2, [\dots, [X_c, X_{c+1}] \dots]] = 0 \quad \text{for all } X_1, \dots, X_{c+1} \in \mathfrak{g}$$

A nonzero abelian Lie algebra is nilpotent of class 1: there  $\mathfrak{g}^1 = \mathfrak{g} \neq 0$  but  $\mathfrak{g}^2 = [\mathfrak{g}, \mathfrak{g}] = 0$ , so the least  $k$  with  $\mathfrak{g}^k = 0$  is  $k = 2$  and the class is  $k - 1 = 1$  (class 0 is reserved for the zero algebra). A nilpotent algebra of class 2 has  $\mathfrak{g}^2 \neq 0$  but  $\mathfrak{g}^3 = [\mathfrak{g}, \mathfrak{g}^2] = 0$ , meaning  $[\mathfrak{g}, \mathfrak{g}]$  is central; the Heisenberg algebra of example 3.5.5 is the smallest such, and its lower central series is computed below. The equivalence between  $\mathfrak{g}^{c+1} = 0$  and the vanishing of every length- $(c + 1)$  iterated bracket is a direct unwinding of the recursion  $\mathfrak{g}^{k+1} = [\mathfrak{g}, \mathfrak{g}^k]$ : the term  $\mathfrak{g}^k$  is spanned by left-normed brackets  $[X_1, [X_2, [\dots, X_k] \dots]]$  of length  $k$ , so  $\mathfrak{g}^{c+1} = 0$  says exactly that all such brackets of length  $c + 1$  vanish, and the inclusion  $[\mathfrak{g}^i, \mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$  of proposition 5.2.2 promotes this to arbitrarily-associated brackets.

Comparing the two chains term by term settles the relationship between the two structural notions.

Since  $\mathfrak{g}^{(k)} \subseteq \mathfrak{g}^{k+1}$ , once the lower central series reaches zero so does the derived series, one step later at worst. Nilpotency therefore implies solvability. The converse fails, and the failure is not exotic: the upper-triangular matrices, solvable by example 5.1.7, are not nilpotent, a point the next proposition proves and the running matrix example makes concrete.

**Proposition 5.2.4: Basic Properties of Nilpotent Lie Algebras**

Let  $\mathfrak{g}$  be a finite-dimensional Lie algebra.

1. If  $\mathfrak{g}$  is nilpotent then it is solvable. The converse fails:  $\mathfrak{b}_n$  (example 5.1.7) is solvable but not nilpotent for  $n \geq 2$ .
2. Every subalgebra and every quotient of a nilpotent Lie algebra is nilpotent.
3. If  $\mathfrak{g} \neq 0$  is nilpotent, then its center  $\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} \mid [X, Y] = 0 \text{ for all } Y \in \mathfrak{g}\}$  is nonzero.

**Proof:**

1. *Nilpotent implies solvable.* If  $\mathfrak{g}^k = 0$  then by proposition 5.2.2(2),  $\mathfrak{g}^{(k-1)} \subseteq \mathfrak{g}^k = 0$ , so  $\mathfrak{g}^{(k-1)} = 0$  and  $\mathfrak{g}$  is solvable in the sense of definition 5.1.5. That the converse fails is exhibited by  $\mathfrak{b}_n$ ; the computation is example 5.2.5 below, whose lower central series stabilizes at the nonzero ideal  $\mathfrak{n}_n$  and never reaches zero.

2. *Subalgebras and quotients.* Let  $\mathfrak{h} \subseteq \mathfrak{g}$  be a subalgebra. A straightforward induction shows  $\mathfrak{h}^k \subseteq \mathfrak{g}^k$ : the base case is  $\mathfrak{h}^1 = \mathfrak{h} \subseteq \mathfrak{g} = \mathfrak{g}^1$ , and if  $\mathfrak{h}^k \subseteq \mathfrak{g}^k$  then  $\mathfrak{h}^{k+1} = [\mathfrak{h}, \mathfrak{h}^k] \subseteq [\mathfrak{g}, \mathfrak{g}^k] = \mathfrak{g}^{k+1}$ . Hence if  $\mathfrak{g}^k = 0$  then  $\mathfrak{h}^k = 0$ , and  $\mathfrak{h}$  is nilpotent.

For a quotient, let  $\pi: \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{a}$  be the projection onto the quotient by an ideal  $\mathfrak{a}$  (definition 5.1.1). Since  $\pi$  is a surjective homomorphism,  $\pi([\mathfrak{g}, \mathfrak{g}^k]) = [\pi(\mathfrak{g}), \pi(\mathfrak{g}^k)]$ , and an induction gives  $(\mathfrak{g}/\mathfrak{a})^k = \pi(\mathfrak{g}^k)$ : the base case is  $(\mathfrak{g}/\mathfrak{a})^1 = \mathfrak{g}/\mathfrak{a} = \pi(\mathfrak{g})$ , and the inductive step reads

$$(\mathfrak{g}/\mathfrak{a})^{k+1} = [\mathfrak{g}/\mathfrak{a}, (\mathfrak{g}/\mathfrak{a})^k] = [\pi(\mathfrak{g}), \pi(\mathfrak{g}^k)] = \pi([\mathfrak{g}, \mathfrak{g}^k]) = \pi(\mathfrak{g}^{k+1})$$

Hence if  $\mathfrak{g}^k = 0$  then  $(\mathfrak{g}/\mathfrak{a})^k = \pi(0) = 0$ , so the quotient is nilpotent.

3. *Nonzero center.* Let  $\mathfrak{g} \neq 0$  be nilpotent, and let  $k \geq 2$  be the least index with  $\mathfrak{g}^k = 0$ ; such  $k$  exists because  $\mathfrak{g}^1 = \mathfrak{g} \neq 0$ , and  $k \geq 2$  since  $\mathfrak{g}^1 \neq 0$ . Then  $\mathfrak{g}^{k-1} \neq 0$  by minimality, while

$$[\mathfrak{g}, \mathfrak{g}^{k-1}] = \mathfrak{g}^k = 0$$

which says precisely that every element of  $\mathfrak{g}^{k-1}$  brackets to zero against all of  $\mathfrak{g}$ , that is  $\mathfrak{g}^{k-1} \subseteq \mathfrak{z}(\mathfrak{g})$ . Since  $\mathfrak{g}^{k-1} \neq 0$ , the center is nonzero, completing the proof.

The third statement is the structural heart of nilpotency. The last nonzero term of the lower central series is central, so a nilpotent Lie algebra always has a nontrivial center to quotient by; the quotient is again nilpotent by part (2), of one lower class, and one may recurse. This is the precise sense in which a nilpotent Lie algebra is built by *iterated central extensions*: repeatedly quotienting by a central ideal reduces  $\mathfrak{g}$  to 0. One such chain of central ideals is the lower central series itself,

$$\mathfrak{g} = \mathfrak{g}_0 \supseteq \mathfrak{g}_1 \supseteq \cdots \supseteq \mathfrak{g}_r = 0, \quad \mathfrak{g}_i = \mathfrak{g}^{i+1}$$

whose terms are the lower-central-series ideals rather than the successively quotiented centers, yet which is still a central chain: each successive quotient  $\mathfrak{g}_i/\mathfrak{g}_{i+1}$  is central in  $\mathfrak{g}/\mathfrak{g}_{i+1}$ , since  $[\mathfrak{g}, \mathfrak{g}^{i+1}] \subseteq \mathfrak{g}^{i+2}$ . Solvability, by contrast, only guarantees the successive quotients are *abelian*, not central. The distinction, central versus abelian, is exactly the extra strength nilpotency carries over solvability, and it is why Engel's theorem in section 5.3 can force a full flag of ideals where Lie's theorem in section 5.4 produces only a flag of subspaces. The nonvanishing of the center also rules out  $\mathfrak{sl}_2$  from ever being nilpotent, since  $\mathfrak{sl}_2$  has trivial center; nilpotency is a condition satisfied only by algebras far from the semisimple ones.

The canonical example of a nilpotent Lie algebra, and the one that fixes the intuition for the whole subject, is the algebra of strictly upper-triangular matrices. Its lower central series can be computed by hand, and each bracket literally pushes the nonzero entries one step further above the diagonal until they run off the top of the matrix. Set against the upper-triangular algebra  $\mathfrak{b}_n$  of example 5.1.7, the computation displays the difference between solvable and nilpotent in coordinates.

**Example 5.2.5: The Strictly Upper-Triangular Algebra  $\mathfrak{n}_n$**

Let  $\mathfrak{n}_n \subseteq \mathfrak{gl}_n(\mathbb{C})$  be the subalgebra of strictly upper-triangular matrices, those with zeros on and below the main diagonal. It is a subalgebra: a product of two strictly upper-triangular matrices is again strictly upper-triangular, so the commutator is too, and  $\dim \mathfrak{n}_n = \binom{n}{2}$ . For  $0 \leq d \leq n-1$  write

$$\mathfrak{n}_n(d) = \text{span} \{E_{ij} \mid j - i \geq d\}$$

for the span of the matrix units  $E_{ij}$  lying at least  $d$  steps above the diagonal, so that  $\mathfrak{n}_n(1) = \mathfrak{n}_n$ , the chain  $\mathfrak{n}_n(1) \supseteq \mathfrak{n}_n(2) \supseteq \cdots \supseteq \mathfrak{n}_n(n-1) \supseteq \mathfrak{n}_n(n) = 0$  descends by one super-diagonal at each step, and  $\mathfrak{n}_n(n) = 0$  because no entry sits  $n$  or more steps above the diagonal in an  $n \times n$  matrix.

The claim is that the lower central series is exactly this chain:

$$\mathfrak{n}_n^k = \mathfrak{n}_n(k) \quad \text{for } 1 \leq k \leq n \quad (5.3)$$

The computation rests on the multiplication rule for matrix units,  $E_{ij}E_{kl} = \delta_{jk}E_{il}$ , which gives the bracket

$$[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{li}E_{kj}$$

Bracketing a unit  $E_{ab}$  with  $a < b$  (a generator of  $\mathfrak{n}_n$ ) against a unit  $E_{cd}$  with  $d - c \geq k$  (a generator of  $\mathfrak{n}_n(k)$ ) produces  $\delta_{bc}E_{ad} - \delta_{da}E_{cb}$ . In the first term  $b = c$  forces  $d - a = (d - c) + (b - a) \geq k + 1$ , so  $E_{ad} \in \mathfrak{n}_n(k + 1)$ ; in the second  $d = a$  forces  $b - c = (b - a) + (d - c) \geq k + 1$ , so  $E_{cb} \in \mathfrak{n}_n(k + 1)$ . Hence  $[\mathfrak{n}_n, \mathfrak{n}_n(k)] \subseteq \mathfrak{n}_n(k + 1)$ . Conversely every generator  $E_{il}$  of  $\mathfrak{n}_n(k + 1)$ , with  $l - i \geq k + 1$ , is produced: choosing any  $j$  with  $i < j$  and  $l - j \geq k$  (possible since  $l - i \geq k + 1$ ), one has  $[E_{ij}, E_{jl}] = E_{il}$  with  $E_{ij} \in \mathfrak{n}_n$  and  $E_{jl} \in \mathfrak{n}_n(k)$ . Therefore  $[\mathfrak{n}_n, \mathfrak{n}_n(k)] = \mathfrak{n}_n(k + 1)$ , and starting from  $\mathfrak{n}_n^1 = \mathfrak{n}_n = \mathfrak{n}_n(1)$  the recursion  $\mathfrak{n}_n^{k+1} = [\mathfrak{n}_n, \mathfrak{n}_n^k]$  yields (5.3) by induction.

In particular  $\mathfrak{n}_n^n = \mathfrak{n}_n(n) = 0$ , so  $\mathfrak{n}_n$  is nilpotent of class  $n - 1$ . For  $n = 3$  the series reads

$$\mathfrak{n}_3^1 = \mathfrak{n}_3 \text{ (dim 3)}, \quad \mathfrak{n}_3^2 = \text{span}\{E_{13}\} \text{ (dim 1)}, \quad \mathfrak{n}_3^3 = 0$$

each bracket raising the super-diagonal by one until the top corner  $E_{13}$  is all that survives, and then vanishing.

This is the same algebra as the Heisenberg algebra  $\mathfrak{h}_3$  of example 3.5.5, under the identification  $P = E_{12}$ ,  $Q = E_{23}$ ,  $C = E_{13}$ : its only nontrivial bracket is  $[P, Q] = C$ , its center is  $\text{span}\{C\} = \mathfrak{n}_3^2$ , and it is the smallest nonabelian nilpotent Lie algebra, of dimension 3 and class 2. The nonzero center predicted by proposition 5.2.4(3) is precisely the last nonzero term  $\mathfrak{n}_3^2 = \text{span}\{C\}$ .

The contrast with  $\mathfrak{b}_n$  is the point of the example. For the full upper-triangular algebra  $\mathfrak{b}_n$  the derived series terminates (example 5.1.7), so  $\mathfrak{b}_n$  is solvable, but its lower central series does not reach zero. Concretely  $\mathfrak{b}_n^2 = [\mathfrak{b}_n, \mathfrak{b}_n] = \mathfrak{n}_n$ , and thereafter the diagonal part of  $\mathfrak{b}_n$  keeps the series from shrinking: writing  $H = \text{diag}(h_1, \dots, h_n)$  for a diagonal matrix, one computes  $[H, E_{ij}] = (h_i - h_j)E_{ij}$ , so for  $i < j$  and a choice of  $H$  with  $h_i \neq h_j$  the bracket recovers  $E_{ij}$ . Thus  $[\mathfrak{b}_n, \mathfrak{n}_n] = \mathfrak{n}_n$ , and the lower central series stabilizes,

$$\mathfrak{b}_n^1 = \mathfrak{b}_n, \quad \mathfrak{b}_n^k = \mathfrak{n}_n \text{ for all } k \geq 2$$

never reaching zero when  $n \geq 2$ , since  $\mathfrak{n}_n \neq 0$ . For  $n = 3$  the dimensions are 6, 3, 3, 3,  $\dots$ , a chain that drops once and then freezes at  $\mathfrak{n}_3$ , whereas the derived series 6, 3, 1, 0 runs to zero. The obstruction is exactly the diagonal torus: it is abelian, so the derived series kills it in one step, but bracketing against it never diminishes  $\mathfrak{n}_n$ , so the lower central series cannot pass below  $\mathfrak{n}_n$ . Solvability tolerates such a nonzero stable floor; nilpotency does not.

Nilpotency admits a reading that will organize the next section. The condition  $\mathfrak{g}^{c+1} = 0$  says every left-normed bracket  $[X_1, [X_2, [\dots, X_{c+1}] \dots]]$  of length  $c+1$  vanishes, which is the statement that the operators  $\text{ad}X_1, \dots, \text{ad}X_c$ , applied in succession to any  $X_{c+1}$ , eventually annihilate it. Nilpotency of the algebra is thus a statement about the operators  $\text{ad}X$  collectively driving everything to zero, an iterated centrality: each application of an  $\text{ad}X$  moves deeper into the filtration, and after  $c$  steps one has fallen out the bottom. The infinitesimal form of this observation, that  $\mathfrak{g}$  is nilpotent precisely when every single operator  $\text{ad}X$  is a nilpotent endomorphism, is Engel's theorem, proved in section 5.3. One direction is immediate and recorded in the exercises: if  $\mathfrak{g}$  is nilpotent of class  $c$  then  $(\text{ad}X)^c = 0$  for every  $X$ , since  $(\text{ad}X)^c Y$  is a length- $(c+1)$  iterated bracket. The content of Engel's theorem is the converse, that the nilpotence of each  $\text{ad}X$  separately forces the collective nilpotence of the algebra.

### Exercise 5.2.1

1. Prove that a finite-dimensional Lie algebra  $\mathfrak{g}$  is nilpotent if and only if the lower central series reaches 0, if and only if there is a finite chain of ideals

$$\mathfrak{g} = \mathfrak{a}_0 \supseteq \mathfrak{a}_1 \supseteq \dots \supseteq \mathfrak{a}_m = 0$$

of  $\mathfrak{g}$  with  $[\mathfrak{g}, \mathfrak{a}_i] \subseteq \mathfrak{a}_{i+1}$  for every  $i$ . (Such a chain is a *central chain*: each step drops into the next term under bracketing with all of  $\mathfrak{g}$ .)

2. Show that if  $\mathfrak{g}$  is nilpotent of class  $c$ , then  $\text{ad}X$  is a nilpotent operator for every  $X \in \mathfrak{g}$ , with  $(\text{ad}X)^c = 0$ . (This is the easy direction of Engel's theorem, section 5.3.) Verify it explicitly on the Heisenberg algebra  $\mathfrak{n}_3$  of example 5.2.5 by writing out  $\text{ad}P$ ,  $\text{ad}Q$ ,  $\text{ad}C$  as  $3 \times 3$  matrices in the basis  $P, Q, C$  and checking each squares to zero.
3. Let  $\mathfrak{g}_1$  and  $\mathfrak{g}_2$  be nilpotent Lie algebras. Prove that the direct sum  $\mathfrak{g}_1 \oplus \mathfrak{g}_2$ , with bracket  $[(X_1, X_2), (Y_1, Y_2)] = ([X_1, Y_1], [X_2, Y_2])$ , is nilpotent, and that its class is the maximum of the two classes.
4. Let  $\mathfrak{z} = \mathfrak{z}(\mathfrak{g})$  be the center of  $\mathfrak{g}$ . Prove that  $\mathfrak{g}$  is nilpotent if and only if  $\mathfrak{g}/\mathfrak{z}$  is nilpotent. (This is the recursive "peel off the center" structure of nilpotent algebras; contrast with solvability, where the analogous statement uses the derived algebra rather than the center.)
5. Give an example of a solvable Lie algebra  $\mathfrak{g}$  that is not nilpotent, other than  $\mathfrak{b}_n$ , and identify the term at which its lower central series stabilizes above zero. (Hint: the two-dimensional nonabelian Lie algebra with basis  $X, Y$  and  $[X, Y] = Y$  is the smallest example.)

6. Using the inclusion  $[\mathfrak{g}^i, \mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$  of proposition 5.2.2, prove that if  $\mathfrak{g}$  is nilpotent then every iterated bracket of length  $c + 1$  vanishes regardless of how the brackets are associated, where  $c$  is the nilpotency class. Deduce that the nilpotency class defined via the lower central series agrees with the longest nonvanishing iterated bracket.

## 5.3 Engel's Theorem

The lower central series of section 5.2 measures nilpotency from the outside, by iterating the bracket against all of  $\mathfrak{g}$  until the chain reaches zero. Engel's theorem replaces this global condition with a pointwise one, phrased entirely in terms of a single operator attached to each element. The bridge is the adjoint representation  $\text{ad}$  of definition 2.5.3: to each  $x \in \mathfrak{g}$  it assigns the endomorphism  $\text{ad}x: y \mapsto [x, y]$  of  $\mathfrak{g}$ , and nilpotency of the algebra turns out to be detectable from the nilpotency of these operators one at a time. One direction is already in hand. The lower central series of definition 5.2.1 makes iterated brackets vanish once  $\mathfrak{g}$  is nilpotent, so  $(\text{ad}x)^k = 0$  for  $k$  large and every  $\text{ad}x$  is a nilpotent operator; this is the easy implication recorded in exercise 5.2.1 (2). Engel's theorem is the converse, and it is the harder half: from the nilpotence of each  $\text{ad}x$  separately one must reconstruct the vanishing of the whole lower central series.

The converse cannot be a formal manipulation of the series, because the hypothesis says nothing about how the operators  $\text{ad}x$  interact for different  $x$ . What it does say is that  $\text{ad}(\mathfrak{g})$  is a Lie subalgebra of  $\mathfrak{gl}(\mathfrak{g})$  consisting of nilpotent operators, and the crux of Engel's theorem is a statement of pure linear algebra about any such algebra of operators: a Lie algebra of nilpotent endomorphisms of a nonzero space always kills a common vector. That linear-algebra fact, proved first, is then applied to  $\text{ad}(\mathfrak{g})$  acting on  $\mathfrak{g}$  itself to run the induction that collapses the lower central series. The pattern will recur in section 5.4: there a solvable algebra of operators is shown to have a common eigenvector, and Lie's theorem follows by the same passage from a pointwise spectral fact to a global structural one.

### Definition 5.3.1: ad-nilpotent Element

Let  $\mathfrak{g}$  be a Lie algebra over a field  $k$ . An element  $x \in \mathfrak{g}$  is *ad-nilpotent* if the operator  $\text{ad}x: \mathfrak{g} \rightarrow \mathfrak{g}, y \mapsto [x, y]$ , is nilpotent, that is,  $(\text{ad}x)^m = 0$  for some  $m \geq 1$ .

The definition isolates exactly the property that exercise 5.2.1 (2) produced as a consequence of nilpotency. It is a condition on one element at a time, referring only to the operator  $\text{ad}x$  and not to the algebra as a whole, and this locality is what makes it a usable hypothesis. The strictly upper-triangular matrices supply the guiding picture.

### Example 5.3.2: ad-nilpotence in $\mathfrak{n}_n$

Let  $\mathfrak{n}_n \subseteq \mathfrak{gl}_n(\mathbb{C})$  be the strictly upper-triangular matrices of example 5.2.5, and take the elementary matrix  $E_{12} \in \mathfrak{n}_3$  (a single 1 in the (1, 2) slot). The operator  $\text{ad}E_{12}$  acts on the basis of  $\mathfrak{n}_3$  given by  $E_{12}, E_{13}, E_{23}$  by

$$[E_{12}, E_{12}] = 0, \quad [E_{12}, E_{13}] = 0, \quad [E_{12}, E_{23}] = E_{13}$$

using  $[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{li}E_{kj}$ . Thus  $\text{ad}E_{12}$  sends  $E_{23} \mapsto E_{13} \mapsto 0$  and annihilates the rest, so  $(\text{ad}E_{12})^2 = 0$  and  $E_{12}$  is ad-nilpotent. The same holds for every element of  $\mathfrak{n}_n$ : nilpotency of the matrix  $X$  forces nilpotency of the operator  $\text{ad}X = L_X - R_X$  (left minus right multiplication), since  $L_X$  and  $R_X$  commute and are each nilpotent when  $X$  is, so their difference is nilpotent by the binomial expansion. Every element of  $\mathfrak{n}_n$  is therefore ad-nilpotent, in agreement with the

nilpotency of  $\mathfrak{n}_n$  established in example 5.2.5.

The example already exhibits the two faces of the coming theorem. On the one hand  $\mathfrak{n}_n$  is nilpotent as an algebra, so all its elements are ad-nilpotent; on the other, its elements are literally nilpotent matrices, and they share the common null vector  $e_1$  of  $\mathbb{C}^n$ , the first standard basis vector, which every strictly upper-triangular matrix kills. The theorem will show these two features are two aspects of one phenomenon. The technical heart is the second: a common null vector for a whole algebra of nilpotent operators.

A single nilpotent operator  $N$  on a nonzero space  $V$  has a nonzero kernel, since  $N^m = 0$  with  $N^{m-1} \neq 0$  produces a nonzero vector  $N^{m-1}v$  killed by  $N$ . The content of the next lemma is that this survives passage from one operator to a whole Lie algebra of them: the kernels can be arranged to overlap in a common nonzero vector. Neither commutativity nor any relation among the operators is assumed beyond closure under the bracket, and that closure is exactly what makes the induction go.

**Lemma 5.3.3: Common Null Vector for Nilpotent Operators**

Let  $V$  be a nonzero finite-dimensional vector space over a field  $k$ , and let  $\mathfrak{g} \subseteq \mathfrak{gl}(V)$  be a Lie subalgebra all of whose elements are nilpotent endomorphisms of  $V$ . Then there is a nonzero vector  $v \in V$  with  $Xv = 0$  for every  $X \in \mathfrak{g}$ .

Before the proof, the mechanism deserves a word, because the induction is on  $\dim \mathfrak{g}$  and every ingredient is arranged to make the algebra one dimension smaller act on a strictly smaller space. The engine has three moving parts. First, a nilpotent operator on  $\mathfrak{gl}(V)$  arises from the adjoint action of a nilpotent operator on  $V$ , so  $\mathfrak{g}$  acts on itself by nilpotent operators, and a maximal proper subalgebra  $\mathfrak{h}$  acquires a nonzero eigenvector in  $\mathfrak{g}/\mathfrak{h}$  for that action, forcing  $\mathfrak{h}$  to be an ideal of codimension one. Second, the inductive hypothesis applied to  $\mathfrak{h}$  on  $V$  produces a nonzero common null space  $W$  of  $\mathfrak{h}$ . Third, choosing  $z \in \mathfrak{g}$  outside  $\mathfrak{h}$ , the ideal property makes  $W$  invariant under  $z$ , and the nilpotent operator  $z|_W$  has a null vector, which is the common null vector sought.

**Proof:**

The argument is by induction on  $\dim \mathfrak{g}$ . If  $\dim \mathfrak{g} = 0$  there is nothing to prove, and if  $\dim \mathfrak{g} = 1$ , say  $\mathfrak{g} = kX$  with  $X$  nilpotent, then  $X$  has a nonzero kernel as noted above, and any nonzero  $v \in \ker X$  works. Assume the statement for all Lie algebras of nilpotent operators of dimension less than  $n := \dim \mathfrak{g}$ , with  $n \geq 2$ .

*Step 1:  $\mathfrak{g}$  acts on itself by nilpotent operators.* For  $X \in \mathfrak{g} \subseteq \mathfrak{gl}(V)$ , the endomorphism  $\text{ad}X$  of  $\mathfrak{g}$  is the restriction to  $\mathfrak{g}$  of  $\text{ad}X$  on all of  $\mathfrak{gl}(V)$ , and on  $\mathfrak{gl}(V)$  one has  $\text{ad}X = L_X - R_X$ , where  $L_X(A) = XA$  and  $R_X(A) = AX$  are left and right multiplication. These two operators commute, and each is nilpotent because  $X$  is: if  $X^m = 0$  then  $L_X^m = R_X^m = 0$ . A difference of two commuting nilpotent operators is nilpotent, by the binomial theorem  $(L_X - R_X)^{2m} = \sum_i \binom{2m}{i} (-1)^i L_X^{2m-i} R_X^i$ , in which every term has  $L_X^{2m-i} = 0$  or  $R_X^i = 0$ . Hence  $\text{ad}X$  is a nilpotent operator on  $\mathfrak{gl}(V)$ , and its restriction to the invariant subspace  $\mathfrak{g}$  is nilpotent as well. Thus  $\text{ad}(\mathfrak{g})$  is a Lie algebra of nilpotent operators on the vector space  $\mathfrak{g}$ .

*Step 2: a maximal proper subalgebra is an ideal of codimension one.* Since  $n \geq 2$ ,  $\mathfrak{g}$  has proper subalgebras (for instance 0), so choose  $\mathfrak{h} \subsetneq \mathfrak{g}$  a proper subalgebra of maximal dimension. The subalgebra  $\mathfrak{h}$  acts on  $\mathfrak{g}$  by  $\text{ad}$ , and it preserves both  $\mathfrak{h}$  (as  $\mathfrak{h}$  is a subalgebra) and hence the quotient space  $\mathfrak{g}/\mathfrak{h}$ . This gives a representation

$$\overline{\text{ad}}: \mathfrak{h} \rightarrow \mathfrak{gl}(\mathfrak{g}/\mathfrak{h}), \quad \overline{\text{ad}}(H)(y + \mathfrak{h}) = [H, y] + \mathfrak{h}$$

By Step 1 each  $\text{ad}H$  is nilpotent on  $\mathfrak{g}$ , so each  $\overline{\text{ad}}(H)$  is nilpotent on the quotient  $\mathfrak{g}/\mathfrak{h}$ . Now  $\dim \mathfrak{h} < n$ , so the inductive hypothesis applies to the Lie algebra  $\overline{\text{ad}}(\mathfrak{h}) \subseteq \mathfrak{gl}(\mathfrak{g}/\mathfrak{h})$  acting on the nonzero space  $\mathfrak{g}/\mathfrak{h}$  (nonzero because  $\mathfrak{h}$  is proper): there is a nonzero coset  $\overline{y} = y + \mathfrak{h}$  with  $\overline{\text{ad}}(H)\overline{y} = 0$  for all  $H \in \mathfrak{h}$ . Unwrapping,  $y \in \mathfrak{g} \setminus \mathfrak{h}$  satisfies

$$[H, y] \in \mathfrak{h} \quad \text{for every } H \in \mathfrak{h} \quad (5.4)$$

Consider the subspace  $\mathfrak{h}' = \mathfrak{h} + ky \subseteq \mathfrak{g}$ . It is a subalgebra: brackets within  $\mathfrak{h}$  stay in  $\mathfrak{h}$ , while  $[H, y] \in \mathfrak{h} \subseteq \mathfrak{h}'$  by (5.4), and  $[y, y] = 0$ . Since  $y \notin \mathfrak{h}$ , the subalgebra  $\mathfrak{h}'$  strictly contains  $\mathfrak{h}$ , and maximality of  $\mathfrak{h}$  forces  $\mathfrak{h}' = \mathfrak{g}$ . Therefore  $\mathfrak{h}$  has codimension one in  $\mathfrak{g}$ , and (5.4) says precisely that  $[\mathfrak{h}, y] \subseteq \mathfrak{h}$ ; combined with  $[\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h}$  and  $\mathfrak{g} = \mathfrak{h} + ky$ , this gives  $[\mathfrak{g}, \mathfrak{h}] \subseteq \mathfrak{h}$ , so  $\mathfrak{h}$  is an ideal of  $\mathfrak{g}$ .

*Step 3: the common null space of  $\mathfrak{h}$  is nonzero and  $\mathfrak{g}$ -invariant.* As a Lie algebra of nilpotent operators on  $V$  of dimension  $\dim \mathfrak{h} = n - 1 < n$ , the ideal  $\mathfrak{h}$  falls under the inductive hypothesis on the original space  $V$ : there is a nonzero vector killed by every element of  $\mathfrak{h}$ . Collect all of them into

$$W = \{w \in V \mid Hw = 0 \text{ for every } H \in \mathfrak{h}\}$$

which is a linear subspace of  $V$ , nonzero by the inductive hypothesis. The claim is that  $W$  is invariant under all of  $\mathfrak{g}$ , not merely under  $\mathfrak{h}$ . Let  $X \in \mathfrak{g}$  and  $w \in W$ ; the task is to show  $Xw \in W$ , that is,  $H(Xw) = 0$  for every  $H \in \mathfrak{h}$ . Write

$$H(Xw) = X(Hw) + [H, X]w$$

from  $HX = XH + [H, X]$  in the associative algebra  $\mathfrak{gl}(V)$ . Now  $Hw = 0$  because  $w \in W$ , so the first term vanishes. For the second,  $[H, X] \in \mathfrak{h}$  because  $\mathfrak{h}$  is an ideal of  $\mathfrak{g}$  (Step 2), and therefore  $[H, X]w = 0$  since  $w \in W$ . Hence  $H(Xw) = 0$  for every  $H \in \mathfrak{h}$ , so  $Xw \in W$ . Thus  $XW \subseteq W$  for every  $X \in \mathfrak{g}$ , and  $W$  is  $\mathfrak{g}$ -invariant.

*Step 4: an eigenvector of  $z|_W$  completes the descent.* Because  $\mathfrak{h}$  has codimension one, choose  $z \in \mathfrak{g}$  with  $z \notin \mathfrak{h}$ , so that  $\mathfrak{g} = \mathfrak{h} \oplus kz$  as vector spaces. By Step 3 the operator  $z$  preserves  $W$ , so it restricts to an endomorphism  $z|_W: W \rightarrow W$ . This restriction is nilpotent, being the restriction of the nilpotent operator  $z$  on  $V$  to an invariant subspace. A nilpotent operator on the nonzero space  $W$  has a nonzero kernel, so there is a nonzero  $v \in W$  with  $zv = 0$ . This  $v$  is the common null vector: every  $X \in \mathfrak{g}$  is  $X = H + \lambda z$  with  $H \in \mathfrak{h}$  and  $\lambda \in k$ , and then

$$Xv = Hv + \lambda zv = 0 + 0 = 0$$

since  $v \in W$  kills  $\mathfrak{h}$  and  $zv = 0$ . Thus  $Xv = 0$  for all  $X \in \mathfrak{g}$ , and  $v \neq 0$ , completing the proof.

The lemma is the linear-algebra core of the whole section, and the induction is worth reading twice, because it runs on two dimensions at once: the algebra  $\mathfrak{g}$  shrinks to the codimension-one ideal  $\mathfrak{h}$ , while the space  $V$  shrinks to the common null space  $W$ . The hypothesis that  $\mathfrak{g}$  be a Lie subalgebra, not merely a set of nilpotent operators, is what powers Step 2, where the self-action by  $\text{ad}$  manufactures the codimension-one ideal; without closure under the bracket there would be no such ideal, and the descent would have no first step. The conclusion is geometric: the nonzero vector  $v$  spans a line on which every operator in  $\mathfrak{g}$  acts as zero, a common fixed direction for the entire algebra. Iterating this observation produces the flag that Engel's theorem is usually stated with. That iteration is now the content of the main theorem, first in its intrinsic form for a Lie algebra and then in its operator form.

**Theorem 5.3.4: Engel's Theorem**

Let  $\mathfrak{g}$  be a finite-dimensional Lie algebra over an arbitrary field  $k$ . Then  $\mathfrak{g}$  is nilpotent if and only if every element  $x \in \mathfrak{g}$  is ad-nilpotent. Equivalently, in operator form: if  $V \neq 0$  is a finite-dimensional vector space and  $\mathfrak{g} \subseteq \mathfrak{gl}(V)$  is a Lie subalgebra all of whose elements are nilpotent endomorphisms, then there is a basis of  $V$  in which every element of  $\mathfrak{g}$  is strictly upper-triangular; that is, a full flag of subspaces

$$0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_m = V, \quad \dim V_i = i$$

with  $\mathfrak{g} V_i \subseteq V_{i-1}$  for every  $i$ .

**Proof:**

*The intrinsic equivalence, forward direction.* Suppose  $\mathfrak{g}$  is nilpotent, so  $\mathfrak{g}^{k+1} = 0$  for some  $k$  in the lower central series of definition 5.2.1. By definition  $\mathfrak{g}^{k+1} = [\mathfrak{g}, [\mathfrak{g}, [\cdots, [\mathfrak{g}, \mathfrak{g}] \cdots]]$  with  $k+1$  copies of  $\mathfrak{g}$ , so for any  $x, y \in \mathfrak{g}$  the iterated bracket  $(\text{ad} x)^k y = [x, [x, \cdots, [x, y] \cdots]]$  lies in  $\mathfrak{g}^{k+1} = 0$ . Hence  $(\text{ad} x)^k = 0$  and every  $x$  is ad-nilpotent. This is exactly the implication recorded in exercise 5.2.1 (2).

*The intrinsic equivalence, converse direction.* Suppose every  $x \in \mathfrak{g}$  is ad-nilpotent, so  $\text{ad}(\mathfrak{g}) \subseteq \mathfrak{gl}(\mathfrak{g})$  is a Lie subalgebra consisting of nilpotent operators (it is the image of  $\mathfrak{g}$  under the Lie algebra homomorphism  $\text{ad}$  of definition 2.5.3, hence a subalgebra). Apply the operator form, proved next, to  $\text{ad}(\mathfrak{g})$  acting on  $V = \mathfrak{g}$ . It yields a full flag of subspaces

$$0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_m = \mathfrak{g}, \quad \dim V_i = i$$

with  $\text{ad}(\mathfrak{g}) V_i \subseteq V_{i-1}$ , that is  $[\mathfrak{g}, V_i] \subseteq V_{i-1}$  for every  $i$ . In particular each  $V_i$  is an ideal of  $\mathfrak{g}$ , and the descending chain of the flag forces the lower central series down:  $\mathfrak{g}^1 = \mathfrak{g} = V_m$ , and inductively if  $\mathfrak{g}^j \subseteq V_{m-j+1}$  then

$$\mathfrak{g}^{j+1} = [\mathfrak{g}, \mathfrak{g}^j] \subseteq [\mathfrak{g}, V_{m-j+1}] \subseteq V_{m-j}$$

so after  $m$  steps  $\mathfrak{g}^{m+1} \subseteq V_0 = 0$ . Therefore  $\mathfrak{g}$  is nilpotent, which establishes the converse and with it the intrinsic equivalence.

*The operator form, existence of the flag.* Let  $V \neq 0$  and  $\mathfrak{g} \subseteq \mathfrak{gl}(V)$  be a Lie subalgebra of nilpotent operators. The flag is built one step at a time, from the bottom, by repeatedly invoking lemma 5.3.3. By that lemma there is a nonzero  $v_1 \in V$  with  $Xv_1 = 0$  for all  $X \in \mathfrak{g}$ ; set  $V_1 = kv_1$ , a one-dimensional subspace with  $\mathfrak{g} V_1 = 0 = V_0$ . Suppose inductively that a chain  $0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_i$  has been constructed with  $\dim V_j = j$  and  $\mathfrak{g} V_j \subseteq V_{j-1}$  for  $j \leq i$ ; each  $V_j$  is then a  $\mathfrak{g}$ -invariant subspace. If  $V_i = V$  the flag is complete. Otherwise form the nonzero quotient space  $V/V_i$ . Every  $X \in \mathfrak{g}$  preserves  $V_i$ , so it descends to an operator  $\bar{X}$  on  $V/V_i$ , and  $\bar{X}$  is nilpotent because  $X$  is. The map  $X \mapsto \bar{X}$  is a Lie algebra homomorphism, so its image  $\bar{\mathfrak{g}} \subseteq \mathfrak{gl}(V/V_i)$  is a Lie algebra of nilpotent operators on the nonzero space  $V/V_i$ . Lemma 5.3.3 supplies a nonzero coset  $\bar{v} = v + V_i$  with  $\bar{X}\bar{v} = 0$  for all  $X$ , that is  $Xv \in V_i$  for all  $X \in \mathfrak{g}$ . Set  $V_{i+1} = V_i + kv$ . Then  $\dim V_{i+1} = i+1$  since  $v \notin V_i$ , and  $\mathfrak{g} V_{i+1} \subseteq V_i$ : on  $V_i$  this holds by the inductive hypothesis ( $\mathfrak{g} V_i \subseteq V_{i-1} \subseteq V_i$ ), and  $\mathfrak{g} v \subseteq V_i$  by construction. The chain has grown by one dimension. Continuing until the top is reached produces the full flag  $0 = V_0 \subsetneq \cdots \subsetneq V_m = V$  with  $\dim V_i = i$  and  $\mathfrak{g} V_i \subseteq V_{i-1}$ .

Choosing a basis  $v_1, \dots, v_m$  of  $V$  adapted to the flag, so that  $v_1, \dots, v_i$  span  $V_i$ , the condition  $\mathfrak{g} V_i \subseteq V_{i-1}$  says that every  $X \in \mathfrak{g}$  sends  $v_i$  into the span of  $v_1, \dots, v_{i-1}$ , which is exactly the statement that the matrix of  $X$  in this basis is strictly upper-triangular. This completes the operator form, and hence the proof.

Engel's theorem completes the reduction of nilpotency to a pointwise condition. The lower central series is a global object, a descending chain of ideals whose terms are computed from all of  $\mathfrak{g}$  at once, yet its termination is now certified by inspecting each operator  $\text{adx}$  in isolation and asking only whether it is nilpotent. The theorem thereby converts a structural property of the algebra into a spectral property of individual operators, and the conversion runs both ways: nilpotency of  $\mathfrak{g}$  produces nilpotent  $\text{adx}$ , and nilpotent  $\text{adx}$  reconstructs nilpotency of  $\mathfrak{g}$ . The operator form is the geometric shadow of this. A Lie algebra of nilpotent operators is not merely a collection of individually triangularizable matrices; the theorem asserts they can be triangularized *simultaneously*, in a single basis, with zeros forced onto the diagonal. This is the precise sense in which  $\mathfrak{n}_n$  of example 5.2.5 is the universal model.

The place where the hypotheses were used is worth marking, because it is narrower than for the theorems to come. The entire argument, from lemma 5.3.3 through the flag, used only that  $k$  is a field: nilpotent operators have nonzero kernels over any field, the induction on dimension needs no more, and at no point was an eigenvalue extracted or a polynomial factored. Engel's theorem therefore holds over an arbitrary field, with no assumption of algebraic closure and no restriction on the characteristic. This is the sharp contrast with Lie's theorem of section 5.4, where the passage from nilpotent to a general solvable algebra forces one to find a common *eigenvector* rather than a common null vector, and eigenvalues exist only once the field is algebraically closed; the characteristic-zero hypothesis enters there as well, for a reason that has no analogue here. The strict upper-triangular form of Engel and the (merely) upper-triangular form of Lie stand on opposite sides of exactly this divide.

The flag makes the reconstruction of the concrete model explicit. For the running strictly-upper-triangular algebra, the theorem recovers the standard flag directly.

**Example 5.3.5: The Standard Flag for  $\mathfrak{n}_n$**

Take  $V = \mathbb{C}^n$  with standard basis  $e_1, \dots, e_n$ , and let  $\mathfrak{n}_n \subseteq \mathfrak{gl}_n(\mathbb{C})$  be the strictly upper-triangular matrices of example 5.2.5, a Lie algebra of nilpotent operators. The subspaces

$$V_i = \text{span}(e_1, \dots, e_i), \quad i = 0, 1, \dots, n$$

form a full flag, and a strictly upper-triangular  $X$  satisfies  $Xe_i \in V_{i-1}$  for each  $i$ : the  $i$ th column of  $X$  has nonzero entries only in rows  $1, \dots, i-1$ . Hence  $\mathfrak{n}_n V_i \subseteq V_{i-1}$ , and this is the flag produced by theorem 5.3.4. The bottom step recovers the common null vector of lemma 5.3.3:  $\mathfrak{n}_n V_1 = 0$  says every element of  $\mathfrak{n}_n$  kills  $e_1$ , which is the common null vector observed after example 5.3.2. The example shows Engel's theorem is exact rather than merely qualitative: it does not just assert that some flag exists, it returns, for the model algebra, the flag one would write down by hand.

A first structural payoff of the theorem is the existence of a center, obtained without leaving the algebra  $\mathfrak{g}$  at all. When every  $\text{adx}$  is nilpotent, the operator form applied to  $\text{ad}(\mathfrak{g})$  on  $\mathfrak{g}$  supplies a nonzero common null vector, and a common null vector for  $\text{ad}(\mathfrak{g})$  is by definition a nonzero central element. This reproves the nonzero-center statement of proposition 5.2.4 from the operator side, and it is the form in which the fact will be used repeatedly in the structure theory.

**Corollary 5.3.6: Nonzero Center of a Nilpotent Lie Algebra**

Let  $\mathfrak{g} \neq 0$  be a finite-dimensional Lie algebra all of whose elements are ad-nilpotent (in particular, any nonzero nilpotent Lie algebra). Then the center  $\mathfrak{z}(\mathfrak{g}) = \{z \in \mathfrak{g} \mid [z, y] = 0 \text{ for all } y \in \mathfrak{g}\}$  is nonzero. Moreover any nonzero ideal  $\mathfrak{a} \trianglelefteq \mathfrak{g}$  meets the center nontrivially:  $\mathfrak{a} \cap \mathfrak{z}(\mathfrak{g}) \neq 0$ .

**Proof:**

Apply the operator form of theorem 5.3.4 to the Lie algebra  $\text{ad}(\mathfrak{g}) \subseteq \mathfrak{gl}(\mathfrak{g})$ , whose elements are nilpotent by hypothesis, acting on the nonzero space  $V = \mathfrak{g}$ . It provides a nonzero  $z \in \mathfrak{g}$  with  $(\text{ad}x)(z) = 0$  for all  $x \in \mathfrak{g}$ , that is  $[x, z] = 0$  for all  $x$ , so  $z \in \mathfrak{z}(\mathfrak{g})$  and the center is nonzero.

For the second statement, let  $\mathfrak{a}$  be a nonzero ideal. Since  $\mathfrak{a}$  is an ideal,  $[\mathfrak{g}, \mathfrak{a}] \subseteq \mathfrak{a}$ , so each  $\text{ad}x$  ( $x \in \mathfrak{g}$ ) restricts to an operator  $\text{ad}x|_{\mathfrak{a}}$  on  $\mathfrak{a}$ , and this restriction is nilpotent because  $\text{ad}x$  is. The image  $\{\text{ad}x|_{\mathfrak{a}} \mid x \in \mathfrak{g}\} \subseteq \mathfrak{gl}(\mathfrak{a})$  is a Lie algebra of nilpotent operators on the nonzero space  $\mathfrak{a}$ , so lemma 5.3.3 yields a nonzero  $z \in \mathfrak{a}$  with  $(\text{ad}x)(z) = [x, z] = 0$  for all  $x \in \mathfrak{g}$ . Then  $z \in \mathfrak{a} \cap \mathfrak{z}(\mathfrak{g})$  and  $z \neq 0$ , completing the proof.

The corollary is the induction lever for nilpotent algebras. Passing to  $\mathfrak{g}/\mathfrak{z}(\mathfrak{g})$  drops the dimension while staying within the class, since quotients of nilpotent algebras are nilpotent by proposition 5.2.4, so a nonzero center guarantees the quotient is strictly smaller and arguments by induction on dimension have a base to descend on. The refinement that every nonzero ideal meets the center will be used when splitting off central subalgebras: any minimal ideal of a nilpotent algebra is in fact central, hence one-dimensional. These consequences flow from the same common-null-vector lemma, now read on the adjoint action rather than on an external representation, which is the recurring move of the subject.

**Exercise 5.3.1**

1. Show that the conclusion of lemma 5.3.3 can fail if  $\mathfrak{g}$  is not assumed to be a Lie subalgebra. Concretely, exhibit a single non-nilpotent operator on  $\mathbb{C}^2$  with no nonzero kernel, and explain why this does not contradict the lemma. Then exhibit two nilpotent operators on  $\mathbb{C}^2$  that span a subspace (not closed under the bracket) with no common null vector, showing that the closure hypothesis, not just nilpotence of the individual operators, is doing work.
2. Prove directly, without invoking the full flag, that if every  $\text{ad}x$  is nilpotent then  $\mathfrak{g}$  has a nonzero center, by applying lemma 5.3.3 to  $\text{ad}(\mathfrak{g})$ . Then deduce Engel's theorem (the converse direction) by induction on  $\dim \mathfrak{g}$ , using the quotient  $\mathfrak{g}/\mathfrak{z}(\mathfrak{g})$ .
3. Let  $\mathfrak{h}_3 \subseteq \mathfrak{n}_3$  be the Heisenberg algebra, spanned by  $E_{12}, E_{23}, E_{13}$  with the single nonzero bracket  $[E_{12}, E_{23}] = E_{13}$ . Verify that every element is ad-nilpotent, compute the lower central series to confirm nilpotency, and exhibit explicitly the flag of ideals in  $\mathfrak{g} = \mathfrak{h}_3$  produced by theorem 5.3.4 for the adjoint action.
4. Let  $\mathfrak{b}_n \subseteq \mathfrak{gl}_n(\mathbb{C})$  be the upper-triangular matrices of example 5.1.7. Show that a diagonal matrix  $D = \text{diag}(d_1, \dots, d_n) \in \mathfrak{b}_n$  with distinct entries is *not* ad-nilpotent, and conclude that  $\mathfrak{b}_n$  is not nilpotent, in agreement with example 5.1.7. Why does this not contradict the operator form of Engel's theorem applied to  $\mathfrak{b}_n$  acting on  $\mathbb{C}^n$ ?
5. Using corollary 5.3.6, prove that a minimal nonzero ideal  $\mathfrak{a}$  of a nilpotent Lie algebra  $\mathfrak{g}$  is contained in the center and is one-dimensional.

## 5.4 Lie's Theorem

Engel's theorem (theorem 5.3.4) governs a Lie algebra of nilpotent operators: such an algebra has a common null vector, and so a full flag of invariant subspaces on which every operator is strictly upper triangular. Lie's theorem is the solvable analogue. A solvable algebra of operators need not consist of nilpotent endomorphisms, so "common null vector" has no chance of surviving, but its natural weakening does: a solvable algebra has a common *eigenvector*, and hence a full flag on which every operator is *upper* triangular, not necessarily strictly. Where Engel triangularizes with zeros on the diagonal, Lie triangularizes and lets the diagonal record the eigenvalues. This is the second of the two triangularization theorems that open the structure theory, and between them they pin down exactly what solvability and nilpotency buy at the level of concrete matrices.

The price of passing from null vectors to eigenvectors is two hypotheses that Engel did not need, both of which are necessary. First, the field must be algebraically closed, because an eigenvector requires an eigenvalue to exist, and over  $\mathbb{R}$  a rotation has none. Second, the field must have characteristic zero, for a subtler reason that the proof will isolate precisely: a trace-counting step divides by the dimension of a subspace, and in characteristic  $p$  that dimension can be a multiple of  $p$ , collapsing the argument. Both failures are exhibited by explicit examples at the end of the section. Throughout, the field is  $k$ , algebraically closed of characteristic zero, and the reader may take  $k = \mathbb{C}$  without losing anything;  $V$  is a nonzero finite-dimensional vector space over  $k$ .

The heart of the matter is an invariance property of weight spaces. Given an ideal  $\mathfrak{a}$  of a Lie algebra of operators and a linear functional  $\lambda: \mathfrak{a} \rightarrow k$ , the vectors on which every  $A \in \mathfrak{a}$  acts as the scalar  $\lambda(A)$  form the *weight space* attached to  $\lambda$ . The following lemma says that once such a weight space is nonzero, the whole algebra preserves it. This is the mechanism that will let the induction in Lie's theorem restrict from  $V$  to a smaller invariant subspace without losing the eigenvector it is building.

### Definition 5.4.1: Weight and Weight Space

Let  $\mathfrak{a} \subseteq \mathfrak{gl}(V)$  be a Lie algebra of operators and  $\lambda: \mathfrak{a} \rightarrow k$  a linear functional (a *weight* of  $\mathfrak{a}$ ). The associated *weight space* is

$$V_\lambda = \{v \in V \mid Av = \lambda(A)v \text{ for all } A \in \mathfrak{a}\}$$

the set of simultaneous eigenvectors of  $\mathfrak{a}$  whose eigenvalue for each  $A$  is  $\lambda(A)$ , together with 0.

A weight space is the simultaneous-eigenspace notion for a whole algebra of operators. For a single operator  $A$  with eigenvalue  $c$ , the weight space of the one-dimensional algebra  $kA$  under  $\lambda(A) = c$  is the ordinary eigenspace  $\ker(A - c)$ . The linearity of  $\lambda$  is forced: if  $v \neq 0$  lies in  $V_\lambda$ , then  $A \mapsto \lambda(A)$  is the composite of  $A \mapsto Av$  with the scalar-extraction  $cv \mapsto c$ , both linear, so a nonzero weight space determines its weight uniquely as a linear functional on  $\mathfrak{a}$ . A common eigenvector for  $\mathfrak{a}$  is precisely a nonzero element of some  $V_\lambda$ , so producing a common eigenvector is the same as producing a weight whose weight space is nonzero.

Consider the upper-triangular algebra  $\mathfrak{b}_2 \subseteq \mathfrak{gl}_2(k)$  acting on  $V = k^2$ . The first basis vector  $e_1$  is a common eigenvector: a matrix  $\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$  sends  $e_1$  to  $ae_1$ , so  $e_1 \in V_\lambda$  for the weight  $\lambda\left(\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}\right) = a$  reading off the top-left entry. This weight is a linear functional on  $\mathfrak{b}_2$ , and the line  $ke_1$  is its weight space. That  $e_1$  is a common eigenvector, and that the associated diagonal entry is a Lie-algebra homomorphism to  $k$ , is Lie's theorem in miniature for  $\mathfrak{b}_2$ ; the theorem asserts this pattern for every solvable algebra once  $k$  is algebraically closed of characteristic zero.

**5.4.2 Note: the weight vanishes on brackets inside  $\mathfrak{a}$**  If  $V_\lambda \neq 0$  for an ideal  $\mathfrak{a}$ , then  $\lambda$  annihilates  $[\mathfrak{a}, \mathfrak{a}]$ : for  $A, B \in \mathfrak{a}$  and  $0 \neq v \in V_\lambda$ , the operator  $[A, B]$  acts on  $v$  as  $AB - BA$ , giving  $\lambda(A)\lambda(B)v - \lambda(B)\lambda(A)v = 0$ , so  $\lambda([A, B])v = 0$  forces  $\lambda([A, B]) = 0$ . The content of the invariance lemma below is the stronger statement that  $\lambda$  vanishes on  $[\mathfrak{g}, \mathfrak{a}]$ , brackets of the ambient algebra with  $\mathfrak{a}$ , which is what makes  $V_\lambda$  invariant under all of  $\mathfrak{g}$ .

The lemma that carries the theorem is the invariance of  $V_\lambda$  under the ambient algebra. The subtlety is that  $x \in \mathfrak{g}$  need not commute with  $\mathfrak{a}$ , so it is not obvious that  $x$  maps a simultaneous eigenvector of  $\mathfrak{a}$  to another one. The proof builds an  $x$ -cyclic subspace, shows  $\mathfrak{a}$  acts on it by upper-triangular matrices with  $\lambda$  on the diagonal, and extracts the invariance from a trace computation. This is the one place where characteristic zero is used, and the proof flags it.

**Lemma 5.4.3: The Invariance Lemma**

Let  $\mathfrak{g} \subseteq \mathfrak{gl}(V)$  be a solvable Lie algebra over  $k$ , let  $\mathfrak{a} \trianglelefteq \mathfrak{g}$  be an ideal, and let  $\lambda: \mathfrak{a} \rightarrow k$  be a weight with nonzero weight space  $V_\lambda$ . Then  $V_\lambda$  is invariant under  $\mathfrak{g}$ ; that is,  $xV_\lambda \subseteq V_\lambda$  for every  $x \in \mathfrak{g}$ . Equivalently,  $\lambda([x, A]) = 0$  for all  $x \in \mathfrak{g}$  and  $A \in \mathfrak{a}$ .

**Proof:**

Fix  $x \in \mathfrak{g}$  and a nonzero  $v \in V_\lambda$ . The argument produces an  $x$ -stable subspace containing  $v$ , on which every  $A \in \mathfrak{a}$  is upper triangular with constant diagonal  $\lambda(A)$ , and reads the equation  $\lambda([x, A]) = 0$  off the trace of  $[x, A]$  on that subspace. The invariance  $xV_\lambda \subseteq V_\lambda$  then follows.

*Step 1: the  $x$ -cyclic flag.* Consider the subspaces

$$W_0 = 0, \quad W_i = \text{span}_k\{v, xv, x^2v, \dots, x^{i-1}v\} \quad (i \geq 1)$$

so  $W_1 = kv \subseteq W_2 \subseteq \dots$  is an increasing chain, each  $W_i$  spanned by the first  $i$  vectors of the orbit of  $v$  under powers of  $x$ . Since  $V$  is finite-dimensional the chain stabilizes; let  $n$  be the least index with  $W_n = W_{n+1}$ , so that  $v, xv, \dots, x^{n-1}v$  are linearly independent and  $x^n v \in W_n$ . Then  $\dim W_n = n$ , the space  $W_n$  is  $x$ -invariant by construction (each  $xW_i \subseteq W_{i+1}$ , and  $xW_n \subseteq W_n$  because  $x^n v \in W_n$ ), and  $v \in W_n$ .

*Step 2: every  $A \in \mathfrak{a}$  is upper triangular on  $W_n$  with diagonal  $\lambda(A)$ .* The claim is that, in the ordered basis  $v, xv, \dots, x^{n-1}v$ ,

$$Ax^i v \equiv \lambda(A)x^i v \pmod{W_i} \quad (A \in \mathfrak{a}, 0 \leq i \leq n-1) \quad (5.5)$$

that is,  $A$  carries  $x^i v$  to  $\lambda(A)x^i v$  plus a combination of the earlier basis vectors  $v, \dots, x^{i-1}v$ . In particular each  $W_i$  is  $A$ -invariant, and  $A|_{W_n}$  is upper triangular with every diagonal entry equal to  $\lambda(A)$ .

The proof of (5.5) is by induction on  $i$ . For  $i = 0$ :  $Av = \lambda(A)v$  since  $v \in V_\lambda$ , and  $W_0 = 0$ , so the congruence holds. Assume it holds for all indices up to  $i-1$  and all  $A \in \mathfrak{a}$ . Then

$$Ax^i v = Ax(x^{i-1}v) = xA(x^{i-1}v) + [A, x](x^{i-1}v)$$

Treat the two terms separately. By the inductive hypothesis applied to  $A$  at index  $i-1$ ,  $A(x^{i-1}v) = \lambda(A)x^{i-1}v + w$  with  $w \in W_{i-1}$ ; applying  $x$  gives  $xA(x^{i-1}v) = \lambda(A)x^i v + xw$ , and  $xw \in xW_{i-1} \subseteq W_i$ , so this term is  $\lambda(A)x^i v$  modulo  $W_i$ . For the second term,  $[A, x] = -[x, A] \in \mathfrak{a}$

because  $\mathfrak{a}$  is an ideal of  $\mathfrak{g}$  and  $x \in \mathfrak{g}$ ; the inductive hypothesis applied to the element  $[A, x] \in \mathfrak{a}$  at index  $i-1$  gives  $[A, x](x^{i-1}v) \equiv \lambda([A, x])x^{i-1}v \pmod{W_{i-1}}$ , so in particular  $[A, x](x^{i-1}v) \in W_i$ . Adding,  $Ax^i v \equiv \lambda(A)x^i v \pmod{W_i}$ , which is (5.5) at index  $i$ . The induction is complete, and it shows simultaneously that each  $W_i$  is  $A$ -invariant for every  $A \in \mathfrak{a}$ .

*Step 3: the trace computation and the characteristic-zero step.* Consider the commutator  $[x, A]$  for a fixed  $A \in \mathfrak{a}$ . Since  $\mathfrak{a}$  is an ideal,  $[x, A] \in \mathfrak{a}$ , so (5.5) applies to it: on  $W_n$ , the operator  $[x, A]$  is upper triangular with every diagonal entry equal to  $\lambda([x, A])$ . Its trace on  $W_n$  is therefore the sum of  $n$  copies of that entry,

$$\mathrm{tr}([x, A]|_{W_n}) = n \lambda([x, A]) \quad (5.6)$$

On the other hand,  $W_n$  is invariant under  $x$  (Step 1) and under  $A$  (Step 2), so both  $x|_{W_n}$  and  $A|_{W_n}$  are endomorphisms of  $W_n$ , and the commutator of their restrictions is the restriction of their commutator:  $[x, A]|_{W_n} = [x|_{W_n}, A|_{W_n}]$ . The trace of any commutator of endomorphisms of a finite-dimensional space vanishes, since  $\mathrm{tr}(PQ) = \mathrm{tr}(QP)$ , so

$$\mathrm{tr}([x, A]|_{W_n}) = \mathrm{tr}(x|_{W_n} A|_{W_n} - A|_{W_n} x|_{W_n}) = 0 \quad (5.7)$$

Comparing (5.6) and (5.7) gives  $n \lambda([x, A]) = 0$ . Here is the one point where the field matters:  $n = \dim W_n$  is a positive integer, and in characteristic zero it is invertible in  $k$ , so dividing by  $n$  yields

$$\lambda([x, A]) = 0$$

In characteristic  $p$  the integer  $n$  could be a multiple of  $p$ , hence zero in  $k$ , and the conclusion would fail; this is exactly the crack that the characteristic- $p$  counterexample at the end of the section exploits.

*Step 4: invariance of  $V_\lambda$ .* It remains to deduce  $x V_\lambda \subseteq V_\lambda$ . Let  $A \in \mathfrak{a}$  and  $w \in V_\lambda$ . Then

$$A(xw) = x(Aw) + [A, x]w = x \lambda(A)w + [A, x]w = \lambda(A)(xw) + \lambda([A, x])w$$

using  $Aw = \lambda(A)w$  and, in the last step, that  $[A, x] \in \mathfrak{a}$  acts on  $w \in V_\lambda$  as the scalar  $\lambda([A, x])$ . Step 3 gives  $\lambda([A, x]) = -\lambda([x, A]) = 0$ , so  $A(xw) = \lambda(A)(xw)$ . As  $A \in \mathfrak{a}$  was arbitrary,  $xw \in V_\lambda$ . Hence  $x V_\lambda \subseteq V_\lambda$ , and  $V_\lambda$  is  $\mathfrak{g}$ -invariant, completing the proof.

The lemma is the engine of Lie's theorem, and it is worth isolating what it accomplishes. A weight space  $V_\lambda$  of an ideal  $\mathfrak{a}$  is manifestly invariant under  $\mathfrak{a}$  itself, by definition. The lemma promotes this to invariance under the entire ambient algebra  $\mathfrak{g}$ , at the cost of the trace computation. Once  $V_\lambda$  is  $\mathfrak{g}$ -invariant, the whole algebra  $\mathfrak{g}$  acts on it, and one may look for a common eigenvector inside  $V_\lambda$  using the remaining directions of  $\mathfrak{g}$  beyond  $\mathfrak{a}$ . This is the reduction step that makes the induction on  $\dim \mathfrak{g}$  run. Solvability enters only through the existence of the codimension-one ideal that the induction requires; the lemma itself needs  $\mathfrak{a}$  to be an ideal, but not  $\mathfrak{g}$  to be solvable, and its proof used solvability nowhere.

With the invariance lemma in place, the theorem is a clean induction. A solvable Lie algebra has a codimension-one ideal, because  $[\mathfrak{g}, \mathfrak{g}]$  is a proper subspace for a nonzero solvable algebra, and any subspace of codimension one containing  $[\mathfrak{g}, \mathfrak{g}]$  is automatically an ideal. Peeling off a one-dimensional complement to such an ideal splits the algebra as  $\mathfrak{h} \oplus kz$ , and the eigenvector for  $\mathfrak{g}$  is assembled from an eigenvector for  $\mathfrak{h}$  and then a single eigenvalue of  $z$  on the invariant weight space.

**Theorem 5.4.4: Lie's Theorem**

Let  $k$  be algebraically closed of characteristic zero,  $V \neq 0$  a finite-dimensional vector space over  $k$ , and  $\mathfrak{g} \subseteq \mathfrak{gl}(V)$  a solvable Lie algebra. Then  $\mathfrak{g}$  has a common eigenvector: there exist  $0 \neq v \in V$  and a weight  $\lambda: \mathfrak{g} \rightarrow k$  with  $xv = \lambda(x)v$  for all  $x \in \mathfrak{g}$ . Consequently there is a full flag

$$0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_n = V, \quad \dim V_i = i$$

of  $\mathfrak{g}$ -invariant subspaces; equivalently, in a suitable basis every element of  $\mathfrak{g}$  is upper triangular, so  $\mathfrak{g}$  is simultaneously triangularizable.

**Proof:**

The common eigenvector is produced by induction on  $\dim \mathfrak{g}$ ; the flag is then a short deduction.

*Step 1: the base case.* If  $\dim \mathfrak{g} = 0$ , then  $\mathfrak{g} = 0$  and every nonzero  $v \in V$  is a common eigenvector with  $\lambda = 0$ . If  $\dim \mathfrak{g} = 1$ , write  $\mathfrak{g} = kx$ . Since  $k$  is algebraically closed, the single operator  $x$  has an eigenvector  $v \neq 0$ , say  $xv = cv$ ; then  $v$  is a common eigenvector for  $\mathfrak{g}$  with weight  $\lambda(x) = c$ . Algebraic closure is used here and nowhere else in the induction.

*Step 2: a codimension-one ideal.* Assume  $\dim \mathfrak{g} \geq 1$  and the result for all solvable subalgebras of smaller dimension acting on any nonzero space. Since  $\mathfrak{g}$  is solvable and nonzero,  $[\mathfrak{g}, \mathfrak{g}] \neq \mathfrak{g}$ : if  $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$ , the derived series (definition 5.1.3) would be constant and never reach 0, contradicting solvability (definition 5.1.5). The quotient  $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$  is therefore a nonzero abelian Lie algebra, in which every subspace is an ideal. Choose a subspace of codimension one in  $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$  and let  $\mathfrak{h}$  be its preimage in  $\mathfrak{g}$ . Then  $\mathfrak{h}$  is a subspace of codimension one containing  $[\mathfrak{g}, \mathfrak{g}]$ , and any such subspace is an ideal: for  $x \in \mathfrak{g}$  and  $y \in \mathfrak{h}$ ,  $[x, y] \in [\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{h}$ . Being a subalgebra of the solvable algebra  $\mathfrak{g}$ , the ideal  $\mathfrak{h}$  is itself solvable (proposition 5.1.6), and  $\dim \mathfrak{h} = \dim \mathfrak{g} - 1$ . Fix any  $z \in \mathfrak{g} \setminus \mathfrak{h}$ , so that

$$\mathfrak{g} = \mathfrak{h} \oplus kz$$

as vector spaces.

*Step 3: a common eigenvector for  $\mathfrak{h}$ .* By the inductive hypothesis applied to  $\mathfrak{h} \subseteq \mathfrak{gl}(V)$ , there is a common eigenvector for  $\mathfrak{h}$ , hence a weight  $\lambda: \mathfrak{h} \rightarrow k$  with  $V_\lambda \neq 0$ , where

$$V_\lambda = \{w \in V \mid Aw = \lambda(A)w \text{ for all } A \in \mathfrak{h}\}$$

is the weight space of definition 5.4.1.

*Step 4: restrict to  $V_\lambda$  and diagonalize  $z$  there.* The algebra  $\mathfrak{h}$  is an ideal of  $\mathfrak{g}$  (Step 2), and  $\mathfrak{g}$  is solvable, so the invariance lemma (lemma 5.4.3) applies:  $V_\lambda$  is invariant under all of  $\mathfrak{g}$ . In particular  $zV_\lambda \subseteq V_\lambda$ , so  $z$  restricts to an operator on the nonzero space  $V_\lambda$ . Since  $k$  is algebraically closed,  $z|_{V_\lambda}$  has an eigenvector: there is  $0 \neq v \in V_\lambda$  and  $c \in k$  with  $zv = cv$ . This  $v$  is the sought common eigenvector for  $\mathfrak{g}$ . Indeed  $v \in V_\lambda$  gives  $Av = \lambda(A)v$  for all  $A \in \mathfrak{h}$ , and  $zv = cv$ ; since every element of  $\mathfrak{g} = \mathfrak{h} \oplus kz$  is  $A + tz$  with  $A \in \mathfrak{h}$  and  $t \in k$ , one has

$$(A + tz)v = \lambda(A)v + tcv = (\lambda(A) + tc)v$$

so  $v$  is a common eigenvector, with weight  $\tilde{\lambda}(A + tz) = \lambda(A) + tc$  extending  $\lambda$  from  $\mathfrak{h}$  to  $\mathfrak{g}$ . This weight is linear, being  $\lambda$  on the first summand and  $t \mapsto tc$  on the second. The induction is complete.

*Step 5: the invariant flag.* The flag follows by induction on  $\dim V = m$ , using the common eigenvector at each stage. For  $m = 1$  take  $0 \subsetneq V$ . For  $m > 1$ , the common eigenvector  $v$  spans a one-dimensional  $\mathfrak{g}$ -invariant subspace  $V_1 = kv$ . The algebra  $\mathfrak{g}$  acts on the quotient  $\bar{V} = V/V_1$ , which has dimension  $m - 1$ , and the image of  $\mathfrak{g}$  in  $\mathfrak{gl}(\bar{V})$  is a homomorphic image of a solvable algebra, hence solvable (proposition 5.1.6). By the inductive hypothesis on dimension of the space,  $\bar{V}$  carries a full flag  $0 = \bar{V}_0 \subsetneq \bar{V}_1 \subsetneq \cdots \subsetneq \bar{V}_{m-1} = \bar{V}$  of subspaces invariant under the image of  $\mathfrak{g}$ . Pulling back along the quotient map  $V \rightarrow \bar{V}$ , and prepending  $V_0 = 0$ , produces a full flag  $0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_m = V$  of  $\mathfrak{g}$ -invariant subspaces with  $\dim V_i = i$ . Choosing a basis  $v_1, \dots, v_m$  adapted to this flag, so that  $v_1, \dots, v_i$  span  $V_i$ , every  $x \in \mathfrak{g}$  maps  $V_i$  into itself and so is upper triangular in this basis. Thus  $\mathfrak{g}$  is simultaneously triangularizable, completing the proof.

Lie's theorem is the exact solvable counterpart of Engel's theorem, and the comparison is the quickest way to hold both in mind. Engel takes an algebra of nilpotent operators and produces a common null vector, a full flag on which every element is *strictly* upper triangular, and it works over any field. Lie takes a solvable algebra and produces a common eigenvector, a full flag on which every element is upper triangular but not necessarily strictly, and it requires the field to be algebraically closed of characteristic zero. The diagonal entries that Engel forces to zero are, in Lie's theorem, the values of the weight  $\lambda$ , a Lie-algebra homomorphism  $\mathfrak{g} \rightarrow k$  reading each element's eigenvalue on the flag. That the diagonal is a homomorphism, and not just a function, is the residue of lemma 5.4.3: a bracket  $[x, y]$  acts by  $\lambda([x, y])$  on the top line of the flag, and that value is 0, so the strictly-upper-triangular part carries all the bracketing. This last observation is precisely the input to the corollary below.

The abstract content of the theorem, stripped of the choice of representation, is that a solvable Lie algebra over  $\mathbb{C}$  admits a filtration by ideals with one-dimensional successive quotients. Applying Lie's theorem to the adjoint representation makes this concrete and yields the structural corollary that separates solvable from semisimple: the derived algebra of a solvable Lie algebra is nilpotent.

#### Corollary 5.4.5: Corollaries of Lie's Theorem

Let  $\mathfrak{g}$  be a finite-dimensional solvable Lie algebra over  $k$ , algebraically closed of characteristic zero.

1. There is a chain of ideals  $0 = \mathfrak{g}_0 \subsetneq \mathfrak{g}_1 \subsetneq \cdots \subsetneq \mathfrak{g}_n = \mathfrak{g}$  with  $\dim \mathfrak{g}_i = i$  and each  $\mathfrak{g}_i$  an ideal of  $\mathfrak{g}$ .
2. The derived algebra  $[\mathfrak{g}, \mathfrak{g}]$  is nilpotent. Equivalently,  $\text{ad}x$  is a nilpotent operator for every  $x \in [\mathfrak{g}, \mathfrak{g}]$ .

#### Proof:

Apply Lie's theorem to the adjoint representation  $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$  of definition 2.5.3, whose image  $\text{ad}\mathfrak{g} \subseteq \mathfrak{gl}(\mathfrak{g})$  is a solvable Lie algebra, being a homomorphic image of the solvable algebra  $\mathfrak{g}$  (proposition 5.1.6). Here  $V = \mathfrak{g}$  is the representation space.

1. By theorem 5.4.4 applied to  $\text{ad}\mathfrak{g}$  acting on  $\mathfrak{g}$ , there is a full flag  $0 = \mathfrak{g}_0 \subsetneq \mathfrak{g}_1 \subsetneq \cdots \subsetneq \mathfrak{g}_n = \mathfrak{g}$  of subspaces invariant under  $\text{ad}\mathfrak{g}$ , with  $\dim \mathfrak{g}_i = i$ . Invariance under  $\text{ad}\mathfrak{g}$  means  $[\mathfrak{g}, \mathfrak{g}_i] = (\text{ad}\mathfrak{g})(\mathfrak{g}_i) \subseteq \mathfrak{g}_i$ , which is exactly the statement that each  $\mathfrak{g}_i$  is an ideal of  $\mathfrak{g}$ . This proves (1).
2. Fix a basis of  $\mathfrak{g}$  adapted to the flag of (1). In this basis every  $\text{ad}x$  is upper triangular, by theorem 5.4.4. For  $x, y \in \mathfrak{g}$  the operator  $\text{ad}[x, y] = [\text{ad}x, \text{ad}y]$  is the commutator of two upper-triangular matrices, hence *strictly* upper triangular: the diagonal of a commutator

of upper-triangular matrices vanishes, since the  $(i, i)$  entry of a product  $UW$  of upper triangular matrices is  $\sum_j u_{ij}w_{ji}$ , and  $u_{ij}w_{ji}$  can be nonzero only when  $i \leq j$  (from  $U$  upper triangular) and  $j \leq i$  (from  $W$ ), forcing  $j = i$ , so the  $(i, i)$  entry of  $[U, W]$  is  $u_{ii}w_{ii} - w_{ii}u_{ii} = 0$  when  $U, W$  are upper triangular. Therefore  $\text{adz}$  is strictly upper triangular for every  $z \in [\mathfrak{g}, \mathfrak{g}]$ , being a linear combination of the  $\text{ad}[x, y]$ . A strictly-upper-triangular matrix is nilpotent, so  $\text{adz}$  is nilpotent for every  $z \in [\mathfrak{g}, \mathfrak{g}]$ .

It remains to pass from “ $\text{adz}$  nilpotent for  $z \in [\mathfrak{g}, \mathfrak{g}]$ ” to “ $[\mathfrak{g}, \mathfrak{g}]$  is nilpotent as a Lie algebra”. Set  $\mathfrak{k} = [\mathfrak{g}, \mathfrak{g}]$ . For  $z \in \mathfrak{k}$ , the operator  $\text{ad}_{\mathfrak{g}}z$  is nilpotent on  $\mathfrak{g}$ , so its restriction  $\text{ad}_{\mathfrak{k}}z = (\text{ad}_{\mathfrak{g}}z)|_{\mathfrak{k}}$ , which is defined because  $\mathfrak{k}$  is an ideal and so  $\text{ad}_{\mathfrak{g}}z$  preserves  $\mathfrak{k}$ , is also nilpotent. Thus every element of  $\mathfrak{k}$  is  $\text{ad}$ -nilpotent in  $\mathfrak{k}$  (definition 5.3.1), and Engel's theorem (theorem 5.3.4) gives that  $\mathfrak{k} = [\mathfrak{g}, \mathfrak{g}]$  is nilpotent, completing the proof.

The corollary is the structural payoff, and it draws the line the whole chapter has been approaching. Solvability is a weaker condition than nilpotency, as the upper-triangular algebra  $\mathfrak{b}_n$  shows:  $\mathfrak{b}_n$  is solvable but not nilpotent (example 5.1.7). Lie's theorem measures exactly how far solvability falls short. It says that solvability fails to be nilpotency only in the diagonal directions, because the entire bracket  $[\mathfrak{g}, \mathfrak{g}]$ , where all the nonabelian structure lives, is nilpotent. A solvable algebra is thus a nilpotent algebra  $[\mathfrak{g}, \mathfrak{g}]$  with an abelian quotient  $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$  stacked on top; the abelian part carries the eigenvalues, the nilpotent part carries the brackets. Over  $\mathbb{C}$  this reads off the concrete matrix picture directly: a solvable Lie algebra of operators is one that can be put into upper-triangular form, and the class of such algebras is exactly the triangularizable ones. This is the structural fact that isolates the opposite extreme. Semisimple algebras, the subject of ??, are those with no nonzero solvable ideal at all, hence no invariant flag of this kind, and the theory splits cleanly along this divide: every finite-dimensional Lie algebra is an extension of a semisimple algebra by its maximal solvable ideal, the radical, and Lie's theorem has just described the internal structure of the solvable part in full.

Both hypotheses on the field are needed, and pinning down where each fails is the surest way to remember why they appear. Over  $\mathbb{R}$ , a common eigenvector may simply not exist: the rotation algebra  $\mathbb{R} \cdot \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  is abelian, hence solvable, yet the rotation has no real eigenvector. In characteristic  $p$ , the abstract statement fails for a different reason, traced precisely to the division by  $n$  in lemma 5.4.3: there is a  $p$ -dimensional solvable, even nilpotent, representation of a two-dimensional solvable algebra with no common eigenvector. The exercises construct both examples in detail and extract the positive consequences that survive: a one-dimensional invariant subspace in every representation, and the classification of the smallest solvable algebras.

### Exercise 5.4.1

1. Over  $\mathbb{R}$ , let  $\mathfrak{g} = \mathbb{R} \cdot R \subseteq \mathfrak{gl}_2(\mathbb{R})$  where  $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ . Show that  $\mathfrak{g}$  is solvable but has no common eigenvector in  $\mathbb{R}^2$ , so the conclusion of Lie's theorem fails over  $\mathbb{R}$ . Which hypothesis of theorem 5.4.4 is violated, and at which step of the proof does the argument break? Verify that after extending scalars to  $\mathbb{C}$  the theorem is restored, and exhibit the common eigenvector in  $\mathbb{C}^2$ .
2. Fix a prime  $p$  and work over a field  $k$  of characteristic  $p$ . On  $V = k^p$  with basis  $e_0, \dots, e_{p-1}$ , let  $X$  be the cyclic shift  $Xe_i = e_{i+1}$  (indices mod  $p$ , so  $Xe_{p-1} = e_0$ ) and let  $H$  be the diagonal operator  $He_i = ie_i$ . Compute  $[H, X]$  and show it equals  $X$ , so that  $\mathfrak{g} = kX + kH$  is a two-dimensional solvable Lie algebra. Prove that  $\mathfrak{g}$  has no common eigenvector in  $V$ . Identify the exact step of the proof of theorem 5.4.4 (via lemma 5.4.3) that fails, and pinpoint where characteristic  $p$  enters.

3. Deduce from theorem 5.4.4 that every finite-dimensional representation  $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$  of a solvable Lie algebra  $\mathfrak{g}$  over  $k$  (algebraically closed, characteristic zero), with  $V \neq 0$ , has a one-dimensional  $\mathfrak{g}$ -invariant subspace. Conversely, show that if every finite-dimensional nonzero representation of  $\mathfrak{g}$  has a one-dimensional invariant subspace, must  $\mathfrak{g}$  be solvable? Decide the question, using the standard representation of  $\mathfrak{sl}_2(\mathbb{C})$  on  $\mathbb{C}^2$  to inform your answer.
4. Prove that the only simple Lie algebra that is also solvable is impossible: no Lie algebra is both simple and solvable. Deduce that a solvable Lie algebra with no proper nonzero ideals is one-dimensional (necessarily abelian), and conversely that every one-dimensional Lie algebra is abelian, solvable, and has no proper nonzero ideals.
5. Let  $\mathfrak{g} \subseteq \mathfrak{gl}(V)$  be solvable over  $k$  (algebraically closed, characteristic zero) and let  $\lambda: \mathfrak{g} \rightarrow k$  be the weight of the common eigenvector produced by theorem 5.4.4. Show directly that  $\lambda$  is a Lie-algebra homomorphism, that is  $\lambda([\mathfrak{g}, \mathfrak{g}]) = 0$ , and use this to give a second proof that every  $x \in [\mathfrak{g}, \mathfrak{g}]$  acts on  $V$  as a nilpotent operator (equivalently, vanishes on the associated graded of the flag of theorem 5.4.4).

## 5.5 The Radical and the Nilradical

The three preceding sections built the machinery of solvability from the inside: the derived series measures how a Lie algebra decays toward zero under repeated bracketing, Engel's theorem reads nilpotency off the operators  $\text{ad}x$ , and Lie's theorem triangularizes a solvable algebra of matrices. All of this describes what it means for an algebra to be solvable. The present section turns the question outward. Given an arbitrary finite-dimensional Lie algebra  $\mathfrak{g}$ , which need be neither solvable nor anything close to it, how much solvability does it contain? The answer is a single canonical ideal, the largest solvable ideal, and its extraction is the pivot on which the entire structure theory of Part II turns. Isolating this ideal splits every Lie algebra into two pieces of opposite character: a solvable part that the tools of this chapter already control, and a quotient with no solvable part at all, the semisimple algebras that ?? takes up.

The existence of a largest solvable ideal is not obvious. A Lie algebra can carry many solvable ideals at once, and to speak of a single largest one requires that they combine rather than compete. The mechanism is the last clause of proposition 5.1.6: the sum of two solvable ideals is again a solvable ideal. Iterating this over all solvable ideals produces one ideal containing every other, and that ideal is what the theory names.

### Proposition 5.5.1: Existence of a Maximal Solvable Ideal

Let  $\mathfrak{g}$  be a finite-dimensional Lie algebra. The sum of all solvable ideals of  $\mathfrak{g}$  is itself a solvable ideal. Consequently  $\mathfrak{g}$  possesses a unique maximal solvable ideal, one that contains every solvable ideal of  $\mathfrak{g}$ .

#### **Proof:**

Let  $\mathfrak{r}$  denote the sum of all solvable ideals of  $\mathfrak{g}$ , that is the set of finite sums  $a_1 + \cdots + a_k$  with each  $a_i$  lying in some solvable ideal. This is an ideal: a sum of ideals is an ideal, since it is a subspace and  $[\mathfrak{g}, \mathfrak{a} + \mathfrak{b}] \subseteq [\mathfrak{g}, \mathfrak{a}] + [\mathfrak{g}, \mathfrak{b}] \subseteq \mathfrak{a} + \mathfrak{b}$  whenever  $\mathfrak{a}, \mathfrak{b}$  are ideals. The task is to show  $\mathfrak{r}$  is solvable.

Because  $\mathfrak{g}$  is finite-dimensional, the ascending chain of partial sums must stabilize: there are finitely many solvable ideals  $\mathfrak{a}_1, \dots, \mathfrak{a}_m$  with

$$\mathfrak{r} = \mathfrak{a}_1 + \mathfrak{a}_2 + \dots + \mathfrak{a}_m$$

Indeed, choosing solvable ideals one at a time so that each strictly enlarges the running sum, dimension forces the process to halt after at most  $\dim \mathfrak{g}$  steps, and the resulting sum already contains every solvable ideal (any further one would either enlarge the sum, contradicting maximality of the running dimension, or be absorbed). It therefore suffices to prove that a sum of finitely many solvable ideals is solvable, which follows by induction on  $m$  from the two-ideal case. For  $m = 1$  there is nothing to prove. For the inductive step, write  $\mathfrak{r} = (\mathfrak{a}_1 + \dots + \mathfrak{a}_{m-1}) + \mathfrak{a}_m$ ; the first summand is solvable by the inductive hypothesis and is an ideal,  $\mathfrak{a}_m$  is a solvable ideal, and the last clause of proposition 5.1.6 states that the sum of two solvable ideals is solvable. Hence  $\mathfrak{r}$  is solvable.

Uniqueness of a maximal solvable ideal is now immediate. The ideal  $\mathfrak{r}$  is solvable and contains every solvable ideal of  $\mathfrak{g}$  by construction, so it is the largest one; any other maximal solvable ideal would be contained in  $\mathfrak{r}$  and, being maximal, would equal it, completing the proof.

The argument is entirely a bookkeeping consequence of one algebraic fact, that solvable ideals add to solvable ideals, together with the finite dimension that keeps the sum from running away. What makes the conclusion carry weight is that it singles out a piece of  $\mathfrak{g}$  with no arbitrary choice involved: the maximal solvable ideal is determined by  $\mathfrak{g}$  alone, not by any basis, generating set, or auxiliary decomposition. This canonicity is what allows the ideal to be given a name and treated as an invariant of the algebra.

#### Definition 5.5.2: The Radical

The *radical* of a finite-dimensional Lie algebra  $\mathfrak{g}$ , written  $\text{rad} \mathfrak{g}$ , is its unique maximal solvable ideal, the ideal produced by proposition 5.5.1. It contains every solvable ideal of  $\mathfrak{g}$ .

The radical collects all of the solvability in  $\mathfrak{g}$  into one ideal. Parallel to it, and contained in it, sits a second canonical ideal that collects all of the nilpotency. The largest nilpotent ideal exists on grounds analogous to those for the radical, though the argument that a sum of two nilpotent ideals is again nilpotent differs from its solvable counterpart and is given after the definition.

#### Definition 5.5.3: The Nilradical

The *nilradical* of a finite-dimensional Lie algebra  $\mathfrak{g}$ , written  $\text{nil} \mathfrak{g}$ , is its unique maximal nilpotent ideal: the sum of all nilpotent ideals of  $\mathfrak{g}$ , which is again a nilpotent ideal and contains every nilpotent ideal.

That the nilradical exists is proved as proposition 5.5.1 was, once one knows that the sum of two nilpotent ideals is a nilpotent ideal. Here the argument is not the extension argument used for solvability, because an extension of a nilpotent algebra by a nilpotent ideal need not be nilpotent, as the two-dimensional non-abelian algebra  $\langle x, y \mid [x, y] = x \rangle$  shows. What replaces it uses that both summands are ideals of the same ambient algebra. Let  $\mathfrak{a}$  and  $\mathfrak{b}$  be nilpotent ideals of  $\mathfrak{g}$ , with lower central series terminating at steps  $p$  and  $q$  respectively. Because both are ideals,  $[\mathfrak{a}, \mathfrak{b}] \subseteq \mathfrak{a} \cap \mathfrak{b}$ , so in an iterated bracket of  $p + q$  elements drawn from  $\mathfrak{a} \cup \mathfrak{b}$  the factors may be regrouped so that either at least  $p$  come from  $\mathfrak{a}$  or at least  $q$  come from  $\mathfrak{b}$ ; the first case lands the bracket in the  $p$ -th term of

the lower central series of  $\mathfrak{a}$ , which is zero, and the second lands it in the  $q$ -th term of  $\mathfrak{b}$ 's, also zero. Hence the  $(p+q)$ -th term of the lower central series of  $\mathfrak{a} + \mathfrak{b}$  vanishes, so  $\mathfrak{a} + \mathfrak{b}$  is nilpotent, and it is an ideal as a sum of ideals. The summation argument of proposition 5.5.1 then produces a unique maximal nilpotent ideal. Since every nilpotent ideal is solvable by proposition 5.2.4, the nilradical is a solvable ideal, and the maximality of the radical among solvable ideals gives the containment

$$\text{nil } \mathfrak{g} \subseteq \text{rad } \mathfrak{g} \quad (5.8)$$

The containment (5.8) can be strict. The two ideals measure different failures of semisimplicity, one solvable and one nilpotent, and the gap between them appears already for the two-dimensional non-abelian algebra  $\mathfrak{g} = \langle x, y \mid [x, y] = x \rangle$ : it is solvable, so  $\text{rad } \mathfrak{g} = \mathfrak{g}$ , while its nilradical is the abelian ideal  $\mathbb{C}x$ , strictly smaller (the computation is carried out in exercise 5.5.1 (5)). That  $\mathfrak{g}$  is solvable without being nilpotent is the same distinction seen in the upper-triangular algebra  $\mathfrak{b}_n$  of example 5.1.7 against its strictly-upper-triangular ideal  $\mathfrak{n}_n$  of example 5.2.5.

A worked case grounds both definitions at once. The full matrix algebra  $\mathfrak{gl}_n$  is the standard test object of Part II, and its radical and nilradical can be computed by hand.

**Example 5.5.4: The Radical of  $\mathfrak{gl}_n$**

Let  $\mathfrak{gl}_n = \mathfrak{gl}_n(\mathbb{C})$  and let  $\mathfrak{z} = \mathbb{C}I$  be its center, the scalar matrices, which is the center of  $\mathfrak{gl}_n$  because a matrix commuting with every matrix is scalar. The center is an abelian ideal, hence solvable, so  $\mathfrak{z} \subseteq \text{rad } \mathfrak{gl}_n$ . The claim is that this containment is an equality,

$$\text{rad } \mathfrak{gl}_n = \mathbb{C}I$$

To see there is nothing more, use the decomposition  $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{C}I$  into traceless matrices and scalars, an identity of ideals recorded in theorem 1.4.3. The intersection  $\text{rad } \mathfrak{gl}_n \cap \mathfrak{sl}_n$  is a solvable ideal of  $\mathfrak{sl}_n$ . For  $n \geq 2$  the algebra  $\mathfrak{sl}_n$  is simple in the sense of definition 5.5.5 below, a standard fact proved for general  $n$  through the root-space decomposition of ?? and verified directly for  $n = 2$  in example 5.5.6; its only ideals are then 0 and  $\mathfrak{sl}_n$ . A solvable ideal cannot be all of  $\mathfrak{sl}_n$ , since a simple algebra is not solvable (its derived algebra equals itself), so  $\text{rad } \mathfrak{gl}_n \cap \mathfrak{sl}_n = 0$ . Any  $X \in \text{rad } \mathfrak{gl}_n$  therefore has its  $\mathfrak{sl}_n$ -component in  $\text{rad } \mathfrak{gl}_n \cap \mathfrak{sl}_n = 0$ , leaving only its scalar part, so  $\text{rad } \mathfrak{gl}_n \subseteq \mathbb{C}I$ . Combined with the reverse containment,  $\text{rad } \mathfrak{gl}_n = \mathbb{C}I$ . The nilradical here coincides with the radical. The center  $\mathbb{C}I$  is an abelian ideal, hence nilpotent as a Lie algebra, so it is a nilpotent ideal and  $\mathbb{C}I \subseteq \text{nil } \mathfrak{gl}_n$ ; the containment (5.8) then closes the circle, giving

$$\text{nil } \mathfrak{gl}_n = \text{rad } \mathfrak{gl}_n = \mathbb{C}I$$

so that for  $\mathfrak{gl}_n$  the two ideals agree.

The example shows the radical detecting exactly the scalar directions of  $\mathfrak{gl}_n$ , the part of the algebra that commutes with everything; here the radical and nilradical coincide because that solvable part is abelian, and an abelian ideal is nilpotent. The gap between the two ideals in (5.8) therefore does not open for  $\mathfrak{gl}_n$ ; it opens when the radical carries a semisimple direction that fails to be nilpotent, as in the non-abelian solvable algebras of the exercises. The computation also isolates the object the whole theory is built to study. The traceless algebra  $\mathfrak{sl}_n$  had radical zero, and an algebra whose radical is zero is precisely one with no solvable part to strip away. This is the definition that organizes the rest of the book.

**Definition 5.5.5: Semisimple and Simple Lie Algebras**

A finite-dimensional Lie algebra  $\mathfrak{g}$  is *semisimple* if its radical is zero,  $\text{rad } \mathfrak{g} = 0$ ; equivalently, if  $\mathfrak{g}$  has no nonzero solvable ideal. It is *simple* if it is non-abelian and its only ideals are 0 and  $\mathfrak{g}$  itself.

Semisimplicity is defined here by a negation:  $\mathfrak{g}$  is semisimple when it contains no solvable ideal other than the zero ideal, so that the radical, the receptacle for all solvability, is empty. Simplicity is the stronger condition that  $\mathfrak{g}$  have no proper nonzero ideal at all, together with the non-abelian requirement that excludes the one-dimensional algebra (whose only ideals are trivially 0 and itself, but which carries no interesting structure). A simple Lie algebra is semisimple: its only ideals are 0 and  $\mathfrak{g}$ , and  $\mathfrak{g}$  is not solvable because  $[\mathfrak{g}, \mathfrak{g}]$  is a nonzero ideal, hence all of  $\mathfrak{g}$ , so the derived series never descends; the radical, being a solvable ideal, cannot be  $\mathfrak{g}$  and so is 0. The simple algebras are thus the atoms among the semisimple ones, and ?? will show that every semisimple algebra is a direct sum of simple ideals.

The first example is the one the reader has carried since Part I. The algebra  $\mathfrak{sl}_2$  is the smallest semisimple Lie algebra, and it is in fact simple.

**Example 5.5.6:  $\mathfrak{sl}_2(\mathbb{C})$  Is Semisimple**

The algebra  $\mathfrak{sl}_2 = \mathfrak{sl}_2(\mathbb{C})$  has the standard basis

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

with the bracket relations  $[h, e] = 2e$ ,  $[h, f] = -2f$ , and  $[e, f] = h$ , computed directly from the matrix commutator. To see  $\mathfrak{sl}_2$  is semisimple it suffices to show  $\text{rad } \mathfrak{sl}_2 = 0$ , and the cleanest route is to show  $\mathfrak{sl}_2$  has no proper nonzero ideal, so that it is simple, whence semisimple. Let  $\mathfrak{a} \neq 0$  be an ideal and pick  $0 \neq x = \alpha h + \beta e + \gamma f \in \mathfrak{a}$ . Applying  $\text{ad } e$  twice,

$$[e, x] = -2\alpha e + \gamma h, \quad [e, [e, x]] = [e, -2\alpha e + \gamma h] = \gamma[e, h] = -2\gamma e$$

and applying  $\text{ad } f$  twice symmetrically gives  $[f, [f, x]] = -2\beta f$ . If  $\gamma \neq 0$  then  $[e, [e, x]] = -2\gamma e \in \mathfrak{a}$  shows  $e \in \mathfrak{a}$ ; then  $[f, e] = -h \in \mathfrak{a}$  and  $[e, f] = h$ , so also  $f = -[h, f]/2 \in \mathfrak{a}$  (from  $[h, f] = -2f$ ), and  $\mathfrak{a} = \mathfrak{sl}_2$ . If instead  $\gamma = 0$  but  $\beta \neq 0$ , the symmetric computation puts  $f \in \mathfrak{a}$  and the same closure gives  $\mathfrak{a} = \mathfrak{sl}_2$ . If  $\beta = \gamma = 0$  but  $\alpha \neq 0$ , then  $h \in \mathfrak{a}$ , and  $[h, e] = 2e$ ,  $[h, f] = -2f$  put  $e, f \in \mathfrak{a}$ , again  $\mathfrak{a} = \mathfrak{sl}_2$ . In every case a nonzero ideal is all of  $\mathfrak{sl}_2$ , so  $\mathfrak{sl}_2$  is simple, hence semisimple.

The verification that  $\text{rad } \mathfrak{sl}_2 = 0$  was reduced to the sharper fact that  $\mathfrak{sl}_2$  has no proper nonzero ideal, and the computation is a model for the general phenomenon: bracketing against  $e$  and  $f$  propagates any nonzero element out to a spanning set. This is why  $\mathfrak{sl}_2$  is the running example of the semisimple theory, the smallest place where root spaces, the Cartan subalgebra  $\mathbb{C}h$ , and the two root directions  $\mathbb{C}e, \mathbb{C}f$  are visible. By contrast the ambient algebra  $\mathfrak{gl}_2$  is not semisimple: it splits as  $\mathfrak{sl}_2 \oplus \mathbb{C}I$ , and the scalar summand  $\mathbb{C}I$  is a nonzero abelian, hence solvable, ideal, so  $\text{rad } \mathfrak{gl}_2 = \mathbb{C}I \neq 0$ . More generally  $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{C}I$  is never semisimple, its radical being the center  $\mathbb{C}I$  computed in example 5.5.4; the presence of the central direction is exactly the obstruction.

The radical was constructed to be the largest solvable ideal, and its defining role is to make the quotient by it semisimple. Passing to  $\mathfrak{g}/\text{rad } \mathfrak{g}$  removes all of the solvability from  $\mathfrak{g}$  at one stroke, and the resulting algebra has no solvability left. This is the structural statement toward which the section

has been building.

**Proposition 5.5.7: The Quotient by the Radical Is Semisimple**

For any finite-dimensional Lie algebra  $\mathfrak{g}$ , the quotient  $\mathfrak{g}/\text{rad}\mathfrak{g}$  is semisimple.

**Proof:**

Write  $\mathfrak{r} = \text{rad}\mathfrak{g}$  and let  $\pi: \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{r}$  be the quotient map, a Lie algebra homomorphism by definition 5.1.1. The claim is that  $\text{rad}(\mathfrak{g}/\mathfrak{r}) = 0$ . Suppose instead that  $\mathfrak{g}/\mathfrak{r}$  has a nonzero solvable ideal  $\bar{\mathfrak{s}}$ . Its preimage  $\mathfrak{s} = \pi^{-1}(\bar{\mathfrak{s}})$  is an ideal of  $\mathfrak{g}$ : for  $x \in \mathfrak{g}$  and  $s \in \mathfrak{s}$  one has  $\pi([x, s]) = [\pi(x), \pi(s)] \in \bar{\mathfrak{s}}$  because  $\bar{\mathfrak{s}}$  is an ideal and  $\pi(s) \in \bar{\mathfrak{s}}$ , so  $[x, s] \in \pi^{-1}(\bar{\mathfrak{s}}) = \mathfrak{s}$ .

This ideal  $\mathfrak{s}$  is solvable. It sits in the short exact sequence

$$0 \longrightarrow \mathfrak{r} \longrightarrow \mathfrak{s} \xrightarrow{\pi} \bar{\mathfrak{s}} \longrightarrow 0$$

since  $\ker(\pi|_{\mathfrak{s}}) = \mathfrak{s} \cap \mathfrak{r} = \mathfrak{r}$  (because  $\mathfrak{r} = \pi^{-1}(0) \subseteq \pi^{-1}(\bar{\mathfrak{s}}) = \mathfrak{s}$ ) and  $\pi(\mathfrak{s}) = \bar{\mathfrak{s}}$ . Thus  $\mathfrak{s}$  is an extension of the solvable algebra  $\bar{\mathfrak{s}}$  by the solvable ideal  $\mathfrak{r}$ , and an extension of a solvable algebra by a solvable ideal is solvable by proposition 5.1.6. So  $\mathfrak{s}$  is a solvable ideal of  $\mathfrak{g}$ .

By maximality of the radical,  $\mathfrak{s} \subseteq \mathfrak{r}$ . But  $\mathfrak{r} \subseteq \mathfrak{s}$  already, so  $\mathfrak{s} = \mathfrak{r}$ , whence  $\bar{\mathfrak{s}} = \pi(\mathfrak{s}) = \pi(\mathfrak{r}) = 0$ , contradicting  $\bar{\mathfrak{s}} \neq 0$ . Therefore  $\mathfrak{g}/\mathfrak{r}$  has no nonzero solvable ideal, that is  $\text{rad}(\mathfrak{g}/\mathfrak{r}) = 0$ , and  $\mathfrak{g}/\mathfrak{r}$  is semisimple, as we sought to show.

The mechanism of the proof is the correspondence between ideals of a quotient and ideals of  $\mathfrak{g}$  lying above the radical: a nonzero solvable ideal of  $\mathfrak{g}/\text{rad}\mathfrak{g}$  would pull back to a solvable ideal of  $\mathfrak{g}$  strictly larger than  $\text{rad}\mathfrak{g}$ , which the maximality of the radical forbids. The radical is thus exactly the obstruction to semisimplicity, and dividing it out removes the obstruction and nothing more.

This is the first structural dichotomy of the subject. Every finite-dimensional Lie algebra  $\mathfrak{g}$  fits into a short exact sequence

$$0 \longrightarrow \text{rad}\mathfrak{g} \longrightarrow \mathfrak{g} \longrightarrow \mathfrak{g}/\text{rad}\mathfrak{g} \longrightarrow 0$$

whose left term is solvable and whose right term is semisimple. Every Lie algebra is, in this sense, an extension of a semisimple algebra by a solvable one, and the two ends are governed by disjoint theories: the solvable end by Engel and Lie, the machinery of this chapter, and the semisimple end by the root-space theory to come. The sequence does more than exist. The Levi decomposition of ?? will show it splits, exhibiting  $\mathfrak{g}$  as a semidirect product  $\mathfrak{g} = \mathfrak{s} \ltimes \text{rad}\mathfrak{g}$  of the radical with a semisimple subalgebra  $\mathfrak{s}$  isomorphic to  $\mathfrak{g}/\text{rad}\mathfrak{g}$ , so that the classification of all Lie algebras reduces to the classification of the semisimple ones together with their solvable extensions.

Semisimplicity has been defined here negatively, as the absence of solvable ideals, and at this stage that is all the reader can be told. The definition is usable, as the computations for  $\mathfrak{sl}_2$  and  $\mathfrak{gl}_n$  show, but it gives no positive handle on what a semisimple algebra looks like from the inside. The recharacterization comes in ??, where the Killing form, an invariant symmetric bilinear form built from the traces of the operators  $\text{ad}x \text{ad}y$ , converts the negative condition  $\text{rad}\mathfrak{g} = 0$  into the positive and computable condition that the Killing form be nondegenerate. That equivalence, Cartan's criterion for semisimplicity, is what makes the semisimple algebras tractable: nondegeneracy of a bilinear form is checked by a single determinant, whereas the absence of solvable ideals is a statement about every ideal at once. With the nondegenerate form in hand, ?? proves that a semisimple algebra

decomposes as a direct sum of simple ideals, closing the gap between the semisimple algebras and their atoms.

### Exercise 5.5.1

1. Verify in full that  $\text{rad}\mathfrak{gl}_n(\mathbb{C}) = \mathbb{C}I$ , the scalar matrices, and identify the semisimple quotient  $\mathfrak{gl}_n/\text{rad}\mathfrak{gl}_n$ . Using  $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{C}I$  from theorem 1.4.3, apply the second isomorphism theorem to identify  $\mathfrak{gl}_n/\mathbb{C}I$  with  $\mathfrak{sl}_n/(\mathfrak{sl}_n \cap \mathbb{C}I)$ , describe precisely what  $\mathfrak{sl}_n \cap \mathbb{C}I$  is over  $\mathbb{C}$ , and conclude that  $\mathfrak{gl}_n/\text{rad}\mathfrak{gl}_n \cong \mathfrak{sl}_n$ . Explain why the characteristic-zero hypothesis is what makes  $\mathfrak{sl}_n \cap \mathbb{C}I = 0$ , so that the projective special linear algebra  $\mathfrak{psl}_n = \mathfrak{sl}_n/\mathfrak{z}(\mathfrak{sl}_n)$  coincides with  $\mathfrak{sl}_n$  here.
2. Give a complete proof that  $\mathfrak{sl}_2(\mathbb{C})$  is simple by showing directly that it has no ideal other than 0 and itself, following the strategy of example 5.5.6 but writing out every case ( $\gamma \neq 0$ ,  $\gamma = 0 \neq \beta$ ,  $\beta = \gamma = 0 \neq \alpha$ ). Conclude that  $\mathfrak{sl}_2(\mathbb{C})$  is semisimple and that its radical and nilradical are both zero.
3. Prove that a Lie algebra of dimension 1 or 2 is never semisimple. Treat the one-dimensional case (necessarily abelian) and both two-dimensional cases (the abelian algebra, and the unique non-abelian one with basis  $x, y$  and  $[x, y] = x$ ), in each case exhibiting a nonzero solvable ideal that is the whole radical.
4. Prove that the radical is preserved by every automorphism: if  $\varphi: \mathfrak{g} \rightarrow \mathfrak{g}$  is a Lie algebra automorphism, then  $\varphi(\text{rad}\mathfrak{g}) = \text{rad}\mathfrak{g}$ . Deduce the same for the nilradical. (Hint: an automorphism carries solvable ideals to solvable ideals bijectively, hence carries the maximal one to the maximal one.)
5. Exhibit a Lie algebra for which the containment  $\text{nil}\mathfrak{g} \subsetneq \text{rad}\mathfrak{g}$  is strict, and one for which it is an equality. For the strict case, use the two-dimensional non-abelian algebra of exercise 5.5.1 (3); for the equality case, use the strictly-upper-triangular algebra  $\mathfrak{n}_n$  of example 5.2.5. Compute both ideals in each case.
6. Show that although  $\mathfrak{g}/\text{rad}\mathfrak{g}$  is always semisimple, the analogous quotient by the nilradical,  $\mathfrak{g}/\text{nil}\mathfrak{g}$ , need not be semisimple. Use  $\mathfrak{g} = \mathfrak{b}_2$ , the  $2 \times 2$  upper-triangular algebra of example 5.1.7, computing  $\text{nil}\mathfrak{b}_2$ , the quotient, and its radical.



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# *Index*

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- Action
  - transitive, 120
- Ad (adjoint representation), 67
- ad (adjoint representation of the algebra), 68
- Adjoint Representation
  - and the bracket, 69
  - for a matrix group, 71
  - is a homomorphism, 68
  - of the group, 67
  - of the Lie algebra, 68
- Ado's Theorem, 98
- Baker-Campbell-Hausdorff Formula, 102
  - commuting exponentials, 107
  - convergence, 103
- Bracket
  - as the adjoint action, 69
  - of a matrix Lie algebra, 28
- Cartan's Theorem
  - closed subgroup, 81
- Category
  - of Lie groups, 45
- Closed Subgroup Theorem, 81
  - Lie algebra of the subgroup, 84
- Conjugation
  - in a Lie group, 42
- Connected Lie Group
  - classification by central quotients, 100
- Continuous Homomorphism is Smooth, 86
- Coset Space
  - as homogeneous space, 120
- Covering Group, 96
  - classification, 132
  - examples, 134
- Covering Homomorphism
  - kernel is discrete and central, 131
- Determinant
  - differential is the trace, 65
- Differential
  - of a Lie group homomorphism, 61
- Discrete Normal Subgroup
  - is central, 131
- Distribution
  - left-invariant, 88
- Double Cover  $SU(2) \rightarrow SO(3)$ , 36
- Equivariant Map, 112
- Exponential Map
  - agreement with the matrix exponential, 58
  - as a bridge to the Lie algebra, 30
  - local diffeomorphism, 57
  - naturality, 63
  - of a Lie group, 55
  - onto  $SU(2)$ , 34
  - properties, 56
- Fiber Bundle
  - principal, 137
- Flag Manifold
  - as a homogeneous space, 127
- Fundamental Group

- of  $SO(3)$ , 38
  - of a Lie group is abelian, 131
  - subgroups index covers, 132
- General Linear Group, 4
  - as a manifold, 4
  - connected components, 5
  - its intrinsic Lie algebra, 52
- Grassmannian
  - as a homogeneous space, 127
- Group Action
  - by conjugation, 113
  - by left translation, 113
  - of  $GL_n$  on  $\mathbb{R}^n$ , 113
  - of  $O(n)$  on the sphere, 113
  - smooth, 112
- Heisenberg Algebra, 108
- Homogeneous Space, 120
  - as coset space, 121
  - as principal bundle, 137
  - isotropy representation, 139
  - quotient manifold structure, 114
  - reductive, 140
  - spheres, 125
- Homomorphism
  - continuous implies smooth, 86
  - determined by its differential, 64
  - integration of, 92
- Hopf Fibration, 129
  - as principal bundle, 138
- Identity Component, 46
- Irrational Line, 79
- Isotropy Representation, 139
  - for  $SU(2)/U(1)$ , 140
- Left-Invariant Distribution, 88
- Left-Invariant Vector Field, 48
  - isomorphism with  $T_e G$ , 49
- Lie Algebra
  - abstract, 29
  - is a real vector space, 24
  - of a closed subgroup, 84
  - of a Lie group, 49
  - of a Lie group  $Lie(G)$ , 50
  - of a matrix group, 24
  - of the classical groups, 26
- Lie Algebra Homomorphism
  - from a group homomorphism, 61
- Lie Functor, 65
- Lie Group, 42
  - generated by a neighborhood of the identity, 57
  - homomorphism, 45
- Lie Group-Lie Algebra Correspondence
  - equivalence of categories, 98
  - on morphisms, 93
- Lie Subalgebra, 76
  - from a Lie subgroup, 77
- Lie Subgroup, 76
  - dense (irrational line), 79
  - embedded, 76
  - immersed, 76
  - Lie algebra of, 77
- Lie's Integration Theorem
  - subalgebras integrate, 90
- Lie's Second Theorem, 92
- Lie's Third Theorem, 98
- Local Isomorphism
  - from the bracket, 106
- Matrix Exponential, 15
  - algebraic properties, 16
  - convergence, 16
  - local diffeomorphism, 20
  - of commuting matrices, 18
- Matrix Logarithm, 19
- Naturality
  - of the bracket under a homomorphism, 61
  - of the exponential map, 63
- One-Parameter Subgroup, 22
  - classification, 22
  - correspondence with the Lie algebra, 54
  - of a Lie group, 54
- Operator Norm, 15
- Orbit, 112
  - as immersed submanifold, 123
  - of a compact group, 123
- Orbit-Stabilizer Theorem
  - smooth, 121
- Orthogonal Group, 9
  - as a manifold, 9
- Principal Bundle, 137
  - Hopf fibration, 138

- Projective Space
  - as a homogeneous space, 127
- Quotient Manifold, 114
  - universal property, 117
- Reductive Decomposition, 140
- Reductive Homogeneous Space, 140
- Representation
  - isotropy, 139
- Special Linear Group, 8
  - as a manifold, 8
  - universal cover of  $SL_2(\mathbb{R})$ , 134
- Special Orthogonal Group, 9
  - as  $SU(2)/\{\pm I\}$ , 36
  - connectedness, 12
- Special Unitary Group, 10
  - action on the sphere, 122
  - and the Hopf fibration, 129
  - as a 3-sphere, 33
  - as a closed subgroup, 85
- Sphere
  - as  $O(n)/O(n-1)$ , 117
  - as a homogeneous space, 125
- Spin Group, 134
- Stabilizer, 112
- Subalgebra
  - integrates to a subgroup, 90
- Symplectic Group, 11
  - as a manifold, 12
- Torus
  - irrational line on, 79
- Translation
  - left and right, 42
- Unitary Group, 10
  - as a manifold, 10
  - connectedness, 12
- Universal Cover
  - of a Lie group, 96
- Universal Property
  - of the quotient  $G/H$ , 117
- Vector Field
  - left-invariant, 48
  - completeness, 51



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